

KFM Components Pass 18

Idea Index, Category Atlas, and Expansion Dossier

Kansas Frontier Matrix

2026-05-11 UTC

Contents

1 Document Metadata	1
1.1 Truth Posture Key	1
1.2 Current-Session Evidence Boundary	1
2 Delta Since Pass 17	2
2.1 Prior baseline carry-forward summary	2
2.2 Pass 18 category delta	2
2.3 Delta interpretation	3
3 Source Ledger	4
3.1 Direct Pass 18 idea-card sources	4
3.2 Complete Phase 1 PDF source ledger	6
4 Executive Determination	14
4.1 Determination table	14
5 Source Synthesis	15
5.1 Source-family weighting from Phase 5	15
6 Structural Rationale	17
6.1 Dependency-order category rationale	17
7 Phase 4 Master Category Map — Pass 18	18
7.1 Category notes	20
8 Detailed Idea Chapters	23
8.1 DOC - Doctrine, Truth Posture, and Responsibility Boundaries	23
8.2 REP - Geographic Representation, Scale, and Spatial Reference	28
8.3 SRC - Source Admission, Connectors, and External-Service Boundaries	51
8.4 MOD - Data Modeling, Domain Semantics, Temporal Structure, and SQL	81
8.5 EVD - Evidence, Provenance, Receipts, Identity, and Attestation	125
8.6 POL - Policy, Rights, Ethics, Sensitivity, and Public-Safety Controls	157
8.7 VAL - Validation, QA, Testing, Observability, and Reliability	169
8.8 ANA - Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation	191
8.9 FIE - Field Capture, Remote Sensing, 3D, and Archaeological Interpretation	290
8.10 REL - Catalog Closure, Publication, Release, Rollback, and Recompile Discipline	345
8.11 API - Governed API, Resource Ontology, HTTP, and Service Contracts	386
8.12 MAP - Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries	430
8.13 UIX - Trust-Visible UI, Evidence Drawer, Focus Mode, Story Nodes, and Review Surfaces	448
8.14 PLN - Planning, Participation, Resilience, and Decision-Support Applications	460
9 Cross-Cutting Themes	494

9.1	P18-XCT-001 — Representation is a governed claim surface, not neutral display	494
9.2	P18-XCT-002 — Temporal structure is a first-class evidence dimension	494
9.3	P18-XCT-003 — Source admission and connector behavior must be policy-aware	494
9.4	P18-XCT-004 — Analysis and ML outputs remain interpretive derivatives	494
9.5	P18-XCT-005 — Field, remote-sensing, and 3D capture need evidentiary humility	495
9.6	P18-XCT-006 — APIs and renderers are trust membranes, not truth stores	495
9.7	P18-XCT-007 — Planning support requires participatory, equity, and scenario context	495
9.8	P18-XCT-008 — Validation, observability, and rollback are part of publication truth	496
10	Overlaps and Gaps	497
10.1	P18-OVL-001 — Source admission overlaps with evidence receipts	497
10.2	P18-OVL-002 — Temporal modeling overlaps with release and correction	497
10.3	P18-OVL-003 — Renderer governance overlaps with API contract design	497
10.4	P18-OVL-004 — Analytics overlaps with planning and policy exposure	497
10.5	P18-OVL-005 — 3D/field capture overlaps with archaeology, hazards, and sensitivity	497
11	Gaps	498
11.1	P18-GAP-001 — Mounted repo behavior remains unverified	498
11.2	P18-GAP-002 — Current source rights and version-sensitive facts were not rechecked in Phase 6	498
11.3	P18-GAP-003 — Five hundred cards need later prioritization into buildable epics	498
11.4	P18-GAP-004 — Schema-home and path placement must remain doctrine-governed	498
11.5	P18-GAP-005 — Source adaptations from textbooks need KFM-specific acceptance criteria . .	498
12	Weakly Supported Material	499
12.1	P18-WEAK-001	499
12.2	P18-WEAK-002	499
12.3	P18-WEAK-003	499
12.4	P18-WEAK-004	499
12.5	P18-WEAK-005	499
12.6	P18-WEAK-006	499
13	Expansion Agenda	500
13.1	P18-AG-001 — Finalize source-ledger and rights-verification backlog	500
13.2	P18-AG-002 — Package 500 cards into category chapters without re-rendering the PDF yet . .	500
13.3	P18-AG-003 — Derive buildable epics from the 14-category map	500
13.4	P18-AG-004 — Create representation and fitness-for-use acceptance criteria	500
13.5	P18-AG-005 — Create temporal-support and correction-support criteria	500
13.6	P18-AG-006 — Define public-safe policy profiles for sensitive domains	500
13.7	P18-AG-007 — Prepare final PDF QA and self-check plan	501
14	Open Questions Seed Fragment	502
15	Final Self-Check	505
16	Appendix A - Prior Baseline Reference	506
16.1	Prior baseline category count reference	506

KFM Components Pass 18

Idea Index, Category Atlas, and Expansion Dossier

A cumulative evidence-first synthesis succeeding Pass 17

Prepared 2026-05-11 UTC

Evidence-first | fail-closed | cite-or-abstain | map-first | time-aware | auditable | reversible

CONFIRMED: rendered in Phase 8 from a single concatenated Markdown file on disk.

UNKNOWN: current mounted-repo implementation maturity remains unverified.

1 Document Metadata

Field	Value
Truth label	CONFIRMED
Document title	KFM Components Pass 18 - Idea Index, Category Atlas, and Expansion Dossier
Pass designation	Pass 18
Succeeds	Pass 17
Run date UTC	2026-05-11
Phase 7 generated at UTC	2026-05-11T13:24:21Z
Build workspace	/mnt/data/kfm_pass_18/
Final PDF status	CONFIRMED not rendered in Phase 7
Source extraction status	CONFIRMED Phase 1 extracted all 40 PDFs with 0 quarantined extraction failures
Prior baseline handling	CONFIRMED sidecar-only; body not inlined
Phase 3 inventory count	CONFIRMED 500 records
Phase 4 category count	CONFIRMED 14 categories
Phase 5 idea-card count	CONFIRMED 500 Markdown cards
Phase 6 synthesis fragments	CONFIRMED cross-cutting, overlaps/gaps, weak-material, expansion-agenda, and open-question fragments
Implementation maturity	UNKNOWN

1.1 Truth Posture Key

CONFIRMED: Verified in this pass from attached files, generated artifacts, hashes, indexes, validation receipts, or current filesystem evidence.

PROPOSED: A synthesis, organizing interpretation, prioritization, future placement, or implementation direction consistent with evidence but not verified as implemented.

UNKNOWN: Not verified strongly enough in this pass, usually because no mounted repository, runtime logs, dashboards, CI results, branch evidence, or deployed services were inspected.

NEEDS VERIFICATION: Checkable before publication, source activation, implementation, or operational use, but not yet checked strongly enough here.

DENY / ABSTAIN / ERROR: Finite outcome labels used for KFM policy, evidence, and runtime boundaries; they are not rhetorical emphasis.

1.2 Current-Session Evidence Boundary

CONFIRMED: Phase 7 uses persisted artifacts from Phases 0-6 and does not read all source PDFs into a prompt.

CONFIRMED: Phase 7 writes Markdown and JSON fragments only; it does not render the final PDF.

UNKNOWN: The final typography, TOC, page numbers, embedded fonts, and visual PDF QA remain unverified until Phase 8 renders and checks the final PDF.

2 Delta Since Pass 17

CONFIRMED: Pass 17 reconciled baseline contains 83 prior ideas, 83 unique prior idea headings, and 10 prior categories.

CONFIRMED: Phase 2 assigned an initial RETAINED carry state to every prior idea in `prior_index.json`; the prior-pass body was not copied into this build.

CONFIRMED: Pass 18 adds 500 persisted idea cards from the Phase 3-5 source inventory: 163 NEW cards and 337 EXPANDED cards.

PROPOSED: The main substantive delta is that Pass 18 widens the prior governance/security/pipeline baseline into representation, GIS analysis, environmental and planning use cases, 3D/field capture, API design, temporal data modeling, and MapLibre artifact governance.

UNKNOWN: No Pass 18 fragment confirms current repository implementation, route behavior, deployment behavior, or public release.

2.1 Prior baseline carry-forward summary

Prior category	Truth label	Prior ideas	Carry-forward state
DOC	CONFIRMED	8	RETAINED sidecar baseline
PIP	CONFIRMED	10	RETAINED sidecar baseline
CAT	CONFIRMED	7	RETAINED sidecar baseline
EVD	CONFIRMED	7	RETAINED sidecar baseline
POL	CONFIRMED	7	RETAINED sidecar baseline
SEC	CONFIRMED	12	RETAINED sidecar baseline
AIG	CONFIRMED	8	RETAINED sidecar baseline
DAT	CONFIRMED	13	RETAINED sidecar baseline
OBS	CONFIRMED	5	RETAINED sidecar baseline
MDP	CONFIRMED	6	RETAINED sidecar baseline

2.2 Pass 18 category delta

Order	Category	Category name	Pass 18 cards	NEW	EXPANDED
1	DOC	Doctrine, Truth Posture, and Responsibility Boundaries	5	2	3
2	REP	Geographic Representation, Scale, and Spatial Reference	25	8	17
3	SRC	Source Admission, Connectors, and External-Service Boundaries	32	11	21
4	MOD	Data Modeling, Domain Semantics, Temporal Structure, and SQL	47	10	37
5	EVD	Evidence, Provenance, Receipts, Identity, and Attestation	34	11	23
6	POL	Policy, Rights, Ethics, Sensitivity, and Public-Safety Controls	12	6	6
7	VAL	Validation, QA, Testing, Observability, and Reliability	23	4	19
8	ANA	Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation	108	43	65

Order	Category	Category name	Pass 18 cards	NEW	EXPANDED
9	FIE	Field Capture, Remote Sensing, 3D, and Archaeological Interpretation	59	23	36
10	REL	Catalog Closure, Publication, Release, Rollback, and Recompile Discipline	43	6	37
11	API	Governed API, Resource Ontology, HTTP, and Service Contracts	47	9	38
12	MAP	Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer	18	7	11
13	UIX	Boundaries Trust-Visible UI, Evidence Drawer, Focus Mode, Story Nodes, and Review Surfaces	12	4	8
14	PLN	Planning, Participation, Resilience, and Decision-Support Applications	35	19	16

2.3 Delta interpretation

PROPOSED: ANA, FIE, API, MOD, and REL form the largest Pass 18 expansion surface, so the final dossier should explain how analysis, field/3D capture, APIs, temporal models, and release discipline become governed carriers of evidence rather than root truth.

PROPOSED: DOC, POL, and UIX are smaller in record count but remain structurally important because they constrain how public-facing interpretation, sensitive material, and trust-visible UI states should behave.

NEEDS VERIFICATION: Any claim that a Pass 18 idea has already been implemented must be checked against a mounted repository, tests, emitted receipts, release manifests, or runtime evidence before being upgraded.

3 Source Ledger

CONFIRMED: Phase 1 hashed and extracted 40 PDF sources; no extraction failure was quarantined.

CONFIRMED: Phase 0/1 classification recorded 11 candidate synthesis/reference sources, 23 governing-vocabulary sources, 5 prior-pass lineage-context sources, and 1 prior baseline source.

CONFIRMED: Phase 5 card index cites 11 source IDs as direct Pass 18 idea-card sources.

PROPOSED: Doctrine files remain governing vocabulary and placement/trust context; they are not re-synthesized as new idea targets in Pass 18.

NEEDS VERIFICATION: Current operational endpoint behavior, external source rights, package versions, and repository implementation facts are outside Phase 8 verification.

3.1 Direct Pass 18 idea-card sources

3.1.1 SRC-P18-004 - AI_Concepts_Using_Python.pdf

- **CONFIRMED:** Cards citing this source: 54.
- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Pages extracted: 435; text extraction status: ok.
- **CONFIRMED:** SHA-256 prefix: cb6275b4bf2c44e5.

3.1.2 SRC-P18-011 - GIS in Sustainable Urban Planning and Management.pdf

- **CONFIRMED:** Cards citing this source: 53.
- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Pages extracted: 365; text extraction status: ok.
- **CONFIRMED:** SHA-256 prefix: 9b4e13779ccf282e.

3.1.3 SRC-P18-009 - Domain-Driven Design Reference.pdf

- **CONFIRMED:** Cards citing this source: 47.
- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Pages extracted: 59; text extraction status: ok.
- **CONFIRMED:** SHA-256 prefix: 4406daa99ff0e3d5.

3.1.4 SRC-P18-039 - Master MapLibre Components-Functions-Features.pdf

- **CONFIRMED:** Cards citing this source: 47.
- **CONFIRMED:** Phase 0 role: candidate_synthesis_source_specialized_atlas.
- **CONFIRMED:** Pages extracted: 554; text extraction status: ok.
- **CONFIRMED:** SHA-256 prefix: 309cf67311059c54.

3.1.5 SRC-P18-002 - a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf

- **CONFIRMED:** Cards citing this source: 46.
- **CONFIRMED:** Phase 0 role: candidate_synthesis_source.
- **CONFIRMED:** Pages extracted: 321; text extraction status: ok.
- **CONFIRMED:** SHA-256 prefix: 0e75fd2bfbe9024b.

3.1.6 SRC-P18-006 - Designing Great Web APIs.pdf

- **CONFIRMED:** Cards citing this source: 46.
- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Pages extracted: 45; text extraction status: ok.
- **CONFIRMED:** SHA-256 prefix: 160655602b709eef.

3.1.7 SRC-P18-007 - developing-time-oriented-database-applications-in-sql.pdf

- **CONFIRMED:** Cards citing this source: 45.
- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Pages extracted: 528; text extraction status: ok.
- **CONFIRMED:** SHA-256 prefix: 63e7657d379daf3d.

3.1.8 SRC-P18-005 - Archaeological 3D GIS.pdf

- **CONFIRMED:** Cards citing this source: 43.
- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Pages extracted: 177; text extraction status: ok.
- **CONFIRMED:** SHA-256 prefix: 27d60483276de20b.

3.1.9 SRC-P18-010 - Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf

- **CONFIRMED:** Cards citing this source: 42.
- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Pages extracted: 141; text extraction status: ok.
- **CONFIRMED:** SHA-256 prefix: 0ef2469bee053ab1.

3.1.10 SRC-P18-003 - Advanced-SQL-Concepts.pdf

- **CONFIRMED:** Cards citing this source: 40.
- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Pages extracted: 112; text extraction status: ok.
- **CONFIRMED:** SHA-256 prefix: 5f489ba2ab4cc154.

3.1.11 SRC-P18-012 - instant-web-scraping-with-java.pdf

- **CONFIRMED:** Cards citing this source: 37.
- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Pages extracted: 72; text extraction status: ok.
- **CONFIRMED:** SHA-256 prefix: b48a86014e094268.

3.2 Complete Phase 1 PDF source ledger

3.2.1 SRC-P18-001 - # Kansas Frontier Matrix Implementation Reference.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 20; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: d948332b6c5bfcdd956cf6264f7bcb88d6881ac00ca8afc2534a02d288d4b3c2.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-001__kansas_frontier_matrix_implementation_reference.txt.

3.2.2 SRC-P18-002 - a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf

- **CONFIRMED:** Phase 0 role: candidate_synthesis_source.
- **CONFIRMED:** Synthesis target in Phase 0 classification: yes.
- **CONFIRMED:** Page count: 321; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 0e75fd2bfb9024bc659f0720e7c385b965921d1a35ae0f4ff27e1681b604b8c.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-002__a_primer_of_gis_fundamental_geographic_and_cartographic_concepts.txt.

3.2.3 SRC-P18-003 - Advanced-SQL-Concepts.pdf

- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Synthesis target in Phase 0 classification: yes.
- **CONFIRMED:** Page count: 112; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 5f489ba2ab4cc154f7d7d404773277c94ac88e89adae89fb367452f93e885ee0.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-003__advanced_sql_concepts.txt.

3.2.4 SRC-P18-004 - AI_Concepts_Using_Python.pdf

- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Synthesis target in Phase 0 classification: yes.
- **CONFIRMED:** Page count: 435; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: cb6275b4bf2c44e5fc56b166b1161ceb859a1de0562462814ad5b8ad6fd111b6.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-004__ai_concepts_using_python.txt.

3.2.5 SRC-P18-005 - Archaeological 3D GIS.pdf

- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Synthesis target in Phase 0 classification: yes.
- **CONFIRMED:** Page count: 177; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 27d60483276de20be77c592222da11819841b7022f756c37e583ecb720649cd2.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-005__archaeological_3d_gis.txt.

3.2.6 SRC-P18-006 - Designing Great Web APIs.pdf

- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Synthesis target in Phase 0 classification: yes.
- **CONFIRMED:** Page count: 45; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 160655602b709eef6cacb0bc3c6ef13639e16533f992be717a9fdd0431686c6b.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-006__designing_great_web_apis.txt.

3.2.7 SRC-P18-007 - developing-time-oriented-database-applications-in-sql.pdf

- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Synthesis target in Phase 0 classification: yes.
- **CONFIRMED:** Page count: 528; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 63e7657d379daf3df56a3a404c78555fab8f1238db388eacf0570710cf8e7108.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-007__developing_time_oriented_database_applications_in_sql.txt.

3.2.8 SRC-P18-008 - Directory Rules.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 22; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 759de4fcb51cf0f55896089e397d9c47481d60d9fb80ac9a44d47b2f60a0a335.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-008__directory_rules.txt.

3.2.9 SRC-P18-009 - Domain-Driven Design Reference.pdf

- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Synthesis target in Phase 0 classification: yes.
- **CONFIRMED:** Page count: 59; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 4406daa99ff0e3d58757d62d40358c9fd745f95137e99f14602659d0c3f54e55.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-009__domain_driven_design_reference.txt.

3.2.10 SRC-P18-010 - Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf

- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Synthesis target in Phase 0 classification: yes.
- **CONFIRMED:** Page count: 141; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 0ef2469bee053ab10f11525c6495e9ad7d7d37f8311f26f667a9dcf13dce1cdd.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-010__earth_space_and_environmental_science_explorations_with_arcgis_pro_ed2.txt.

3.2.11 SRC-P18-011 - GIS in Sustainable Urban Planning and Management.pdf

- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Synthesis target in Phase 0 classification: yes.
- **CONFIRMED:** Page count: 365; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 9b4e13779ccf282eefab4965b94f834ba4b8236388a1ccccfca8228136379e5b.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-011__gis_in_sustainable_urban_planning_and_management.txt.

3.2.12 SRC-P18-012 - instant-web-scraping-with-java.pdf

- **CONFIRMED:** Phase 0 role: reference_source_candidate.
- **CONFIRMED:** Synthesis target in Phase 0 classification: yes.
- **CONFIRMED:** Page count: 72; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: b48a86014e094268fdaae87ac560b1a94de4998294403b7a7afb30199448d4a1.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-012__instant_web_scraping_with_java.txt.

3.2.13 SRC-P18-013 - Kansas Frontier Matrix Definitive Greenfield Building Plan_v1_1.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 28; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: b28b0cba512b95c8c106a143acb65c89d384843ff313420b2429daa51f8fcf78.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-013__kansas_frontier_matrix_definitive_greenfield_building_plan_v1_1.txt.

3.2.14 SRC-P18-014 - Kansas Frontier Matrix Pipeline Living Implementation Manual_v0.3.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 30; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 43d0c6fea4cc64edb87238a13ac49b639934a82dcef0fab2ef49217add0ba8cf.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-014__kansas_frontier_matrix_pipeline_living_implementation_manual_v0_3.txt.

3.2.15 SRC-P18-015 - KFM Agriculture Domain Implementation Dossier_REVISED_2026-04-21.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 48; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 0bf41c729bc2139d8c519f994ba59fe6e7e2b640b302475a7c88549d3796d411.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-015__kfm_agriculture_domain_implementation_dossier_revised_2026_04_21.txt.

3.2.16 SRC-P18-016 - KFM_Archaeology_Architecture_Plan_PDF_Only.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 51; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 585d7e927a4accf80bbd271ffcdf207b0c45c158a9780a98d9419d9b0d19223f.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-016__kfm_archaeology_architecture_plan_pdf_only.txt.

3.2.17 SRC-P18-017 - KFM_Atmosphere_Air_PDF_Only_Architecture_Report_2026-04-21.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 52; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 69d0d5e9a698866867707570aa1b5c23e9471ecc7f9696459de685f2312da00b.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-017__kfm_atmosphere_air_pdf_only_architecture_report_2026_04_21.txt.

3.2.18 SRC-P18-018 - KFM_Components_Pass_10_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf

- **CONFIRMED:** Phase 0 role: prior_pass_lineage_context.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 125; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 8b7cd350451791b6cbc39c93bea31ed01458f0fb3bb3b4909bcf5dc19d293ed3.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-018__kfm_components_pass_10_idea_index_category_atlas_and_expansion_dossier.txt.

3.2.19 SRC-P18-019 - KFM_Components_Pass_11_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf

- **CONFIRMED:** Phase 0 role: prior_pass_lineage_context.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 123; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: b1c6de0c6eb3c80fa58674c6a6b33448666addfbf01ea3ff4404f61891c34b3f.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-019__kfm_components_pass_11_part_2_idea_index_category_atlas_and_expansion_dossier.txt.

3.2.20 SRC-P18-020 - KFM_Components_Pass_13_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf

- **CONFIRMED:** Phase 0 role: prior_pass_lineage_context.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 88; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 71fe46e2d137c8bd2abfd32282841287ba8af60125f7884a824f41c1e6cf0108.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-020__kfm_components_pass_13_part_2_idea_index_category_atlas_and_expansion_dossier.txt.

3.2.21 SRC-P18-021 - kfm_encyclopedia.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 82; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: cc899a7a57cbadb5870709be07d9b0dbfd01712cd794d63dc4d640485970419a.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-021__kfm_encyclopedia.txt.

3.2.22 SRC-P18-022 - KFM_Fauna_Architecture_PDF_Only_Report.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 37; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 47309b51122866974f567a929153e78407557e4548449800acb6613980cec40f.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-022__kfm_fauna_architecture_pdf_only_report.txt.

3.2.23 SRC-P18-023 - KFM_Flora_Architecture_PDF_Only_Implementation_Blueprint.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 36; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 5507d694e6500df99abd4b352a2f410fc06a241615ccffbc0c9ecc620e26506f.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-023__kfm_flora_architecture_pdf_only_implementation_blueprint.txt.

3.2.24 SRC-P18-024 - KFM_Geology_Natural_Resources_Architecture_PDF_Only_Report_2026-04-21.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 42; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: d334f43df8fd74f17115cc0f51861cf8238c9cb99d37adaf95f5e4e1655fdf51.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-024__kfm_geology_natural_resources_architecture_pdf_only_report_2026_04_21.txt.

3.2.25 SRC-P18-025 - KFM_Governed_AI_Extended_Pro_Source_Ledger_PDF_Only_Architecture_Report_2026-04-20.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 36; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 0abbd5fcb54c21d63602487fb945ad2e2a3d1ff2a190251f56f91e7294ee5d75.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-025__kfm_governed_ai_extended_pro_source_ledger_pdf_only_architecture_report_2026_04_.txt.

3.2.26 SRC-P18-026 - kfm_habitat_architecture_pdf_only_blueprint_2026-04-21.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 28; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 82626239b2f12690f20ab73fbaca96594c7ef40ae6ede2f612fe14ae6e9007d2.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-026__kfm_habitat_architecture_pdf_only_blueprint_2026_04_21.txt.

3.2.27 SRC-P18-027 - KFM_Habitat_Fauna_Thin_Slice_Extended_Pro_Blueprint.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 20; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: a95594c12d1dd5fb9fe7f16461ab89e90a1eaf9c379c077c1159686c14e5d48e.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-027__kfm_habitat_fauna_thin_slice_extended_pro_blueprint.txt.

3.2.28 SRC-P18-028 - kfm_hazards_extended_pro_pdf_only_blueprint.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 26; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 52bd2f3e85a0fc12b04473a126bef74323deda498664ddad362d437fafb1d7a6.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-028__kfm_hazards_extended_pro_pdf_only_blueprint.txt.

3.2.29 SRC-P18-029 - KFM_Hydrology_Extended_Pro_PDF_Only_Reference_Report_2026-04-21.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 43; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 0e7c3531bd8d94d080a71ae92ffd866dd48c0c9fedc918b446cb6c8e6c0bd0b8.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-029__kfm_hydrology_extended_pro_pdf_only_reference_report_2026_04_21.txt.

3.2.30 SRC-P18-030 - KFM_MapLibre_Operating_Architecture_Governed_UI_AI_Interaction_Manual_REVISED.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 22; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 77f56ec1ab632b76c7728cfb250330271b7dc8948db95c8c0594c92ad9ca6b36.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-030__kfm_maplibre_operating_architecture_governed_ui_ai_interaction_manual_revised.txt.

3.2.31 SRC-P18-031 - KFM_Pass_15_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf

- **CONFIRMED:** Phase 0 role: prior_pass_lineage_context.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 48; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 1c62fdb3d141068d8bb1c170aee4a8acfc89aec53cee2569eb7de04c77594c75.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-031__kfm_pass_15_part_2_idea_index_category_atlas_and_expansion_dossier.txt.

3.2.32 SRC-P18-032 - KFM_Pass_16_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf

- **CONFIRMED:** Phase 0 role: prior_pass_lineage_context.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 97; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 05cae979949edcd65e1d2de7589c8371e8c01b14e7c3a516a5c35a920d47e312.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-032__kfm_pass_16_part_2_idea_index_category_atlas_and_expansion_dossier.txt.

3.2.33 SRC-P18-033 - KFM_Pass_17_Part_2_Idea_Index_Category_Atlas_and_Expansion_Dossier.pdf

- **CONFIRMED:** Phase 0 role: prior_pass_baseline.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 60; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: bc602237ff1082089aba786c397b67c6d5dc2a0d0b6b10629e45769f80786ed2.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-033__kfm_pass_17_part_2_idea_index_category_atlas_and_expansion_dossier.txt.

3.2.34 SRC-P18-034 - KFM_People_Genealogy_DNA_Land_Ownership_Architecture_Blueprint.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 30; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 6833e9ecf8ba7302deb841a3d7c5bb89619dfdd4956864fcce8f9561c3d75744.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-034__kfm_people_genealogy_dna_land_ownership_architecture_blueprint.txt.

3.2.35 SRC-P18-035 - KFM_Roads_Rail_Trade_Routes_PDF_Only_Architecture_Plan_2026-04-21.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 35; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 8a0c22ab452ea5b9fd4662fa129d02a281b55b135bf5a8674d158e1ff425d3b1.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-035__kfm_roads_rail_trade_routes_pdf_only_architecture_plan_2026_04_21.txt.

3.2.36 SRC-P18-036 - kfm_settlements_infrastructure_extended_pro_plan_2026-04-21.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 43; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 5b6a6124c91297ed4f38769b7e23171241ad1b805314fe74984b97cb4a1da8e1.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-036__kfm_settlements_infrastructure_extended_pro_plan_2026_04_21.txt.

3.2.37 SRC-P18-037 - kfm_soil_architecture_extended_pro_pdf_only_report.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 25; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 7c2d498212b9ad56f3ba37bf91f841e9f328794e8aa4940f8f665a4116c5aaea.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-037__kfm_soil_architecture_extended_pro_pdf_only_report.txt.

3.2.38 SRC-P18-038 - KFM_Whole_UI_Governed_AI_Expansion_Report.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 23; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: b0766ae7c55a7b59850b34cf7f48e199108c822c377817d01b63c81f805a7232.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-038__kfm_whole_ui_governed_ai_expansion_report.txt.

3.2.39 SRC-P18-039 - Master MapLibre Components-Functions-Features.pdf

- **CONFIRMED:** Phase 0 role: candidate_synthesis_source_specialized_atlas.
- **CONFIRMED:** Synthesis target in Phase 0 classification: yes.
- **CONFIRMED:** Page count: 554; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 309cf67311059c549e144ae9961b2f49eddf1caab8739a51b47ae88c2f5c1c90.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-039__master_maplibre_components_functions_features.txt.

3.2.40 SRC-P18-040 - Ollama & Ubuntu Information.pdf

- **CONFIRMED:** Phase 0 role: governing_vocabulary_not_synthesis_target.
- **CONFIRMED:** Synthesis target in Phase 0 classification: no.
- **CONFIRMED:** Page count: 66; text quality flag: ok; quarantine status: not_quarantined.
- **CONFIRMED:** SHA-256: 378fec4c010b918d840f8e4a886ca6c3587ecaf0787e7c840b11f1f03fc43e6d.
- **CONFIRMED:** Extracted text path: /mnt/data/kfm_pass_18/text/SRC-P18-040__ollama_ubuntu_information.txt.

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4 Executive Determination

CONFIRMED: Pass 18 has a complete 500-card inventory, 14-category atlas, Phase 6 synthesis fragments, and validation receipts through Phase 6.

CONFIRMED: The largest Phase 4 categories by card count are ANA (108), FIE (59), MOD (47), API (47), REL (43).

CONFIRMED: Phase 5 validation reports 500 card files present, 500 aggregate card-index records, 0 missing card files after integrity repair, and 0 post-repair SHA-256 mismatches.

PROPOSED: The center of gravity of Pass 18 is governed interpretation: how KFM should move from source admission and evidence-backed data structures into maps, APIs, UI explanations, 3D/field capture, planning indicators, and analytical summaries without letting any derivative become sovereign truth.

PROPOSED: The strongest editorial determination is that KFM needs explicit fitness-for-use language for representation, temporal support, analysis/model outputs, public-safe geospatial derivatives, and API/UI evidence payloads.

PROPOSED: The final dossier should treat GIS textbooks, API/SQL/DDD references, AI/Python references, archaeology/3D GIS references, web-scraping material, and the MapLibre atlas as source support for design ideas that must be adapted into KFM's governance membrane.

UNKNOWN: Actual repo surfaces remain unverified: route names, package manager, source registries, schema files, policy bundles, workflows, dashboards, runtime logs, emitted proof packs, release manifests, and deployment state.

NEEDS VERIFICATION: Before public or operational source activation, KFM must verify source rights, current API behavior, licensing, sensitivity posture, tool versions, runtime security, and release controls.

4.1 Determination table

Area	Truth label	Determination
Baseline	CONFIRMED	Pass 17 is the prior baseline and remains sidecar-only.
New source corpus	CONFIRMED	Pass 18 uses 11 direct synthesis/reference source IDs in the 500-card Phase 5 index.
Category atlas	CONFIRMED	Phase 4 assigned all 500 Pass 18 records to 14 dependency-ordered categories.
Main synthesis	PROPOSED	Pass 18 extends KFM doctrine into governed representation, analysis, API/UI, temporal modeling, field/3D capture, and planning support.
Publication status	CONFIRMED	No final PDF has been rendered through Phase 7.
Implementation status	UNKNOWN	Repo implementation maturity cannot be claimed from current artifacts.

5 Source Synthesis

CONFIRMED: The Pass 18 direct synthesis sources include GIS/cartographic representation, environmental GIS exercises, sustainable urban planning GIS, archaeological 3D GIS, advanced SQL, temporal database design, domain-driven design, web API design, AI/Python concepts, Java web scraping, and a cumulative MapLibre components atlas.

PROPOSED: The source corpus is best read as an applied-expansion corpus rather than a single-domain corpus: it translates KFM's prior evidence, policy, and release doctrine into representation choices, analytical methods, source-connector behavior, API contracts, map artifacts, and planning decision contexts.

PROPOSED: The GIS/cartography sources support the proposition that maps are authored representations with scale, projection, symbol, omission, and audience choices; KFM should make those choices inspectable in layer and release metadata.

PROPOSED: The urban planning and environmental GIS sources support the proposition that planning, hazards, vulnerability, climate, water, and field-data workflows become policy-relevant when they appear as public decision-support surfaces.

PROPOSED: The archaeological 3D GIS source supports the proposition that 3D models, field capture, sub-surface interpretation, and visualization should be separated into captured data, interpreted claims, review state, and public-safe derivatives.

PROPOSED: The SQL, temporal database, DDD, and API sources support a KFM pattern in which bounded contexts, resource contracts, valid/transaction time, recursive/hierarchical queries, and service boundaries are treated as governance and maintainability surfaces.

PROPOSED: The AI/Python and analytics source material supports using models, indicators, clustering, regression, and evaluation methods as bounded interpretive derivatives, not as independent evidence.

PROPOSED: The web-scraping/source-acquisition source material supports treating scrapers, connectors, retries, errors, and persistence as governed intake behavior with receipts and quarantine outcomes.

PROPOSED: The MapLibre atlas supports keeping MapLibre, PMTiles, COGs, renderer state, Story Nodes, Evidence Drawer, and Focus Mode downstream of released evidence, manifests, policy decisions, and rollback targets.

UNKNOWN: The source corpus does not prove that these patterns exist in the current KFM repository.

NEEDS VERIFICATION: Source extraction quality, embedded visual/table details, and version-sensitive operational facts must be rechecked where final prose makes high-stakes or operational claims.

5.1 Source-family weighting from Phase 5

Source family	Truth label	Card support	Synthesis role
AI / data analysis concepts	CONFIRMED	54	PROPOSED bounded model and analysis governance.
Sustainable urban planning GIS	CONFIRMED	53	PROPOSED planning, participation, indicators, resilience framing.
DDD and bounded contexts	CONFIRMED	47	PROPOSED domain language and responsibility-boundary framing.
MapLibre cumulative atlas	CONFIRMED	47	PROPOSED renderer, tile, manifest, UI, and release-governance framing.
GIS/cartography primer	CONFIRMED	46	PROPOSED representation and scale-governance framing.
Web API design	CONFIRMED	46	PROPOSED governed resource ontology and API contract framing.
Temporal database design	CONFIRMED	45	PROPOSED valid-time, transaction-time, and correction-support framing.
Archaeological 3D GIS	CONFIRMED	43	PROPOSED field/3D/interpretation/sensitivity framing.
Environmental ArcGIS exercises	CONFIRMED	42	PROPOSED hazard, field-data, water, climate, and workflow framing.

Source family	Truth label	Card support	Synthesis role
Advanced SQL	CONFIRMED	40	PROPOSED analytic query and validation framing.
Java web scraping	CONFIRMED	37	PROPOSED connector, error-handling, and persistence framing.

6 Structural Rationale

CONFIRMED: Phase 4 derived 14 categories and assigned all 500 idea records to exactly one category each.

PROPOSED: The category order is a dependency order rather than a reading-order or source-file order: doctrine and representation come before source admission, modeling, evidence, policy, validation, analysis, field capture, release, API, map artifacts, UI, and planning use.

PROPOSED: This structure preserves KFM's core invariant by ensuring that public-facing planning, map, API, UI, and AI interpretation categories appear downstream of evidence, policy, validation, and release categories.

PROPOSED: The atlas deliberately keeps analysis and planning large enough to reflect the Pass 18 corpus while keeping policy, UI, and documentation categories visible even when their card counts are smaller.

CONFIRMED: Phase 6 identified 8 cross-cutting themes, 5 overlap groups, 5 gap records, 6 weak-material records, 7 expansion-agenda items, and 80 open-question seed records.

UNKNOWN: The category map does not establish repo folder placement, schema authority, test status, or runtime behavior.

6.1 Dependency-order category rationale

Order	Category	Truth label	Rationale
1	DOC	PROPOSED	Establish truth posture and responsibility boundaries before any downstream claim.
2	REP	PROPOSED	Make representation choices visible before maps are treated as outputs.
3	SRC	PROPOSED	Admit sources through governed connectors before modeling or publication.
4	MOD	PROPOSED	Define domain semantics and temporal structure before evidence and APIs depend on them.
5	EVD	PROPOSED	Attach provenance, receipts, identity, and attestation before public exposure.
6	POL	PROPOSED	Apply rights, ethics, sensitivity, and safety controls before validation/release.
7	VAL	PROPOSED	Test and observe artifacts before analysis or publication relies on them.
8	ANA	PROPOSED	Keep analysis and ML as interpretation over validated evidence.
9	FIE	PROPOSED	Treat field, remote-sensing, 3D, and archaeology as capture and interpretation surfaces requiring review.
10	REL	PROPOSED	Close catalogs, release objects, rollback paths, and recompile discipline before API/UI exposure.
11	API	PROPOSED	Expose governed resources after release-state and evidence envelopes exist.
12	MAP	PROPOSED	Deliver tiles, rasters, vectors, and renderer artifacts as downstream carriers.
13	UIX	PROPOSED	Present trust-visible UI, Evidence Drawer, Focus Mode, and review surfaces after governed data flows.
14	PLN	PROPOSED	Use decision-support outputs only after representation, evidence, policy, analysis, and public-surface controls are visible.

7 Phase 4 Master Category Map — Pass 18

Created: 2026-05-11T02:28:57Z

CONFIRMED: Phase 4 read 500 persisted Phase 3 inventory records.

PROPOSED: The 14-category atlas below is the dependency-ordered Pass 18 category map for Phase 5 card writing.

UNKNOWN: Implementation maturity, repo route names, tests, runtime behavior, and deployment state remain unverified.

Order	Code	Category	Phase 3 idea count	NEW	EXPANDED	Boundary
1	DOC	Doctrine, Truth Posture, and Responsibility Boundaries	5	2	3	Root governance: truth labels, lifecycle law, responsibility boundaries, bounded implementation claims, public-safe defaults, and core KFM invariants.
2	REP	Geographic Representation, Scale, and Spatial Reference	25	8	17	Map and geographic information semantics: representation choices, projection, datum, scale, generalization, cartographic communication, topology, and spatial model families.
3	SRC	Source Admission, Connectors, and External-Service Boundaries	32	11	21	Admission and connector discipline: source descriptors, web scraping, APIs, field intake, external services, source terms, source-role, freshness, and stewardship checks.
4	MOD	Data Modeling, Domain Semantics, Temporal Structure, and SQL	47	10	37	Structural semantics: bounded contexts, domain language, temporal structure, SQL/query semantics, object boundaries, and analytical interfaces.
5	EVD	Evidence, Provenance, Receipts, Identity, and Attestation	34	11	23	Evidence support: provenance, receipts, identity, attestation, source-linked evidence, and auditable production/transform context.

Order	Code	Category	Phase 3 idea count	NEW	EXPANDED	Boundary
6	POL	Policy, Rights, Ethics, Sensitivity, and Public-Safety Controls	12	6	6	Exposure control: policy, rights, ethics, sensitivity, consent, access, redaction, generalization, and deny-by-default publication posture.
7	VAL	Validation, QA, Testing, Observability, and Reliability	23	4	19	Assurance controls: validation, QA, tests, fixtures, visual regression, observability, reliability, and reproducibility checks.
8	ANA	Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation	108	43	65	Derived reasoning: analysis, indicators, statistics, machine-learning methods, model interpretation, uncertainty, parameters, and limitations.
9	FIE	Field Capture, Remote Sensing, 3D, and Archaeological Interpretation	59	23	36	Capture and interpretation-heavy evidence: field collection, remote sensing, 3D/2.5D modeling, archaeology, volumes, visibility, and method provenance.
10	REL	Catalog Closure, Publication, Release, Rollback, and Recompile Discipline	43	6	37	Release control: catalog closure, publication, release manifests, rollback, correction, recompile discipline, and citable published state.
11	API	Governed API, Resource Ontology, HTTP, and Service Contracts	47	9	38	Governed service interfaces: resource ontology, HTTP semantics, finite outcomes, service contracts, API documentation, and exposure boundaries.
12	MAP	Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries	18	7	11	Renderable carriers: map artifacts, tiles, raster/vector delivery, renderer boundaries, style/layer manifests, and downstream map delivery.

Order	Code	Category	Phase 3 idea count	NEW	EXPANDED	Boundary
13	UIX	Trust-Visible UI, Evidence Drawer, Focus Mode, Story Nodes, and Review Surfaces	12	4	8	Trust-visible interaction: Evidence Drawer, Focus Mode, Story Nodes, review surfaces, stale/denied/error states, and bounded user inspection.
14	PLN	Planning, Participation, Resilience, and Decision-Support Applications	35	19	16	Application-facing decision support: planning, participation, resilience, VGI, indicators, scenarios, and urban/community interpretation.

7.1 Category notes

7.1.1 1. DOC — Doctrine, Truth Posture, and Responsibility Boundaries

Status: CONFIRMED source support for grouped ideas; PROPOSED category synthesis.

Definition: Root governance: truth labels, lifecycle law, responsibility boundaries, bounded implementation claims, public-safe defaults, and core KFM invariants.

Top source support: SRC-P18-009 (5).

Example records: KFM-P18-INV-169, KFM-P18-INV-188, KFM-P18-INV-191, KFM-P18-INV-396, KFM-P18-INV-416.

7.1.2 2. REP — Geographic Representation, Scale, and Spatial Reference

Status: CONFIRMED source support for grouped ideas; PROPOSED category synthesis.

Definition: Map and geographic information semantics: representation choices, projection, datum, scale, generalization, cartographic communication, topology, and spatial model families.

Top source support: SRC-P18-002 (19), SRC-P18-009 (2), SRC-P18-010 (1), SRC-P18-039 (1), SRC-P18-004 (1).

Example records: KFM-P18-INV-001, KFM-P18-INV-002, KFM-P18-INV-004, KFM-P18-INV-051, KFM-P18-INV-052, KFM-P18-INV-055, KFM-P18-INV-076, KFM-P18-INV-077.

7.1.3 3. SRC — Source Admission, Connectors, and External-Service Boundaries

Status: CONFIRMED source support for grouped ideas; PROPOSED category synthesis.

Definition: Admission and connector discipline: source descriptors, web scraping, APIs, field intake, external services, source terms, source-role, freshness, and stewardship checks.

Top source support: SRC-P18-012 (16), SRC-P18-009 (4), SRC-P18-002 (4), SRC-P18-039 (3), SRC-P18-010 (2).

Example records: KFM-P18-INV-008, KFM-P18-INV-036, KFM-P18-INV-039, KFM-P18-INV-044, KFM-P18-INV-068, KFM-P18-INV-072, KFM-P18-INV-149, KFM-P18-INV-215.

7.1.4 4. MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL

Status: CONFIRMED source support for grouped ideas; PROPOSED category synthesis.

Definition: Structural semantics: bounded contexts, domain language, temporal structure, SQL/query semantics, object boundaries, and analytical interfaces.

Top source support: SRC-P18-007 (30), SRC-P18-009 (14), SRC-P18-003 (1), SRC-P18-039 (1), SRC-P18-005 (1).

Example records: KFM-P18-INV-043, KFM-P18-INV-092, KFM-P18-INV-093, KFM-P18-INV-094, KFM-P18-INV-097, KFM-P18-INV-106, KFM-P18-INV-119, KFM-P18-INV-120.

7.1.5 5. EVD — Evidence, Provenance, Receipts, Identity, and Attestation

Status: CONFIRMED source support for grouped ideas; PROPOSED category synthesis.

Definition: Evidence support: provenance, receipts, identity, attestation, source-linked evidence, and auditable production/transform context.

Top source support: SRC-P18-012 (9), SRC-P18-009 (6), SRC-P18-011 (5), SRC-P18-010 (4), SRC-P18-005 (3).

Example records: KFM-P18-INV-037, KFM-P18-INV-085, KFM-P18-INV-096, KFM-P18-INV-112, KFM-P18-INV-129, KFM-P18-INV-148, KFM-P18-INV-168, KFM-P18-INV-173.

7.1.6 6. POL — Policy, Rights, Ethics, Sensitivity, and Public-Safety Controls

Status: CONFIRMED source support for grouped ideas; PROPOSED category synthesis.

Definition: Exposure control: policy, rights, ethics, sensitivity, consent, access, redaction, generalization, and deny-by-default publication posture.

Top source support: SRC-P18-010 (4), SRC-P18-004 (4), SRC-P18-002 (2), SRC-P18-011 (2).

Example records: KFM-P18-INV-005, KFM-P18-INV-056, KFM-P18-INV-123, KFM-P18-INV-208, KFM-P18-INV-298, KFM-P18-INV-314, KFM-P18-INV-341, KFM-P18-INV-358.

7.1.7 7. VAL — Validation, QA, Testing, Observability, and Reliability

Status: CONFIRMED source support for grouped ideas; PROPOSED category synthesis.

Definition: Assurance controls: validation, QA, tests, fixtures, visual regression, observability, reliability, and reproducibility checks.

Top source support: SRC-P18-012 (6), SRC-P18-010 (5), SRC-P18-039 (4), SRC-P18-009 (3), SRC-P18-002 (2).

Example records: KFM-P18-INV-038, KFM-P18-INV-046, KFM-P18-INV-067, KFM-P18-INV-078, KFM-P18-INV-125, KFM-P18-INV-126, KFM-P18-INV-131, KFM-P18-INV-147.

7.1.8 8. ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation

Status: CONFIRMED source support for grouped ideas; PROPOSED category synthesis.

Definition: Derived reasoning: analysis, indicators, statistics, machine-learning methods, model interpretation, uncertainty, parameters, and limitations.

Top source support: SRC-P18-004 (42), SRC-P18-003 (34), SRC-P18-009 (8), SRC-P18-007 (7), SRC-P18-002 (6).

Example records: KFM-P18-INV-003, KFM-P18-INV-013, KFM-P18-INV-018, KFM-P18-INV-021, KFM-P18-INV-022, KFM-P18-INV-024, KFM-P18-INV-026, KFM-P18-INV-027.

7.1.9 9. FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation

Status: CONFIRMED source support for grouped ideas; PROPOSED category synthesis.

Definition: Capture and interpretation-heavy evidence: field collection, remote sensing, 3D/2.5D modeling, archaeology, volumes, visibility, and method provenance.

Top source support: SRC-P18-005 (35), SRC-P18-010 (16), SRC-P18-002 (4), SRC-P18-007 (1), SRC-P18-011 (1).

Example records: KFM-P18-INV-006, KFM-P18-INV-007, KFM-P18-INV-016, KFM-P18-INV-017, KFM-P18-INV-019, KFM-P18-INV-020, KFM-P18-INV-053, KFM-P18-INV-057.

7.1.10 10. REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline

Status: CONFIRMED source support for grouped ideas; PROPOSED category synthesis.

Definition: Release control: catalog closure, publication, release manifests, rollback, correction, recompile discipline, and citable published state.

Top source support: SRC-P18-039 (11), SRC-P18-010 (7), SRC-P18-002 (6), SRC-P18-004 (5), SRC-P18-011 (4).

Example records: KFM-P18-INV-009, KFM-P18-INV-023, KFM-P18-INV-047, KFM-P18-INV-058, KFM-P18-INV-075, KFM-P18-INV-102, KFM-P18-INV-110, KFM-P18-INV-118.

7.1.11 11. API — Governed API, Resource Ontology, HTTP, and Service Contracts

Status: CONFIRMED source support for grouped ideas; PROPOSED category synthesis.

Definition: Governed service interfaces: resource ontology, HTTP semantics, finite outcomes, service contracts, API documentation, and exposure boundaries.

Top source support: SRC-P18-006 (39), SRC-P18-009 (4), SRC-P18-012 (4).

Example records: KFM-P18-INV-025, KFM-P18-INV-040, KFM-P18-INV-041, KFM-P18-INV-042, KFM-P18-INV-066, KFM-P18-INV-095, KFM-P18-INV-098, KFM-P18-INV-099.

7.1.12 12. MAP — Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries

Status: CONFIRMED source support for grouped ideas; PROPOSED category synthesis.

Definition: Renderable carriers: map artifacts, tiles, raster/vector delivery, renderer boundaries, style/layer manifests, and downstream map delivery.

Top source support: SRC-P18-039 (17), SRC-P18-002 (1).

Example records: KFM-P18-INV-045, KFM-P18-INV-050, KFM-P18-INV-073, KFM-P18-INV-074, KFM-P18-INV-180, KFM-P18-INV-181, KFM-P18-INV-182, KFM-P18-INV-183.

7.1.13 13. UIX — Trust-Visible UI, Evidence Drawer, Focus Mode, Story Nodes, and Review Surfaces

Status: CONFIRMED source support for grouped ideas; PROPOSED category synthesis.

Definition: Trust-visible interaction: Evidence Drawer, Focus Mode, Story Nodes, review surfaces, stale/denied/error states, and bounded user inspection.

Top source support: SRC-P18-039 (9), SRC-P18-006 (2), SRC-P18-007 (1).

Example records: KFM-P18-INV-048, KFM-P18-INV-049, KFM-P18-INV-063, KFM-P18-INV-115, KFM-P18-INV-179, KFM-P18-INV-185, KFM-P18-INV-187, KFM-P18-INV-250.

7.1.14 14. PLN — Planning, Participation, Resilience, and Decision-Support Applications

Status: CONFIRMED source support for grouped ideas; PROPOSED category synthesis.

Definition: Application-facing decision support: planning, participation, resilience, VGI, indicators, scenarios, and urban/community interpretation.

Top source support: SRC-P18-011 (35).

Example records: KFM-P18-INV-010, KFM-P18-INV-011, KFM-P18-INV-012, KFM-P18-INV-014, KFM-P18-INV-015, KFM-P18-INV-060, KFM-P18-INV-062, KFM-P18-INV-086.

8 Detailed Idea Chapters

CONFIRMED: This section includes 500 persisted Phase 5 idea cards. Each card preserves stable ID, status label, carry-forward state, category, normalized statement, detailed explanation, why it matters, related ideas, dependencies, tensions, expansion directions, open questions, future work, and source attribution.

PROPOSED: Category ordering follows the dependency-ordered Phase 4 atlas.

UNKNOWN: Card prose does not prove current KFM repository implementation, route behavior, tests, dashboards, deployment state, or public release.

8.1 DOC - Doctrine, Truth Posture, and Responsibility Boundaries

Status: CONFIRMED source support for grouped Phase 3 ideas; PROPOSED category synthesis

Definition: Root governance: truth labels, lifecycle law, responsibility boundaries, bounded implementation claims, public-safe defaults, and core KFM invariants.

CONFIRMED: This chapter contains 5 Pass 18 idea cards assigned to DOC.

Idea ID	Title	Carry state	Status	Source IDs
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KFM-P18-INV-169 | Core-domain prioritization around inspectable claims | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-188 | Shared governance kernel across bounded domain lanes | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-191 | Domain vision and highlighted core for inspectable claims | NEW | PROPOSED | SRC-P18-009 |

KFM-P18-INV-396 | Responsibility layers as KFM large-scale structure | NEW | PROPOSED | SRC-P18-009 |

KFM-P18-INV-416 | Core-domain focus before generic subdomain expansion | EXPANDED | PROPOSED | SRC-P18-009 |

8.1.1 KFM-P18-INV-169 — Core-domain prioritization around inspectable claims

Idea card metadata - Stable ID: KFM-P18-INV-169 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 169 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: DOC — Doctrine, Truth Posture, and Responsibility Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should prioritize the inspectable-claim membrane as its core domain and treat tiles, indexes, dashboards, and provider integrations as supporting or generic subdomains unless evidence proves otherwise.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should prioritize the inspectable-claim membrane as its core domain and treat tiles, indexes, dashboards, and provider integrations as supporting or generic subdomains unless evidence proves otherwise. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps scarce design and review attention focused on evidence, policy, release, correction, and rollback rather than implementation conveniences.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-POL-006.

Dependencies - PROPOSED: Domain vision statement - PROPOSED: Architecture decision register - PROPOSED: Roadmap prioritization

Tensions - PROPOSED: Some support components, such as validators, are also trust-critical and cannot be treated as commodity.

Expansion directions - PROPOSED: Classify object families as core, supporting, generic, or experimental during Phase 4 category derivation.

Open questions - NEEDS VERIFICATION: Which current KFM object families are core enough to receive separate steward ownership?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 47; basis: DDD Core Domain pattern warning that the essence of a domain model can be obscured by many necessary components; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-001, KFM-IDX-EVD-001, KFM-IDX-POL-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.1.2 KFM-P18-INV-188 — Shared governance kernel across bounded domain lanes

Idea card metadata - Stable ID: KFM-P18-INV-188 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 188 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: DOC — Doctrine, Truth Posture, and Responsibility Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should keep shared governance objects small, explicit, and jointly controlled across domain lanes rather than duplicating EvidenceBundle, DecisionEnvelope, receipts, and release objects per lane.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should keep shared governance objects small, explicit, and jointly controlled across domain lanes rather than duplicating EvidenceBundle, DecisionEnvelope, receipts, and release objects per lane. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reduces divergence in the trust membrane while preserving domain-specific bounded contexts.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-006. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-PIP-004.

Dependencies - PROPOSED: Core governance schema set - PROPOSED: Schema-home ADR - PROPOSED: Cross-domain CI

Tensions - PROPOSED: A shared kernel can become too large and freeze domain lanes.

Expansion directions - PROPOSED: Define the minimum shared governance kernel and require ADRs for changes.

Open questions - NEEDS VERIFICATION: Which object families belong in the shared kernel versus lane-specific profiles?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 38; basis: Shared Kernel pattern with explicit boundary, small shared subset, consultation before changes, and continuous integration; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-006, KFM-IDX-EVD-001, KFM-IDX-PIP-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.1.3 KFM-P18-INV-191 — Domain vision and highlighted core for inspectable claims

Idea card metadata - Stable ID: KFM-P18-INV-191 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 191 - Status label: PROPOSED - Carry-forward state: NEW - Category: DOC — Doctrine, Truth Posture, and Responsibility Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should maintain a concise domain-vision and highlighted-core statement that keeps inspectable claims, evidence closure, policy posture, and rollback as the project center of gravity.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should maintain a concise domain-vision and highlighted-core statement that keeps inspectable claims, evidence closure, policy posture, and rollback as the project center of gravity. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It gives future contributors a compact way to distinguish core KFM work from adjacent mapping, visualization, or AI features.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-DOC-007. - CONFIRMED baseline link: KFM-IDX-MDP-003.

Dependencies - PROPOSED: Doctrine summary - PROPOSED: Category map - PROPOSED: ADR/changelog linkage

Tensions - PROPOSED: A highlighted core can become slogan-like if not backed by contracts and tests.

Expansion directions - PROPOSED: Add a one-page core-domain statement to front matter and category rationale in later phases.

Open questions - NEEDS VERIFICATION: Should the Pass 18 dossier include this as doctrine synthesis or expansion agenda?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 49, 50, 52; basis: Domain Vision Statement, Highlighted Core, and Segregated Core patterns for keeping core model value visible and separated from supporting elements; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.1.4 KFM-P18-INV-396 — Responsibility layers as KFM large-scale structure

Idea card metadata - Stable ID: KFM-P18-INV-396 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 396 - Status label: PROPOSED - Carry-forward state: NEW - Category: DOC — Doctrine, Truth Posture, and Responsibility Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM architecture documentation should name responsibility layers—evidence, policy, catalog, release, API, UI, AI, and operations—so cross-domain changes are evaluated against the layer they affect.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM architecture documentation should name responsibility layers—evidence, policy, catalog, release, API, UI, AI, and operations—so cross-domain changes are evaluated against the layer they affect. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents topic-centric design from hiding which governance layer owns a change.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-006. - CONFIRMED baseline link: KFM-IDX-MDP-003.

Dependencies - PROPOSED: responsibility layer - PROPOSED: architecture register - PROPOSED: change impact - PROPOSED: ADR

Tensions - PROPOSED: Layer names can become bureaucratic if not tied to decisions and tests.

Expansion directions - PROPOSED: Create a responsibility_layer field in change-impact summaries.

Open questions - NEEDS VERIFICATION: Which layers are mandatory for every KFM PR impact statement?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 50, 51, 52; basis: Domain-Driven Design Reference lists Responsibility Layers, Knowledge Level, and Pluggable Component Framework under large-scale structure.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.1.5 KFM-P18-INV-416 — Core-domain focus before generic subdomain expansion

Idea card metadata - Stable ID: KFM-P18-INV-416 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 416 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: DOC — Doctrine, Truth Posture, and Responsibility Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM expansion should identify the core domain value of each lane before investing equally in generic supporting subdomains.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM expansion should identify the core domain value of each lane before investing equally in generic supporting subdomains. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Governed systems can sprawl; a core-domain statement helps prioritize scarce implementation, validation, and review effort.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-004. - CONFIRMED baseline link: KFM-IDX-PIP-001.

Dependencies - PROPOSED: domain vision statement - PROPOSED: category map - PROPOSED: thin-slice selection - PROPOSED: review priority

Tensions - PROPOSED: Some generic governance infrastructure is foundational and cannot always wait.

Expansion directions - PROPOSED: Require a domain vision statement in Phase 4 category rationale.

Open questions - NEEDS VERIFICATION: Which Pass 18 categories are core KFM differentiators versus supporting subdomains?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 40, 41, 42; basis: Domain-Driven Design Reference defines Core Domain, Generic Subdomains, and Domain Vision Statement as strategic design tools.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-004, KFM-IDX-PIP-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2 REP - Geographic Representation, Scale, and Spatial Reference

Status: CONFIRMED source support for grouped Phase 3 ideas; PROPOSED category synthesis

Definition: Geographic information and map representation semantics: projection, datum, scale, symbolization, generalization, and spatial reference limits.

CONFIRMED: This chapter contains 25 Pass 18 idea cards assigned to REP.

Idea ID	Title	Carry state	Status	Source IDs
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KFM-P18-INV-001 | Representation choices as governance objects | NEW | PROPOSED | SRC-P18-002 |

KFM-P18-INV-002 | Fitness-for-use quality posture for maps and layers | NEW | PROPOSED | SRC-P18-002 |

KFM-P18-INV-004 | Projection and coordinate choices as auditable metadata | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-051 | Projection distortion as a release metadata obligation | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-052 | Accuracy and precision split for observations and geometry | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-055 | Administrative boundaries as governed assertions | NEW | PROPOSED | SRC-P18-002 |

KFM-P18-INV-076 | Scale as a release-profile constraint | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-077 | Symbol and convention choices as inspectable communication risk | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-101 | Spatial data infrastructure as governance substrate | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-122 | Anticorruption and open-host services for external model boundaries | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-127 | Locational reference systems as legal and cultural context | NEW | PROPOSED | SRC-P18-002 |

KFM-P18-INV-167 | Layered architecture protects domain truth from UI and storage leakage | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-174 | XY table-to-point conversion as coordinate-creation evidence | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-225 | Datum, projection, and coordinate uncertainty travel with locations | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-253 | Digital Earth-style global context remains derivative context | NEW | PROPOSED | SRC-P18-002 |

KFM-P18-INV-325 | Map cultures and implicit values as representation review | NEW | PROPOSED | SRC-P18-002 |

KFM-P18-INV-332 | Dynamic H3 generalization for sensitive raster and event layers | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-352 | Map misuse review for public-facing visualizations | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-370 | Clustering labels as exploratory groupings, not discovered truth | NEW | PROPOSED | SRC-P18-004 |

KFM-P18-INV-376 | Sinton time-space-attribute triad as claim-scope metadata | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-377 | Measurement-scale class as aggregation and legend policy | NEW | PROPOSED | SRC-P18-002 |

KFM-P18-INV-401 | Cartographic omission register for public maps | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-426 | Projection distortion disclosure for statewide map claims | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-462 | Period-boundary predicate disclosure for temporal claims | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-476 | Map-scale generalization disclosure | EXPANDED | PROPOSED | SRC-P18-002 |

8.2.1 KFM-P18-INV-001 — Representation choices as governance objects

Idea card metadata - Stable ID: KFM-P18-INV-001 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 1 - Status label: PROPOSED - Carry-forward state: NEW - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should treat every map and geographic layer as a choice-bearing representation rather than a neutral copy of territory.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat every map and geographic layer as a choice-bearing representation rather than a neutral copy of territory. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents downstream map users from confusing visual presence with evidentiary completeness.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-007. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: LayerManifest or equivalent representation metadata

Tensions - PROPOSED: A readable map often simplifies; KFM still needs to expose what was simplified.

Expansion directions - PROPOSED: Add representation_intent and known_omissions fields to map/layer review notes.

Open questions - NEEDS VERIFICATION: Where should representation-purpose metadata live relative to catalog and layer manifests?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 14, 15, 17, 18; basis: geographic and cartographic representation / communication concepts; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.2 KFM-P18-INV-002 — Fitness-for-use quality posture for maps and layers

Idea card metadata - Stable ID: KFM-P18-INV-002 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 2 - Status label: PROPOSED - Carry-forward state: NEW - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should assess map and layer quality against intended purpose and audience rather than treating quality as a single absolute score.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should assess map and layer quality against intended purpose and audience rather than treating quality as a single absolute score. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It gives reviewers a principled way to allow generalized public layers without overstating precision.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: Declared intended_use and prohibited_use metadata

Tensions - PROPOSED: Users may infer precision from polished cartography.

Expansion directions - PROPOSED: Create a purpose-bound quality rubric for public map products.

Open questions - NEEDS VERIFICATION: Should fitness-for-use be validated at layer level, release level, or both?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 32; basis: fitness-for-use quality discussion; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.3 KFM-P18-INV-004 — Projection and coordinate choices as auditable metadata

Idea card metadata - Stable ID: KFM-P18-INV-004 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 4 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED

Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should make coordinate reference, projection, transformation, and scale decisions explicit in evidence and layer outputs.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should make coordinate reference, projection, transformation, and scale decisions explicit in evidence and layer outputs. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Projection and transformation choices affect measurement, overlay, and legal/interpretive confidence.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-001. - CONFIRMED baseline link: KFM-IDX-DAT-004.

Dependencies - PROPOSED: CRS metadata in processed artifacts and EvidenceBundle references

Tensions - PROPOSED: Public maps often reproject for usability while evidence may require source CRS fidelity.

Expansion directions - PROPOSED: Add CRS/projection transform receipts for spatial joins and map exports.

Open questions - NEEDS VERIFICATION: Should transform receipts be required for all cross-source overlays?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 10, 15, 20; basis: projection and representation foundations; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-001, KFM-IDX-DAT-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.4 KFM-P18-INV-051 — Projection distortion as a release metadata obligation

Idea card metadata - Stable ID: KFM-P18-INV-051 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 51 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should publish projection, coordinate-reference, transformation, and distortion choices as reviewable release metadata for spatial layers.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea

should be treated as a governed design candidate: KFM should publish projection, coordinate-reference, transformation, and distortion choices as reviewable release metadata for spatial layers. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents users from treating map position, area, direction, or distance as equally reliable at every scale and projection.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-DOC-008.

Dependencies - PROPOSED: LayerManifest CRS/projection metadata - PROPOSED: release review checklist

Tensions - PROPOSED: Usable maps may require projections that distort some properties.

Expansion directions - PROPOSED: Add projection_purpose, distortion_warning, and transform_chain fields to layer/release metadata.

Open questions - NEEDS VERIFICATION: Which layer types require a full transform chain versus a simpler CRS declaration?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 86, 113; basis: projection transformation and locational/coordinate reference discussion; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-DOC-008 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.5 KFM-P18-INV-052 — Accuracy and precision split for observations and geometry

Idea card metadata - Stable ID: KFM-P18-INV-052 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 52 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should keep accuracy and precision separate in spatial evidence, especially for surveyed, GPS-derived, digitized, or remotely sensed geometry.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should keep accuracy and precision separate in spatial evidence, especially for surveyed, GPS-derived, digitized, or remotely sensed geometry. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A precise coordinate can still be wrong; a generalized public geometry can be appropriate if its accuracy and intended use are clear.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: GeometryQuality metadata - PROPOSED: observation method fields

Tensions - PROPOSED: Public generalization can reduce precision intentionally while preserving fitness for use.

Expansion directions - PROPOSED: Create accuracy_class and precision_class fields with source-method provenance.

Open questions - NEEDS VERIFICATION: Should geometry quality be attached to every feature or inherited from dataset-level metadata?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 150, 161; basis: surveying/GPS/digitization and accuracy-versus-precision discussion; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-001, KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.6 KFM-P18-INV-055 — Administrative boundaries as governed assertions

Idea card metadata - Stable ID: KFM-P18-INV-055 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 55 - Status label: PROPOSED - Carry-forward state: NEW - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should model administrative spaces and boundaries as evidence-bound legal or governance assertions, not neutral background geometry.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should model administrative spaces and boundaries as evidence-bound legal or governance assertions, not neutral background geometry. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: County lines, municipal limits, parcels, districts, and jurisdictions carry authority and consequences; they need source, validity, and correction lineage.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: AdministrativeBoundaryAssertion - PROPOSED: valid_time/source_role fields

Tensions - PROPOSED: Administrative boundaries may be used as basemaps even when their legal status is material.

Expansion directions - PROPOSED: Create a `boundary_authority_role` and `legal_status_basis` field family.

Open questions - NEEDS VERIFICATION: Which administrative boundaries are stable enough for basemap use and which require full assertion handling?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — [a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf](#); pages 246; basis: administrative GI and maps in governance, commerce, defense, and taxation; Phase 1 role: `candidate_synthesis_source`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.7 KFM-P18-INV-076 — Scale as a release-profile constraint

Idea card metadata - Stable ID: KFM-P18-INV-076 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 76 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should treat map scale and generalization as release-profile constraints rather than purely cartographic styling choices.

Detailed explanation - CONFIRMED: Source SRC-P18-002 ([a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf](#)) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat map scale and generalization as release-profile constraints rather than purely cartographic styling choices. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps users from treating a county-scale, generalized, or thematic layer as fit for parcel-scale or exact-location decisions.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-006.

Dependencies - PROPOSED: LayerManifest scale metadata - PROPOSED: Release profile registry

Tensions - PROPOSED: Generalization improves readability but may weaken claim precision.

Expansion directions - PROPOSED: Define `scale_denominator_range`, `generalization_method`, and `prohibited_use` fields in layer metadata.

Open questions - NEEDS VERIFICATION: Should scale limits be enforced at UI zoom levels, evidence statements, or both?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — [a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf](#); pages 21, 24, 33, 54; basis: scale, map type, data combination, and cartographic representation discussion; Phase 1 role: `candidate_synthesis_source`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-POL-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.8 KFM-P18-INV-077 — Symbol and convention choices as inspectable communication risk

Idea card metadata - Stable ID: KFM-P18-INV-077 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 77 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should record symbol, color, and convention choices that materially affect interpretation of a public map.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record symbol, color, and convention choices that materially affect interpretation of a public map. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents trust-visible maps from communicating unsupported certainty, danger, priority, or authority through visual language alone.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-007. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: StyleManifest review fields - PROPOSED: Accessibility and bias review

Tensions - PROPOSED: Neutral-looking map styling can still imply threat, safety, or authority.

Expansion directions - PROPOSED: Add `symbol_intent`, `color_semantics`, `accessibility_check`, and `bias_review_note` to style review artifacts.

Open questions - NEEDS VERIFICATION: Which symbol changes require governance review versus normal design review?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 15, 23, 30, 32; basis: map symbols, conventions, cultural bias, and misuse discussion; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-007, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.9 KFM-P18-INV-101 — Spatial data infrastructure as governance substrate

Idea card metadata - Stable ID: KFM-P18-INV-101 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 101 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should treat spatial data infrastructure decisions as governance decisions over reference systems, interoperability, ownership, and lifecycle, not merely as GIS plumbing.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat spatial data infrastructure decisions as governance decisions over reference systems, interoperability, ownership, and lifecycle, not merely as GIS plumbing. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes the infrastructure that allows layers to interoperate visible enough for audit, correction, and release review.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-006. - CONFIRMED baseline link: KFM-IDX-CAT-002.

Dependencies - PROPOSED: Directory Rules responsibility roots - PROPOSED: LayerManifest metadata - PROPOSED: Coordinate reference registry

Tensions - PROPOSED: Infrastructure choices can look technical while silently controlling authority and interoperability.

Expansion directions - PROPOSED: Add an SDI posture field to source and layer manifests.

Open questions - NEEDS VERIFICATION: Which KFM register owns cross-domain CRS, datum, and locational-system policy?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 6, 53, 113, 114; basis: spatial data infrastructure, coordinate systems, projection/datum, and reference-system discussion; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-006, KFM-IDX-CAT-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.10 KFM-P18-INV-122 — Anticorruption and open-host services for external model boundaries

Idea card metadata - Stable ID: KFM-P18-INV-122 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 122 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should use anticorruption layers for external source models and open-host/published-language services for stable cross-context integration.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should use anticorruption layers for external source models and open-host/published-language services for stable cross-context integration. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps vendor/source schemas from contaminating KFM canonical concepts while still enabling interoperable APIs.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-006. - CONFIRMED baseline link: KFM-IDX-CAT-005.

Dependencies - PROPOSED: Connector adapter contract - PROPOSED: Published language schema

Tensions - PROPOSED: Adapters can become hidden canonical stores if not bounded.

Expansion directions - PROPOSED: Add adapter_role and source_model_to_kfm_mapping receipts.

Open questions - NEEDS VERIFICATION: Which external source models need explicit anticorruption adapters first?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 34, 42, 43; basis: anticorruption layer, open-host service, and published language discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-006, KFM-IDX-CAT-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.11 KFM-P18-INV-127 — Locational reference systems as legal and cultural context

Idea card metadata - Stable ID: KFM-P18-INV-127 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 127 - Status label: PROPOSED - Carry-forward state: NEW - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should distinguish coordinate position from locational-reference systems such as survey, administrative, or culturally meaningful location descriptions.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should distinguish coordinate position from locational-reference systems such as survey, administrative, or culturally meaningful location descriptions. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture,

review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents coordinates from overwriting legal, archival, or stewarded location evidence.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-006.

Dependencies - PROPOSED: Place registry - PROPOSED: CRS/datum registry - PROPOSED: Land/assertion evidence models

Tensions - PROPOSED: Coordinate precision can create false authority when the source evidence is locational rather than surveyed.

Expansion directions - PROPOSED: Add locational_reference_type fields to place and land assertions.

Open questions - NEEDS VERIFICATION: Where should non-coordinate locational systems be registered?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 113, 114, 116, 120, 121; basis: locational and coordinate systems discussion including land-survey and coordinate-reference examples; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.12 KFM-P18-INV-167 — Layered architecture protects domain truth from UI and storage leakage

Idea card metadata - Stable ID: KFM-P18-INV-167 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 167 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should keep domain truth logic, policy decisions, UI rendering, database persistence, and application orchestration in separable layers with explicit contracts.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should keep domain truth logic, policy decisions, UI rendering, database persistence, and application orchestration in separable layers with explicit contracts. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents a UI component, database query, or tile renderer from quietly becoming the source of truth for a claim.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-007. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: Contract surface - PROPOSED: Policy layer - PROPOSED: Renderer boundary

Tensions - PROPOSED: Some validation logic must run near data stores for performance but still needs domain ownership.

Expansion directions - PROPOSED: Map each trust-critical rule to a layer owner and test home.

Open questions - NEEDS VERIFICATION: Which rules may be duplicated in UI for affordance while remaining non-authoritative?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 17; basis: DDD Layered Architecture pattern warning against domain logic diffused into UI widgets or database scripts; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-007, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.13 KFM-P18-INV-174 — XY table-to-point conversion as coordinate-creation evidence

Idea card metadata - Stable ID: KFM-P18-INV-174 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 174 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should treat XY-to-point conversion as a coordinate-creation transform with declared X/Y fields, CRS, precision, source table, and validation checks.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat XY-to-point conversion as a coordinate-creation transform with declared X/Y fields, CRS, precision, source table, and validation checks. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents point layers from appearing as direct observations when they are generated from tabular coordinate fields under specific assumptions.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-004. - CONFIRMED baseline link: KFM-IDX-CAT-007.

Dependencies - PROPOSED: CRS metadata - PROPOSED: Coordinate validation - PROPOSED: Transform receipt

Tensions - PROPOSED: Manual or legacy tables may have swapped axes, missing CRS, or inconsistent precision.

Expansion directions - PROPOSED: Add fixtures for axis swap, out-of-range coordinates, and missing CRS.

Open questions - NEEDS VERIFICATION: Which coordinate conversions require manual reviewer approval?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 104; basis: ArcGIS exercise displaying earthquake table XY data as points using longitude and latitude fields; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-004, KFM-IDX-CAT-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.14 KFM-P18-INV-225 — Datum, projection, and coordinate uncertainty travel with locations

Idea card metadata - Stable ID: KFM-P18-INV-225 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 225 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM location evidence should carry coordinate system, projection/datum, scale, positional uncertainty, and georeferencing method where these affect claim precision or spatial joins.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM location evidence should carry coordinate system, projection/datum, scale, positional uncertainty, and georeferencing method where these affect claim precision or spatial joins. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It avoids false precision in historical, field, address, and transformed spatial records.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: Coordinate metadata - PROPOSED: Datum/projection field - PROPOSED: Positional uncertainty - PROPOSED: Spatial join validator

Tensions - PROPOSED: A precise-looking coordinate can be wrong by meaningful distances if datum, scale, or georeferencing method is missing.

Expansion directions - PROPOSED: Add coordinate_support fields for CRS, datum, source method, precision, uncertainty radius, and transformation history.

Open questions - NEEDS VERIFICATION: What minimum coordinate metadata should be required before public map exposure?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 32, 67, 71, 132; basis: projections and coordinate systems as quality choices, unprojected-coordinate distortion, datum-related positional uncertainty, and georeferencing locality descriptions; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-EVD-001, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.15 KFM-P18-INV-253 — Digital Earth-style global context remains derivative context

Idea card metadata - Stable ID: KFM-P18-INV-253 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 253 - Status label: PROPOSED - Carry-forward state: NEW - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should treat global georeferenced visualization context as a useful derivative frame, not as authoritative evidence for local claims.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat global georeferenced visualization context as a useful derivative frame, not as authoritative evidence for local claims. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps broad visual context from outranking local source authority, scale limits, or evidence bundles.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-007. - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-CAT-005.

Dependencies - PROPOSED: Context layer manifest - PROPOSED: scale/resolution caveat - PROPOSED: EvidenceBundle resolution

Tensions - PROPOSED: Global context can make low-resolution data look locally precise.

Expansion directions - PROPOSED: Add context_only and maximum_supported_scale fields for global visualization layers.

Open questions - NEEDS VERIFICATION: Which global context layers should be allowed in public map basemaps without EvidenceBundle click-through?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 242, 243; basis: Digital Earth, variable scales and resolutions, global georeferencing, global environmental research; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.16 KFM-P18-INV-325 — Map cultures and implicit values as representation review

Idea card metadata - Stable ID: KFM-P18-INV-325 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 325 - Status label: PROPOSED - Carry-forward state: NEW - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM map-review workflows should identify cultural values, implicit conventions, and potential misuse risks in geographic representation before public release.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM map-review workflows should identify cultural values, implicit conventions, and potential misuse risks in geographic representation before public release. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes map publication sensitive to social meaning and not only technical accuracy.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: representation review - PROPOSED: audience/context label - PROPOSED: misuse-risk note

Tensions - PROPOSED: Technical correctness does not guarantee culturally safe or non-misleading representation.

Expansion directions - PROPOSED: Add cultural_representation_review to public map release checklist.

Open questions - NEEDS VERIFICATION: Which map themes require community or steward review before release?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 232; basis: GIS primer states maps and GI are complex symbolic vehicles whose representation choices reflect cultural issues, values, and implicit conventions; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.17 KFM-P18-INV-332 — Dynamic H3 generalization for sensitive raster and event layers

Idea card metadata - Stable ID: KFM-P18-INV-332 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 332 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should support dynamic H3 generalization policies for sensitive raster, SAR, HydroGNSS, and event layers when exact or fine-grid exposure is unsafe.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should support dynamic H3 generalization policies for sensitive raster, SAR, HydroGNSS, and event layers when exact or fine-grid exposure is unsafe. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It gives reviewers a deterministic way to reduce spatial precision without hiding the transformation.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-004. - CONFIRMED baseline link: KFM-IDX-POL-002.

Dependencies - PROPOSED: H3 resolution policy - PROPOSED: sensitivity label - PROPOSED: generalization receipt - PROPOSED: LayerManifest

Tensions - PROPOSED: Over-generalization can erase local significance; under-generalization can leak protected locations.

Expansion directions - PROPOSED: Define `h3_generalization_profile` with input/output resolution and reason codes.

Open questions - NEEDS VERIFICATION: Should H3 generalization be allowed for cultural-sensitive data or only environmental rasters?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 627; basis: Master MapLibre v1.9 delta names dynamic H3 generalization as a governance control for HydroGNSS and sensitive data.; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-004, KFM-IDX-POL-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.18 KFM-P18-INV-352 — Map misuse review for public-facing visualizations

Idea card metadata - Stable ID: KFM-P18-INV-352 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 352 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM publication review should include a cartographic misuse check that asks whether selection, framing, scale, symbology, or omission could mislead users about evidence strength.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. -

CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - **PROPOSED:** In KFM, this idea should be treated as a governed design candidate: KFM publication review should include a cartographic misuse check that asks whether selection, framing, scale, symbology, or omission could mislead users about evidence strength. - **PROPOSED:** Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - **PROPOSED:** Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - **UNKNOWN:** No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - **NEEDS VERIFICATION:** Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - **PROPOSED:** Evidence-first publication can still mislead if the visual encoding overstates precision, authority, or completeness.

Related ideas - **CONFIRMED** baseline link: KFM-IDX-POL-003. - **CONFIRMED** baseline link: KFM-IDX-DOC-007.

Dependencies - **PROPOSED:** style review - **PROPOSED:** release review - **PROPOSED:** scale metadata - **PROPOSED:** symbology policy

Tensions - **PROPOSED:** Misuse checks can become subjective unless anchored in explicit review criteria.

Expansion directions - **PROPOSED:** Add `cartographic_misuse_check` to map release self-checks.

Open questions - **NEEDS VERIFICATION:** Who signs off on map communication risk for high-consequence layers?

Future work - **PROPOSED:** Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - **CONFIRMED** source attribution: SRC-P18-002 — `a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf`; pages 219, 232, 234; basis: A Primer of GIS includes a chapter on Map Cultures, Misuses, and GI and discusses selective/framing choices in cartography.; Phase 1 role: candidate_synthesis_source.

Carry-forward note - **CONFIRMED:** Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-003, KFM-IDX-DOC-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - **CONFIRMED:** Required schema fields are present in this Markdown card. - **CONFIRMED:** No prior-pass body text is inlined. - **CONFIRMED:** Narrative claims use explicit truth labels. - **UNKNOWN:** Repository implementation status remains unverified.

8.2.19 KFM-P18-INV-370 — Clustering labels as exploratory groupings, not discovered truth

Idea card metadata - Stable ID: KFM-P18-INV-370 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 370 - Status label: PROPOSED - Carry-forward state: NEW - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-004

Normalized statement - **PROPOSED:** KFM should mark unsupervised clustering outputs as exploratory groupings whose labels, cluster count, and initialization choices remain reviewable and non-authoritative unless independently validated.

Detailed explanation - **CONFIRMED:** Source SRC-P18-004 (`AI_Concepts_Using_Python.pdf`) supports the source concept recorded for this idea in the Phase 3 inventory. - **CONFIRMED:** Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - **PROPOSED:** In KFM, this idea should be treated as a governed design candidate: KFM should mark unsupervised clustering outputs as exploratory groupings whose labels, cluster count, and initialization choices remain reviewable and non-authoritative unless independently validated. - **PROPOSED:** Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - **PROPOSED:** Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - **UNKNOWN:** No current-session repository

evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents visually plausible clusters from being mistaken for authoritative domain categories.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-007. - CONFIRMED baseline link: KFM-IDX-AIG-006.

Dependencies - PROPOSED: cluster count - PROPOSED: initialization method - PROPOSED: evaluation metric - PROPOSED: review state

Tensions - PROPOSED: Exploratory clusters can still influence policy if displayed without caveats.

Expansion directions - PROPOSED: Add clustering_output_role to model-assisted layer manifests.

Open questions - NEEDS VERIFICATION: What validation is needed before a cluster label becomes a public claim?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 132, 135; basis: AI Concepts Using Python describes K-Means clustering as unsupervised grouping and notes limitations such as predefined k and sensitivity to initialization.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.20 KFM-P18-INV-376 — Sinton time-space-attribute triad as claim-scope metadata

Idea card metadata - Stable ID: KFM-P18-INV-376 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 376 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM spatial claims should record which dimension is controlled and which is measured—time, space, or attribute—so representation choices are visible in EvidenceBundle scope metadata.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM spatial claims should record which dimension is controlled and which is measured—time, space, or attribute—so representation choices are visible in EvidenceBundle scope metadata. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes clear whether a claim is about a fixed place over changing time, a fixed time over space, or an attribute sampled across a surface.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: EvidenceBundle scope - PROPOSED: spatial representation model - PROPOSED: attribute schema - PROPOSED: time scope

Tensions - PROPOSED: The triad may be too abstract unless translated into simple layer metadata.

Expansion directions - PROPOSED: Add representation_dimension fields to layer and evidence profiles.

Open questions - NEEDS VERIFICATION: Which KFM claim families need full time/space/attribute scope declaration?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 50, 51, 61; basis: A Primer of GIS discusses measurement frameworks and Sinton's time, space, and attribute approach, including vector and raster distinctions.; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-012, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.21 KFM-P18-INV-377 — Measurement-scale class as aggregation and legend policy

Idea card metadata - Stable ID: KFM-P18-INV-377 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 377 - Status label: PROPOSED - Carry-forward state: NEW - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should record whether mapped attributes are nominal, ordinal, interval, ratio, or otherwise classified before allowing aggregation, color ramps, ranking, or summary statistics.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record whether mapped attributes are nominal, ordinal, interval, ratio, or otherwise classified before allowing aggregation, color ramps, ranking, or summary statistics. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A ratio variable can be averaged or totaled in ways a nominal class cannot; the legend and analytics must respect the measurement class.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: attribute registry - PROPOSED: legend grammar - PROPOSED: aggregation validator - PROPOSED: summary-stat rule

Tensions - PROPOSED: Some spatial attributes combine measurement classes or have domain-specific encodings.

Expansion directions - PROPOSED: Add measurement_scale_class to schema fields used in public analytics.

Open questions - NEEDS VERIFICATION: How should KFM handle mixed or custom measurement classes?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 50, 56, 60; basis: A Primer of GIS discusses Stevens-style measurement classes and review questions about nominal, interval, ordinal, and ratio measurements.; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.22 KFM-P18-INV-401 — Cartographic omission register for public maps

Idea card metadata - Stable ID: KFM-P18-INV-401 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 401 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM public map layers should record purposeful omissions, simplifications, and emphasis choices so viewers can inspect what the map chose not to show.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM public map layers should record purposeful omissions, simplifications, and emphasis choices so viewers can inspect what the map chose not to show. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Omission and simplification can change a public interpretation as much as included features; an omission register makes cartographic selectivity auditable.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-POL-002.

Dependencies - PROPOSED: LayerManifest - PROPOSED: map-purpose statement - PROPOSED: cartographic review - PROPOSED: EvidenceBundle scope

Tensions - PROPOSED: Too much omission metadata may overwhelm ordinary map users.

Expansion directions - PROPOSED: Add an omission_register section to public LayerManifest and Evidence Drawer payloads.

Open questions - NEEDS VERIFICATION: Which omissions are material enough to disclose in a public layer?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 54, 57, 59; basis: A Primer of GIS treats map representation as selective and discusses generalization, map purpose, quality, legends, and filtered details.; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-012, KFM-IDX-POL-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.23 KFM-P18-INV-426 — Projection distortion disclosure for statewide map claims

Idea card metadata - Stable ID: KFM-P18-INV-426 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 426 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM map releases should disclose projection choice and material distortion risks when statewide or regional spatial claims could be affected by transformation choices.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM map releases should disclose projection choice and material distortion risks when statewide or regional spatial claims could be affected by transformation choices. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Projection choices shape what distances, areas, and directions can be interpreted safely; public claims need the same disclosure discipline as evidence claims.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-CAT-001.

Dependencies - PROPOSED: CRS metadata - PROPOSED: LayerManifest - PROPOSED: cartographic review - PROPOSED: EvidenceBundle spatial scope

Tensions - PROPOSED: Most viewers do not want projection theory in ordinary popups.

Expansion directions - PROPOSED: Add a projection_disclosure block to LayerManifest and release review checklists.

Open questions - NEEDS VERIFICATION: Which KFM layers require full distortion disclosure versus a short CRS note?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 79, 80, 94; basis: A Primer of GIS discusses projections as transformations from the spherical earth to a plane and treats projection choice as a cartographic decision with consequences for map communication.; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-012, KFM-IDX-CAT-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.24 KFM-P18-INV-462 — Period-boundary predicate disclosure for temporal claims

Idea card metadata - Stable ID: KFM-P18-INV-462 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 462 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM temporal claims using intervals should disclose period-boundary semantics, including whether periods are closed, half-open, overlapping, meeting, during, starting, or finishing.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM temporal claims using intervals should disclose period-boundary semantics, including whether periods are closed, half-open, overlapping, meeting, during, starting, or finishing. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Temporal claims can change meaning depending on boundary convention and overlap logic.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-CAT-005.

Dependencies - PROPOSED: temporal schema - PROPOSED: time interval - PROPOSED: claim scope - PROPOSED: validation fixture

Tensions - PROPOSED: Different databases encode period endpoints differently.

Expansion directions - PROPOSED: Record interval_convention in temporal schemas and public explanations.

Open questions - NEEDS VERIFICATION: Should KFM standardize half-open periods for all release intervals?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 112, 113, 117, 126, 127; basis: The temporal database source explains periods as anchored durations and lists predicates such as before, meets, overlaps, during, starts, and finishes using interval boundary rules.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-001, KFM-IDX-CAT-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.2.25 KFM-P18-INV-476 — Map-scale generalization disclosure

Idea card metadata - Stable ID: KFM-P18-INV-476 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 476 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REP — Geographic Representation, Scale, and Spatial Reference - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory and Phase 4 dependency ordering; subject to final synthesis review. - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM public map layers should disclose the map scale or zoom context at which generalization choices are valid, instead of treating all simplified geometry as equally precise.

Detailed explanation - CONFIRMED: The GIS/cartography source supports the card by treating geographic information and maps as selective representations shaped by scale, classification, coordinate systems, and communication choices. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and source basis: The GIS primer connects map scale with representation choices and describes generalization as filtering and emphasizing details for a map purpose. - CONFIRMED: Phase 4 assigned this idea to REP — Geographic Representation, Scale, and Spatial Reference before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM public map layers should disclose the map scale or zoom context at which generalization choices are valid, instead of treating all simplified geometry as equally precise. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: The source confirms the conceptual basis listed above; it does not prove KFM has implemented the proposed schema field, validator, policy, layer metadata, API route, UI control, or release gate.

Why it matters - PROPOSED: Scale-aware disclosure prevents readers from over-interpreting generalized features as survey-grade positions.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-001. - CONFIRMED baseline link: KFM-IDX-DOC-008.

Dependencies - PROPOSED: map scale - PROPOSED: cartographic representation - PROPOSED: LayerManifest - PROPOSED: EvidenceBundle

Tensions - PROPOSED: Too much scale metadata can make simple layer legends hard to read.

Expansion directions - PROPOSED: Add `scale_validity` and `generalization_method` fields to public layer manifests.

Open questions - NEEDS VERIFICATION: Which existing layer families need scale-dependent caveats in Phase 5 cards?

Future work - PROPOSED: Convert this idea into a fixture-backed design note during later synthesis only after source support, rights posture, implementation home, policy impact, validator shape, and rollback consequence are checked. - PROPOSED: Preserve the idea as a candidate in the Phase 8 baseline-for-next-pass sidecar so later passes can retain, expand, supersede, or retire it with explicit evidence.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 42, 57, 207; basis: The GIS primer connects map scale with representation choices and describes generalization as filtering and emphasizing details for a map purpose.; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card expands or sharpens prior Pass 17 baseline links without re-authoring the prior-pass body.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - CONFIRMED: Source attribution is present and points to the Phase 3 source record. - UNKNOWN: Repository implementation status remains unverified.

8.3 SRC - Source Admission, Connectors, and External-Service Boundaries

Status: CONFIRMED source support for grouped Phase 3 ideas; PROPOSED category synthesis

Definition: Source descriptors, scraping/connectors, field intake, external services, acquisition-method provenance, and adapter boundaries.

CONFIRMED: This chapter contains 32 Pass 18 idea cards assigned to SRC.

Idea ID	Title	Carry state	Status	Source IDs
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KFM-P18-INV-008 | Scenario and resource maps require metadata-aware interpretation | NEW | PROPOSED | SRC-P18-010 |

KFM-P18-INV-036 | Scraping as a governed fallback acquisition method | NEW | PROPOSED | SRC-P18-012 |

KFM-P18-INV-039 | Ajax and form workflows as high-risk acquisition surfaces | NEW | PROPOSED | SRC-P18-012 |

KFM-P18-INV-044 | Anticorruption layers for external source models | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-068 | Set-operator provenance for cross-source reconciliation | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-072 | MLT and emerging renderer formats as pilot-only until verified | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-149 | Scaled scraping needs source-budget and identity controls | NEW | PROPOSED | SRC-P18-012 |

KFM-P18-INV-215 | Connector error handling quarantines missing or broken web data | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-217 | Rate-limit and concurrency budgets for web watchers | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-235 | Anti-corruption layers for external source adapters | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-242 | Metadata-standard conformance as a catalog gate | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-249 | Dynamic source watchers open reviewed PRs, not publications | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-251 | Spatial data infrastructure access constraints as source-admission evidence | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-273 | Scraped data persistence must retain source, query, and sanitization context | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-281 | Connector error handling emits explicit failure dispositions | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-282 | Reusable scraper packages need source-specific adapter boundaries | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-286 | Scraper concurrency requires source politeness and shared-state controls | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-303 | Activity-step inventory before resource-route design | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-307 | Anticorruption adapters protect KFM from upstream source models | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-335 | IoT-style telemetry feeds as event sources with command boundaries | NEW | PROPOSED | SRC-P18-006 |

KFM-P18-INV-337 | Scraper error handling should quarantine missing or broken web data | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-371 | Scraper persistence should preserve source captures and retrieval context | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-372 | Scraper concurrency as a source-stewardship budget | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-402 | Normalization-form checks for spatial attribute tables | NEW | PROPOSED | SRC-P18-002 |

KFM-P18-INV-422 | Distributed scraper coordination requires source load budgets | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-423 | Scraper undercover behavior is disallowed unless explicitly reviewed | NEW | PROPOSED | SRC-P18-012 |

KFM-P18-INV-428 | Spatial data infrastructure governance as source-sharing context | NEW | PROPOSED | SRC-P18-002 |

KFM-P18-INV-440 | Anticorruption layers for external source adapters | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-447 | Scraper threading budgets as rate-limit and stewardship controls | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-453 | Spatial data infrastructure constraints as source-admission metadata | NEW | PROPOSED | SRC-P18-002 |

KFM-P18-INV-472 | Reusable scraper tool packages as adapter boundary | NEW | PROPOSED | SRC-P18-012 |

KFM-P18-INV-473 | Dynamic source watchers open PRs rather than publishing directly | EXPANDED | PROPOSED | SRC-P18-039 |

8.3.1 KFM-P18-INV-008 — Scenario and resource maps require metadata-aware interpretation

Idea card metadata - Stable ID: KFM-P18-INV-008 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 8 - Status label: PROPOSED - Carry-forward state: NEW - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should distinguish observed environmental resource layers from scenario, projection, or model layers and expose the distinction in metadata and UI.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should distinguish observed environmental resource layers from scenario, projection, or model layers and expose the distinction in metadata and UI. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Scenario layers are useful, but they are not observations or released facts about the present.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-009. - CONFIRMED baseline link: KFM-IDX-CAT-001.

Dependencies - PROPOSED: Knowledge-character labels; model/scenario metadata fields

Tensions - PROPOSED: Scenario visualizations can look as concrete as observed layers.

Expansion directions - PROPOSED: Define model_observation_scenario labels for map popups and Evidence Drawer.

Open questions - NEEDS VERIFICATION: Which scenario metadata is mandatory before a map layer can be public?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 109, 110, 123, 124, 127; basis: water-resource and climate scenario exercises with metadata; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.2 KFM-P18-INV-036 — Scraping as a governed fallback acquisition method

Idea card metadata - Stable ID: KFM-P18-INV-036 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 36 - Status label: PROPOSED - Carry-forward state: NEW - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM should allow web scraping only as a fallback source-acquisition method with explicit source terms, scope, rate, and evidence controls.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should allow web scraping only as a fallback source-acquisition method with explicit source terms, scope, rate, and evidence controls. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents opportunistic scraping from bypassing source authority, rights, or lifecycle controls.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-008. - CONFIRMED baseline link: KFM-IDX-POL-003.

Dependencies - PROPOSED: SourceDescriptor - PROPOSED: terms/robots review - PROPOSED: rate budget - PROPOSED: RunReceipt - PROPOSED: quarantine path

Tensions - PROPOSED: The scraping reference is technical; source legality and rights remain NEEDS VERIFICATION for any real target.

Expansion directions - PROPOSED: Create `scrape_allowed=false` as default in source descriptors.

Open questions - NEEDS VERIFICATION: Which source classes may be scraped at all, and which require manual steward approval?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — `instant-web-scraping-with-java.pdf`; pages 10; basis: scraping scope, redirects, rate limiting, Ajax, forms, and data-after-fetch questions; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.3 KFM-P18-INV-039 — Ajax and form workflows as high-risk acquisition surfaces

Idea card metadata - Stable ID: KFM-P18-INV-039 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 39 - Status label: PROPOSED - Carry-forward state: NEW - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM should treat Ajax pages, form submission, proxy rotation, and undercover scraping tactics as high-risk acquisition paths requiring explicit denial or steward review.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (`instant-web-scraping-with-java.pdf`) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat Ajax pages, form submission, proxy rotation, and undercover scraping tactics as high-risk acquisition paths requiring explicit denial or steward review. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps KFM aligned with source integrity and rights rather than normalizing evasive acquisition behavior.

Related ideas - CONFIRMED baseline link: KFM-IDX-SEC-009. - CONFIRMED baseline link: KFM-IDX-POL-003.

Dependencies - PROPOSED: Source terms review - PROPOSED: deny-by-default policy - PROPOSED: manual steward approval - PROPOSED: audit log

Tensions - PROPOSED: Some official sites expose data through forms; blanket denial may block legitimate access.

Expansion directions - PROPOSED: Add `interactive_source_access_risk` to source descriptor review.

Open questions - NEEDS VERIFICATION: Which interactive acquisition methods are acceptable when the source is public but lacks an API?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 42, 46, 50; basis: form submission, Ajax scraping, tests/proxies, and undercover sections; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.4 KFM-P18-INV-044 — Anticorruption layers for external source models

Idea card metadata - Stable ID: KFM-P18-INV-044 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 44 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should use anticorruption layers and published languages to translate external source models into KFM contracts without importing source-specific assumptions as canon.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should use anticorruption layers and published languages to translate external source models into KFM contracts without importing source-specific assumptions as canon. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It preserves KFM's canonical object semantics while still using valuable external data.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-008. - CONFIRMED baseline link: KFM-IDX-CAT-001.

Dependencies - PROPOSED: Source adapter contract - PROPOSED: published schema language - PROPOSED: translation tests - PROPOSED: source-role registry

Tensions - PROPOSED: A translation layer can hide loss or distortion unless transform receipts are explicit.

Expansion directions - PROPOSED: Require transform_receipt for every source-model-to-KFM-model conversion.

Open questions - NEEDS VERIFICATION: When should KFM conform to an upstream standard versus wrap it with an anticorruption adapter?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 41, 42, 43; basis: anticorruption layer, open-host service, and published language patterns; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-008, KFM-IDX-CAT-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.5 KFM-P18-INV-068 — Set-operator provenance for cross-source reconciliation

Idea card metadata - Stable ID: KFM-P18-INV-068 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 68 - Status label: PROPOSED - Carry-forward state: NEW - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM should record when analytical results come from UNION, UNION ALL, INTERSECT, EXCEPT/MINUS, or equivalent set reconciliation operations.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record when analytical results come from UNION, UNION ALL, INTERSECT, EXCEPT/MINUS, or equivalent set reconciliation operations. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Source comparison and crosswalk outputs can change materially depending on duplicate handling and inclusion/exclusion semantics.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-005. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: QueryPlanReceipt - PROPOSED: reconciliation_method field

Tensions - PROPOSED: SQL-specific operators may need translation into dataframe, graph, or ETL equivalents.

Expansion directions - PROPOSED: Add set_operation_type and duplicate_policy to reconciliation outputs.

Open questions - NEEDS VERIFICATION: Should INTERSECT/EXCEPT-derived claims receive a different confidence label than direct-source claims?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 21; basis: set operators, DISTINCT, duplicate handling, and vertical combination discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.6 KFM-P18-INV-072 — MLT and emerging renderer formats as pilot-only until verified

Idea card metadata - Stable ID: KFM-P18-INV-072 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 72 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from

CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should treat MapLibre Tile and other emerging renderer formats as pilot candidates until tooling, renderer support, validation, and fallback behavior are verified.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat MapLibre Tile and other emerging renderer formats as pilot candidates until tooling, renderer support, validation, and fallback behavior are verified. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: New formats can improve performance, but premature adoption can weaken validation, accessibility, or rollback if toolchains are not mature.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-SEC-004.

Dependencies - PROPOSED: format_adoption_adr - PROPOSED: fallback renderer path - PROPOSED: compatibility tests

Tensions - PROPOSED: Performance pressure can push adoption before governance parity is ready.

Expansion directions - PROPOSED: Create experimental_format_status with pilot, fallback, validation, and approval fields.

Open questions - NEEDS VERIFICATION: What evidence is sufficient to promote MLT from pilot to default for any layer family?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 3, 4, 14; basis: MapLibre ecosystem watchlist, MLT pilot status, Native Vulkan/Martin notes, and verification posture; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-SEC-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.7 KFM-P18-INV-149 — Scaled scraping needs source-budget and identity controls

Idea card metadata - Stable ID: KFM-P18-INV-149 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 149 - Status label: PROPOSED - Carry-forward state: NEW - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM should require source-budget, concurrency, identity, and shared-state controls before threaded or distributed scraping is admitted as a connector strategy.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded

this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should require source-budget, concurrency, identity, and shared-state controls before threaded or distributed scraping is admitted as a connector strategy. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, run-time routing, or release state.

Why it matters - PROPOSED: It reduces legal, ethical, and data-integrity risk from aggressive acquisition fallback methods.

Related ideas - CONFIRMED baseline link: KFM-IDX-SEC-003. - CONFIRMED baseline link: KFM-IDX-PIP-003.

Dependencies - PROPOSED: SourceDescriptor terms review - PROPOSED: Connector throttle policy - PROPOSED: RunReceipt

Tensions - PROPOSED: Speed and coverage can conflict with source terms and evidence integrity.

Expansion directions - PROPOSED: Add max_concurrency, rate_limit_policy, and identity_policy to scraping source descriptors.

Open questions - NEEDS VERIFICATION: Should KFM ever allow distributed scraping for public-source intake?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 50, 54, 59, 63; basis: threading and RMI scraping sections and warnings about shared state; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.8 KFM-P18-INV-215 — Connector error handling quarantines missing or broken web data

Idea card metadata - Stable ID: KFM-P18-INV-215 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 215 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM web connectors should catch missing elements, broken links, server failures, malformed URLs, and unexpected page states, then emit quarantine records rather than silently skipping or fabricating data.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM web connectors should catch missing elements, broken links, server failures, malformed URLs, and unexpected page states, then emit quarantine records rather than silently skipping or fabricating data. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository

evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It supports fail-closed source intake and prevents silent evidence loss.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-002. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-SEC-010.

Dependencies - PROPOSED: Connector error envelope - PROPOSED: Quarantine record - PROPOSED: Retry policy - PROPOSED: Source receipt

Tensions - PROPOSED: Over-aggressive retries can look like successful intake if failures are hidden.

Expansion directions - PROPOSED: Add connector_error_record fixtures for missing element, 404/500, malformed URL, redirect loop, and empty response.

Open questions - NEEDS VERIFICATION: Which connector failures should produce ERROR versus QUARANTINE versus ABSTAIN?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 31, 39; basis: handling web scraping errors, catching exceptions for missing elements, broken links, server failures, malformed URLs, and redirects; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-002, KFM-IDX-EVD-001, KFM-IDX-SEC-010 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.9 KFM-P18-INV-217 — Rate-limit and concurrency budgets for web watchers

Idea card metadata - Stable ID: KFM-P18-INV-217 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 217 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM web watchers should record rate-limit posture, concurrency/threading budget, politeness constraints, and provider terms before using threaded or distributed fetching.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM web watchers should record rate-limit posture, concurrency/threading budget, politeness constraints, and provider terms before using threaded or distributed fetching. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents watcher throughput from violating source terms or degrading source reliability.

Related ideas - CONFIRMED baseline link: KFM-IDX-SEC-010. - CONFIRMED baseline link: KFM-IDX-PIP-008. - CONFIRMED baseline link: KFM-IDX-OBS-005.

Dependencies - PROPOSED: Source terms review - PROPOSED: Concurrency budget - PROPOSED: Rate-limit receipt - PROPOSED: Fetch scheduler

Tensions - PROPOSED: More concurrency can improve speed while weakening rights, reliability, and operational ethics.

Expansion directions - PROPOSED: Add watcher_budget fields for max_parallelism, delay, retry, backoff, source_terms_url, and observed throttling.

Open questions - NEEDS VERIFICATION: Which sources permit automated fetching, and what limits should fail closed?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 10, 59; basis: rate limiter and redirect challenges, plus faster scraping through threading and shared resources; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-SEC-010, KFM-IDX-PIP-008, KFM-IDX-OBS-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.10 KFM-P18-INV-235 — Anti-corruption layers for external source adapters

Idea card metadata - Stable ID: KFM-P18-INV-235 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 235 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM connectors should translate external source models through anti-corruption adapters instead of letting source-specific assumptions reshape canonical KFM objects.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM connectors should translate external source models through anti-corruption adapters instead of letting source-specific assumptions reshape canonical KFM objects. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It preserves KFM source-role and evidence semantics across heterogeneous APIs, GIS services, and document formats.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-008. - CONFIRMED baseline link: KFM-IDX-DOC-003. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: Source descriptor - PROPOSED: Adapter schema - PROPOSED: Canonical object contract - PROPOSED: Mapping receipt

Tensions - PROPOSED: Over-translation can erase source-specific caveats needed for evidence review.

Expansion directions - PROPOSED: Add `adapter_mapping_receipt` fields for source fields, canonical fields, dropped fields, caveats, and unmapped values.

Open questions - NEEDS VERIFICATION: Which source-specific fields must remain visible even after canonical normalization?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 41; basis: anti-corruption layer protects downstream model intent from upstream or legacy-system model corruption; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-008, KFM-IDX-DOC-003, KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.11 KFM-P18-INV-242 — Metadata-standard conformance as a catalog gate

Idea card metadata - Stable ID: KFM-P18-INV-242 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 242 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM catalog records should run metadata-standard conformance checks and record missing requirements before public release.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM catalog records should run metadata-standard conformance checks and record missing requirements before public release. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes catalog completeness a validation result rather than an editorial afterthought.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-001. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-MDP-003.

Dependencies - PROPOSED: Catalog record - PROPOSED: Metadata profile - PROPOSED: Conformance validator - PROPOSED: Missing-field report

Tensions - PROPOSED: Different standards can require different fields and vocabulary mappings.

Expansion directions - PROPOSED: Add `catalog_metadata_conformance_report` fields for profile, `missing_required_fields`, warnings, and release decision.

Open questions - NEEDS VERIFICATION: Which metadata standards are mandatory for each artifact family: STAC, DCAT, FGDC, ISO, or KFM profile?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 124, 128, 129; basis: editing layer metadata and verifying FGDC CSDGM and ISO 19115 metadata-standard compliance and missing requirements; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-001, KFM-IDX-EVD-001, KFM-IDX-MDP-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.12 KFM-P18-INV-249 — Dynamic source watchers open reviewed PRs, not publications

Idea card metadata - Stable ID: KFM-P18-INV-249 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 249 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM dynamic source watchers should create reviewed diffs or pull requests when material changes occur, not publish directly from watcher output.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM dynamic source watchers should create reviewed diffs or pull requests when material changes occur, not publish directly from watcher output. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It preserves promotion as a governed state transition rather than a background synchronization side effect.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-003. - CONFIRMED baseline link: KFM-IDX-PIP-008. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Watcher receipt - PROPOSED: Material-change diff - PROPOSED: PR review - PROPOSED: Promotion gate

Tensions - PROPOSED: Automation can still pressure reviewers if every source change becomes noisy.

Expansion directions - PROPOSED: Add watcher_material_change_report with no-op, changed, failed, quarantined, and reviewer-required outcomes.

Open questions - NEEDS VERIFICATION: Which watcher changes are material enough to open a PR versus emit a no-op receipt?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 20; basis: dynamic source watchers use conditional GET and automation to open PRs when material changes occur; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-003, KFM-IDX-PIP-008, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.13 KFM-P18-INV-251 — Spatial data infrastructure access constraints as source-admission evidence

Idea card metadata - Stable ID: KFM-P18-INV-251 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 251 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should record access terms, participation constraints, cost-recovery posture, and metadata availability for spatial data infrastructure sources before admitting them into governed intake.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record access terms, participation constraints, cost-recovery posture, and metadata availability for spatial data infrastructure sources before admitting them into governed intake. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes source availability, rights, and institutional constraints part of source governance rather than a hidden acquisition nuisance.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-001. - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: SourceDescriptor - PROPOSED: rights/access review - PROPOSED: source-admission decision

Tensions - PROPOSED: Useful public data may be technically available but legally, institutionally, or financially constrained.

Expansion directions - PROPOSED: Add `source_access_profile` fields for `metadata_only`, `registration_required`, `fee_required`, `redistribution_allowed`, and `public_safety_exception_notes`.

Open questions - NEEDS VERIFICATION: Which access constraints should block KFM intake versus block only public redistribution?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 241, 242; basis: spatial data infrastructure, metadata access, cost recovery, participation, public safety, and environmental-protection concerns; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-001, KFM-IDX-POL-001, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass

prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.14 KFM-P18-INV-273 — Scraped data persistence must retain source, query, and sanitization context

Idea card metadata - Stable ID: KFM-P18-INV-273 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 273 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM scraping fallbacks should persist not only extracted values but also source URL, query path, retrieval context, and sanitization outcome.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM scraping fallbacks should persist not only extracted values but also source URL, query path, retrieval context, and sanitization outcome. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps source acquisition auditable and reduces injection and provenance risks.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-001. - CONFIRMED baseline link: KFM-IDX-SEC-004. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Scrape receipt - PROPOSED: raw snapshot or hash - PROPOSED: sanitization result - PROPOSED: database write log

Tensions - PROPOSED: Persisting scraped values without raw/source context destroys auditability.

Expansion directions - PROPOSED: Add `scrape_persistence_receipt` fields for `source_url`, `request_headers_hash`, `raw_snapshot_hash`, `extracted_fields`, `sanitized`, and `storage_target`.

Open questions - NEEDS VERIFICATION: Should KFM store full raw HTML, redacted snapshots, or only hashes for scraped sources?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 33, 34, 35; basis: persisting scraped data, JDBC database connection, stored URLs/titles, and prepared-statement input sanitization; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-001, KFM-IDX-SEC-004, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.15 KFM-P18-INV-281 — Connector error handling emits explicit failure dispositions

Idea card metadata - Stable ID: KFM-P18-INV-281 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 281 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM web-source connectors should catch, classify, and record connection, missing-element, missing-page, and server failures instead of returning silent empty results.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM web-source connectors should catch, classify, and record connection, missing-element, missing-page, and server failures instead of returning silent empty results. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It ensures source intake can quarantine or abstain rather than fabricate absence from a failed scrape.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-003. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: connector run receipt - PROPOSED: failure disposition enum - PROPOSED: quarantine policy

Tensions - PROPOSED: A scrape failure is not evidence that a source lacks the target content.

Expansion directions - PROPOSED: Add failure outcome fixtures for timeout, missing selector, 404, malformed HTML, and server unavailable.

Open questions - NEEDS VERIFICATION: Which connector failures should quarantine the run versus emit a no-change receipt?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 31; basis: The scraping source describes frequent web errors and the need to catch, throw, and handle exceptions when pages, elements, or servers fail.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-003, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.16 KFM-P18-INV-282 — Reusable scraper packages need source-specific adapter boundaries

Idea card metadata - Stable ID: KFM-P18-INV-282 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 282 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM scraping and HTML-source adapters should separate reusable tooling from source-specific extraction logic so tool reuse does not collapse source identity.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM scraping and HTML-source adapters should separate reusable tooling from source-specific extraction logic so tool reuse does not collapse source identity. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It supports maintainable connectors while preserving per-source evidence, rights, selectors, and failure profiles.

Related ideas - CONFIRMED baseline link: KFM-IDX-SRC-002. - CONFIRMED baseline link: KFM-IDX-SEC-004.

Dependencies - PROPOSED: source adapter contract - PROPOSED: shared connector library - PROPOSED: SourceDescriptor

Tensions - PROPOSED: Reusable tooling can hide source-specific caveats if adapter records are too thin.

Expansion directions - PROPOSED: Record selector strategy, parser library, and source-specific extraction assumptions in SourceDescriptor extensions.

Open questions - NEEDS VERIFICATION: How much shared connector code can change before all dependent source receipts need a new spec_hash?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 34, 42; basis: The scraping source describes multi-class scraper packages, reusable tools, and separate scraper packages that communicate with shared tool packages.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-SRC-002, KFM-IDX-SEC-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.17 KFM-P18-INV-286 — Scraper concurrency requires source politeness and shared-state controls

Idea card metadata - Stable ID: KFM-P18-INV-286 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 286 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM threaded or distributed scraping should require concurrency limits, source politeness rules, shared-state safety, and per-thread receipt attribution.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM

threaded or distributed scraping should require concurrency limits, source politeness rules, shared-state safety, and per-thread receipt attribution. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents performance tuning from becoming uncontrolled source load or unverifiable acquisition order.

Related ideas - CONFIRMED baseline link: KFM-IDX-SEC-007. - CONFIRMED baseline link: KFM-IDX-PIP-008.

Dependencies - PROPOSED: source rate-limit policy - PROPOSED: thread/run attribution - PROPOSED: distributed connector receipt

Tensions - PROPOSED: Throughput gains may conflict with source terms, rate limits, and reproducibility.

Expansion directions - PROPOSED: Add `max_concurrency` and `politeness_delay` fields to web SourceDescriptors.

Open questions - NEEDS VERIFICATION: Should distributed scraping be denied unless an explicit source permission or API endpoint exists?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — `instant-web-scraping-with-java.pdf`; pages 59, 62, 66; basis: The scraping source describes threading and RMI for faster scraping, including parallel page retrieval and remote implementations.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-SEC-007, KFM-IDX-PIP-008 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.18 KFM-P18-INV-303 — Activity-step inventory before resource-route design

Idea card metadata - Stable ID: KFM-P18-INV-303 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 303 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM API route proposals should derive candidate resources from user activities and activity steps before naming endpoints or URL structures.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (`Designing Great Web APIs.pdf`) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM API route proposals should derive candidate resources from user activities and activity steps before naming endpoints or URL structures. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents route design from following topic names instead of governed user tasks.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-007. - CONFIRMED baseline link: KFM-IDX-DAT-001.

Dependencies - PROPOSED: activity inventory - PROPOSED: participant labels - PROPOSED: resource ontology draft

Tensions - PROPOSED: Some KFM resources are evidence objects, not user-visible tasks, so both task and trust-object views must align.

Expansion directions - PROPOSED: Require activity_step_basis for new public API routes.

Open questions - NEEDS VERIFICATION: Which routes are trust-object routes versus task-workflow routes?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 33; basis: Web API source describes identifying activities and activity steps, then deriving business entities and candidate API resources from them.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-007, KFM-IDX-DAT-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.19 KFM-P18-INV-307 — Anticorruption adapters protect KFM from upstream source models

Idea card metadata - Stable ID: KFM-P18-INV-307 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 307 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM source adapters should translate external or legacy models defensively so upstream shapes do not overwrite KFM lane semantics.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM source adapters should translate external or legacy models defensively so upstream shapes do not overwrite KFM lane semantics. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps source ingestion from importing incompatible meanings as canonical truth.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-001. - CONFIRMED baseline link: KFM-IDX-SEC-004.

Dependencies - PROPOSED: SourceDescriptor - PROPOSED: adapter contract - PROPOSED: semantic mapping table

Tensions - PROPOSED: Over-normalization may lose source nuance; under-translation may corrupt KFM semantics.

Expansion directions - PROPOSED: Add adapter_semantic_mapping and lost_or_changed_meaning notes to source descriptors.

Open questions - NEEDS VERIFICATION: When should KFM expose original source terminology beside normalized vocabulary?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 41; basis: DDD reference describes an anticorruption layer as a defensive translation layer when upstream models do not fit downstream needs.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-001, KFM-IDX-SEC-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.20 KFM-P18-INV-335 — IoT-style telemetry feeds as event sources with command boundaries

Idea card metadata - Stable ID: KFM-P18-INV-335 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 335 - Status label: PROPOSED - Carry-forward state: NEW - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should treat IoT-style telemetry APIs as event feeds with explicit source, cadence, command/response boundaries, and provenance rather than as manually curated observations.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat IoT-style telemetry APIs as event feeds with explicit source, cadence, command/response boundaries, and provenance rather than as manually curated observations. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It helps KFM handle sensor-like and device-origin feeds without collapsing them into human observation claims.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-008. - CONFIRMED baseline link: KFM-IDX-SEC-009.

Dependencies - PROPOSED: telemetry source descriptor - PROPOSED: cadence - PROPOSED: command boundary - PROPOSED: event receipt

Tensions - PROPOSED: Command-capable APIs create safety and authorization risk beyond passive data ingestion.

Expansion directions - PROPOSED: Create telemetry_event_source guidance for sensor/device source descriptors.

Open questions - NEEDS VERIFICATION: Should KFM ever send commands, or only ingest telemetry from external systems?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 2, 3; basis: Designing Great Web APIs notes that IoT devices broadcast telemetry data and receive commands through APIs.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.21 KFM-P18-INV-337 — Scraper error handling should quarantine missing or broken web data

Idea card metadata - Stable ID: KFM-P18-INV-337 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 337 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM web acquisition should quarantine missing code, broken links, failed selectors, and unexpected responses instead of silently skipping or fabricating scraped data.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM web acquisition should quarantine missing code, broken links, failed selectors, and unexpected responses instead of silently skipping or fabricating scraped data. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It aligns fallback web acquisition with KFM fail-closed truth posture.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-POL-007.

Dependencies - PROPOSED: error class - PROPOSED: quarantine destination - PROPOSED: retry policy - PROPOSED: source receipt

Tensions - PROPOSED: Too many quarantines can create backlog; silent skipping creates invisible data loss.

Expansion directions - PROPOSED: Add negative scraping fixtures for broken link, missing selector, and unexpected HTML cases.

Open questions - NEEDS VERIFICATION: What errors are retryable, quarantined, or permanent DENY outcomes?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 22; basis: Instant Web Scraping with Java contains a dedicated error-handling recipe for missing code, broken links, and unexpected situations.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-004, KFM-IDX-POL-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN:

Repository implementation status remains unverified.

8.3.22 KFM-P18-INV-371 — Scraper persistence should preserve source captures and retrieval context

Idea card metadata - Stable ID: KFM-P18-INV-371 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 371 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM scraping or web-source connectors should persist source captures, retrieval metadata, and parsing results separately so later audits can distinguish fetched content from derived records.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM scraping or web-source connectors should persist source captures, retrieval metadata, and parsing results separately so later audits can distinguish fetched content from derived records. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It supports replay, correction, and rights review when a scraped source changes or disappears.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-006. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: capture artifact - PROPOSED: retrieval receipt - PROPOSED: parser version - PROPOSED: derived record

Tensions - PROPOSED: Stored captures may carry rights or privacy obligations.

Expansion directions - PROPOSED: Define `scape_capture_receipt` and `parser_output_receipt` separation.

Open questions - NEEDS VERIFICATION: When should source captures be quarantined instead of preserved in WORK?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 11, 42; basis: Instant Web Scraping with Java describes persisting data from the web for later retrieval and explains that console output is insufficient unless data is stored for useful applications.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-006, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.23 KFM-P18-INV-372 — Scraper concurrency as a source-stewardship budget

Idea card metadata - Stable ID: KFM-P18-INV-372 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 372 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM web watchers should treat threading, distributed scraping, and remote invocation as governed concurrency budgets constrained by source terms, rate limits, and stewardship posture.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM web watchers should treat threading, distributed scraping, and remote invocation as governed concurrency budgets constrained by source terms, rate limits, and stewardship posture. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps acquisition performance from becoming abusive, unstable, or terms-violating.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-003. - CONFIRMED baseline link: KFM-IDX-PIP-005.

Dependencies - PROPOSED: rate-limit policy - PROPOSED: thread budget - PROPOSED: source terms - PROPOSED: backoff strategy

Tensions - PROPOSED: Concurrency pressure can conflict with source stewardship and robots/terms constraints.

Expansion directions - PROPOSED: Add max_concurrency and steward_contact fields to web source descriptors.

Open questions - NEEDS VERIFICATION: Which source families require explicit steward approval before parallel fetching?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 8, 10, 11; basis: Instant Web Scraping with Java includes faster scraping through threading and RMI, and distinguishes larger-scale scraping from simple scripts.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-003, KFM-IDX-PIP-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.24 KFM-P18-INV-402 — Normalization-form checks for spatial attribute tables

Idea card metadata - Stable ID: KFM-P18-INV-402 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 402 - Status label: PROPOSED - Carry-forward state: NEW - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM processed spatial tables should pass normalization-style checks that separate entity keys, dependent attributes, and relationship fields before publication.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM processed spatial tables should pass normalization-style checks that separate entity keys, dependent attributes, and relationship fields before publication. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Poorly normalized spatial tables blur feature identity, relationship meaning, and repeated attributes, making downstream joins and evidence claims harder to audit.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012.

Dependencies - PROPOSED: schema registry - PROPOSED: feature key rule - PROPOSED: relationship field model - PROPOSED: validator

Tensions - PROPOSED: Some denormalized public tiles are valid derivatives even when canonical tables remain normalized.

Expansion directions - PROPOSED: Create a normalization_check validator for processed tabular geospatial artifacts.

Open questions - NEEDS VERIFICATION: Which denormalizations are allowed for released PMTiles or GeoParquet derivatives?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 138, 144, 145; basis: A Primer of GIS explains databases as central to geographic information, with keys, relationships, and dependency levels in attribute design.; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.25 KFM-P18-INV-422 — Distributed scraper coordination requires source load budgets

Idea card metadata - Stable ID: KFM-P18-INV-422 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 422 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM distributed or threaded scraping jobs should declare source load budgets, concurrency limits, and stop conditions before activation.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category

before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM distributed or threaded scraping jobs should declare source load budgets, concurrency limits, and stop conditions before activation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Fast or distributed crawling can harm sources or violate terms; governed acquisition must prioritize stewardship over throughput.

Related ideas - CONFIRMED baseline link: KFM-IDX-SEC-011. - CONFIRMED baseline link: KFM-IDX-PIP-006.

Dependencies - PROPOSED: SourceDescriptor - PROPOSED: robots/terms check - PROPOSED: rate limit policy - PROPOSED: run receipt

Tensions - PROPOSED: Some urgent source refreshes may need higher temporary budgets under approval.

Expansion directions - PROPOSED: Add `concurrency_limit` and `source_load_budget` fields to web-source connectors.

Open questions - NEEDS VERIFICATION: Who approves temporary source-load budget changes?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — `instant-web-scraping-with-java.pdf`; pages 50, 54; basis: Instant Web Scraping with Java includes recipes on faster scraping through threading and RMI after earlier sections on robust scraping and error handling.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-SEC-011, KFM-IDX-PIP-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.26 KFM-P18-INV-423 — Scraper undercover behavior is disallowed unless explicitly reviewed

Idea card metadata - Stable ID: KFM-P18-INV-423 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 423 - Status label: PROPOSED - Carry-forward state: NEW - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM should deny disguised, evasive, or undercover scraping behavior by default unless a legal, ethical, and steward-reviewed source-admission decision explicitly permits it.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (`instant-web-scraping-with-java.pdf`) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should deny disguised, evasive, or undercover scraping behavior by default unless a legal, ethical, and steward-reviewed source-admission decision explicitly permits it. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Techniques that hide crawler identity may undermine source rights, terms, trust, and steward relationships even when technically feasible.

Related ideas - CONFIRMED baseline link: KFM-IDX-SEC-012. - CONFIRMED baseline link: KFM-IDX-POL-003.

Dependencies - PROPOSED: source ethics review - PROPOSED: legal terms check - PROPOSED: security review - PROPOSED: deny policy

Tensions - PROPOSED: Some benign user-agent configuration is normal and not evasive.

Expansion directions - PROPOSED: Create a scraper_evasion_policy with DENY default and explicit exceptions.

Open questions - NEEDS VERIFICATION: How should KFM distinguish legitimate user-agent declaration from evasive behavior?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 39; basis: Instant Web Scraping with Java includes a section on going undercover, making it a source-adaptor risk family for KFM governance review.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.27 KFM-P18-INV-428 — Spatial data infrastructure governance as source-sharing context

Idea card metadata - Stable ID: KFM-P18-INV-428 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 428 - Status label: PROPOSED - Carry-forward state: NEW - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM SourceDescriptors should record whether a source participates in a spatial data infrastructure, clearinghouse, or institutional sharing arrangement.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM SourceDescriptors should record whether a source participates in a spatial data infrastructure, clearinghouse, or institutional sharing arrangement. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Source-sharing institutions affect authority, update cadence, access terms, and steward expectations.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-004. - CONFIRMED baseline link: KFM-IDX-POL-003.

Dependencies - PROPOSED: SourceDescriptor - PROPOSED: rights profile - PROPOSED: steward registry - PROPOSED: source-role registry

Tensions - PROPOSED: Infrastructure participation is not the same as source authority.

Expansion directions - PROPOSED: Add `source_infrastructure_context` to the source ledger.

Open questions - NEEDS VERIFICATION: Which SDI/clearinghouse fields should be mandatory versus advisory?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — [a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf](#); pages 251, 317, 320; basis: A Primer of GIS discusses spatial data infrastructures and the governance context around geographic information sharing.; Phase 1 role: `candidate_synthesis_source`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.28 KFM-P18-INV-440 — Anticorruption layers for external source adapters

Idea card metadata - Stable ID: KFM-P18-INV-440 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 440 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM connectors should translate external source models through anticorruption layers before internal governed contracts consume them.

Detailed explanation - CONFIRMED: Source SRC-P18-009 ([Domain-Driven Design Reference.pdf](#)) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM connectors should translate external source models through anticorruption layers before internal governed contracts consume them. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: External schemas and vocabularies should not leak into canonical KFM objects as unreviewed authority.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-004. - CONFIRMED baseline link: KFM-IDX-DAT-013.

Dependencies - PROPOSED: connector adapter - PROPOSED: `SourceDescriptor` - PROPOSED: published language - PROPOSED: schema registry

Tensions - PROPOSED: Over-translation can hide source-specific caveats.

Expansion directions - PROPOSED: Add `anticorruption_layer_note` to source-adapter documentation.

Open questions - NEEDS VERIFICATION: Which source fields should be preserved verbatim even after translation?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 3, 41, 42; basis: Domain-Driven Design Reference defines Anticorruption Layer as a context-mapping pattern.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-004, KFM-IDX-DAT-013 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.29 KFM-P18-INV-447 — Scraper threading budgets as rate-limit and stewardship controls

Idea card metadata - Stable ID: KFM-P18-INV-447 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 447 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM web source adapters should cap concurrency and record threading decisions as source-stewardship controls.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM web source adapters should cap concurrency and record threading decisions as source-stewardship controls. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Fast scraping can harm upstream services or violate norms even when technically possible.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-003. - CONFIRMED baseline link: KFM-IDX-SEC-004.

Dependencies - PROPOSED: source adapter - PROPOSED: rate-limit policy - PROPOSED: run receipt - PROPOSED: stewardship review

Tensions - PROPOSED: Concurrency needs vary by source and may change over time.

Expansion directions - PROPOSED: Add concurrency_budget to SourceDescriptor and adapter run receipts.

Open questions - NEEDS VERIFICATION: Should source stewards be able to set hard per-source concurrency limits?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 8, 58, 59; basis: Instant Web Scraping with Java discusses faster scraping through threading.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-003, KFM-IDX-SEC-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.30 KFM-P18-INV-453 — Spatial data infrastructure constraints as source-admission metadata

Idea card metadata - Stable ID: KFM-P18-INV-453 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 453 - Status label: PROPOSED - Carry-forward state: NEW - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM source admission should record spatial-data-infrastructure constraints such as participation limits, charging/cost recovery, sharing barriers, and institutional access obligations.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM source admission should record spatial-data-infrastructure constraints such as participation limits, charging/cost recovery, sharing barriers, and institutional access obligations. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A source can be technically useful yet operationally constrained by institutional terms, cost, or participation rules.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-CAT-003.

Dependencies - PROPOSED: SourceDescriptor - PROPOSED: rights review - PROPOSED: source registry - PROPOSED: release gate

Tensions - PROPOSED: Cost/access metadata may change independently of data content.

Expansion directions - PROPOSED: Add source-access posture to SourceDescriptor review fields.

Open questions - NEEDS VERIFICATION: How should KFM represent sources that are public-interest but fee-gated?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 253; basis: The GIS primer discusses SDI data sharing difficulty, cost allocation, participation issues, cost recovery, and administrative concerns.; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.31 KFM-P18-INV-472 — Reusable scraper tool packages as adapter boundary

Idea card metadata - Stable ID: KFM-P18-INV-472 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 472 - Status label: PROPOSED - Carry-forward state: NEW - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM scraping adapters should separate reusable tooling packages from source-specific scraper packages, with tests that prove shared tools work across adapters.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM scraping adapters should separate reusable tooling packages from source-specific scraper packages, with tests that prove shared tools work across adapters. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A clean adapter boundary reduces drift and makes source-specific behavior easier to review and retire.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-006. - CONFIRMED baseline link: KFM-IDX-PIP-010.

Dependencies - PROPOSED: adapter package - PROPOSED: shared scraper tools - PROPOSED: unit tests - PROPOSED: source-specific fixture

Tensions - PROPOSED: Shared tools can hide source-specific obligations unless each adapter still has a descriptor.

Expansion directions - PROPOSED: Record shared-tool version and source-adapter link in receipts.

Open questions - NEEDS VERIFICATION: How should shared scraper utilities be retired without breaking source adapters?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 34, 42, 46; basis: The Java scraping source describes multi-class scrapers, separate tool packages reused across websites, and unit tests to ensure scraper tools continue to work together; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.3.32 KFM-P18-INV-473 — Dynamic source watchers open PRs rather than publishing directly

Idea card metadata - Stable ID: KFM-P18-INV-473 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 473 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: SRC — Source Admission, Connectors, and External-Service Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM dynamic source watchers should detect material change and open reviewed PRs or review packets rather than directly publishing refreshed artifacts.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM dynamic source watchers should detect material change and open

reviewed PRs or review packets rather than directly publishing refreshed artifacts. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Watchers can improve freshness without bypassing review, policy, and release controls.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-008. - CONFIRMED baseline link: KFM-IDX-DOC-002.

Dependencies - PROPOSED: watcher - PROPOSED: conditional GET - PROPOSED: review packet - PROPOSED: promotion gate

Tensions - PROPOSED: Too many watcher PRs can overwhelm reviewers if materiality thresholds are weak.

Expansion directions - PROPOSED: Define watcher material-change thresholds and PR payload minimums.

Open questions - NEEDS VERIFICATION: Which source families can use automatic PR emission safely?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 20; basis: The Master MapLibre atlas records that dynamic source watchers use conditional GET and GitHub automation to open PRs only when material change exists.; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-008, KFM-IDX-DOC-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4 MOD - Data Modeling, Domain Semantics, Temporal Structure, and SQL

Status: CONFIRMED source support for grouped Phase 3 ideas; PROPOSED category synthesis

Definition: Domain models, temporal structure, DDD boundaries, SQL-backed state semantics, shared kernels, and object-family discipline.

CONFIRMED: This chapter contains 47 Pass 18 idea cards assigned to MOD.

Idea ID	Title	Carry state	Status	Source IDs
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KFM-P18-INV-043 | Context maps for KFM bounded-context integration | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-092 | Tracking logs preserve reconstructable transaction history | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-093 | Open-ended “now” periods need explicit semantics | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-094 | Temporal partitioning and vacuuming are governance decisions | NEW | PROPOSED | SRC-P18-007 |

KFM-P18-INV-097 | Validator interfaces should be intention-revealing and assertion-backed | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-106 | LEAD/LAG as temporal change-detection primitives | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-119 | Temporal vacuuming as retention and semantics risk | NEW | PROPOSED | SRC-P18-007 |

KFM-P18-INV-120 | Domain events for governed state transitions | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-145 | Tracking-log after-images for report-time reconstruction | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-146 | Published Language and Context Map as public contract guardrails | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-170 | Period predicates for temporal overlap semantics | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-171 | Valid-time and transaction-time orthogonality | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-172 | Bitemporal sequenced modifications need complexity disclosure | NEW | PROPOSED | SRC-P18-007 |

KFM-P18-INV-186 | Contract-backed map registries and telemetry events | NEW | PROPOSED | SRC-P18-039 |

KFM-P18-INV-189 | Context maps for cross-domain translation and influence | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-193 | Temporal referential integrity over validity periods | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-194 | Now-relative validity and forever sentinel risk | NEW | PROPOSED | SRC-P18-007 |

KFM-P18-INV-195 | Temporal partitioning requires consistency receipts | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-196 | Temporal dialect fixtures for period predicates | NEW | PROPOSED | SRC-P18-007 |

KFM-P18-INV-234 | Assertions and invariants as machine-checkable governance | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-237 | Tracking-log variants need replay semantics and after-image policy | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-239 | Transaction-time state tables as audit truth for repo-independent replay | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-266 | Temporal design should separate static schema design from temporal annotation | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-269 | Factories enforce valid creation of receipts and envelopes | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-270 | Separate Ways prevents forced integration of unrelated lanes | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-287 | Now-time handling must be explicit in temporal records | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-288 | Temporal keys and uniqueness cross valid periods | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-289 | Temporal referential integrity spans historical state | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-290 | Temporal duplicate elimination is canonicalization, not cosmetic cleanup | NEW | PROPOSED | SRC-P18-007 |

KFM-P18-INV-291 | Bitemporal tables separate world validity from database correction time | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-304 | Domain events carry the meaning of state change | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-305 | Assertions and invariants as machine-checkable governance contracts | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-308 | Valid-time and transaction-time roles must be declared per claim family | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-309 | Bitemporal modification complexity needs release limitation labels | NEW | PROPOSED | SRC-P18-007 |

KFM-P18-INV-349 | Temporal table design should declare whether history is current-state, valid-time, transaction-time, or bitemporal | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-374 | Domain events as auditable change facts | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-375 | Context maps as cross-lane dependency governance | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-389 | Site geodatabase structures as field investigation schemas | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-392 | Temporal uniqueness constraints include validity semantics | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-394 | Temporal migration separates static design from history annotation | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-395 | Lane-specific ubiquitous language as anti-drift glossary | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-439 | Temporal vacuuming policy for archival retention versus deletion | NEW | PROPOSED | SRC-P18-007 |

KFM-P18-INV-463 | Transaction-time presence periods for audit history | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-489 | Valid-time and transaction-time split for public claim history | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-490 | Temporal referential-integrity gate for history tables | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-491 | Coalesced temporal views as derived artifacts | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-492 | Continuous-integration discipline inside bounded contexts | EXPANDED | PROPOSED | SRC-P18-009 |

8.4.1 KFM-P18-INV-043 — Context maps for KFM bounded-context integration

Idea card metadata - Stable ID: KFM-P18-INV-043 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 43 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should maintain context maps between domain lanes, API surfaces, data lifecycle stores, and external source systems.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should maintain context maps between domain lanes, API surfaces, data lifecycle stores, and external source systems. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps hydrology, fauna, people/DNA, MapLibre, AI, and catalog surfaces related without collapsing their models.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-006. - CONFIRMED baseline link: KFM-IDX-MDP-003.

Dependencies - PROPOSED: Context map document - PROPOSED: bounded-context register - PROPOSED: integration relationship labels

Tensions - PROPOSED: Context maps can become stale if not tied to ADRs and tests.

Expansion directions - PROPOSED: Add context_map_ref to domain architecture chapters.

Open questions - NEEDS VERIFICATION: Which KFM object families are shared kernel versus published language?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 35, 36; basis: bounded context and context map definitions and guidance; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-006, KFM-IDX-MDP-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.2 KFM-P18-INV-092 — Tracking logs preserve reconstructable transaction history

Idea card metadata - Stable ID: KFM-P18-INV-092 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 92 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should require tracking-log design choices to be explicit when reconstructability or report consistency depends on transaction history.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should require tracking-log design choices to be explicit when reconstructability or report consistency depends on transaction history. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It supports replay, correction, and audit without pretending all historical states remain recoverable by default.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-EVD-002.

Dependencies - PROPOSED: TrackingLog policy - PROPOSED: Report snapshot manifest - PROPOSED: Reconstruction validator

Tensions - PROPOSED: A log can preserve evidence poorly if image strategy and reconstruction rules are inconsistent.

Expansion directions - PROPOSED: Define log_image_strategy, operation_code, transaction_id, and reconstructability_scope fields.

Open questions - NEEDS VERIFICATION: When should a KFM release snapshot be favored over reconstructing from transaction logs?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 17, 18, 242, 259, 260, 261, 262; basis: report inconsistency, tracking logs, before/after images, operation codes, and reconstruction algorithms; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-004, KFM-IDX-EVD-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.3 KFM-P18-INV-093 — Open-ended “now” periods need explicit semantics

Idea card metadata - Stable ID: KFM-P18-INV-093 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 93 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from

CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should define explicit semantics for open-ended current periods such as now, current, active, and unknown-end.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should define explicit semantics for open-ended current periods such as now, current, active, and unknown-end. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents active records from being treated as timeless truth or from breaking reproducibility during later correction.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-008. - CONFIRMED baseline link: KFM-IDX-CAT-007.

Dependencies - PROPOSED: Temporal vocabulary - PROPOSED: Valid-time schema - PROPOSED: Release-time policy

Tensions - PROPOSED: “Current” depends on observation time, source time, transaction time, and release time.

Expansion directions - PROPOSED: Add `temporal_open_end_policy` and `now_resolution_time` fields to temporal claim schemas.

Open questions - NEEDS VERIFICATION: Should KFM use symbolic now, bounded intervals, or release-resolved end times in public APIs?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 12, 13, 17, 18; basis: handling now, current-state extraction, prior-state queries, and date-column complexity; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-008, KFM-IDX-CAT-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.4 KFM-P18-INV-094 — Temporal partitioning and vacuuming are governance decisions

Idea card metadata - Stable ID: KFM-P18-INV-094 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 94 - Status label: PROPOSED - Carry-forward state: NEW - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should treat temporal partitioning, archival-store vacuuming, and purging as governance decisions with semantic consequences.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3

inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat temporal partitioning, archival-store vacuuming, and purging as governance decisions with semantic consequences. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes storage optimization visible as a potential evidence and audit tradeoff.

Related ideas - CONFIRMED baseline link: KFM-IDX-SEC-004. - CONFIRMED baseline link: KFM-IDX-PIP-004.

Dependencies - PROPOSED: Retention policy - PROPOSED: Archive manifest - PROPOSED: Vacuuming receipt

Tensions - PROPOSED: Purging old states can improve performance while weakening auditability.

Expansion directions - PROPOSED: Define `retention_class`, `purge_reason`, `vacuuming_specification`, and `lost_query_capability` fields.

Open questions - NEEDS VERIFICATION: What KFM object families must never be vacuumed without external archival proof?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 287, 288, 293, 294, 295, 298; basis: current/archive temporal partitioning and vacuuming semantics discussion; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.5 KFM-P18-INV-097 — Validator interfaces should be intention-revealing and assertion-backed

Idea card metadata - Stable ID: KFM-P18-INV-097 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 97 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM validators should expose intention-revealing interfaces, predictable side effects, and explicit assertions about outcomes.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM validators should expose intention-revealing interfaces, predictable side effects, and explicit assertions about outcomes. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes policy and validation code reviewable without requiring reviewers to infer intent from hidden implementation details.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-SEC-008.

Dependencies - PROPOSED: Validator registry - PROPOSED: Assertion contract - PROPOSED: No-side-effect policy

Tensions - PROPOSED: Validators that mutate inputs can make pass/fail results unreproducible.

Expansion directions - PROPOSED: Add `validator_intent`, `input_contract`, `output_assertions`, and `side_effect_policy` fields.

Open questions - NEEDS VERIFICATION: Should KFM require validators to be pure functions except for receipt emission?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 21, 27, 28, 29, 32, 51; basis: service assertions, intention-revealing interfaces, side-effect-free functions, and cohesive mechanisms; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-001, KFM-IDX-SEC-008 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.6 KFM-P18-INV-106 — LEAD/LAG as temporal change-detection primitives

Idea card metadata - Stable ID: KFM-P18-INV-106 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 106 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM should treat LEAD/LAG-style navigation calculations as explicit temporal comparisons with period, sort, and missing-row caveats.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat LEAD/LAG-style navigation calculations as explicit temporal comparisons with period, sort, and missing-row caveats. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It supports transparent change detection for hydrology, air, agriculture, and release deltas.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-003. - CONFIRMED baseline link: KFM-IDX-OBS-004.

Dependencies - PROPOSED: Temporal ordering policy - PROPOSED: Observation panel schema

Tensions - PROPOSED: Previous-row logic can be wrong if rows are missing, duplicated, or sorted by the wrong timestamp.

Expansion directions - PROPOSED: Add `previous_observation_basis` and `temporal_sort_key` fields.

Open questions - NEEDS VERIFICATION: How should KFM expose missing-prior-row caveats in derived deltas?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 13, 24; basis: LEAD/LAG and advanced analytic function discussion; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-003, KFM-IDX-OBS-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.7 KFM-P18-INV-119 — Temporal vacuuming as retention and semantics risk

Idea card metadata - Stable ID: KFM-P18-INV-119 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 119 - Status label: PROPOSED - Carry-forward state: NEW - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should treat temporal vacuuming and archival partition pruning as governed retention decisions because they can alter later query meaning.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat temporal vacuuming and archival partition pruning as governed retention decisions because they can alter later query meaning. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, run-time routing, or release state.

Why it matters - PROPOSED: It keeps storage management from silently weakening auditability and correction history.

Related ideas - CONFIRMED baseline link: KFM-IDX-SEC-010. - CONFIRMED baseline link: KFM-IDX-EVD-007.

Dependencies - PROPOSED: Retention policy - PROPOSED: Archive manifest - PROPOSED: Rollback target

Tensions - PROPOSED: Operational storage pressure may conflict with long-term evidence reconstruction.

Expansion directions - PROPOSED: Add `temporal_retention_policy` with prohibited deletion classes.

Open questions - NEEDS VERIFICATION: Which historical records are never eligible for vacuuming?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 293, 294; basis: transaction-time vacuuming, entity/temporal vacuuming, and semantics warning; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.8 KFM-P18-INV-120 — Domain events for governed state transitions

Idea card metadata - Stable ID: KFM-P18-INV-120 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 120 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should model consequential transitions—intake accepted, quarantine assigned, promotion denied, release published, correction issued—as domain events that experts care about.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should model consequential transitions—intake accepted, quarantine assigned, promotion denied, release published, correction issued—as domain events that experts care about. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes lifecycle history explainable without reverse-engineering state from current records.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-001. - CONFIRMED baseline link: KFM-IDX-EVD-005.

Dependencies - PROPOSED: Event schema - PROPOSED: PromotionReceipt - PROPOSED: Correction record

Tensions - PROPOSED: Too many low-value events can obscure the few that matter for governance.

Expansion directions - PROPOSED: Define a governed_event_type enum and event EvidenceRef requirements.

Open questions - NEEDS VERIFICATION: Which events are mandatory for release audit?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 20; basis: Domain Events definition and lifecycle-state explanation; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-001, KFM-IDX-EVD-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.9 KFM-P18-INV-145 — Tracking-log after-images for report-time reconstruction

Idea card metadata - Stable ID: KFM-P18-INV-145 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 145 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain

Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should use tracking-log or receipt-equivalent after-image records where report-time reconstruction is required for audits, corrections, or disputed releases.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should use tracking-log or receipt-equivalent after-image records where report-time reconstruction is required for audits, corrections, or disputed releases. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It supports answering why two reports differed and which state was valid at a release time.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-010. - CONFIRMED baseline link: KFM-IDX-OBS-003.

Dependencies - PROPOSED: RunReceipt - PROPOSED: CorrectionNotice - PROPOSED: RollbackReference

Tensions - PROPOSED: Full after-image retention may conflict with privacy, cost, or minimization policies.

Expansion directions - PROPOSED: Add retention-class options for after-image-like records.

Open questions - NEEDS VERIFICATION: Which lifecycle phases require after-image storage?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 219, 220, 235, 236, 237, 248; basis: tracking log and after-image reconstruction discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-010, KFM-IDX-OBS-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.10 KFM-P18-INV-146 — Published Language and Context Map as public contract guardrails

Idea card metadata - Stable ID: KFM-P18-INV-146 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 146 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should use Published Language and Context Map practices to keep cross-lane API terms, evidence terms, and public-facing vocabulary stable and inspectable.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category

before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should use Published Language and Context Map practices to keep cross-lane API terms, evidence terms, and public-facing vocabulary stable and inspectable. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps hydrology, fauna, archaeology, AI, map, and release teams from using the same term with different authority semantics.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-006. - CONFIRMED baseline link: KFM-IDX-MDP-004.

Dependencies - PROPOSED: Domain glossary - PROPOSED: Context map - PROPOSED: OpenAPI contract

Tensions - PROPOSED: Local domain nuance must be preserved without fragmenting public vocabulary.

Expansion directions - PROPOSED: Maintain cross-lane term mappings and anti-translation notes.

Open questions - NEEDS VERIFICATION: Which terms require project-wide published-language control?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 28, 29, 36; basis: Domain-Driven Design context map and published language patterns; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-006, KFM-IDX-MDP-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.11 KFM-P18-INV-170 — Period predicates for temporal overlap semantics

Idea card metadata - Stable ID: KFM-P18-INV-170 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 170 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should define temporal predicate semantics for before, meets, overlaps, during, starts, and finishes when joining claims, events, releases, and source observations.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should define temporal predicate semantics for before, meets, overlaps, during, starts, and finishes when joining claims, events, releases, and source observations. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reduces ambiguity in questions such as whether a hazard, permit, ownership assertion, observation, and release were simultaneous, adjacent, or merely overlapping.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-008. - CONFIRMED baseline link: KFM-IDX-CAT-004.

Dependencies - PROPOSED: Temporal contract - PROPOSED: Bitemporal query tests - PROPOSED: Claim relationship vocabulary

Tensions - PROPOSED: Closed, open, and closed-open interval choices can affect edge cases.

Expansion directions - PROPOSED: Create interval-boundary fixtures for exact adjacency, overlap, containment, and null/open-ended periods.

Open questions - NEEDS VERIFICATION: Which period representation should KFM standardize for public APIs and database queries?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 117, 126; basis: temporal SQL period predicate tables covering before, meets, overlaps, during, starts, finishes, and related operations; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-008, KFM-IDX-CAT-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.12 KFM-P18-INV-171 — Valid-time and transaction-time orthogonality

Idea card metadata - Stable ID: KFM-P18-INV-171 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 171 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should keep real-world valid time, source/observation time, system transaction time, and release time orthogonal where claims may be corrected or replayed.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should keep real-world valid time, source/observation time, system transaction time, and release time orthogonal where claims may be corrected or replayed. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents corrections to database knowledge from rewriting what was true, observed, believed, or published at a different time.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-010. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-DOC-002.

Dependencies - PROPOSED: Temporal schema - PROPOSED: Correction model - PROPOSED: ReleaseManifest

Tensions - PROPOSED: Maintaining multiple time dimensions increases query and explanation complexity.

Expansion directions - PROPOSED: Add public explanation labels for valid-time, recorded-time, release-time, and correction-time.

Open questions - NEEDS VERIFICATION: Which KFM lanes require full bitemporal storage versus lighter release-history receipts?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 277; basis: transaction-time state table chapter stating valid-time tables model changes in reality while transaction-time tables model changes in the database; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-010, KFM-IDX-EVD-001, KFM-IDX-DOC-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.13 KFM-P18-INV-172 — Bitemporal sequenced modifications need complexity disclosure

Idea card metadata - Stable ID: KFM-P18-INV-172 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 172 - Status label: PROPOSED - Carry-forward state: NEW - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should document when bitemporal modifications are optimized for representation size versus operational simplicity, and should include tests for the chosen strategy.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should document when bitemporal modifications are optimized for representation size versus operational simplicity, and should include tests for the chosen strategy. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents correction-safe temporal tables from becoming inscrutable or inconsistent because modification logic was too complex to audit.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-006. - CONFIRMED baseline link: KFM-IDX-MDP-005.

Dependencies - PROPOSED: Temporal modification tests - PROPOSED: Schema ADR - PROPOSED: Correction receipt

Tensions - PROPOSED: Simpler representations may store more rows, while optimized representations may be harder to verify.

Expansion directions - PROPOSED: Compare simple and minimized bitemporal modification strategies on fixtures.

Open questions - NEEDS VERIFICATION: Where should KFM accept extra rows to keep correction logic inspectable?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 323; basis: bitemporal modification discussion showing sequenced insertion/deletion complexity and representation-minimization tradeoff; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.14 KFM-P18-INV-186 — Contract-backed map registries and telemetry events

Idea card metadata - Stable ID: KFM-P18-INV-186 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 186 - Status label: PROPOSED - Carry-forward state: NEW - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM map registries, render hints, UI telemetry, and layer events should share schema-backed contracts so UI behavior remains inspectable and testable.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM map registries, render hints, UI telemetry, and layer events should share schema-backed contracts so UI behavior remains inspectable and testable. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It closes a gap between backend trust contracts and frontend map behavior.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-008. - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-OBS-002.

Dependencies - PROPOSED: ViewRegistry schema - PROPOSED: Telemetry schema - PROPOSED: LayerManifest - PROPOSED: Frontend validator

Tensions - PROPOSED: Telemetry can drift into surveillance or analytics overreach without policy constraints.

Expansion directions - PROPOSED: Add telemetry minimization and schema validation gates to map-shell tests.

Open questions - NEEDS VERIFICATION: Which UI events are necessary for trust-state auditing and which should be denied?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 14, 15; basis: Pydantic/JSON Schema-backed UI registries, map configs, telemetry events, and analytics events; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.15 KFM-P18-INV-189 — Context maps for cross-domain translation and influence

Idea card metadata - Stable ID: KFM-P18-INV-189 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 189 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should maintain context maps that identify domain models, translation points, shared kernels, partnerships, and upstream/downstream influence among domain lanes and platform services.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should maintain context maps that identify domain models, translation points, shared kernels, partnerships, and upstream/downstream influence among domain lanes and platform services. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It helps prevent hydrology, roads, settlements, ecology, and AI/UI terms from bleeding into each other without translation.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-004. - CONFIRMED baseline link: KFM-IDX-DAT-001. - CONFIRMED baseline link: KFM-IDX-MDP-003.

Dependencies - PROPOSED: Domain context register - PROPOSED: Translation contract - PROPOSED: ADR index - PROPOSED: Review ownership

Tensions - PROPOSED: Too much mapping can become documentation overhead if not linked to validators and contracts.

Expansion directions - PROPOSED: Add context-map fragments to the category atlas and future architecture chapters.

Open questions - NEEDS VERIFICATION: Where should KFM store context-map metadata without creating a parallel contract home?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 36, 37; basis: Context Map and Partnership patterns for bounded contexts, points of contact, explicit translation, and coordinated integration; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-004, KFM-IDX-DAT-001, KFM-IDX-MDP-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.16 KFM-P18-INV-193 — Temporal referential integrity over validity periods

Idea card metadata - Stable ID: KFM-P18-INV-193 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 193 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM temporal claims should validate referential integrity across valid-time and transaction-time periods, not only across static identifiers.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM temporal claims should validate referential integrity across valid-time and transaction-time periods, not only across static identifiers. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, run-time routing, or release state.

Why it matters - PROPOSED: It prevents a claim from referencing an entity, geography, owner, source, or status outside the period in which that reference is valid.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-DAT-001. - CONFIRMED baseline link: KFM-IDX-CAT-007.

Dependencies - PROPOSED: Temporal schema - PROPOSED: Referenced object lifecycle - PROPOSED: Integrity validator

Tensions - PROPOSED: Temporal integrity expressions can be complex and dialect-sensitive.

Expansion directions - PROPOSED: Add temporal referential-integrity fixtures for geography versions, ownership periods, and source authority intervals.

Open questions - NEEDS VERIFICATION: Which KFM relationships require full bitemporal integrity versus current-only checks?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 351, 352, 353; basis: valid-time and bitemporal foreign-key integrity constraints using sequenced/current checks over valid and transaction periods; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-001, KFM-IDX-DAT-001, KFM-IDX-CAT-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.17 KFM-P18-INV-194 — Now-relative validity and forever sentinel risk

Idea card metadata - Stable ID: KFM-P18-INV-194 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 194 - Status label: PROPOSED - Carry-forward state: NEW - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should make now-relative validity and forever-style end-time sentinels explicit so public users do not confuse current-known state with permanent truth.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should make now-relative validity and forever-style end-time sentinels explicit so public users do not confuse current-known state with permanent truth. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps “current” records from silently becoming indefinite claims.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-DAT-001. - CONFIRMED baseline link: KFM-IDX-OBS-003.

Dependencies - PROPOSED: Temporal field convention - PROPOSED: Current-state projection - PROPOSED: Staleness policy

Tensions - PROPOSED: Sentinel dates are convenient but can look like real dates in public records.

Expansion directions - PROPOSED: Add explicit `current_known`, `valid_until_unknown`, and sentinel-date handling to temporal schema notes.

Open questions - NEEDS VERIFICATION: Should KFM use sentinel dates, nullable ends, explicit status values, or separate current projections?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 230, 355; basis: discussion of CURRENT DATE, forever-style valid end values, current stores, and the passage-of-time effects on temporal tables; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.18 KFM-P18-INV-195 — Temporal partitioning requires consistency receipts

Idea card metadata - Stable ID: KFM-P18-INV-195 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 195 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM may use current/history or bitemporal partitioning for performance, but each partition strategy should emit consistency receipts and disclose how records enter, leave, or replicate across stores.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM may use current/history or bitemporal partitioning for performance, but each partition strategy should emit consistency receipts and disclose how records enter, leave, or replicate across stores. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents performance-oriented projections from becoming hidden alternate truth stores.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-002. - CONFIRMED baseline link: KFM-IDX-PIP-009. - CONFIRMED baseline link: KFM-IDX-OBS-003.

Dependencies - PROPOSED: Partition strategy - PROPOSED: Consistency validator - PROPOSED: Projection receipt

Tensions - PROPOSED: A current projection can change with time alone, even without a new source event.

Expansion directions - PROPOSED: Require partition strategy metadata and projection receipts for temporal stores.

Open questions - NEEDS VERIFICATION: Which temporal projections are allowed to support public APIs, and with what stale-state labels?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 230, 354, 355; basis: temporal partitioning into current/history or bitemporal stores, including replication and application consistency responsibilities; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-002, KFM-IDX-PIP-009, KFM-IDX-OBS-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.19 KFM-P18-INV-196 — Temporal dialect fixtures for period predicates

Idea card metadata - Stable ID: KFM-P18-INV-196 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 196 - Status label: PROPOSED - Carry-forward state: NEW - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM temporal validators should test period predicates, constructors, overlaps, meets, starts, finishes, and difference operations against the actual SQL dialect or implementation used.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3

inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM temporal validators should test period predicates, constructors, overlaps, meets, starts, finishes, and difference operations against the actual SQL dialect or implementation used. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It avoids false confidence in temporal logic when a database lacks native period support or implements operations differently.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-001. - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-CAT-007.

Dependencies - PROPOSED: Temporal predicate library - PROPOSED: Database dialect decision - PROPOSED: Fixture matrix

Tensions - PROPOSED: Portability can fail silently when temporal operations are emulated in SQL.

Expansion directions - PROPOSED: Create dialect-specific temporal predicate tests before relying on time-range joins.

Open questions - NEEDS VERIFICATION: Which database engine and temporal feature set will the mounted repo actually use?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 115, 126, 127, 128; basis: period predicates and DBMS-specific period-operation equivalents for overlaps, meets, starts, finishes, constructors, and casts; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.20 KFM-P18-INV-234 — Assertions and invariants as machine-checkable governance

Idea card metadata - Stable ID: KFM-P18-INV-234 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 234 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM validators should encode postconditions and invariants for key object families so reviewers do not have to infer side effects from procedural code.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM validators should encode postconditions and invariants for key object families so reviewers do not have to infer side effects from procedural code. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. -

UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - **NEEDS VERIFICATION:** Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - **PROPOSED:** It turns trust-bearing behavior into checkable contracts rather than hidden runtime assumptions.

Related ideas - **CONFIRMED** baseline link: KFM-IDX-DOC-001. - **CONFIRMED** baseline link: KFM-IDX-POL-005. - **CONFIRMED** baseline link: KFM-IDX-EVD-001.

Dependencies - **PROPOSED:** Validator contract - **PROPOSED:** Object-family invariant list - **PROPOSED:** Fixture suite

Tensions - **PROPOSED:** Some invariants are policy decisions and need governance review, not just engineering implementation.

Expansion directions - **PROPOSED:** Create invariant registries for EvidenceBundle, ReleaseManifest, PromotionDecision, LayerManifest, and CorrectionNotice.

Open questions - **NEEDS VERIFICATION:** Which invariants are global KFM law versus domain-lane policy?

Future work - **PROPOSED:** Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - **CONFIRMED** source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 20, 29; basis: domain events explain meaningful state changes; assertions state postconditions and invariants of classes and aggregates; Phase 1 role: reference_source_candidate.

Carry-forward note - **CONFIRMED:** Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-001, KFM-IDX-POL-005, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - **CONFIRMED:** Required schema fields are present in this Markdown card. - **CONFIRMED:** No prior-pass body text is inlined. - **CONFIRMED:** Narrative claims use explicit truth labels. - **UNKNOWN:** Repository implementation status remains unverified.

8.4.21 KFM-P18-INV-237 — Tracking-log variants need replay semantics and after-image policy

Idea card metadata - Stable ID: KFM-P18-INV-237 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 237 - Status label: **PROPOSED** - Carry-forward state: **EXPANDED** - Category: **MOD** — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: **PROPOSED** assignment derived from **CONFIRMED** Phase 3 inventory; subject to Phase 5 card-level review - Source support label: **CONFIRMED** - Implementation maturity: **UNKNOWN** — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-007

Normalized statement - **PROPOSED:** KFM change logs should specify whether they store before-images, after-images, both images, or operations, and how those records reconstruct state for audit replay.

Detailed explanation - **CONFIRMED:** Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - **CONFIRMED:** Phase 3 inventory recorded this idea with source support label **CONFIRMED** and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - **PROPOSED:** In KFM, this idea should be treated as a governed design candidate: KFM change logs should specify whether they store before-images, after-images, both images, or operations, and how those records reconstruct state for audit replay. - **PROPOSED:** Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - **PROPOSED:** Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - **UNKNOWN:** No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - **NEEDS VERIFICATION:** Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - **PROPOSED:** It makes rollback and audit replay feasible without ambiguity about what a logged change means.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-PIP-009. - CONFIRMED baseline link: KFM-IDX-CAT-006.

Dependencies - PROPOSED: Change log schema - PROPOSED: Replay plan - PROPOSED: Rollback target - PROPOSED: Transaction-time fields

Tensions - PROPOSED: Operation logs may be compact but harder to replay consistently than after-image logs.

Expansion directions - PROPOSED: Define `change_log_mode` and `replay_semantics` fields for correction, release, and rollback records.

Open questions - NEEDS VERIFICATION: Which KFM logs must be replayable versus merely auditable?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 262, 277; basis: tracking logs can record before-images, after-images, both, or operations; after-images support transaction-time reconstruction; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-001, KFM-IDX-PIP-009, KFM-IDX-CAT-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.22 KFM-P18-INV-239 — Transaction-time state tables as audit truth for repo-independent replay

Idea card metadata - Stable ID: KFM-P18-INV-239 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 239 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM audit stores should preserve transaction-time periods so reviewers can reconstruct what the system believed at a past audit or release moment.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM audit stores should preserve transaction-time periods so reviewers can reconstruct what the system believed at a past audit or release moment. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It supports proof, correction, and rollback investigations when published truth changes over time.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-PIP-009. - CONFIRMED baseline link: KFM-IDX-CAT-006.

Dependencies - PROPOSED: Transaction-time columns - PROPOSED: Release-time reference - PROPOSED: Audit query fixtures

Tensions - PROPOSED: Transaction time records system history, not necessarily real-world validity.

Expansion directions - PROPOSED: Create audit_query fixtures for what-was-published, what-was-known, and what-was-corrected questions.

Open questions - NEEDS VERIFICATION: How should KFM expose transaction-time history without confusing it with valid-time claim history?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 277, 278, 337; basis: transaction-time state table represents a sequence of snapshot states over time with timestamped periods; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-001, KFM-IDX-PIP-009, KFM-IDX-CAT-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.23 KFM-P18-INV-266 — Temporal design should separate static schema design from temporal annotation

Idea card metadata - Stable ID: KFM-P18-INV-266 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 266 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM temporal schema design should first clarify the nontemporal domain model, then annotate entities, relationships, attributes, and keys with temporal semantics.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM temporal schema design should first clarify the nontemporal domain model, then annotate entities, relationships, attributes, and keys with temporal semantics. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reduces early schema complexity and makes temporal decisions explicit rather than implicit in ad hoc date columns.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-PIP-003. - CONFIRMED baseline link: KFM-IDX-MDP-003.

Dependencies - PROPOSED: Domain model - PROPOSED: temporal annotation ADR - PROPOSED: schema validator

Tensions - PROPOSED: Delaying temporal decisions too long can also miss requirements for history and correction.

Expansion directions - PROPOSED: Add temporal_annotation_register with valid_time, transaction_time, source_time, release_time, and correction_time fields per object family.

Open questions - NEEDS VERIFICATION: Which KFM object families need temporal annotation before first implementation?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 343, 344; basis: temporal application design first models current reality, then augments entities, relationships, attributes, and keys with temporal semantics; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-PIP-003, KFM-IDX-MDP-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.24 KFM-P18-INV-269 — Factories enforce valid creation of receipts and envelopes

Idea card metadata - Stable ID: KFM-P18-INV-269 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 269 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM object creation for receipts, bundles, envelopes, and manifests should use controlled constructors or factories that enforce invariants before records exist.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM object creation for receipts, bundles, envelopes, and manifests should use controlled constructors or factories that enforce invariants before records exist. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, run-time routing, or release state.

Why it matters - PROPOSED: It prevents partially valid trust objects from being passed downstream as if they were complete.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-PIP-003. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: Factory or constructor contract - PROPOSED: schema validation - PROPOSED: invariant tests

Tensions - PROPOSED: Factory discipline can be bypassed if raw JSON/YAML is hand-authored without validation.

Expansion directions - PROPOSED: Add factory acceptance tests for EvidenceBundle, DecisionEnvelope, RunReceipt, ReleaseManifest, and PromotionReceipt.

Open questions - NEEDS VERIFICATION: Should all trust objects be authored only through validators/factories, or can manual draft records exist in WORK?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 18; basis: factories encapsulate complex object creation, create aggregates/value objects as a piece, and enforce invariants; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-001, KFM-IDX-PIP-003, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.25 KFM-P18-INV-270 — Separate Ways prevents forced integration of unrelated lanes

Idea card metadata - Stable ID: KFM-P18-INV-270 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 270 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should allow bounded contexts to remain separate when integration would add more ambiguity than value.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should allow bounded contexts to remain separate when integration would add more ambiguity than value. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It protects domain specificity and prevents shared schemas from becoming a Big Ball of Mud.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-MDP-003. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: Context map - PROPOSED: ADR - PROPOSED: shared-kernel decision

Tensions - PROPOSED: Too much separation can duplicate governance objects or impede cross-domain evidence.

Expansion directions - PROPOSED: Add context_integration_decision with shared_kernel, anticorruption_layer, separate_ways, and rationale fields.

Open questions - NEEDS VERIFICATION: Which KFM domain pairs should explicitly choose Separate Ways until a source-backed relation emerges?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 37, 38; basis: Separate Ways cuts unrelated bounded contexts loose; Big Ball of Mud arises when boundaries are inconsistent and models mix; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-001, KFM-IDX-MDP-003, KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.26 KFM-P18-INV-287 — Now-time handling must be explicit in temporal records

Idea card metadata - Stable ID: KFM-P18-INV-287 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 287 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM temporal schemas should model open-ended or current validity explicitly instead of treating “now” as an ordinary static date.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM temporal schemas should model open-ended or current validity explicitly instead of treating “now” as an ordinary static date. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It avoids ambiguity in current-state claims, stale data behavior, and release-time interpretation.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-007. - CONFIRMED baseline link: KFM-IDX-DAT-012.

Dependencies - PROPOSED: valid_time interval fields - PROPOSED: release_time fields - PROPOSED: stale-state policy

Tensions - PROPOSED: Current validity changes with time even when the stored record is unchanged.

Expansion directions - PROPOSED: Add open_interval_semantics to temporal schema notes.

Open questions - NEEDS VERIFICATION: Should KFM represent “current” as null end, infinity, or explicit open interval by schema family?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 143; basis: The temporal SQL source identifies Handling Now as a design issue within defining state tables.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-007, KFM-IDX-DAT-012 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.27 KFM-P18-INV-288 — Temporal keys and uniqueness cross valid periods

Idea card metadata - Stable ID: KFM-P18-INV-288 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 288 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain

Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM identity and uniqueness checks should account for temporal intervals so records can be unique within valid-time scopes without erasing history.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM identity and uniqueness checks should account for temporal intervals so records can be unique within valid-time scopes without erasing history. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It protects KFM from false duplicates and false conflicts in time-aware domain lanes.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-002. - CONFIRMED baseline link: KFM-IDX-CAT-007.

Dependencies - PROPOSED: temporal key schema - PROPOSED: interval overlap validator - PROPOSED: identity strategy

Tensions - PROPOSED: Stable identity across time can conflict with changing names, geometries, or classifications.

Expansion directions - PROPOSED: Add overlap and temporal uniqueness validators for time-aware fixtures.

Open questions - NEEDS VERIFICATION: Which KFM object families require uniqueness by valid time versus release time?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 141, 143; basis: The temporal SQL source treats temporal keys and reexamined uniqueness as core state-table design concerns.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-002, KFM-IDX-CAT-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.28 KFM-P18-INV-289 — Temporal referential integrity spans historical state

Idea card metadata - Stable ID: KFM-P18-INV-289 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 289 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM temporal references should verify that linked records are valid over compatible periods, not merely that a referenced ID exists somewhere in history.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM temporal references should verify that linked records are valid over compatible periods, not merely that a referenced ID exists somewhere in history. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, run-time routing, or release state.

Why it matters - PROPOSED: It makes cross-domain historical claims safer when geography, ownership, status, or source authority changes over time.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-007. - CONFIRMED baseline link: KFM-IDX-EVD-004.

Dependencies - PROPOSED: temporal relationship schema - PROPOSED: valid-time compatibility validator - PROPOSED: EvidenceBundle references

Tensions - PROPOSED: A relationship may be true for one period and false for another while both endpoints persist.

Expansion directions - PROPOSED: Add relationship interval validation for temporal edges.

Open questions - NEEDS VERIFICATION: How should KFM handle evidence that supports a relation but not its exact start/end period?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 150, 151, 152; basis: The temporal SQL source treats referential integrity as a temporal state-table issue with dedicated discussion beyond ordinary keys.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-007, KFM-IDX-EVD-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.29 KFM-P18-INV-290 — Temporal duplicate elimination is canonicalization, not cosmetic cleanup

Idea card metadata - Stable ID: KFM-P18-INV-290 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 290 - Status label: PROPOSED - Carry-forward state: NEW - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM temporal processing should coalesce or eliminate duplicate temporal states only through documented rules that preserve evidence and interval semantics.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM temporal processing should coalesce or eliminate duplicate temporal states only through documented rules that preserve evidence and interval semantics. - PROPOSED: Phase 3 scope

note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents redundant intervals from cluttering outputs while protecting temporal meaning and provenance.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-002. - CONFIRMED baseline link: KFM-IDX-PIP-004.

Dependencies - PROPOSED: temporal canonicalization rule - PROPOSED: coalescing receipt - PROPOSED: query validator

Tensions - PROPOSED: Coalescing can simplify display but may hide separate evidence events if not recorded.

Expansion directions - PROPOSED: Add coalescing receipts that list merged interval IDs and evidence references.

Open questions - NEEDS VERIFICATION: Should coalesced public outputs retain individual source-event links in the Evidence Drawer?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 182, 183, 184; basis: The temporal SQL source has a dedicated discussion of eliminating duplicates within sequenced queries and temporal state-table querying.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.30 KFM-P18-INV-291 — Bitemporal tables separate world validity from database correction time

Idea card metadata - Stable ID: KFM-P18-INV-291 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 291 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM audit-critical records should distinguish valid time from transaction or correction time so past claims and later corrections can both be inspected.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM audit-critical records should distinguish valid time from transaction or correction time so past claims and later corrections can both be inspected. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support

confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It supports correction lineage, rollback, and retrospective analysis without rewriting history.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-007. - CONFIRMED baseline link: KFM-IDX-PIP-006. - CONFIRMED baseline link: KFM-IDX-EVD-005.

Dependencies - PROPOSED: bitemporal schema profile - PROPOSED: correction notice - PROPOSED: release manifest

Tensions - PROPOSED: Bitemporal completeness increases schema complexity but is essential where correction history matters.

Expansion directions - PROPOSED: Define which KFM object families require bitemporal storage versus receipt-only history.

Open questions - NEEDS VERIFICATION: Should public APIs expose transaction time directly or only through correction lineage cards?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 299, 300, 301, 307; basis: The temporal SQL source introduces bitemporal tables and separates valid-time and transaction-time state-table concepts.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-007, KFM-IDX-PIP-006, KFM-IDX-EVD-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.31 KFM-P18-INV-304 — Domain events carry the meaning of state change

Idea card metadata - Stable ID: KFM-P18-INV-304 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 304 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should use domain events to record why a governed object changed, not only that its state or attributes changed.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should use domain events to record why a governed object changed, not only that its state or attributes changed. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It lets review, correction, promotion, denial, and rollback histories preserve domain meaning.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: domain event schema - PROPOSED: actor and time fields - PROPOSED: state transition receipt

Tensions - PROPOSED: Event streams can become noisy unless event types remain domain-significant.

Expansion directions - PROPOSED: Add event_meaning and transition_reason fields to promotion and correction event records.

Open questions - NEEDS VERIFICATION: Which KFM lifecycle changes deserve domain events rather than log-only entries?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 20; basis: DDD reference says audit trails trace state but are not usually suited for program logic, while domain events capture happenings that domain experts care about and the causes of state changes.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-004, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.32 KFM-P18-INV-305 — Assertions and invariants as machine-checkable governance contracts

Idea card metadata - Stable ID: KFM-P18-INV-305 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 305 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM service contracts, validators, and aggregate modifiers should declare post-conditions and invariants in tests or documentation when they cannot be encoded directly.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM service contracts, validators, and aggregate modifiers should declare post-conditions and invariants in tests or documentation when they cannot be encoded directly. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes governance behavior reviewable and testable rather than hidden in implementation paths.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-007. - CONFIRMED baseline link: KFM-IDX-MDP-002.

Dependencies - PROPOSED: contract assertions - PROPOSED: unit tests - PROPOSED: validator outcome fixtures

Tensions - PROPOSED: Assertions must be precise enough to test without becoming too rigid for domain evolution.

Expansion directions - PROPOSED: Add `invariant_assertions` sections to core KFM object-family contracts.

Open questions - NEEDS VERIFICATION: Which invariants are doctrine-level and which are lane-specific?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 29; basis: DDD reference says operations should state post-conditions and invariants, and if assertions cannot be coded directly they should be written into automated unit tests, documentation, or diagrams.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-007, KFM-IDX-MDP-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.33 KFM-P18-INV-308 — Valid-time and transaction-time roles must be declared per claim family

Idea card metadata - Stable ID: KFM-P18-INV-308 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 308 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM claim families should declare whether they model reality over valid time, table history over transaction time, or both.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM claim families should declare whether they model reality over valid time, table history over transaction time, or both. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents a record history from being mistaken for a history of reality, or vice versa.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-004. - CONFIRMED baseline link: KFM-IDX-PIP-004.

Dependencies - PROPOSED: temporal role enum - PROPOSED: claim family registry - PROPOSED: Evidence-Bundle temporal fields

Tensions - PROPOSED: Both time axes may be necessary but expensive to model and explain.

Expansion directions - PROPOSED: Add `temporal_role = valid_time | transaction_time | bitemporal | user_defined_time` to object-family registry.

Open questions - NEEDS VERIFICATION: Which KFM domains need bitemporal support at first release?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 47, 273; basis: Temporal SQL source distinguishes valid-time tables, transaction-time tracking logs, and bitemporal tables, and notes that valid time and transaction time are orthogonal.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-004, KFM-IDX-PIP-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.34 KFM-P18-INV-309 — Bitemporal modification complexity needs release limitation labels

Idea card metadata - Stable ID: KFM-P18-INV-309 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 309 - Status label: PROPOSED - Carry-forward state: NEW - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM bitemporal updates and corrections should disclose representation-minimization choices and complexity when release history depends on them.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM bitemporal updates and corrections should disclose representation-minimization choices and complexity when release history depends on them. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes correction-safe histories honest about how temporal representation was maintained.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-010. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: bitemporal update receipt - PROPOSED: representation policy - PROPOSED: temporal regression tests

Tensions - PROPOSED: Minimized representations are compact but can make updates harder to audit.

Expansion directions - PROPOSED: Add representation_policy and temporal_update_strategy fields to bitemporal receipts.

Open questions - NEEDS VERIFICATION: Should KFM prefer simpler non-minimized bitemporal history for auditability?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 307, 323; basis: Temporal SQL source illustrates current valid-time/current transaction-time insertion in a bitemporal table and notes that optimized sequenced insertions can minimize rows but add substantial complexity and cost.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.35 KFM-P18-INV-349 — Temporal table design should declare whether history is current-state, valid-time, transaction-time, or bitemporal

Idea card metadata - Stable ID: KFM-P18-INV-349 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 349 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM claim-family design should declare whether its records are current-state only, valid-time state, transaction-time state, tracking-log, or bitemporal before schema and constraint design.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM claim-family design should declare whether its records are current-state only, valid-time state, transaction-time state, tracking-log, or bitemporal before schema and constraint design. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents ad hoc date columns from undermining keys, constraints, correction, and public explanation.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-008. - CONFIRMED baseline link: KFM-IDX-PIP-010.

Dependencies - PROPOSED: claim family - PROPOSED: temporal type - PROPOSED: valid-time field - PROPOSED: transaction-time field - PROPOSED: constraint rule

Tensions - PROPOSED: A richer temporal model increases implementation burden and must be justified by claim significance.

Expansion directions - PROPOSED: Require temporal_type in claim-family schema notes.

Open questions - NEEDS VERIFICATION: Which KFM domains can remain current-state only without weakening correction history?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 111, 143, 277, 343; basis: Developing Time-Oriented Database Applications in SQL separates valid-time, transaction-time, bitemporal tables, tracking logs, querying, modification, and design chapters.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-008, KFM-IDX-PIP-010 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.36 KFM-P18-INV-374 — Domain events as auditable change facts

Idea card metadata - Stable ID: KFM-P18-INV-374 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 374 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should represent consequential domain changes as domain events that domain experts care about, separate from storage updates and downstream projections.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should represent consequential domain changes as domain events that domain experts care about, separate from storage updates and downstream projections. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It gives KFM a language for change facts that can feed receipts, review queues, and public correction history.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-010. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: bounded context - PROPOSED: event schema - PROPOSED: aggregate state - PROPOSED: projection

Tensions - PROPOSED: Not every implementation update is a domain event; over-eventing can create noise.

Expansion directions - PROPOSED: Define a domain_event_candidate checklist for KFM lanes.

Open questions - NEEDS VERIFICATION: Which KFM changes deserve domain events rather than run-only receipts?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 20; basis: Domain-Driven Design Reference defines Domain Events as something that happened that domain experts care about.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-010, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.37 KFM-P18-INV-375 — Context maps as cross-lane dependency governance

Idea card metadata - Stable ID: KFM-P18-INV-375 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 375 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should maintain context maps that show bounded-context relationships among domain lanes, shared governance kernels, external source adapters, and public API languages.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should maintain context maps that show bounded-context relationships among domain lanes, shared governance kernels, external source adapters, and public API languages. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes dependencies and translation boundaries inspectable before cross-lane models drift into hidden coupling.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-006. - CONFIRMED baseline link: KFM-IDX-MDP-003.

Dependencies - PROPOSED: bounded context - PROPOSED: shared kernel - PROPOSED: anticorruption layer - PROPOSED: published language

Tensions - PROPOSED: A context map can become stale unless tied to ADR and schema changes.

Expansion directions - PROPOSED: Create a context_map_update requirement for cross-lane schema changes.

Open questions - NEEDS VERIFICATION: Should context maps live in front matter, ADRs, or domain chapter appendices?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 35, 36; basis: Domain-Driven Design Reference describes bounded contexts and Context Map as a large-scale view of model development across a project and integrations.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-006, KFM-IDX-MDP-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.38 KFM-P18-INV-389 — Site geodatabase structures as field investigation schemas

Idea card metadata - Stable ID: KFM-P18-INV-389 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 389 - Status label: PROPOSED - Carry-forward state: NEW - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM field archaeology and 3D-investigation lanes should treat site geodatabase schemes as domain-specific schemas requiring documentation, review, and migration discipline.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this

idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM field archaeology and 3D-investigation lanes should treat site geodatabase schemes as domain-specific schemas requiring documentation, review, and migration discipline. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A field geodatabase encodes what can be observed, related, and later interpreted; it is part of the evidence model.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-004. - CONFIRMED baseline link: KFM-IDX-MDP-003.

Dependencies - PROPOSED: site schema - PROPOSED: field record - PROPOSED: 3D boundary model - PROPOSED: migration note

Tensions - PROPOSED: Legacy geodatabases may not map cleanly to KFM schema authority.

Expansion directions - PROPOSED: Create a geodatabase-to-KFM schema migration profile.

Open questions - NEEDS VERIFICATION: How should KFM preserve legacy site geodatabase semantics?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 66, 67, 75; basis: Archaeological 3D GIS includes site geodatabase schemes and 3D GIS systems developed to support field investigation.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.39 KFM-P18-INV-392 — Temporal uniqueness constraints include validity semantics

Idea card metadata - Stable ID: KFM-P18-INV-392 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 392 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM temporal tables should define uniqueness and primary-key constraints over identity plus validity semantics rather than treating dates as ordinary appended columns.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM temporal tables should define uniqueness and primary-key constraints over identity plus validity semantics rather than treating dates as ordinary appended columns. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION:

Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Time-aware evidence can duplicate or contradict itself unless uniqueness rules know which periods overlap or supersede.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-011. - CONFIRMED baseline link: KFM-IDX-PIP-010.

Dependencies - PROPOSED: entity identity - PROPOSED: valid period - PROPOSED: temporal key - PROPOSED: integrity validator

Tensions - PROPOSED: Temporal uniqueness rules are more complex than ordinary primary keys.

Expansion directions - PROPOSED: Add temporal_key_profile to time-aware schema records.

Open questions - NEEDS VERIFICATION: Which KFM tables need overlap-denial versus overlap-allowance rules?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 117, 121, 126; basis: Developing Time-Oriented Database Applications in SQL discusses temporal keys, uniqueness, and referential integrity in time-varying tables.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-011, KFM-IDX-PIP-010 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.40 KFM-P18-INV-394 — Temporal migration separates static design from history annotation

Idea card metadata - Stable ID: KFM-P18-INV-394 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 394 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should migrate non-temporal tables into temporal claim families by first validating the static schema, then adding valid-time, transaction-time, or bitemporal annotation deliberately.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should migrate non-temporal tables into temporal claim families by first validating the static schema, then adding valid-time, transaction-time, or bitemporal annotation deliberately. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents teams from adding date columns ad hoc and accidentally breaking keys, constraints, and query meaning.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-006. - CONFIRMED baseline link: KFM-IDX-DAT-011.

Dependencies - PROPOSED: static schema - PROPOSED: temporal annotation - PROPOSED: migration plan - PROPOSED: constraint review

Tensions - PROPOSED: Retrofitting time semantics can expose prior modeling mistakes.

Expansion directions - PROPOSED: Create temporal_migration_checklist for KFM schema evolution.

Open questions - NEEDS VERIFICATION: Which existing claim families should remain non-temporal?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 343, 355, 382; basis: Developing Time-Oriented Database Applications in SQL explains temporal database design as a staged process from conceptual and logical design through application development.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-006, KFM-IDX-DAT-011 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.41 KFM-P18-INV-395 — Lane-specific ubiquitous language as anti-drift glossary

Idea card metadata - Stable ID: KFM-P18-INV-395 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 395 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: Each KFM domain or subsystem lane should maintain a ubiquitous-language glossary that distinguishes domain terms, KFM governance terms, and external-source vocabulary.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: Each KFM domain or subsystem lane should maintain a ubiquitous-language glossary that distinguishes domain terms, KFM governance terms, and external-source vocabulary. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Shared words like observation, event, release, source, and claim can drift across lanes unless vocabulary boundaries are explicit.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-006. - CONFIRMED baseline link: KFM-IDX-MDP-003.

Dependencies - PROPOSED: bounded context - PROPOSED: glossary - PROPOSED: published language - PROPOSED: source vocabulary

Tensions - PROPOSED: Glossaries can become stale unless linked to schemas and ADRs.

Expansion directions - PROPOSED: Add glossary_update_required checks for schema or contract PRs.

Open questions - NEEDS VERIFICATION: Should glossary ownership sit with docs stewards or domain owners?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 3, 6, 36; basis: Domain-Driven Design Reference defines ubiquitous language as language structured around the domain model and used within a bounded context; it also covers published language.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-006, KFM-IDX-MDP-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.42 KFM-P18-INV-439 — Temporal vacuuming policy for archival retention versus deletion

Idea card metadata - Stable ID: KFM-P18-INV-439 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 439 - Status label: PROPOSED - Carry-forward state: NEW - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should define when temporal records may be compacted, archived, or retained, without treating vacuuming as silent deletion of evidence.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should define when temporal records may be compacted, archived, or retained, without treating vacuuming as silent deletion of evidence. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Storage maintenance can destroy audit value unless retention and archival rules are explicit.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-005. - CONFIRMED baseline link: KFM-IDX-EVD-006.

Dependencies - PROPOSED: retention policy - PROPOSED: proof archive - PROPOSED: rollback target - PROPOSED: audit ledger

Tensions - PROPOSED: Privacy obligations may require deletion that conflicts with audit retention.

Expansion directions - PROPOSED: Add temporal_retention_policy to release and correction runbooks.

Open questions - NEEDS VERIFICATION: How should KFM reconcile right-to-erasure obligations with immutable proof packs?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 293, 294, 295, 296, 298; basis: Developing Time-Oriented Database Applications in SQL discusses vacuuming in temporal table contexts.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.43 KFM-P18-INV-463 — Transaction-time presence periods for audit history

Idea card metadata - Stable ID: KFM-P18-INV-463 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 463 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM audit tables should distinguish transaction-time presence periods from valid-time truth periods so corrections can show when a record was stored, changed, or logically deleted.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM audit tables should distinguish transaction-time presence periods from valid-time truth periods so corrections can show when a record was stored, changed, or logically deleted. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A corrected claim needs both what was true in the world and when KFM believed or stored it.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-002. - CONFIRMED baseline link: KFM-IDX-EVD-006.

Dependencies - PROPOSED: audit log - PROPOSED: bitemporal table - PROPOSED: correction record - PROPOSED: rollback target

Tensions - PROPOSED: Transaction time can reveal operational history that may be sensitive or noisy.

Expansion directions - PROPOSED: Add transaction-time examples to correction and rollback fixtures.

Open questions - NEEDS VERIFICATION: Which audit periods are public, steward-only, or internal?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 278, 307, 323; basis: The temporal database source defines transaction-time state tables as rows timestamped by a period of presence and illustrates bitemporal modifications over valid and transaction time.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-002, KFM-IDX-EVD-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.44 KFM-P18-INV-489 — Valid-time and transaction-time split for public claim history

Idea card metadata - Stable ID: KFM-P18-INV-489 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 489 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should separate when a claim was true in the world, when the source asserted it, when KFM ingested it, and when KFM released it.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should separate when a claim was true in the world, when the source asserted it, when KFM ingested it, and when KFM released it. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Temporal separation is essential for corrections, stale-state labels, and historical explanations.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-PIP-010.

Dependencies - PROPOSED: valid time - PROPOSED: transaction time - PROPOSED: claim history - PROPOSED: release manifest

Tensions - PROPOSED: Many sources lack enough timestamps to fully populate all time dimensions.

Expansion directions - PROPOSED: Make temporal_scope fields explicit in Phase 4 categories and Phase 5 idea cards.

Open questions - NEEDS VERIFICATION: Which time dimensions are mandatory for each source role?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 37, 43; basis: The temporal database source distinguishes valid-time state tables, which record enterprise history, from transaction-time state tables, which record when facts are stored in the database.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-001, KFM-IDX-PIP-010 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.45 KFM-P18-INV-490 — Temporal referential-integrity gate for history tables

Idea card metadata - Stable ID: KFM-P18-INV-490 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 490 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM validators should test temporal foreign-key relationships so time-bounded observations do not point to nonexistent or temporally invalid source, geography, or identity records.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM validators should test temporal foreign-key relationships so time-bounded observations do not point to nonexistent or temporally invalid source, geography, or identity records. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A valid-looking relation can be false if the related record did not exist or was not valid during the claimed interval.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: referential integrity - PROPOSED: history table - PROPOSED: sequenced constraint - PROPOSED: validator

Tensions - PROPOSED: Sequenced temporal integrity can be expensive to validate at large scale.

Expansion directions - PROPOSED: Create no-network temporal-integrity fixtures for geography-version and source-version tables.

Open questions - NEEDS VERIFICATION: Should temporal referential-integrity failures be DENY or QUARANTINE by default?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 134, 127; basis: The temporal database source discusses temporal keys and referential-integrity constraints for time-varying tables, including current, nonsequenced, and sequenced constraints.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-004, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.46 KFM-P18-INV-491 — Coalesced temporal views as derived artifacts

Idea card metadata - Stable ID: KFM-P18-INV-491 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 491 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM coalesced temporal layers should be marked as derived artifacts with receipt-backed rules for merging adjacent intervals.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3

inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM coalesced temporal layers should be marked as derived artifacts with receipt-backed rules for merging adjacent intervals. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Coalesced views improve usability but can hide the original observation cadence and should not replace canonical temporal records.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-009. - CONFIRMED baseline link: KFM-IDX-CAT-005.

Dependencies - PROPOSED: coalescing - PROPOSED: duplicate elimination - PROPOSED: temporal view - PROPOSED: derived artifact

Tensions - PROPOSED: Users may prefer simplified intervals even when original change points matter.

Expansion directions - PROPOSED: Record coalescing_rule_hash in temporal release manifests.

Open questions - NEEDS VERIFICATION: Which Phase 5 cards need coalescing-versus-canonical distinctions?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 183, 184; basis: The temporal database source treats coalescing as a way to merge adjacent value-equivalent rows and distinguishes coalescing from duplicate elimination.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-009, KFM-IDX-CAT-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.4.47 KFM-P18-INV-492 — Continuous-integration discipline inside bounded contexts

Idea card metadata - Stable ID: KFM-P18-INV-492 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 492 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MOD — Data Modeling, Domain Semantics, Temporal Structure, and SQL - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM domain lanes should use continuous integration checks to prevent schema, vocabulary, validator, and documentation drift inside each bounded context.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM domain lanes should use continuous integration checks to prevent schema, vocabulary, validator, and documentation drift inside each bounded context. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented,

tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Domain growth can fracture KFM language and trust objects unless each lane has repeatable integration checks.

Related ideas - CONFIRMED baseline link: KFM-IDX-MDP-002. - CONFIRMED baseline link: KFM-IDX-PIP-004.

Dependencies - PROPOSED: bounded context - PROPOSED: continuous integration - PROPOSED: schema test - PROPOSED: validator

Tensions - PROPOSED: Strict CI can slow exploratory source intake if not paired with quarantine paths.

Expansion directions - PROPOSED: Map Phase 4 categories to bounded-context CI obligations.

Open questions - NEEDS VERIFICATION: Which cross-lane shared kernels require extra integration tests?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 12; basis: The DDD reference states that a bounded context must be kept sound and that teams need continuous integration to prevent model fragmentation.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-MDP-002, KFM-IDX-PIP-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5 EVD - Evidence, Provenance, Receipts, Identity, and Attestation

Status: CONFIRMED source support for grouped Phase 3 ideas; PROPOSED category synthesis

Definition: Evidence objects, provenance, receipts, identity, attestation, catalog/proof links, and digest-bearing support for claims.

CONFIRMED: This chapter contains 34 Pass 18 idea cards assigned to EVD.

Idea ID	Title	Carry state	Status	Source IDs
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KFM-P18-INV-037 | HTTP failures, redirects, and rate limits become acquisition evidence | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-085 | Qualitative GIS perception evidence requires interpretive status labels | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-096 | Shared Kernel versus Separate Ways as schema reuse test | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-112 | Extended Matrix as 3D interpretive lineage control | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-129 | Geographic transformations as auditable analysis steps | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-148 | Scraper error handling as acquisition receipt material | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-168 | Domain events need occurrence time, entry time, and actor identity | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-173 | Geodatabase import and summary-stat receipts | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-175 | Headless and form-scraping interaction-path receipts | NEW | PROPOSED | SRC-P18-012 |

KFM-P18-INV-200 | Third-party enrichment services as receipt-bearing transforms | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-202 | Interactive GIS knowledge co-production as review record | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-218 | Form and AJAX interaction receipts for dynamic sources | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-232 | Clustering outputs are exploratory segmentation evidence | NEW | PROPOSED | SRC-P18-004 |

KFM-P18-INV-238 | Sequenced temporal modifications require split-row receipts | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-252 | Digital map libraries preserve original-map rights and scan lineage | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-255 | Missing-value filters in GIS exercises become visible analytic decisions | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-267 | Value objects for immutable evidence components | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-268 | Entities for stable identity across changing attributes | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-283 | Scraped-data persistence requires sanitization evidence | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-297 | Data-poor scenario planning records assumptions as evidence | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-339 | Scraped-data persistence must retain source query and sanitation context | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-345 | Environmental health co-production records as evidence of deliberation | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-348 | As-of report snapshots for temporally changing claims | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-373 | Dynamic-form and Ajax sources require interaction-state receipts | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-381 | Self-joins as relationship-edge construction with role labels | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-398 | Gendered travel perception evidence as protected interpretive layer | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-411 | VR and eye-tracking outputs as experimental evidence | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-414 | Temporal coalescing receipts for interval outputs | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-419 | Credit-consuming external GIS operations as cost-aware workflow evidence | NEW | PROPOSED | SRC-P18-010 |

KFM-P18-INV-435 | Volumetric archaeological measurements as separate evidence objects | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-446 | Form-submission source adapters need authorization and state receipts | NEW | PROPOSED | SRC-P18-012 |

KFM-P18-INV-464 | Entity versus value-object identity scope | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-471 | AJAX dynamic-content capture receipts for scrapers | NEW | PROPOSED | SRC-P18-012 |

KFM-P18-INV-493 | Shared-kernel governance for cross-domain trust objects | EXPANDED | PROPOSED | SRC-P18-009 |

8.5.1 KFM-P18-INV-037 — HTTP failures, redirects, and rate limits become acquisition evidence

Idea card metadata - Stable ID: KFM-P18-INV-037 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 37 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM should record redirects, HTTP failures, missing code, broken links, rate limits, and retry decisions as source-acquisition evidence.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record redirects, HTTP failures, missing code, broken links, rate limits, and retry decisions as source-acquisition evidence. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source

role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes source instability visible and replayable during promotion review.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-002. - CONFIRMED baseline link: KFM-IDX-PIP-005.

Dependencies - PROPOSED: Retrieval receipt - PROPOSED: retry/backoff policy - PROPOSED: dead-letter or quarantine record - PROPOSED: staleness badge

Tensions - PROPOSED: Repeated retries can create load or terms risks if not governed.

Expansion directions - PROPOSED: Add `fetch_error_class` and `redirect_chain` fields to retrieval receipts.

Open questions - NEEDS VERIFICATION: Should failed scrape attempts be cataloged as retrieval episodes or only retained as run receipts?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — `instant-web-scraping-with-java.pdf`; pages 10, 30, 40; basis: redirect and error-handling discussion in scraping recipes; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-002, KFM-IDX-PIP-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.2 KFM-P18-INV-085 — Qualitative GIS perception evidence requires interpretive status labels

Idea card metadata - Stable ID: KFM-P18-INV-085 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 85 - Status label: PROPOSED - Carry-forward state: NEW - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM should treat qualitative GIS outputs such as children's perception maps as interpretive evidence with method, participant, and privacy controls.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (`GIS in Sustainable Urban Planning and Management.pdf`) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat qualitative GIS outputs such as children's perception maps as interpretive evidence with method, participant, and privacy controls. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It lets KFM include lived-experience evidence without pretending that participatory maps are the same as instrument observations.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-005. - CONFIRMED baseline link: KFM-IDX-DAT-010.

Dependencies - PROPOSED: QualitativeEvidence schema - PROPOSED: Participant privacy policy - PROPOSED: Interpretive status label

Tensions - PROPOSED: Participatory evidence is valuable but can expose vulnerable groups or overstate representativeness.

Expansion directions - PROPOSED: Add `qualitative_method`, `participant_class`, `consent_status`, `anonymization_profile`, and `interpretation_scope` fields.

Open questions - NEEDS VERIFICATION: What review is required before child-centered perception maps become public?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 7, 17, 19; basis: children perception, qualitative GIS, child-friendly city, and methodological demonstration entries; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.3 KFM-P18-INV-096 — Shared Kernel versus Separate Ways as schema reuse test

Idea card metadata - Stable ID: KFM-P18-INV-096 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 96 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should distinguish shared kernels from separate-ways domain models before reusing schemas across lanes.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should distinguish shared kernels from separate-ways domain models before reusing schemas across lanes. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents unnecessary cross-domain schema coupling while preserving truly shared trust objects.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-006. - CONFIRMED baseline link: KFM-IDX-CAT-001.

Dependencies - PROPOSED: Schema-home ADR - PROPOSED: Shared object family registry - PROPOSED: Domain model review

Tensions - PROPOSED: Over-sharing schemas creates brittle integration; under-sharing duplicates trust concepts.

Expansion directions - PROPOSED: Add `shared_kernel_justification` and `separate_ways_decision` notes to cross-domain contracts.

Open questions - NEEDS VERIFICATION: Which KFM objects are truly shared kernel: EvidenceBundle, DecisionEnvelope, or more?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 38, 44, 59; basis: Shared Kernel, Separate Ways, and framework interoperability discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-006, KFM-IDX-CAT-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.4 KFM-P18-INV-112 — Extended Matrix as 3D interpretive lineage control

Idea card metadata - Stable ID: KFM-P18-INV-112 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 112 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM archaeological 3D outputs should distinguish reality-based models, interpretive reconstructions, and Extended Matrix-style interpretation links.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM archaeological 3D outputs should distinguish reality-based models, interpretive reconstructions, and Extended Matrix-style interpretation links. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents interpretive reconstructions from being published as captured site evidence.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-011. - CONFIRMED baseline link: KFM-IDX-POL-007.

Dependencies - PROPOSED: 3D artifact manifest - PROPOSED: Archaeology sensitivity policy - PROPOSED: Interpretation review

Tensions - PROPOSED: Interpretive 3D products are persuasive even when evidence support is partial.

Expansion directions - PROPOSED: Add model_role and interpretation_link fields to 3D artifact manifests.

Open questions - NEEDS VERIFICATION: How should KFM encode interpretive confidence in 3D Story Nodes?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 11, 56; basis: extended matrix approach and interpretive/reality-based model discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-011, KFM-IDX-POL-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.5 KFM-P18-INV-129 — Geographic transformations as auditable analysis steps

Idea card metadata - Stable ID: KFM-P18-INV-129 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 129 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should receipt geographic transformations, including projection, datum, resampling, simplification, and coordinate conversion, as auditable analysis steps.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should receipt geographic transformations, including projection, datum, resampling, simplification, and coordinate conversion, as auditable analysis steps. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It lets reviewers reconstruct why a geometry shifted, simplified, or changed scale between source, processed, and published artifacts.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-002. - CONFIRMED baseline link: KFM-IDX-CAT-002.

Dependencies - PROPOSED: TransformReceipt - PROPOSED: CRS registry - PROPOSED: Processed artifact manifests

Tensions - PROPOSED: Too much transform detail can burden readers unless exposed through layered evidence drawers.

Expansion directions - PROPOSED: Create a transformation summary field for public evidence payloads.

Open questions - NEEDS VERIFICATION: What is the minimum transform chain needed for a public claim?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 28, 66, 113, 114; basis: projection and coordinate-transformation discussion; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-002, KFM-IDX-CAT-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.6 KFM-P18-INV-148 — Scraper error handling as acquisition receipt material

Idea card metadata - Stable ID: KFM-P18-INV-148 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 148 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED

Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM should record scraper errors, redirects, rate-limit responses, missing markup, form failures, and retries as acquisition receipt material.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record scraper errors, redirects, rate-limit responses, missing markup, form failures, and retries as acquisition receipt material. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents a failed or partial scrape from being treated as a clean source snapshot.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-002. - CONFIRMED baseline link: KFM-IDX-PIP-005.

Dependencies - PROPOSED: SourceRunReceipt - PROPOSED: ConnectorLog - PROPOSED: Quarantine disposition

Tensions - PROPOSED: Detailed error logs may contain sensitive URLs, tokens, or form data.

Expansion directions - PROPOSED: Define redacted scraper error receipt fields.

Open questions - NEEDS VERIFICATION: Which scraper failures force quarantine?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 10, 11, 22, 31; basis: web scraping source discussion of redirects, rate limits, broken links, and error handling; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-002, KFM-IDX-PIP-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.7 KFM-P18-INV-168 — Domain events need occurrence time, entry time, and actor identity

Idea card metadata - Stable ID: KFM-P18-INV-168 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 168 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM domain events should distinguish when an event happened, when it was entered or observed by KFM, and which actors/entities participated.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM domain events should distinguish when an event happened, when it was entered or observed by KFM, and which actors/entities participated. - PROPOSED: Phase 3 scope note: Inventory-level candidate record;

full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It supports replay, correction, provenance, and out-of-order event handling without collapsing domain time and system time.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-008. - CONFIRMED baseline link: KFM-IDX-CAT-004. - CONFIRMED baseline link: KFM-IDX-DOC-003.

Dependencies - PROPOSED: DomainEvent schema - PROPOSED: RunReceipt - PROPOSED: EvidenceBundle

Tensions - PROPOSED: Some source feeds may not provide all actor or timestamp fields.

Expansion directions - PROPOSED: Add missing-time ABSTAIN and partial-event fixture cases.

Open questions - NEEDS VERIFICATION: How should KFM label inferred actor identity versus source-provided actor identity?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 20; basis: DDD Domain Events pattern describing immutable events with event time, entered time, and involved identities; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-008, KFM-IDX-CAT-004, KFM-IDX-DOC-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.8 KFM-P18-INV-173 — Geodatabase import and summary-stat receipts

Idea card metadata - Stable ID: KFM-P18-INV-173 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 173 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should emit receipts for geodatabase imports, standalone table imports, field renaming, and summary-stat generation before using the output as evidence.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should emit receipts for geodatabase imports, standalone table imports, field renaming, and summary-stat generation before using the output as evidence. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It exposes the workflow steps that turn source files into feature classes, tables, and counts used in public map claims.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-CAT-004. - CONFIRMED baseline link: KFM-IDX-DAT-013.

Dependencies - PROPOSED: Transform receipt - PROPOSED: Field mapping - PROPOSED: Summary-stat metric definition

Tensions - PROPOSED: Commercial GIS workflows can hide intermediate state unless exported into KFM receipts.

Expansion directions - PROPOSED: Define receipt adapters for desktop-GIS-derived imports and summaries.

Open questions - NEEDS VERIFICATION: How should KFM handle field renames imposed by source tools or file-format limits?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 47, 48; basis: ArcGIS wildfire exercise importing shapefiles/tables to a geodatabase and summarizing agency counts; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-001, KFM-IDX-CAT-004, KFM-IDX-DAT-013 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.9 KFM-P18-INV-175 — Headless and form-scraping interaction-path receipts

Idea card metadata - Stable ID: KFM-P18-INV-175 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 175 - Status label: PROPOSED - Carry-forward state: NEW - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM source admission should require interaction-path receipts for headless-browser or form-submission scraping, including selected elements, submitted fields, authentication posture, and success checks.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM source admission should require interaction-path receipts for headless-browser or form-submission scraping, including selected elements, submitted fields, authentication posture, and success checks. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes dynamic acquisition reproducible and policy-reviewable when an API is absent and a browser-like workflow is used.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-008. - CONFIRMED baseline link: KFM-IDX-SEC-011. - CONFIRMED baseline link: KFM-IDX-POL-007.

Dependencies - PROPOSED: SourceDescriptor - PROPOSED: Scraper test fixture - PROPOSED: Access terms review

Tensions - PROPOSED: Authenticated or form-based scraping can violate source terms or expose credentials.

Expansion directions - PROPOSED: Fail closed unless rights, robots/source terms, credential handling, and replay tests are documented.

Open questions - NEEDS VERIFICATION: Are any form-based source acquisitions allowed for KFM without explicit source permission?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 54; basis: web-scraping recipe using HtmlUnit form elements, button clicks, login result checks, and XPath for form interaction; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.10 KFM-P18-INV-200 — Third-party enrichment services as receipt-bearing transforms

Idea card metadata - Stable ID: KFM-P18-INV-200 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 200 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should treat external GIS enrichment and hosted analysis tools as receipt-bearing transforms with account, cost, selected variables, filters, and source caveats recorded before outputs support claims.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat external GIS enrichment and hosted analysis tools as receipt-bearing transforms with account, cost, selected variables, filters, and source caveats recorded before outputs support claims. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents vendor-generated enrichment outputs from entering KFM as unexplained evidence.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-DAT-013. - CONFIRMED baseline link: KFM-IDX-POL-007.

Dependencies - PROPOSED: Transform receipt - PROPOSED: Variable selection log - PROPOSED: NoData filter - PROPOSED: Vendor/source terms review

Tensions - PROPOSED: Hosted enrichment can hide source data lineage and costs unless explicitly recorded.

Expansion directions - PROPOSED: Define receipts for third-party enrichment, selected variables, filters, and summary-stat recalculation.

Open questions - NEEDS VERIFICATION: Are vendor-enriched attributes allowed for public KFM claims, and under what source/rights labels?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 11, 12, 13, 14, 15; basis: ArcGIS Online Enrich Layer workflow with paid account/credit note, selected income variable, enriched attributes, summary statistics, and filtering out 0.00 values; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-001, KFM-IDX-DAT-013, KFM-IDX-POL-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.11 KFM-P18-INV-202 — Interactive GIS knowledge co-production as review record

Idea card metadata - Stable ID: KFM-P18-INV-202 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 202 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM participatory planning sessions should record stakeholder perspectives, social-learning context, and deliberation outputs as review evidence rather than flattening them into a single authoritative layer.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM participatory planning sessions should record stakeholder perspectives, social-learning context, and deliberation outputs as review evidence rather than flattening them into a single authoritative layer. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It gives participatory GIS outcomes a governed evidence trail while avoiding false objectivity.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-POL-003. - CONFIRMED baseline link: KFM-IDX-DOC-006.

Dependencies - PROPOSED: Session record - PROPOSED: Stakeholder role labels - PROPOSED: Consent/review state - PROPOSED: Decision log

Tensions - PROPOSED: Consensus can mask power differences unless stakeholder roles and dissent are preserved.

Expansion directions - PROPOSED: Define ParticipatoryReviewRecord fields for participant role, scope, consensus, dissent, and evidence impact.

Open questions - NEEDS VERIFICATION: How should KFM expose participatory-review evidence without exposing private or sensitive participant data?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 96, 97; basis: knowledge co-production and social learning in interactive

GIS-based approaches, including collaborative dialogue and participant perspectives; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-001, KFM-IDX-POL-003, KFM-IDX-DOC-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.12 KFM-P18-INV-218 — Form and AJAX interaction receipts for dynamic sources

Idea card metadata - Stable ID: KFM-P18-INV-218 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 218 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM dynamic web connectors should record submitted forms, GET/POST parameters, browser/JavaScript execution requirements, and Ajax-derived data paths before outputs support evidence claims.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM dynamic web connectors should record submitted forms, GET/POST parameters, browser/JavaScript execution requirements, and Ajax-derived data paths before outputs support evidence claims. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes dynamic acquisition reproducible and exposes when a connector depends on UI behavior rather than a stable public API.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-PIP-008. - CONFIRMED baseline link: KFM-IDX-SEC-010.

Dependencies - PROPOSED: Interaction receipt - PROPOSED: Form parameters - PROPOSED: Headless/browser runtime - PROPOSED: Terms review

Tensions - PROPOSED: Dynamic sites can change behavior without explicit API versioning.

Expansion directions - PROPOSED: Add dynamic_interaction_receipt fields for forms, headers, script execution, session state, and captured URL traces.

Open questions - NEEDS VERIFICATION: When should KFM deny scraping dynamic pages and require an official API or manual review instead?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 51, 55; basis: submitting forms via GET/POST and scraping Ajax pages by executing client-side JavaScript to access hidden data; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-001, KFM-IDX-PIP-008, KFM-IDX-SEC-010 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.13 KFM-P18-INV-232 — Clustering outputs are exploratory segmentation evidence

Idea card metadata - Stable ID: KFM-P18-INV-232 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 232 - Status label: PROPOSED - Carry-forward state: NEW - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should label clustering-derived groups as exploratory segmentation evidence unless an independent source or review validates the classes as domain truth.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should label clustering-derived groups as exploratory segmentation evidence unless an independent source or review validates the classes as domain truth. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents unsupervised clusters from being mistaken for authoritative categories, habitat types, risk classes, or community labels.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-001. - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: Cluster run receipt - PROPOSED: Feature set - PROPOSED: Distance metric - PROPOSED: Review caveat

Tensions - PROPOSED: Cluster names often invite causal or normative interpretation that the algorithm did not prove.

Expansion directions - PROPOSED: Require cluster_model_receipt fields for k, features, distance metric, initialization, stability, and label provenance.

Open questions - NEEDS VERIFICATION: Can any clustering-derived class be public without a domain-review explanation?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 132, 133, 134; basis: K-Means clustering groups unlabeled data by similarity through centroid assignment and iterative refinement; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.14 KFM-P18-INV-238 — Sequenced temporal modifications require split-row receipts

Idea card metadata - Stable ID: KFM-P18-INV-238 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 238 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM temporal updates should record when a modification splits a validity period into old, changed, and retained intervals.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM temporal updates should record when a modification splits a validity period into old, changed, and retained intervals. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents quiet temporal edits from rewriting history without showing which interval was affected.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-003. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: Temporal update receipt - PROPOSED: Valid-time interval - PROPOSED: Transaction-time stamp - PROPOSED: Integrity validator

Tensions - PROPOSED: Fine-grained temporal edits are hard to inspect if only final rows are shown.

Expansion directions - PROPOSED: Add temporal_update_receipt fields for old_period, affected_period, retained_periods, statements, and integrity outcome.

Open questions - NEEDS VERIFICATION: Should all temporal claim edits require a human-readable interval diff?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 219, 220, 240; basis: sequenced updates split validity periods into updated and retained portions and can require multiple statements; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-003, KFM-IDX-EVD-001, KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.15 KFM-P18-INV-252 — Digital map libraries preserve original-map rights and scan lineage

Idea card metadata - Stable ID: KFM-P18-INV-252 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 252 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM historic-map and scanned-map workflows should separate original paper-map rights, scan provenance, georeferencing, and derived GIS use.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM historic-map and scanned-map workflows should separate original paper-map rights, scan provenance, georeferencing, and derived GIS use. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents a scanned map overlay from being treated as a rights-free or lineage-free GIS layer.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-002. - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: Original map citation - PROPOSED: scan receipt - PROPOSED: georeference annotation - PROPOSED: rights review

Tensions - PROPOSED: The public may see an overlay as modern truth even when it is a historic representation.

Expansion directions - PROPOSED: Add historic_map_scan_receipt fields for original_rights, scan_source, georeference_method, GCP_count, and derivative_use_limits.

Open questions - NEEDS VERIFICATION: How should KFM handle scanned maps whose copyright is clear but georeferencing quality is weak?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 242; basis: digital map libraries, scanning paper maps, copyright constraints, digital format storage, coordinate system referencing; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-002, KFM-IDX-POL-001, KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.16 KFM-P18-INV-255 — Missing-value filters in GIS exercises become visible analytic decisions

Idea card metadata - Stable ID: KFM-P18-INV-255 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 255 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM GIS workflows should record when missing values or missing features are filtered before summary statistics or spatial analysis.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory.

- **CONFIRMED**: Phase 3 inventory recorded this idea with source support label **CONFIRMED** and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - **PROPOSED**: In KFM, this idea should be treated as a governed design candidate: KFM GIS workflows should record when missing values or missing features are filtered before summary statistics or spatial analysis. - **PROPOSED**: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - **PROPOSED**: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - **UNKNOWN**: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - **NEEDS VERIFICATION**: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - **PROPOSED**: It makes data exclusion visible to reviewers and prevents filtered summaries from appearing complete.

Related ideas - **CONFIRMED** baseline link: KFM-IDX-EVD-001. - **CONFIRMED** baseline link: KFM-IDX-DAT-003. - **CONFIRMED** baseline link: KFM-IDX-OBS-004.

Dependencies - **PROPOSED**: Filter condition - **PROPOSED**: summary-stat receipt - **PROPOSED**: missingness note

Tensions - **PROPOSED**: Filtering improves readability but may hide where source coverage failed.

Expansion directions - **PROPOSED**: Add GIS filter_receipt entries for missing_value_rule, features_removed, fields_removed, and impact_on_summary.

Open questions - **NEEDS VERIFICATION**: What missing-value thresholds should force **ABSTAIN** for public summaries?

Future work - **PROPOSED**: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - **CONFIRMED** source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 7; basis: exercise objectives include exploring summary statistics and filtering out missing data values and features; Phase 1 role: reference_source_candidate.

Carry-forward note - **CONFIRMED**: Carry-forward state is **EXPANDED**; this card extends KFM-IDX-EVD-001, KFM-IDX-DAT-003, KFM-IDX-OBS-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - **CONFIRMED**: Required schema fields are present in this Markdown card. - **CONFIRMED**: No prior-pass body text is inlined. - **CONFIRMED**: Narrative claims use explicit truth labels. - **UNKNOWN**: Repository implementation status remains unverified.

8.5.17 KFM-P18-INV-267 — Value objects for immutable evidence components

Idea card metadata - Stable ID: KFM-P18-INV-267 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 267 - Status label: **PROPOSED** - Carry-forward state: **EXPANDED** - Category: **EVD** — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: **PROPOSED** assignment derived from **CONFIRMED** Phase 3 inventory; subject to Phase 5 card-level review - Source support label: **CONFIRMED** - Implementation maturity: **UNKNOWN** — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-009

Normalized statement - **PROPOSED**: KFM should model immutable descriptive components such as coordinates, time intervals, checksums, and source citations as value objects where identity is not conceptually needed.

Detailed explanation - **CONFIRMED**: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - **CONFIRMED**: Phase 3 inventory recorded this idea with source support label **CONFIRMED** and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - **PROPOSED**: In KFM, this idea should be treated as a governed design candidate: KFM should model immutable descriptive components such as coordinates, time intervals, checksums, and source citations as value objects where identity is not conceptually needed. - **PROPOSED**: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - **PROPOSED**: Public or semi-public

use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reduces identity clutter while preserving meaning and validation for evidence-bearing fields.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: Value-object schema pattern - PROPOSED: hashing rule - PROPOSED: side-effect-free validation

Tensions - PROPOSED: Some fields that look like values may later need independent correction history.

Expansion directions - PROPOSED: Define KFM value-object candidates: CoordinateValue, TimeIntervalValue, DigestValue, CitationValue, and PolicyLabelValue.

Open questions - NEEDS VERIFICATION: Which evidence components require identity because they can be separately reviewed or corrected?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 12; basis: value objects describe characteristics, have no conceptual identity, should be immutable, and use side-effect-free operations; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-001, KFM-IDX-EVD-001, KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.18 KFM-P18-INV-268 — Entities for stable identity across changing attributes

Idea card metadata - Stable ID: KFM-P18-INV-268 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 268 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should reserve entity identity for things that must remain distinguishable across changing attributes, histories, representations, or source descriptions.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should reserve entity identity for things that must remain distinguishable across changing attributes, histories, representations, or source descriptions. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It aligns deterministic identifiers with domain meaning rather than only database convenience.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-CAT-006. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Identity policy - PROPOSED: entity schema - PROPOSED: crosswalk register

Tensions - PROPOSED: Over-assigning identity can make derived or transient objects look canonical.

Expansion directions - PROPOSED: Add `entity_identity_decision` fields for `identity_basis`, `sameness_rule`, `source_crosswalks`, and `mutable_attributes`.

Open questions - NEEDS VERIFICATION: Which spatial features are entities, and which are value-object geometry snapshots?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 11; basis: entities require lifecycle continuity, identity, unique distinction, and model-defined sameness; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-001, KFM-IDX-CAT-006, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.19 KFM-P18-INV-283 — Scraped-data persistence requires sanitization evidence

Idea card metadata - Stable ID: KFM-P18-INV-283 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 283 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM connectors that persist scraped records should record sanitization, prepared-statement use, schema target, and write outcome in the run receipt.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM connectors that persist scraped records should record sanitization, prepared-statement use, schema target, and write outcome in the run receipt. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It treats persistence as a security- and provenance-significant event, not a neutral save operation.

Related ideas - CONFIRMED baseline link: KFM-IDX-SEC-002. - CONFIRMED baseline link: KFM-IDX-PIP-003. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: run receipt - PROPOSED: database write audit - PROPOSED: input sanitization policy

Tensions - PROPOSED: Persistence success does not prove extraction correctness or rights for reuse.

Expansion directions - PROPOSED: Add receipt fields for `sanitized_input=true` and `prepared_statement=true` where relevant.

Open questions - NEEDS VERIFICATION: Should scrapers be allowed to persist directly to WORK, or only through a staged extract artifact?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 45; basis: The source warns that web inputs are untrusted and recommends always sanitizing inputs using prepared statements before database insertion.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-SEC-002, KFM-IDX-PIP-003, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.20 KFM-P18-INV-297 — Data-poor scenario planning records assumptions as evidence

Idea card metadata - Stable ID: KFM-P18-INV-297 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 297 - Status label: PROPOSED - Carry-forward state: NEW - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM planning-support scenarios should record data scarcity, assumptions, model choices, and scenario alternatives before using them in policy or map narratives.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM planning-support scenarios should record data scarcity, assumptions, model choices, and scenario alternatives before using them in policy or map narratives. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps useful scenario work transparent when evidence is incomplete or uneven.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-OBS-005.

Dependencies - PROPOSED: scenario manifest - PROPOSED: assumption register - PROPOSED: model run receipt

Tensions - PROPOSED: Data-poor planning may be necessary, but weak data should not be hidden by polished maps.

Expansion directions - PROPOSED: Create scenario_assumption_register fields for missing data, proxy variables, and sensitivity tests.

Open questions - NEEDS VERIFICATION: When should a data-poor scenario be allowed in a public Story Node versus steward-only review?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 22; basis: The sustainable urban GIS source includes a planning support tool for scenario analysis of low-carbon transport plans in a data-poor context.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.21 KFM-P18-INV-339 — Scraped-data persistence must retain source query and sanitation context

Idea card metadata - Stable ID: KFM-P18-INV-339 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 339 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM should persist scraped data only with source URL/query, retrieval time, sanitization step, parser version, and rights/terms context.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should persist scraped data only with source URL/query, retrieval time, sanitization step, parser version, and rights/terms context. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents scraped rows from detaching from the conditions under which they were obtained.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-POL-003.

Dependencies - PROPOSED: source URL - PROPOSED: query parameters - PROPOSED: retrieval time - PROPOSED: parser version - PROPOSED: sanitization receipt - PROPOSED: rights status

Tensions - PROPOSED: Persistence can preserve prohibited content if rights and terms are not checked first.

Expansion directions - PROPOSED: Add scraped_data_receipt minimum fields to source intake validation.

Open questions - NEEDS VERIFICATION: Should scraped candidate data expire automatically when no release path exists?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 33; basis: Instant Web Scraping with Java includes persisting scraped data as an advanced scraping topic following robust acquisition.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-001, KFM-IDX-POL-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN:

Repository implementation status remains unverified.

8.5.22 KFM-P18-INV-345 — Environmental health co-production records as evidence of deliberation

Idea card metadata - Stable ID: KFM-P18-INV-345 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 345 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM interactive GIS workshops for environmental health or planning should preserve participant role, issue framing, deliberation outcome, and evidence changes as review records.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM interactive GIS workshops for environmental health or planning should preserve participant role, issue framing, deliberation outcome, and evidence changes as review records. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It captures deliberation as evidence without converting every workshop output into authoritative fact.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-005. - CONFIRMED baseline link: KFM-IDX-DAT-013.

Dependencies - PROPOSED: participant role - PROPOSED: issue framing - PROPOSED: workshop artifact - PROPOSED: evidence change - PROPOSED: review note

Tensions - PROPOSED: Participant privacy and representativeness must be handled carefully.

Expansion directions - PROPOSED: Add `co_production_review_record` fields to planning-support idea cards.

Open questions - NEEDS VERIFICATION: What consent and anonymization rules apply to workshop maps?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 83; basis: GIS in Sustainable Urban Planning and Management includes a chapter on knowledge co-production and social learning on environmental health issues using interactive GIS approaches.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-005, KFM-IDX-DAT-013 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.23 KFM-P18-INV-348 — As-of report snapshots for temporally changing claims

Idea card metadata - Stable ID: KFM-P18-INV-348 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 348 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED

Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM reports should pin the as-of transaction or release snapshot used to compute a table, map, or statistic when underlying records may change after publication.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM reports should pin the as-of transaction or release snapshot used to compute a table, map, or statistic when underlying records may change after publication. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It lets reviewers reconstruct why two reports about the same period may differ.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-OBS-003.

Dependencies - PROPOSED: as_of time - PROPOSED: release id - PROPOSED: transaction snapshot - PROPOSED: tracking log reference - PROPOSED: report receipt

Tensions - PROPOSED: Snapshot pinning can be confused with real-world valid time unless both are labeled.

Expansion directions - PROPOSED: Add as_of_release_snapshot to report and export receipts.

Open questions - NEEDS VERIFICATION: When is release snapshot sufficient versus full transaction-time history?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 219, 248; basis: Developing Time-Oriented Database Applications in SQL discusses retaining tracking logs and report consistency when data changes over time.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-001, KFM-IDX-OBS-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.24 KFM-P18-INV-373 — Dynamic-form and Ajax sources require interaction-state receipts

Idea card metadata - Stable ID: KFM-P18-INV-373 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 373 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM connectors for dynamic web pages should record form submissions, browser-like JavaScript execution, redirects, and hidden state as interaction receipts before treating results as evidence.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category

before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM connectors for dynamic web pages should record form submissions, browser-like JavaScript execution, redirects, and hidden state as interaction receipts before treating results as evidence. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes dynamic acquisition replayable and distinguishes source data from client-side transformation.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-008. - CONFIRMED baseline link: KFM-IDX-SEC-011.

Dependencies - PROPOSED: form parameters - PROPOSED: browser state - PROPOSED: JS execution context - PROPOSED: interaction receipt

Tensions - PROPOSED: Capturing browser-like state may include cookies, tokens, or sensitive parameters.

Expansion directions - PROPOSED: Define safe redaction rules for interaction-state receipts.

Open questions - NEEDS VERIFICATION: Can dynamic-source connectors operate without headless-browser evidence?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 10, 11, 46; basis: Instant Web Scraping with Java highlights data behind Ajax or forms, submitting forms, and executing JavaScript as a browser would for dynamic pages.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-008, KFM-IDX-SEC-011 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.25 KFM-P18-INV-381 — Self-joins as relationship-edge construction with role labels

Idea card metadata - Stable ID: KFM-P18-INV-381 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 381 - Status label: PROPOSED - Carry-forward state: NEW - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM should treat SQL self-joins that construct manager, parent, predecessor, adjacency, or containment relationships as edge-building operations requiring role labels and temporal constraints.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat SQL self-joins that construct manager, parent, predecessor, adjacency, or containment relationships as edge-building operations requiring role labels and temporal constraints. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION:

Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Self-joins silently create relationship edges; without role and time qualifiers those edges can become ambiguous graph facts.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-007. - CONFIRMED baseline link: KFM-IDX-DAT-010.

Dependencies - PROPOSED: source table - PROPOSED: join condition - PROPOSED: edge role - PROPOSED: valid time - PROPOSED: graph projection

Tensions - PROPOSED: Some self-joins are analytic conveniences rather than domain relationship claims.

Expansion directions - PROPOSED: Add self_join_edge_role fields to graph-building receipts.

Open questions - NEEDS VERIFICATION: When should a self-join result become a graph edge versus a temporary analytic table?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 11; basis: Advanced SQL Concepts gives a self-join example for employees and their managers.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.26 KFM-P18-INV-398 — Gendered travel perception evidence as protected interpretive layer

Idea card metadata - Stable ID: KFM-P18-INV-398 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 398 - Status label: PROPOSED - Carry-forward state: NEW - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM mobility and urban morphology layers should separate observed movement, built-environment measures, and gendered perception evidence, with sensitivity and interpretation labels.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM mobility and urban morphology layers should separate observed movement, built-environment measures, and gendered perception evidence, with sensitivity and interpretation labels. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Perception evidence can reveal lived access and safety concerns that are not visible in network data alone.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-002. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: survey or interview evidence - PROPOSED: built-environment metric - PROPOSED: mobility observation - PROPOSED: sensitivity label

Tensions - PROPOSED: Perception evidence can expose vulnerable groups or be overgeneralized.

Expansion directions - PROPOSED: Define protected perception evidence roles for mobility planning layers.

Open questions - NEEDS VERIFICATION: What privacy controls should govern gendered mobility perception layers?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 154, 155, 156; basis: GIS in Sustainable Urban Planning and Management includes a chapter on city morphology and women's perception of travel.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.27 KFM-P18-INV-411 — VR and eye-tracking outputs as experimental evidence

Idea card metadata - Stable ID: KFM-P18-INV-411 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 411 - Status label: PROPOSED - Carry-forward state: NEW - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should label VR, eye-tracking, and immersive 3D outputs as experimental or interpretive evidence unless reviewed protocols support stronger claims.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should label VR, eye-tracking, and immersive 3D outputs as experimental or interpretive evidence unless reviewed protocols support stronger claims. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Immersive outputs can reveal patterns in human attention or visibility but are sensitive to equipment, participant setup, and model assumptions.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-013. - CONFIRMED baseline link: KFM-IDX-AIG-006.

Dependencies - PROPOSED: experiment protocol - PROPOSED: participant privacy policy - PROPOSED: 3D model provenance - PROPOSED: review state

Tensions - PROPOSED: Participant-derived immersive data may carry privacy obligations beyond normal map interaction telemetry.

Expansion directions - PROPOSED: Add immersive_experiment_status to 3D and Story Node evidence manifests.

Open questions - NEEDS VERIFICATION: Which VR or eye-tracking outputs should be denied from public release by default?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 15, 125, 127; basis: Archaeological 3D GIS includes future-development material and figures involving VR-based eye-tracking and 3D visualization.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.28 KFM-P18-INV-414 — Temporal coalescing receipts for interval outputs

Idea card metadata - Stable ID: KFM-P18-INV-414 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 414 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM interval-based outputs should record whether adjacent or overlapping periods were coalesced, split, or left as separate temporal facts.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM interval-based outputs should record whether adjacent or overlapping periods were coalesced, split, or left as separate temporal facts. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Temporal coalescing can change event counts, duration summaries, and public interpretation of continuity; it must be reviewable.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-003. - CONFIRMED baseline link: KFM-IDX-CAT-002.

Dependencies - PROPOSED: temporal transform receipt - PROPOSED: valid-time schema - PROPOSED: duration policy - PROPOSED: release manifest

Tensions - PROPOSED: Uncoalesced records may be more faithful to source events but less readable.

Expansion directions - PROPOSED: Add coalescing_method and interval_policy fields to temporal outputs.

Open questions - NEEDS VERIFICATION: When should KFM preserve source-split intervals even if they are semantically continuous?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 143, 145, 158; basis: Developing Time-Oriented Database Applications in SQL discusses current/prior state extraction, sequenced queries, duplicate elimination, and temporal table logic.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-003, KFM-IDX-CAT-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.29 KFM-P18-INV-419 — Credit-consuming external GIS operations as cost-aware workflow evidence

Idea card metadata - Stable ID: KFM-P18-INV-419 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 419 - Status label: PROPOSED - Carry-forward state: NEW - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM workflows that rely on external GIS platforms or credit-consuming operations should record cost, dependency, and reproducibility caveats.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM workflows that rely on external GIS platforms or credit-consuming operations should record cost, dependency, and reproducibility caveats. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: External hosted analysis can be valid but may not be reproducible without platform access, credentials, credits, or identical service versions.

Related ideas - CONFIRMED baseline link: KFM-IDX-OBS-004. - CONFIRMED baseline link: KFM-IDX-PIP-006.

Dependencies - PROPOSED: external platform descriptor - PROPOSED: cost/credit receipt - PROPOSED: service version - PROPOSED: replay policy

Tensions - PROPOSED: Educational exercises may intentionally use hosted functions that KFM would replace with governed pipelines.

Expansion directions - PROPOSED: Add external_service_cost_caveat to workflow receipts and weak material.

Open questions - NEEDS VERIFICATION: Which external hosted operations should be replaced before publication-grade KFM use?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 4, 7, 8; basis: The ArcGIS Pro exploration text notes that some exercises require ArcGIS Online with credits and include publishing and enrichment workflows.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.30 KFM-P18-INV-435 — Volumetric archaeological measurements as separate evidence objects

Idea card metadata - Stable ID: KFM-P18-INV-435 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 435 - Status label: PROPOSED - Carry-forward state: NEW - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should model archaeological volume measurements, CT-derived volumes, and deposit-volume estimates as evidence objects distinct from visual meshes.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should model archaeological volume measurements, CT-derived volumes, and deposit-volume estimates as evidence objects distinct from visual meshes. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A volume estimate is an analytical claim, not merely a 3D asset; separating it improves citation, uncertainty, and review.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-DAT-009.

Dependencies - PROPOSED: 3D asset manifest - PROPOSED: measurement receipt - PROPOSED: uncertainty profile - PROPOSED: review state

Tensions - PROPOSED: Volumetric methods vary widely across instruments and processing pipelines.

Expansion directions - PROPOSED: Add volume_measurement_evidence to 3D/archaeology object families.

Open questions - NEEDS VERIFICATION: Which volume methods are admissible for public evidence without specialist review?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 8, 10, 26, 42; basis: Archaeological 3D GIS lists volumes and volumetric data among 3D archaeological methods and examples.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.31 KFM-P18-INV-446 — Form-submission source adapters need authorization and state receipts

Idea card metadata - Stable ID: KFM-P18-INV-446 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 446 - Status label: PROPOSED - Carry-forward state: NEW - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM source adapters that submit forms should record authorization, terms, submitted parameters, and resulting state as retrieval evidence.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM source adapters that submit forms should record authorization, terms, submitted parameters, and resulting state as retrieval evidence. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Form-based retrieval can cross access and terms boundaries; stateful inputs must be auditable.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-003. - CONFIRMED baseline link: KFM-IDX-SEC-004.

Dependencies - PROPOSED: SourceDescriptor - PROPOSED: retrieval receipt - PROPOSED: terms check - PROPOSED: quarantine gate

Tensions - PROPOSED: Some forms prohibit automated submission or require credentials.

Expansion directions - PROPOSED: Add form_submission_receipt fields to source-adaptor patterns.

Open questions - NEEDS VERIFICATION: Which form submissions are categorically denied for KFM connectors?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 8, 11, 51; basis: Instant Web Scraping with Java includes material on submitting basic forms and retrieving data from pages.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.32 KFM-P18-INV-464 — Entity versus value-object identity scope

Idea card metadata - Stable ID: KFM-P18-INV-464 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 464 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM schema design should separate entities that require durable identity from value objects that only describe characteristics or computations and should not receive artificial identity.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM schema design should separate entities that require durable identity from value objects that only describe characteristics or computations and should not receive artificial identity. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that

this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Not every structured value needs a stable ID; over-identification can make the evidence graph harder to correct.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-006. - CONFIRMED baseline link: KFM-IDX-DOC-008.

Dependencies - PROPOSED: schema design - PROPOSED: deterministic ID policy - PROPOSED: object-family registry - PROPOSED: validator

Tensions - PROPOSED: KFM often benefits from deterministic identifiers, but inappropriate IDs can imply unwarranted persistence.

Expansion directions - PROPOSED: Mark object-family records as entity, value_object, or derived_artifact before Phase 5 cards.

Open questions - NEEDS VERIFICATION: Which KFM objects should be immutable values rather than entities?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 18, 19; basis: The DDD reference distinguishes entities, whose identity runs through lifecycle changes, from value objects, where attaching identity can add complexity and muddle the model.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-006, KFM-IDX-DOC-008 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.33 KFM-P18-INV-471 — AJAX dynamic-content capture receipts for scrapers

Idea card metadata - Stable ID: KFM-P18-INV-471 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 471 - Status label: PROPOSED - Carry-forward state: NEW - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM source adapters that depend on dynamic pages should record whether client-side JavaScript or AJAX execution was required, how loading completion was detected, and what browser-emulation tool was used.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM source adapters that depend on dynamic pages should record whether client-side JavaScript or AJAX execution was required, how loading completion was detected, and what browser-emulation tool was used. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A dynamic-page scrape can capture different evidence than a static HTTP fetch, so the execution method must be auditable.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-010. - CONFIRMED baseline link: KFM-IDX-EVD-006.

Dependencies - PROPOSED: source adapter - PROPOSED: browser-emulation tool - PROPOSED: run receipt - PROPOSED: test fixture

Tensions - PROPOSED: Browser automation can increase source load and may violate source terms if not reviewed.

Expansion directions - PROPOSED: Add `dynamic_content_mode` to source-adapter receipts.

Open questions - NEEDS VERIFICATION: Which sources prohibit automated browser execution?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — `instant-web-scraping-with-java.pdf`; pages 55, 58; basis: The Java scraping source explains that Ajax pages can require JavaScript execution and shows waiting for Ajax-loaded content before extraction.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.5.34 KFM-P18-INV-493 — Shared-kernel governance for cross-domain trust objects

Idea card metadata - Stable ID: KFM-P18-INV-493 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 493 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: EVD — Evidence, Provenance, Receipts, Identity, and Attestation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM shared objects such as `EvidenceBundle`, `DecisionEnvelope`, `SourceDescriptor`, and `ReleaseManifest` should be governed as a shared kernel rather than redefined independently by each lane.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (`Domain-Driven Design Reference.pdf`) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM shared objects such as `EvidenceBundle`, `DecisionEnvelope`, `SourceDescriptor`, and `ReleaseManifest` should be governed as a shared kernel rather than redefined independently by each lane. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A fragmented trust vocabulary would break cite-or-abstain behavior across domains.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-006. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: shared kernel - PROPOSED: `EvidenceBundle` - PROPOSED: `DecisionEnvelope` - PROPOSED: `ReleaseManifest`

Tensions - PROPOSED: Shared kernel control can become a bottleneck for domain-specific needs.

Expansion directions - PROPOSED: Define shared-kernel object families explicitly in the Master Category Map.

Open questions - NEEDS VERIFICATION: Which objects are shared-kernel versus domain-local extensions?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 38; basis: The DDD reference describes Shared Kernel as an intimate interdependency that can leverage design work or undermine it if integration overhead is unmanaged.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-006, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.6 POL - Policy, Rights, Ethics, Sensitivity, and Public-Safety Controls

Status: CONFIRMED source support for grouped Phase 3 ideas; PROPOSED category synthesis

Definition: Policy, rights, ethics, sensitivity, access, consent, redaction/generalization, public-safety, and deny-by-default controls.

CONFIRMED: This chapter contains 12 Pass 18 idea cards assigned to POL.

Idea ID	Title	Carry state	Status	Source IDs
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KFM-P18-INV-005 | Map culture and misuse review | NEW | PROPOSED | SRC-P18-002 |

KFM-P18-INV-056 | Drone flight constraints as policy-aware spatial overlays | NEW | PROPOSED | SRC-P18-010 |

KFM-P18-INV-123 | Drone/airspace constraints as policy-aware spatial overlays | NEW | PROPOSED | SRC-P18-010 |

KFM-P18-INV-208 | Participatory qualitative GIS needs consent and minor-safety controls | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-298 | Qualitative GIS perception layers require participant-method context | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-314 | AI data ethics gate for privacy, bias, ownership, and transparency | NEW | PROPOSED | SRC-P18-004 |

KFM-P18-INV-341 | Ethical data-use fields for AI-supporting datasets | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-358 | Drone and airspace exercises as policy-aware spatial-admission patterns | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-403 | Copyright and distribution-rights check before reusing geographic information | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-457 | AI data-ethics register for privacy, bias, ownership, and transparency | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-466 | Drone authorization parameters as capture metadata | NEW | PROPOSED | SRC-P18-010 |

KFM-P18-INV-483 | Transparency and privacy notes for AI-derived explanations | EXPANDED | PROPOSED | SRC-P18-004 |

8.6.1 KFM-P18-INV-005 — Map culture and misuse review

Idea card metadata - Stable ID: KFM-P18-INV-005 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 5 - Status label: PROPOSED - Carry-forward state: NEW - Category: POL — Policy, Rights, Ethics, Sensitivity, and Public-Safety Controls - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should treat map misuse, cultural context, and the power of representation as review topics, not only styling concerns.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat map misuse, cultural context, and the power of representation as review topics, not only styling concerns. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should

remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It helps prevent persuasive maps from laundering weak evidence or culturally sensitive interpretations.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-002. - CONFIRMED baseline link: KFM-IDX-DOC-007.

Dependencies - PROPOSED: Review checklist for public map products

Tensions - PROPOSED: Ethical map review can slow publication but protects trust.

Expansion directions - PROPOSED: Create a map-misuse risk checklist for public releases.

Open questions - NEEDS VERIFICATION: Which releases require cultural/steward review beyond normal policy gates?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 232; basis: map cultures, misuses, and GI; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.6.2 KFM-P18-INV-056 — Drone flight constraints as policy-aware spatial overlays

Idea card metadata - Stable ID: KFM-P18-INV-056 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 56 - Status label: PROPOSED - Carry-forward state: NEW - Category: POL — Policy, Rights, Ethics, Sensitivity, and Public-Safety Controls - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should treat drone/no-fly, buffer, and operational restriction layers as policy-aware overlays with source, time, and use limitations.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat drone/no-fly, buffer, and operational restriction layers as policy-aware overlays with source, time, and use limitations. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Regulated-flight information can affect behavior and liability; it requires freshness, authority, and disclaimer discipline.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: source authority check - PROPOSED: freshness window - PROPOSED: policy disclaimer

Tensions - PROPOSED: A map can aid planning without becoming a legal or real-time aviation authority.

Expansion directions - PROPOSED: Define `regulated_overlay_type` and `not_official_guidance` disclaimers for public layers.

Open questions - NEEDS VERIFICATION: Should KFM deny or abstain from live operational aviation guidance?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 29; basis: B4UFLY drone exercise with geocoding, buffers, spatial selection, and sharing; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.6.3 KFM-P18-INV-123 — Drone/airspace constraints as policy-aware spatial overlays

Idea card metadata - Stable ID: KFM-P18-INV-123 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 123 - Status label: PROPOSED - Carry-forward state: NEW - Category: POL — Policy, Rights, Ethics, Sensitivity, and Public-Safety Controls - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should model drone or airspace advisory layers as policy/context overlays with source, time, and authority caveats rather than generic hazard maps.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should model drone or airspace advisory layers as policy/context overlays with source, time, and authority caveats rather than generic hazard maps. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It illustrates how regulatory or advisory spatial context can affect permitted field activity and public interpretation.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-002. - CONFIRMED baseline link: KFM-IDX-DAT-007.

Dependencies - PROPOSED: Policy overlay manifest - PROPOSED: Authority/source-role registry

Tensions - PROPOSED: Advisory tools may change and are not substitutes for official current flight authorization.

Expansion directions - PROPOSED: Add `overlay_role=policy_advisory` and freshness requirements.

Open questions - NEEDS VERIFICATION: Should KFM admit airspace advisory data only as operational_context?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 29, 30, 31; basis: B4UFLY drone exercise, geocoding, buffer analysis, feature selection, and map/layer packaging; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.6.4 KFM-P18-INV-208 — Participatory qualitative GIS needs consent and minor-safety controls

Idea card metadata - Stable ID: KFM-P18-INV-208 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 208 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: POL — Policy, Rights, Ethics, Sensitivity, and Public-Safety Controls - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM participatory or qualitative GIS workflows involving children or vulnerable groups should require consent, safe-session documentation, and privacy-preserving release transforms.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM participatory or qualitative GIS workflows involving children or vulnerable groups should require consent, safe-session documentation, and privacy-preserving release transforms. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It protects participants while still allowing perception and place-experience evidence to inform planning.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-AIG-004.

Dependencies - PROPOSED: Consent receipt - PROPOSED: Participant protection policy - PROPOSED: Generalized output - PROPOSED: Review state

Tensions - PROPOSED: Participatory evidence can expose sensitive household, school, or movement information.

Expansion directions - PROPOSED: Add minor_participation and qualitative_gis release controls to sensitivity policy.

Open questions - NEEDS VERIFICATION: What qualitative GIS artifacts should be denied public release even when summarized?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 182; basis: qualitative GIS with children, informed consent from legal guardians and children, and safe non-formal learning-space conditions; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-001, KFM-IDX-EVD-001, KFM-IDX-AIG-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.6.5 KFM-P18-INV-298 — Qualitative GIS perception layers require participant-method context

Idea card metadata - Stable ID: KFM-P18-INV-298 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 298 - Status label: PROPOSED - Carry-forward state: NEW - Category: POL — Policy, Rights, Ethics, Sensitivity, and Public-Safety Controls - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM qualitative GIS layers should record participant group, collection method, consent, anonymization, and interpretation limits before map display.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM qualitative GIS layers should record participant group, collection method, consent, anonymization, and interpretation limits before map display. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, run-time routing, or release state.

Why it matters - PROPOSED: It prevents perception data from being flattened into ordinary spatial measurements.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-004. - CONFIRMED baseline link: KFM-IDX-DAT-012.

Dependencies - PROPOSED: consent/anonymization policy - PROPOSED: qualitative method record - PROPOSED: participant protection review

Tensions - PROPOSED: Qualitative spatial evidence is valuable but may involve vulnerable participants and privacy constraints.

Expansion directions - PROPOSED: Add `qualitative_gis_method` and `participant_protection` fields to perception layers.

Open questions - NEEDS VERIFICATION: Should any child-participant perception geometries be public, or only generalized summaries?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 176; basis: The sustainable urban GIS source includes a qualitative GIS methodological investigation of children's perception of a city centre and participatory planning context.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.6.6 KFM-P18-INV-314 — AI data ethics gate for privacy, bias, ownership, and transparency

Idea card metadata - Stable ID: KFM-P18-INV-314 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 314 - Status label: PROPOSED - Carry-forward state: NEW - Category: POL — Policy, Rights, Ethics, Sensitivity, and Public-Safety Controls - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM AI-assisted analyses should pass a data-ethics gate that checks privacy, bias/fairness, ownership, and transparency before outputs are treated as publishable.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM AI-assisted analyses should pass a data-ethics gate that checks privacy, bias/fairness, ownership, and transparency before outputs are treated as publishable. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, run-time routing, or release state.

Why it matters - PROPOSED: It aligns AI use with KFM's evidence-first and rights-aware publication posture.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-001. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: privacy review - PROPOSED: bias/fairness check - PROPOSED: rights/ownership review - PROPOSED: data-use explanation

Tensions - PROPOSED: Ethical checks may be qualitative, but KFM needs auditable outcomes.

Expansion directions - PROPOSED: Add data_ethics_gate outcome to model and AI-response receipts.

Open questions - NEEDS VERIFICATION: What minimum documentation satisfies transparency for AI-assisted public map layers?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 84; basis: AI Concepts source lists key ethical challenges in data usage: privacy, bias and fairness, ownership, and transparency.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.6.7 KFM-P18-INV-341 — Ethical data-use fields for AI-supporting datasets

Idea card metadata - Stable ID: KFM-P18-INV-341 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 341 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: POL — Policy, Rights, Ethics, Sensitivity, and Public-Safety Controls - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM datasets used for AI support should carry purpose, consent or rights basis, bias-risk notes, and sensitive-use restrictions before they influence model outputs.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM datasets used for AI support should carry purpose, consent or rights basis, bias-risk notes, and sensitive-use restrictions before they influence model outputs. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents AI acceleration from weakening rights, consent, and sensitivity rules.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-003. - CONFIRMED baseline link: KFM-IDX-POL-005.

Dependencies - PROPOSED: purpose statement - PROPOSED: rights basis - PROPOSED: consent status - PROPOSED: bias-risk note - PROPOSED: sensitive-use restriction

Tensions - PROPOSED: Ethics labels can become perfunctory if not tied to policy gates and review evidence.

Expansion directions - PROPOSED: Add ai_dataset_ethics_block to weak-material and expansion-agenda fragments.

Open questions - NEEDS VERIFICATION: How should bias risk be measured for sparse Kansas historical/geographic data?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 83; basis: AI Concepts Using Python includes ethical considerations in data usage as part of the role of data in AI; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-003, KFM-IDX-POL-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.6.8 KFM-P18-INV-358 — Drone and airspace exercises as policy-aware spatial-admission patterns

Idea card metadata - Stable ID: KFM-P18-INV-358 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 358 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: POL — Policy, Rights, Ethics, Sensitivity, and Public-Safety Controls - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should treat drone-flight, airspace, or field-capture planning layers as policy-aware admission contexts whose constraints must be checked before data collection is recommended.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat drone-flight, airspace, or field-capture planning layers as policy-aware admission contexts whose constraints must be checked before data collection is recommended. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card

prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Field-capture workflows can expose safety, privacy, aviation, or property constraints before evidence even enters RAW.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-003. - CONFIRMED baseline link: KFM-IDX-PIP-008.

Dependencies - PROPOSED: policy constraint layer - PROPOSED: field plan - PROPOSED: collector role - PROPOSED: admission decision

Tensions - PROPOSED: KFM must not imply operational aviation permission without authoritative source checks.

Expansion directions - PROPOSED: Define pre-collection policy checks for drone-derived evidence.

Open questions - NEEDS VERIFICATION: Which authoritative airspace sources are needed before any drone workflow becomes actionable?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 3, 29; basis: Earth, Space, and Environmental Science Explorations with ArcGIS Pro includes a chapter titled “Want to fly a drone? Know B4UFLY!” with geocoding, buffer analysis, feature selection, and sharing objectives.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-003, KFM-IDX-PIP-008 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.6.9 KFM-P18-INV-403 — Copyright and distribution-rights check before reusing geographic information

Idea card metadata - Stable ID: KFM-P18-INV-403 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 403 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: POL — Policy, Rights, Ethics, Sensitivity, and Public-Safety Controls - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should block reuse of external geographic information until copyright, distribution rights, and intended-use compatibility are recorded as source-admission evidence.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should block reuse of external geographic information until copyright, distribution rights, and intended-use compatibility are recorded as source-admission evidence. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Geographic information can be technically useful but legally or ethically unavailable for public release; rights uncertainty must fail closed.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-003. - CONFIRMED baseline link: KFM-IDX-CAT-004.

Dependencies - PROPOSED: SourceDescriptor - PROPOSED: rights_spdx or NOASSERTION - PROPOSED: policy decision - PROPOSED: release gate

Tensions - PROPOSED: Open educational examples may not carry production-ready license metadata.

Expansion directions - PROPOSED: Add rights_reuse_check to source intake and cite rights uncertainty in weak material.

Open questions - NEEDS VERIFICATION: How should KFM handle educational or textbook-derived examples that are conceptual rather than operational sources?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 164; basis: A Primer of GIS names copyright status and distribution rights as assessment items before existing geographic information is reused.; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-003, KFM-IDX-CAT-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.6.10 KFM-P18-INV-457 — AI data-ethics register for privacy, bias, ownership, and transparency

Idea card metadata - Stable ID: KFM-P18-INV-457 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 457 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: POL — Policy, Rights, Ethics, Sensitivity, and Public-Safety Controls - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM AI-related data intake should include an ethics register that flags privacy, bias/fairness, ownership/permission, and transparency obligations before model use.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM AI-related data intake should include an ethics register that flags privacy, bias/fairness, ownership/permission, and transparency obligations before model use. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Governed AI cannot rely only on technical validation; data rights and fairness risks must be visible before use.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-003. - CONFIRMED baseline link: KFM-IDX-POL-002.

Dependencies - PROPOSED: AI data source - PROPOSED: policy review - PROPOSED: sensitivity register - PROPOSED: model card

Tensions - PROPOSED: Ethics labels can become superficial if not tied to enforcement gates.

Expansion directions - PROPOSED: Create model-input ethics fixtures that force DENY or ABSTAIN for missing rights or bias review.

Open questions - NEEDS VERIFICATION: What minimum fairness evidence is required for spatial classification outputs?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 84, 356, 366; basis: The AI/Python source lists ethical challenges in AI data usage, including privacy, bias and fairness, ownership, transparency, security, and compliance obligations.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-003, KFM-IDX-POL-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.6.11 KFM-P18-INV-466 — Drone authorization parameters as capture metadata

Idea card metadata - Stable ID: KFM-P18-INV-466 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 466 - Status label: PROPOSED - Carry-forward state: NEW - Category: POL — Policy, Rights, Ethics, Sensitivity, and Public-Safety Controls - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM drone-derived field evidence should record authorization parameters, operating area, altitude, date/time, airspace review result, and applicable safety constraints.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM drone-derived field evidence should record authorization parameters, operating area, altitude, date/time, airspace review result, and applicable safety constraints. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Drone imagery may be useful evidence, but acquisition legality and operating constraints are part of admissibility.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-003. - CONFIRMED baseline link: KFM-IDX-EVD-006.

Dependencies - PROPOSED: field capture record - PROPOSED: rights/safety policy - PROPOSED: source descriptor - PROPOSED: run receipt

Tensions - PROPOSED: Regulatory details can change and may require current verification before operational use.

Expansion directions - PROPOSED: Add drone_capture_authorization fields to field-evidence candidate schemas.

Open questions - NEEDS VERIFICATION: What authorization evidence is sufficient for historical drone collections?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 29, 30, 31; basis: The ArcGIS environmental text introduces drone use, airport/airspace constraints, LAANC authorization, and operational parameters such as date/time, flight altitude, and aircraft model.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.6.12 KFM-P18-INV-483 — Transparency and privacy notes for AI-derived explanations

Idea card metadata - Stable ID: KFM-P18-INV-483 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 483 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: POL — Policy, Rights, Ethics, Sensitivity, and Public-Safety Controls - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM AI-derived explanations should carry explicit transparency and privacy notes that separate model reasoning support from evidence, source authority, and release state.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM AI-derived explanations should carry explicit transparency and privacy notes that separate model reasoning support from evidence, source authority, and release state. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Users need to know when a statement is an evidence-backed claim versus a model-assisted explanation.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-003. - CONFIRMED baseline link: KFM-IDX-POL-005.

Dependencies - PROPOSED: transparency - PROPOSED: privacy - PROPOSED: AI output - PROPOSED: policy gate

Tensions - PROPOSED: Transparency notes must avoid exposing private chain-of-thought or sensitive data.

Expansion directions - PROPOSED: Extend runtime_response_envelope metadata with public transparency_note and privacy_basis fields.

Open questions - NEEDS VERIFICATION: How should Phase 7 front matter describe model-assisted sections without overclaiming?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 348; basis: The AI/Python source identifies transparency, explainability, privacy, and protection of sensitive information as responsible AI challenges.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-003, KFM-IDX-POL-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7 VAL - Validation, QA, Testing, Observability, and Reliability

Status: CONFIRMED source support for grouped Phase 3 ideas; PROPOSED category synthesis

Definition: Validation, QA, testing, observability, reliability, release assurance, operational telemetry, and review readiness.

CONFIRMED: This chapter contains 23 Pass 18 idea cards assigned to VAL.

Idea ID	Title	Carry state	Status	Source IDs
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KFM-P18-INV-038 | Scraper tests and modular packages as connector minimums | NEW | PROPOSED | SRC-P18-012 |

KFM-P18-INV-046 | OCI/Cosign publication for PMTiles, COG, and GeoParquet artifacts | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-067 | Scraper persistence and tests before source admission | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-078 | Process representation requires reliability and comparability metadata | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-125 | Accessibility and performance budgets as map-release criteria | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-126 | Topology errors as pre-release geometry defects | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-131 | Join conditions as crosswalk validation controls | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-147 | Side-effect-free validation functions for deterministic gates | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-150 | Style source-layer validation for released map integrity | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-190 | Declarative policy and validation design | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-192 | Cohesive mechanisms for reusable validators and transforms | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-216 | Web connector persistence and test fixtures | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-246 | Web connector tests guard against site-shape drift | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-256 | Guided GIS task workflows can be converted into validation checkpoints | NEW | PROPOSED | SRC-P18-010 |

KFM-P18-INV-338 | Robust scraper architecture as testable connector design | NEW | PROPOSED | SRC-P18-012 |

KFM-P18-INV-344 | GIS accessibility and complexity as product-governance evidence | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-360 | GIS exercise QA prompts as analysis-step receipt cues | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-418 | GIS exercise questions as embedded validation checkpoints | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-443 | Missing-value filters as analysis QA receipts | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-481 | Training-test split receipts for model-assisted layers | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-495 | Environmental GIS lab outputs as QA fixture patterns | NEW | PROPOSED | SRC-P18-010 |

KFM-P18-INV-498 | Durable scraper checkpoint state for source adapters | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-500 | SLO and error-budget gates for public map services | EXPANDED | PROPOSED | SRC-P18-039 |

8.7.1 KFM-P18-INV-038 — Scraper tests and modular packages as connector minimums

Idea card metadata - Stable ID: KFM-P18-INV-038 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 38 - Status label: PROPOSED - Carry-forward state: NEW - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM should require modular connector code and fixture-based tests before any scraper or acquisition bot is admitted to the pipeline.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should require modular connector code and fixture-based tests before any scraper or acquisition bot is admitted to the pipeline. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reduces brittle one-off acquisition scripts that cannot be audited or safely replayed.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-006. - CONFIRMED baseline link: KFM-IDX-POL-007.

Dependencies - PROPOSED: Connector test fixtures - PROPOSED: no-network test mode - PROPOSED: module boundary - PROPOSED: source mock

Tensions - PROPOSED: Tests can pass on fixtures while live source behavior changes.

Expansion directions - PROPOSED: Create a connector_admission_check that denies untested source fetchers.

Open questions - NEEDS VERIFICATION: What is the minimum no-network fixture set for scrape-based connectors?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 36, 50, 60; basis: testing, robust/scalable code, packages, and threaded crawler material; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.2 KFM-P18-INV-046 — OCI/Cosign publication for PMTiles, COG, and GeoParquet artifacts

Idea card metadata - Stable ID: KFM-P18-INV-046 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 46 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should treat PMTiles, COG, and GeoParquet map artifacts as digest-pinned, signed, provenance-linked release assets.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat PMTiles, COG, and GeoParquet map artifacts as digest-pinned, signed, provenance-linked release assets. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes map artifacts auditable and rollback-capable rather than anonymous static files.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-003. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: TileArtifactManifest - PROPOSED: Cosign verification - PROPOSED: PROV record - PROPOSED: release digest - PROPOSED: rollback target

Tensions - PROPOSED: Package/version and hosting details remain NEEDS VERIFICATION before operational use.

Expansion directions - PROPOSED: Add artifact_media_type and cosign_bundle_ref to release manifests.

Open questions - NEEDS VERIFICATION: Should all public map artifacts require OCI references or only high-significance releases?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 10, 14; basis: OCI, PMTiles/GeoParquet/COG, digest, SBOM, PROV, and attestation entries; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-003, KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.3 KFM-P18-INV-067 — Scraper persistence and tests before source admission

Idea card metadata - Stable ID: KFM-P18-INV-067 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 67 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM should require acquisition connectors that scrape or crawl the web to persist intermediate results, handle failures, and pass tests before source admission.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should require acquisition connectors that scrape or crawl the web to persist intermediate results, handle failures, and pass tests before source admission. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, run-time routing, or release state.

Why it matters - PROPOSED: Scraping failures, rate limits, dynamic pages, and changing markup can silently corrupt source evidence unless captured as testable acquisition behavior.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-005. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: SourceAcquisitionPolicy - PROPOSED: connector fixture suite - PROPOSED: failure receipt

Tensions - PROPOSED: Some scraping techniques, especially undercover or high-concurrency tactics, may be denied on policy grounds even if technically possible.

Expansion directions - PROPOSED: Add scraper_admission_review with source terms, robots/rate posture, fixtures, persistence and test results.

Open questions - NEEDS VERIFICATION: Should KFM globally deny proxy rotation and undercover scraping unless a steward exception exists?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 7, 9, 14, 22, 25, 33, 36, 46, 50; basis: scraping recipes including errors, persistence, tests, Ajax, threading; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-005, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.4 KFM-P18-INV-078 — Process representation requires reliability and comparability metadata

Idea card metadata - Stable ID: KFM-P18-INV-078 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 78 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should require reliability and comparability metadata when maps represent processes, change, or dynamic phenomena.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea

should be treated as a governed design candidate: KFM should require reliability and comparability meta-data when maps represent processes, change, or dynamic phenomena. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes change maps, anomaly surfaces, and trend summaries accountable to temporal and sensor comparability rather than only visual plausibility.

Related ideas - CONFIRMED baseline link: KFM-IDX-OBS-002. - CONFIRMED baseline link: KFM-IDX-DAT-013.

Dependencies - PROPOSED: Observation comparability rules - PROPOSED: Temporal metadata - PROPOSED: Sensor/source descriptors

Tensions - PROPOSED: Process maps can imply causality even when only selected moments are represented.

Expansion directions - PROPOSED: Define comparability_basis and process_representation_mode fields for dynamic layers.

Open questions - NEEDS VERIFICATION: How should KFM distinguish observed process, modeled process, and illustrative animation?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 27, 39; basis: dynamic visualization, geostatistics, reliability, and seasonal image comparability discussion; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-OBS-002, KFM-IDX-DAT-013 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.5 KFM-P18-INV-125 — Accessibility and performance budgets as map-release criteria

Idea card metadata - Stable ID: KFM-P18-INV-125 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 125 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM map view registries should store accessibility and performance budgets as release criteria for heavy or consequential map surfaces.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM map view registries should store accessibility and performance budgets as release criteria for heavy or consequential map surfaces. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents technically correct map layers from becoming unusable, inaccessible, or too slow for public review.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-006. - CONFIRMED baseline link: KFM-IDX-OBS-005.

Dependencies - PROPOSED: View registry - PROPOSED: Preview renderer - PROPOSED: Release gate

Tensions - PROPOSED: Strict budgets may require generalization or progressive loading that changes what users see first.

Expansion directions - PROPOSED: Add `a11y_score`, `tile_load_budget`, `route_load_budget`, and `degradation_behavior` checks.

Open questions - NEEDS VERIFICATION: Which budgets are mandatory for public release versus internal review?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 11, 34; basis: ML-064-024 accessibility/performance-budget record and preview-build validation notes; Phase 1 role: `candidate_synthesis_source_specialized_atlas`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-006, KFM-IDX-OBS-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.6 KFM-P18-INV-126 — Topology errors as pre-release geometry defects

Idea card metadata - Stable ID: KFM-P18-INV-126 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 126 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should treat topology defects, invalid shared boundaries, and geometry-model mismatches as pre-release validation defects rather than as cosmetic map issues.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (`a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf`) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat topology defects, invalid shared boundaries, and geometry-model mismatches as pre-release validation defects rather than as cosmetic map issues. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents public layers from carrying broken adjacency, overlap, or network assumptions into downstream maps, joins, and evidence claims.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-004. - CONFIRMED baseline link: KFM-IDX-POL-004.

Dependencies - PROPOSED: Geometry validators - PROPOSED: LayerManifest - PROPOSED: SourceDescriptor geometry model fields

Tensions - PROPOSED: Topology validation may require domain-specific tolerances rather than one universal rule.

Expansion directions - PROPOSED: Define topology fixture classes for polygons, networks, and rasters.

Open questions - NEEDS VERIFICATION: Which geometry defects deny promotion versus require reviewer warning?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 60, 61, 174, 193; basis: topology and vector/raster representation discussion in the GIS primer; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-004, KFM-IDX-POL-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.7 KFM-P18-INV-131 — Join conditions as crosswalk validation controls

Idea card metadata - Stable ID: KFM-P18-INV-131 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 131 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM should record join keys and ON-clause filters used in crosswalks as validation-relevant metadata, not hide them inside query text.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record join keys and ON-clause filters used in crosswalks as validation-relevant metadata, not hide them inside query text. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes crosswalk drift, unmatched rows, and accidental many-to-many expansion inspectable.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-007. - CONFIRMED baseline link: KFM-IDX-PIP-006.

Dependencies - PROPOSED: CrosswalkManifest - PROPOSED: ReconciliationReport

Tensions - PROPOSED: Query optimization can obscure semantic join intent unless explicitly documented.

Expansion directions - PROPOSED: Emit join cardinality and unmatched-row counts in crosswalk receipts.

Open questions - NEEDS VERIFICATION: Which crosswalk joins require steward review?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 20, 21, 42, 84; basis: advanced SQL discussion of joins, ON clauses, and combining related tables; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-007, KFM-IDX-PIP-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.8 KFM-P18-INV-147 — Side-effect-free validation functions for deterministic gates

Idea card metadata - Stable ID: KFM-P18-INV-147 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 147 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM validators should be side-effect-free where practical and should express assertions explicitly so validation results are deterministic, replayable, and auditable.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM validators should be side-effect-free where practical and should express assertions explicitly so validation results are deterministic, replayable, and auditable. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It helps reviewers reproduce why a gate allowed, denied, abstained, or errored.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-004. - CONFIRMED baseline link: KFM-IDX-DOC-008.

Dependencies - PROPOSED: Validator registry - PROPOSED: Policy test fixtures - PROPOSED: RunReceipt

Tensions - PROPOSED: Some validators need external evidence resolution; those calls must be separated from pure checks.

Expansion directions - PROPOSED: Separate evidence-fetch steps from pure validation functions.

Open questions - NEEDS VERIFICATION: Which validators can be purely deterministic offline?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 21, 28, 29, 32; basis: side-effect-free functions, assertions, and declarative design patterns; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-004, KFM-IDX-DOC-008 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.9 KFM-P18-INV-150 — Style source-layer validation for released map integrity

Idea card metadata - Stable ID: KFM-P18-INV-150 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 150 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED

Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should validate that every MapLibre style source-layer, layer id, promoteId, and source URL resolves to a released manifest entry before a style can publish.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should validate that every MapLibre style source-layer, layer id, promoteId, and source URL resolves to a released manifest entry before a style can publish. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents public styles from loading unreleased, stale, wrong-source, or direct-database layers.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-011. - CONFIRMED baseline link: KFM-IDX-POL-004.

Dependencies - PROPOSED: StyleManifest - PROPOSED: LayerManifest - PROPOSED: MapReleaseManifest - PROPOSED: Visual regression tests

Tensions - PROPOSED: External basemap sources may need different handling from KFM claim-bearing layers.

Expansion directions - PROPOSED: Add style-to-manifest consistency tests and deny unreleased source-layer references.

Open questions - NEEDS VERIFICATION: Which external basemap exceptions are allowed, and how are they labeled?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 19, 25, 26, 30; basis: Master MapLibre atlas v1.9 discussion of StyleManifest, source-layer checks, source registry, and visual regression; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-011, KFM-IDX-POL-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.10 KFM-P18-INV-190 — Declarative policy and validation design

Idea card metadata - Stable ID: KFM-P18-INV-190 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 190 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should prefer declarative validation and policy specifications where rules can act as executable, reviewable guarantees rather than hidden procedural checks.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should prefer declarative validation and policy specifications where rules can act as executable, reviewable guarantees rather than hidden procedural checks. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reinforces fail-closed behavior and makes reviewers inspect the rule itself, not only procedural code.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-MDP-002.

Dependencies - PROPOSED: JSON Schema - PROPOSED: OPA/Rego policy - PROPOSED: Conftest or equivalent - PROPOSED: Validator registry

Tensions - PROPOSED: Declarations can be bypassed if runtime and CI do not enforce the same rule bundle.

Expansion directions - PROPOSED: Tie each declared rule to CI fixtures and runtime policy parity checks.

Open questions - NEEDS VERIFICATION: Which KFM checks should remain procedural because they cannot be expressed declaratively?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 32; basis: Declarative Design pattern describing executable specifications, formal guarantees, and risks when developers bypass declarations; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-001, KFM-IDX-PIP-004, KFM-IDX-MDP-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.11 KFM-P18-INV-192 — Cohesive mechanisms for reusable validators and transforms

Idea card metadata - Stable ID: KFM-P18-INV-192 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 192 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should factor generic mechanisms such as hashing, canonicalization, envelope validation, catalog closure, and temporal predicates into reusable frameworks without letting them swallow domain meaning.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should factor generic mechanisms such as hashing, canonicalization, envelope validation, catalog closure, and temporal predicates into reusable frameworks without letting them swallow domain meaning. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5.

- **PROPOSED**: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - **UNKNOWN**: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - **NEEDS VERIFICATION**: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - **PROPOSED**: It allows domain lanes to share reliable machinery while keeping their source roles and claim semantics distinct.

Related ideas - **CONFIRMED** baseline link: KFM-IDX-DOC-008. - **CONFIRMED** baseline link: KFM-IDX-PIP-010. - **CONFIRMED** baseline link: KFM-IDX-CAT-004.

Dependencies - **PROPOSED**: Shared utility package - **PROPOSED**: Validator registry - **PROPOSED**: Domain-lane adapters

Tensions - **PROPOSED**: Over-generalized frameworks can obscure source-specific risk.

Expansion directions - **PROPOSED**: List candidate shared mechanisms separately from domain-specific schemas in Phase 4 categories.

Open questions - **NEEDS VERIFICATION**: Which mechanisms should be shared packages versus per-lane validators?

Future work - **PROPOSED**: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - **CONFIRMED** source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 51; basis: Cohesive Mechanisms pattern for separating complex generic mechanisms from core domain concepts through intention-revealing interfaces; Phase 1 role: reference_source_candidate.

Carry-forward note - **CONFIRMED**: Carry-forward state is **EXPANDED**; this card extends KFM-IDX-DOC-008, KFM-IDX-PIP-010, KFM-IDX-CAT-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - **CONFIRMED**: Required schema fields are present in this Markdown card. - **CONFIRMED**: No prior-pass body text is inlined. - **CONFIRMED**: Narrative claims use explicit truth labels. - **UNKNOWN**: Repository implementation status remains unverified.

8.7.12 KFM-P18-INV-216 — Web connector persistence and test fixtures

Idea card metadata - Stable ID: KFM-P18-INV-216 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 216 - Status label: **PROPOSED** - Carry-forward state: **EXPANDED** - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: **PROPOSED** assignment derived from **CONFIRMED** Phase 3 inventory; subject to Phase 5 card-level review - Source support label: **CONFIRMED** - Implementation maturity: **UNKNOWN** — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-012

Normalized statement - **PROPOSED**: KFM web-derived source intake should persist fetched results and provide connector tests that prove parsing, storage, and failure handling against controlled fixtures.

Detailed explanation - **CONFIRMED**: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - **CONFIRMED**: Phase 3 inventory recorded this idea with source support label **CONFIRMED** and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - **PROPOSED**: In KFM, this idea should be treated as a governed design candidate: KFM web-derived source intake should persist fetched results and provide connector tests that prove parsing, storage, and failure handling against controlled fixtures. - **PROPOSED**: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - **PROPOSED**: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - **UNKNOWN**: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - **NEEDS VERIFICATION**: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - **PROPOSED**: It helps distinguish repeatable source acquisition from ad hoc scraping.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-SEC-010.

Dependencies - PROPOSED: Fetch fixture - PROPOSED: Parser fixture - PROPOSED: Persistence receipt - PROPOSED: Connector test suite

Tensions - PROPOSED: Persistence can store bad data as durably as good data if validation is late.

Expansion directions - PROPOSED: Create no-network fixtures for fetched HTML/JSON, parsed records, persistence receipts, and failed parse cases.

Open questions - NEEDS VERIFICATION: Should KFM allow any live web scraping connector before no-network fixture tests pass?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 11, 42; basis: persisting web-scraped data for later retrieval and using tests/JUnit-style practices in web scraping workflows; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-004, KFM-IDX-EVD-001, KFM-IDX-SEC-010 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.13 KFM-P18-INV-246 — Web connector tests guard against site-shape drift

Idea card metadata - Stable ID: KFM-P18-INV-246 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 246 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM web connectors should include parser, selector, and site-shape tests that fail closed when page structure changes.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM web connectors should include parser, selector, and site-shape tests that fail closed when page structure changes. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents a connector from silently ingesting the wrong HTML element or form response as evidence.

Related ideas - CONFIRMED baseline link: KFM-IDX-SEC-010. - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Parser fixture - PROPOSED: Selector contract - PROPOSED: No-network test - PROPOSED: Quarantine outcome

Tensions - PROPOSED: Tests based on unstable pages can become brittle unless fixtures preserve expected source shape.

Expansion directions - PROPOSED: Add connector_shape_test fixtures with expected selectors, schema output, and missing/changed-page quarantine cases.

Open questions - NEEDS VERIFICATION: What web-source drift should trigger quarantine versus manual review?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 46, 48, 54; basis: JUnit-style tests for scraper code and warning that website forms/page structures can change over time; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-SEC-010, KFM-IDX-PIP-004, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.14 KFM-P18-INV-256 — Guided GIS task workflows can be converted into validation checkpoints

Idea card metadata - Stable ID: KFM-P18-INV-256 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 256 - Status label: PROPOSED - Carry-forward state: NEW - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM can convert tutorial-style step sequences and embedded check questions into repeatable workflow checkpoints for derived GIS products.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM can convert tutorial-style step sequences and embedded check questions into repeatable workflow checkpoints for derived GIS products. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It turns manual GIS pedagogy into auditable workflow validation without pretending the tutorial is implementation proof.

Related ideas - CONFIRMED baseline link: KFM-IDX-MDP-003. - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Workflow recipe - PROPOSED: checkpoint list - PROPOSED: expected output digest or assertion

Tensions - PROPOSED: Educational exercises often depend on proprietary tools and manual actions.

Expansion directions - PROPOSED: Define task_checkpoint_receipt for step_id, expected_artifact, user_answer, assertion_result, and reviewer_note.

Open questions - NEEDS VERIFICATION: Which GIS tasks should be automated, and which should remain reviewer-guided?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 4; basis: technical objectives, step-by-step exercises, and embedded questions that ensure the learner is on track; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.15 KFM-P18-INV-338 — Robust scraper architecture as testable connector design

Idea card metadata - Stable ID: KFM-P18-INV-338 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 338 - Status label: PROPOSED - Carry-forward state: NEW - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM scraper code should be structured into testable adapter components with clear parsing, persistence, retry, and validation responsibilities before a web source is admitted.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM scraper code should be structured into testable adapter components with clear parsing, persistence, retry, and validation responsibilities before a web source is admitted. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps scraping fallback connectors reviewable and replaceable.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-006. - CONFIRMED baseline link: KFM-IDX-MDP-002.

Dependencies - PROPOSED: parser module - PROPOSED: fetch module - PROPOSED: retry policy - PROPOSED: unit tests - PROPOSED: fixture HTML

Tensions - PROPOSED: Over-engineering a scraper can imply source authority it does not have.

Expansion directions - PROPOSED: Create fixture-backed scraper adapter tests for candidate fallback sources.

Open questions - NEEDS VERIFICATION: Which scraped sources deserve full adapter investment versus one-time quarantine analysis?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 25, 36; basis: Instant Web Scraping with Java includes recipes for robust/scalable code and writing tests.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.16 KFM-P18-INV-344 — GIS accessibility and complexity as product-governance evidence

Idea card metadata - Stable ID: KFM-P18-INV-344 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 344 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM should record usability, training burden, and tool-complexity risks when GIS workflows are proposed for public, steward, or planning-support users.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record usability, training burden, and tool-complexity risks when GIS workflows are proposed for public, steward, or planning-support users. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It treats map-first usability as part of governance and adoption, not only UI polish.

Related ideas - CONFIRMED baseline link: KFM-IDX-OBS-003. - CONFIRMED baseline link: KFM-IDX-MDP-002.

Dependencies - PROPOSED: user group - PROPOSED: tool complexity - PROPOSED: training need - PROPOSED: workflow burden - PROPOSED: adoption risk

Tensions - PROPOSED: Lowering complexity can hide evidence detail; exposing everything can overwhelm non-expert users.

Expansion directions - PROPOSED: Add audience_complexity notes to map-release and review-console plans.

Open questions - NEEDS VERIFICATION: How should KFM test whether evidence-rich maps remain usable to non-expert reviewers?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 15, 16; basis: GIS in Sustainable Urban Planning and Management notes that GIS is increasingly available while powerful applications remain complex and that implementation gaps remain.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-OBS-003, KFM-IDX-MDP-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.17 KFM-P18-INV-360 — GIS exercise QA prompts as analysis-step receipt cues

Idea card metadata - Stable ID: KFM-P18-INV-360 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 360 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM analytic notebooks and map workflows should convert embedded QA prompts such as summary-statistic checks and missing-value filters into explicit analysis-step receipt fields.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM analytic notebooks and map workflows should convert embedded QA prompts such as summary-statistic checks and missing-value filters into explicit analysis-step receipt fields. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Pedagogical checkpoints can become governance checkpoints when transformed into receipts and validation evidence.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-010. - CONFIRMED baseline link: KFM-IDX-EVD-006.

Dependencies - PROPOSED: analysis notebook - PROPOSED: summary statistics - PROPOSED: missing-data rule - PROPOSED: run receipt

Tensions - PROPOSED: Human exercise prompts are not automatically sufficient as automated tests.

Expansion directions - PROPOSED: Convert common GIS tutorial checks into machine-readable QA receipt examples.

Open questions - NEEDS VERIFICATION: Which human-in-the-loop checks should become automated validators?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 7; basis: Earth, Space, and Environmental Science Explorations with ArcGIS Pro lists exploring summary statistics and filtering out missing data values and features in a flood-hazard exercise.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-010, KFM-IDX-EVD-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.18 KFM-P18-INV-418 — GIS exercise questions as embedded validation checkpoints

Idea card metadata - Stable ID: KFM-P18-INV-418 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 418 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM training, review, and workflow documentation should use embedded checkpoint questions as lightweight validation evidence for human-in-the-loop GIS procedures.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea

should be treated as a governed design candidate: KFM training, review, and workflow documentation should use embedded checkpoint questions as lightweight validation evidence for human-in-the-loop GIS procedures. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Stepwise GIS work often depends on human choices; embedded checks create a record that the analyst followed and understood the workflow.

Related ideas - CONFIRMED baseline link: KFM-IDX-MDP-004. - CONFIRMED baseline link: KFM-IDX-PIP-005.

Dependencies - PROPOSED: runbook template - PROPOSED: review checklist - PROPOSED: analyst receipt - PROPOSED: training artifact

Tensions - PROPOSED: Question answers are not equivalent to machine validation.

Expansion directions - PROPOSED: Add human_checkpoint_answers to review receipts for manual GIS workflows.

Open questions - NEEDS VERIFICATION: Which manual workflows require reviewer-signoff checkpoints rather than optional training prompts?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 4, 7, 8; basis: Earth, Space, and Environmental Science Explorations with ArcGIS Pro describes exercises with technical objectives, context, step-by-step directions, and embedded questions.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-MDP-004, KFM-IDX-PIP-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.19 KFM-P18-INV-443 — Missing-value filters as analysis QA receipts

Idea card metadata - Stable ID: KFM-P18-INV-443 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 443 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM analysis workflows should record missing-value filters and summary-statistic steps as QA receipts before publishing derived layers.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM analysis workflows should record missing-value filters and summary-statistic steps as QA receipts before publishing derived layers. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION:

Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Filtering missing values changes the population of evidence and can bias outputs if not disclosed.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-005. - CONFIRMED baseline link: KFM-IDX-DAT-012.

Dependencies - PROPOSED: analysis receipt - PROPOSED: DQ report - PROPOSED: summary statistics - PROPOSED: layer manifest

Tensions - PROPOSED: Educational exercises may not map directly to production-grade QA.

Expansion directions - PROPOSED: Create a `missing_value_filter_receipt` for derived analyses.

Open questions - NEEDS VERIFICATION: When should missing values cause ABSTAIN rather than filtered publication?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 7, 18, 21; basis: Earth, Space, and Environmental Science Explorations with ArcGIS Pro includes objectives for filtering missing data values and exploring summary statistics.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-005, KFM-IDX-DAT-012 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.20 KFM-P18-INV-481 — Training-test split receipts for model-assisted layers

Idea card metadata - Stable ID: KFM-P18-INV-481 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 481 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM model-assisted layers should record split method, random seed, stratification choices, and evaluation holdout scope before model outputs can support a public explanation.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM model-assisted layers should record split method, random seed, stratification choices, and evaluation holdout scope before model outputs can support a public explanation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Model evidence is not inspectable unless reviewers can see how performance was evaluated.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-004. - CONFIRMED baseline link: KFM-IDX-DOC-007.

Dependencies - PROPOSED: training data - PROPOSED: test data - PROPOSED: model evaluation - PROPOSED: AIReceipt

Tensions - PROPOSED: Spatial autocorrelation can make ordinary random splits overstate real-world performance.

Expansion directions - PROPOSED: Add spatial_holdout_policy to model-evaluation receipts.

Open questions - NEEDS VERIFICATION: Should spatial-block cross-validation be the default for map-derived models?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 156, 157, 163; basis: The AI/Python source treats training/testing splits as a fundamental machine-learning workflow step and warns that evaluating on training data can give optimistic results.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-004, KFM-IDX-DOC-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.21 KFM-P18-INV-495 — Environmental GIS lab outputs as QA fixture patterns

Idea card metadata - Stable ID: KFM-P18-INV-495 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 495 - Status label: PROPOSED - Carry-forward state: NEW - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM can adapt educational GIS exercise patterns into QA fixtures that state expected layer inputs, transformations, checks, and reviewer questions.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM can adapt educational GIS exercise patterns into QA fixtures that state expected layer inputs, transformations, checks, and reviewer questions. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Structured exercises provide a useful template for reproducible no-network validation tasks.

Related ideas - CONFIRMED baseline link: KFM-IDX-MDP-003. - CONFIRMED baseline link: KFM-IDX-DAT-012.

Dependencies - PROPOSED: GIS exercise - PROPOSED: embedded questions - PROPOSED: fixture - PROPOSED: validation checkpoint

Tensions - PROPOSED: Educational steps may not meet production governance requirements without stronger provenance.

Expansion directions - PROPOSED: Use lab-style objective/checkpoint templates in Phase 6 validation fragments.

Open questions - NEEDS VERIFICATION: Which source families can be tested with synthetic lab-style fixtures?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 69, 85, 86; basis: The ArcGIS environmental text structures chapters as technical exercises with objectives, step-by-step workflows, and embedded validation questions across hazards and field data tasks.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.22 KFM-P18-INV-498 — Durable scraper checkpoint state for source adapters

Idea card metadata - Stable ID: KFM-P18-INV-498 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 498 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM source adapters that scrape or crawl should persist checkpoint, error, and retry state in run receipts or adapter state records, not only in transient process memory.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM source adapters that scrape or crawl should persist checkpoint, error, and retry state in run receipts or adapter state records, not only in transient process memory. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Durable checkpoints make crawler behavior auditable, restartable, and respectful of source limits.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-006. - CONFIRMED baseline link: KFM-IDX-PIP-010.

Dependencies - PROPOSED: scraper - PROPOSED: persistence - PROPOSED: checkpoint - PROPOSED: run receipt

Tensions - PROPOSED: Persisting too much page state can capture data that rights or robots rules do not allow for reuse.

Expansion directions - PROPOSED: Define crawl_checkpoint_receipt and rate_limit_state fields for source adapters.

Open questions - NEEDS VERIFICATION: Which crawl state should be redacted from public receipts?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 31, 42, 59; basis: The Java scraping source covers handling errors, persisting data, and faster scraping through threading as part of robust web acquisition.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-006, KFM-IDX-PIP-010 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.7.23 KFM-P18-INV-500 — SLO and error-budget gates for public map services

Idea card metadata - Stable ID: KFM-P18-INV-500 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 500 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: VAL — Validation, QA, Testing, Observability, and Reliability - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory and Phase 4 dependency ordering; subject to final synthesis review. - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM public map services should include SLO and error-budget checks that can block promotion or trigger rollback without treating telemetry as truth content.

Detailed explanation - CONFIRMED: The Master MapLibre atlas supports the card by recording map-layer trust, StyleManifest/TileArtifactManifest patterns, visual regression, SLOs, PMTiles/COG governance, and MapLibre as downstream renderer. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and source basis: The Master MapLibre atlas carries v1.9 records and anti-pattern notes about SLOs, error budgets, tile latency, and trust-membrane risks for public map services. - CONFIRMED: Phase 4 assigned this idea to VAL — Validation, QA, Testing, Observability, and Reliability before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM public map services should include SLO and error-budget checks that can block promotion or trigger rollback without treating telemetry as truth content. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: The source confirms the conceptual basis listed above; it does not prove KFM has implemented the proposed schema field, validator, policy, layer metadata, API route, UI control, or release gate.

Why it matters - PROPOSED: A layer that is evidence-correct but operationally unreliable can still mislead or fail users at publication time.

Related ideas - CONFIRMED baseline link: KFM-IDX-OBS-003. - CONFIRMED baseline link: KFM-IDX-OBS-004.

Dependencies - PROPOSED: SLO - PROPOSED: error budget - PROPOSED: public map service - PROPOSED: rollback

Tensions - PROPOSED: SLO gates may delay publication of time-sensitive but non-life-safety context layers.

Expansion directions - PROPOSED: Tie map-service SLO checks to release validation and rollback fragments in Phase 6.

Open questions - NEEDS VERIFICATION: Which SLOs are mandatory for static PMTiles versus server-mediated layers?

Future work - PROPOSED: Convert this idea into a fixture-backed design note during later synthesis only after source support, rights posture, implementation home, policy impact, validator shape, and rollback consequence are checked. - PROPOSED: Preserve the idea as a candidate in the Phase 8 baseline-for-next-pass sidecar so later passes can retain, expand, supersede, or retire it with explicit evidence.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 34, 35; basis: The Master MapLibre atlas carries v1.9 records and anti-pattern notes about SLOs, error budgets, tile latency, and trust-membrane risks for public map services.; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card expands or sharpens prior Pass 17 baseline links without re-authoring the prior-pass body.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - CONFIRMED: Source attribution is present and points to the Phase 3 source record. - UNKNOWN: Repository implementation status remains unverified.

8.8 ANA - Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation

Status: CONFIRMED source support for grouped Phase 3 ideas; PROPOSED category synthesis

Definition: Analysis, indicators, statistics, model interpretation, AI/ML method evaluation, and derived metric discipline.

CONFIRMED: This chapter contains 108 Pass 18 idea cards assigned to ANA.

Idea ID	Title	Carry state	Status	Source IDs
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KFM-P18-INV-003 | Things-versus-events distinction for spatial claims | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-013 | Multiple-deprivation and cumulative-burden indicator bundles | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-018 | Separate captured 3D models from interpretive reconstructions | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-021 | Valid-time, transaction-time, and bitemporal truth posture | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-022 | Temporal keys and constraints as first-class validation | NEW | PROPOSED | SRC-P18-007 |

KFM-P18-INV-024 | Bounded contexts and ubiquitous language for KFM lanes | NEW | PROPOSED | SRC-P18-009 |

KFM-P18-INV-026 | Window analytics for released observation panels | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-027 | Recursive CTEs for hierarchies, networks, and calendar scaffolds | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-028 | CTEs as transparent transformation stages | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-029 | Set and join operations as explicit crosswalk semantics | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-030 | Null, aggregation, and COALESCE handling as evidence caveats | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-031 | AI data lifecycle as governed model input boundary | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-032 | Model-type labels for AI/ML outputs | NEW | PROPOSED | SRC-P18-004 |

KFM-P18-INV-033 | Train/test/evaluation discipline for model-assisted claims | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-034 | Bias, transparency, and explainability review for AI-supported surfaces | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-035 | Data quality dimensions as feature-source gates | NEW | PROPOSED | SRC-P18-004 |

KFM-P18-INV-054 | Positions, networks, fields, and transformations as distinct model families | NEW | PROPOSED | SRC-P18-002 |

KFM-P18-INV-061 | TOD indices as composite indicators with mode-shift assumptions | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-064 | Temporal query class labels for public explanations | NEW | PROPOSED | SRC-P18-007 |

KFM-P18-INV-065 | Aggregates, repositories, and domain events for lane boundaries | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-069 | CASE and conditional aggregation as auditable classification rules | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-070 | Missing-data and feature-preprocessing decisions as evidence transforms | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-071 | Algorithm limitation labels for classifications and clusters | NEW | PROPOSED | SRC-P18-004 |

KFM-P18-INV-082 | Global water-resource layers need classification and denominator transparency | NEW | PROPOSED | SRC-P18-010 |

KFM-P18-INV-083 | Raster suitability workflows need criteria weights and exclusion receipts | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-084 | Age-sensitive built-environment walking indicators | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-087 | Water-supply preference models need stated-choice caveats | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-088 | Urban-growth scenarios are planning evidence, not predictions | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-103 | Geostatistical products require uncertainty and model metadata | NEW | PROPOSED | SRC-P18-002 |

KFM-P18-INV-104 | Spatial model family labels for positions, networks, fields, and transformations | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-105 | Window frame semantics for analytical panels | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-107 | Database views as documented analytical contract boundaries | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-108 | Training/testing split reproducibility | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-109 | Confusion-matrix and precision/recall evidence for classification layers | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-111 | AI data lifecycle as a governed intake-to-use lane | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-121 | Aggregates and repositories as consistency boundaries | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-132 | Window ordering and tie-breakers as reproducibility requirements | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-133 | Recursive CTE termination and depth guards | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-134 | Machine-learning workflow failures as abstain triggers | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-135 | Ethical data-use review as AI source-admission gate | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-136 | Learning-mode labels for model-assisted surfaces | NEW | PROPOSED | SRC-P18-004 |

KFM-P18-INV-143 | Sequenced and nonsequenced temporal query labels | NEW | PROPOSED | SRC-P18-007 |

KFM-P18-INV-155 | DISTINCT and NULL semantics as deduplication risk | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-156 | SQL dialect verification for set-operator compatibility | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-157 | COUNT denominator semantics for public metrics | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-158 | Correlation aggregates as relationship evidence, not causality | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-159 | Model-selection rationale for model-assisted layers | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-160 | Hyperparameter tuning receipts for model claims | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-163 | Overfitting and underfitting as public model limitation labels | NEW | PROPOSED | SRC-P18-004 |

KFM-P18-INV-165 | Hands-on modelers as governance review discipline | NEW | PROPOSED | SRC-P18-009 |

KFM-P18-INV-166 | Refactoring-to-insight requires ADR and evidence trace | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-219 | Training/test split and time-leakage guards for model claims | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-220 | Imbalanced classification metrics and confusion-matrix disclosure | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-221 | Feature-engineering and preprocessing receipts for AI-derived layers | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-226 | Window-function analytics preserve row-level evidence | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-227 | Recursive CTE calendars and hierarchy fixtures require bounds | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-229 | Pre-filtered aggregation as a performance and correctness gate | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-230 | Missing-data mechanism and imputation receipts | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-231 | Model-evaluation receipts must prove generalization, not just fit | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-260 | K-means cluster choices require parameter and stability receipts | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-261 | KNN distance metrics and scaling are evidence-critical parameters | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-262 | Linear regression assumptions become model limitation labels | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-263 | Data visualization choices in AI analysis are interpretive outputs | NEW | PROPOSED | SRC-P18-004 |

KFM-P18-INV-265 | Correlated subqueries expose outer-query dependency in analytic receipts | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-284 | Connector unit tests are source-shape regression controls | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-306 | Published languages for interchange without freezing internal models | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-310 | ROLLUP and CUBE subtotal semantics require hierarchy receipts | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-311 | WITH CHECK OPTION as a view-level data-quality guardrail | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-312 | Timestamp-difference metrics require explicit unit conversion | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-313 | Missing-data method receipts distinguish deletion, imputation, and advanced repair | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-315 | Model split choice as an evaluation-fairness record | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-340 | AI data lifecycle as a governed transform chain | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-342 | Train-test split and evaluation records for model-assisted map claims | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-359 | Mars landing suitability as a transferable site-selection exercise | NEW | PROPOSED | SRC-P18-010 |

KFM-P18-INV-368 | AI missing-data handling as an evidence-quality decision | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-369 | Mathematical foundation notes as model explanation metadata | NEW | PROPOSED | SRC-P18-004 |

KFM-P18-INV-379 | Variogram and sampling metadata for interpolated surfaces | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-380 | FIRST_VALUE and LAST_VALUE anchors for released observation panels | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-382 | Subquery-in-FROM outputs as named analytical intermediates | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-383 | SQL performance optimization as reproducibility evidence for large panels | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-384 | Scikit-learn workflow artifacts as model-evidence chain | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-385 | Hybrid learning-mode labels for semi-supervised and reinforcement outputs | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-386 | Visualization artifacts for non-map AI analysis | NEW | PROPOSED | SRC-P18-004 |

KFM-P18-INV-393 | Tracking-log report reconstruction windows | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-399 | Scraper unit tests as source-adaptor regression evidence | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-404 | Transaction boundary receipts for data-changing SQL operations | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-405 | HAVING filters as post-aggregation policy decisions | NEW | PROPOSED | SRC-P18-003 |

KFM-P18-INV-406 | Query-cost and performance budget as reproducibility metadata | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-407 | Cross-validation and overfitting labels for predictive layers | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-408 | Feature scaling and normalization transform receipts | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-409 | AI visualization outputs remain explanatory derivatives | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-415 | Sequenced versus nonsequenced query label in public explanations | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-417 | Knowledge-level models for policy and source-role rules | NEW | PROPOSED | SRC-P18-009 |

KFM-P18-INV-429 | Set-operator merge semantics for multi-source evidence panels | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-430 | NULL-sensitive aggregate metrics for data-quality reports | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-431 | Feature-scaling receipts for model-assisted spatial analysis | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-432 | Confusion-matrix evidence for classification layers | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-433 | Deployment-monitoring records for model-assisted services | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-438 | Bitemporal correction model for released claim history | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-454 | ROLLUP and CUBE subtotal semantics for aggregate releases | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-455 | Database views as governed analytical interfaces | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-456 | Missing-data treatment receipts for AI-derived analyses | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-465 | Aggregate boundaries as validation units | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-477 | Choropleth classification scheme disclosure | NEW | PROPOSED | SRC-P18-002 |

KFM-P18-INV-478 | Recursive CTE traversal for evidence-graph hierarchies | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-479 | Window-frame definitions for rolling public metrics | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-480 | Subtotal semantics for ROLLUP and CUBE outputs | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-482 | Unsupervised cluster labels as provisional hypotheses | NEW | PROPOSED | SRC-P18-004 |

8.8.1 KFM-P18-INV-003 — Things-versus-events distinction for spatial claims

Idea card metadata - Stable ID: KFM-P18-INV-003 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 3 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should keep relatively static things and selected moments in processes as distinct claim families.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should keep relatively static things and selected moments in processes as distinct claim families. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback

target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reduces category drift between stable features, observations, alerts, historical events, and derived temporal snapshots.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-PIP-008.

Dependencies - PROPOSED: Event envelope or observation schema family

Tensions - PROPOSED: Some objects, such as roads or stream channels, are both things and histories of events.

Expansion directions - PROPOSED: Define a KFM claim-family matrix for feature, observation, event, model, and release.

Open questions - NEEDS VERIFICATION: How should KFM name records that are both feature state and event consequence?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 17, 18; basis: things/events and geographic information distinction; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-001, KFM-IDX-PIP-008 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.2 KFM-P18-INV-013 — Multiple-deprivation and cumulative-burden indicator bundles

Idea card metadata - Stable ID: KFM-P18-INV-013 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 13 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM should model composite social/environmental indicators as evidence-bound bundles with transparent inputs, weights, and caveats.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should model composite social/environmental indicators as evidence-bound bundles with transparent inputs, weights, and caveats. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Composite indicators are powerful and risky; opaque weights can produce authoritative-looking but weakly inspectable claims.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-003. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: IndicatorBundle schema; weighting receipt; source descriptors

Tensions - PROPOSED: Indicators communicate quickly but can hide uncertainty and normative choices.

Expansion directions - PROPOSED: Create a composite_indicator_receipt candidate object.

Open questions - NEEDS VERIFICATION: Should indicator weights require review separate from data-source review?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 58, 59, 98; basis: multiple deprivation and cumulative burden cases; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.3 KFM-P18-INV-018 — Separate captured 3D models from interpretive reconstructions

Idea card metadata - Stable ID: KFM-P18-INV-018 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 18 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should label reality-based capture, processed model, and interpretive reconstruction as separate evidence characters.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should label reality-based capture, processed model, and interpretive reconstruction as separate evidence characters. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Reconstructed scenes can be useful storytelling but weak evidence unless their assumptions are inspectable.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-007. - CONFIRMED baseline link: KFM-IDX-DAT-012.

Dependencies - PROPOSED: ModelAssumptionLedger; 3D EvidenceBundle links

Tensions - PROPOSED: Interpretive models are visually persuasive even when evidence is partial.

Expansion directions - PROPOSED: Define knowledge_character values for captured_model, processed_model, and reconstruction.

Open questions - NEEDS VERIFICATION: What citation burden should an interpretive 3D reconstruction carry?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 10, 42, 55; basis: interpretative versus realistic 3D model examples; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.4 KFM-P18-INV-021 — Valid-time, transaction-time, and bitemporal truth posture

Idea card metadata - Stable ID: KFM-P18-INV-021 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 21 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should preserve valid time and transaction/release time as separate dimensions where claims can change or be corrected.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should preserve valid time and transaction/release time as separate dimensions where claims can change or be corrected. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Historical, environmental, and release-correction claims need to say both when something was true and when KFM knew or published it.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-CAT-004. - CONFIRMED baseline link: KFM-IDX-EVD-006.

Dependencies - PROPOSED: Temporal fields in claim/evidence/release schemas

Tensions - PROPOSED: Bitemporal rigor increases schema and query complexity.

Expansion directions - PROPOSED: Add valid_time, observed_time, retrieval_time, transaction_time, release_time, correction_time comparison table.

Open questions - NEEDS VERIFICATION: Which KFM object families require full bitemporal treatment versus simpler time stamps?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 7, 11, 12, 18, 20; basis: valid-time, transaction-time, bitemporal, and temporal-data framing; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-001, KFM-IDX-CAT-004, KFM-IDX-EVD-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.5 KFM-P18-INV-022 — Temporal keys and constraints as first-class validation

Idea card metadata - Stable ID: KFM-P18-INV-022 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 22 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should validate temporal uniqueness, overlap, referential integrity, and interval/period constraints explicitly rather than treating dates as ordinary columns.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should validate temporal uniqueness, overlap, referential integrity, and interval/period constraints explicitly rather than treating dates as ordinary columns. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Without temporal constraints, overlapping or contradictory histories can appear authoritative.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-CAT-007.

Dependencies - PROPOSED: Temporal validation rules; interval/period schema conventions

Tensions - PROPOSED: Legacy sources may have fuzzy or incomplete dates.

Expansion directions - PROPOSED: Define a temporal_constraint_validator inventory entry for later phases.

Open questions - NEEDS VERIFICATION: How should KFM represent fuzzy, approximate, or contested valid-time intervals?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 12, 145; basis: temporal keys, uniqueness, and constraints; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.6 KFM-P18-INV-024 — Bounded contexts and ubiquitous language for KFM lanes

Idea card metadata - Stable ID: KFM-P18-INV-024 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 24 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from

CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should use bounded contexts and published/ubiquitous language to keep domain lanes coherent while allowing governed cross-domain integration.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should use bounded contexts and published/ubiquitous language to keep domain lanes coherent while allowing governed cross-domain integration. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Hydrology, archaeology, land ownership, fauna, and map delivery use overlapping words that can mean different things without context boundaries.

Related ideas - CONFIRMED baseline link: KFM-IDX-MDP-002. - CONFIRMED baseline link: KFM-IDX-DOC-006.

Dependencies - PROPOSED: Domain glossary; lane README; context map

Tensions - PROPOSED: Shared governance objects must remain consistent while domain languages differ.

Expansion directions - PROPOSED: Create a KFM Context Map chapter connecting domain lanes and shared governance kernels.

Open questions - NEEDS VERIFICATION: Which terms require global KFM definitions versus lane-specific meanings?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 6, 8, 36; basis: DDD definitions and context map; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.7 KFM-P18-INV-026 — Window analytics for released observation panels

Idea card metadata - Stable ID: KFM-P18-INV-026 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 26 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM should use window-function-style analytics for running totals, rankings, moving averages, and partitioned comparisons in governed observation panels.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before

Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should use window-function-style analytics for running totals, rankings, moving averages, and partitioned comparisons in governed observation panels. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps analytical comparisons reproducible and tied to declared partitions instead of ad hoc spreadsheet calculations.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-004. - CONFIRMED baseline link: KFM-IDX-OBS-001.

Dependencies - PROPOSED: Analytical view contract - PROPOSED: declared partition/order fields - PROPOSED: release snapshot or query receipt

Tensions - PROPOSED: Powerful aggregate displays may look authoritative even when source coverage is uneven.

Expansion directions - PROPOSED: Define a reusable window_metric_manifest for ranking and trend panels.

Open questions - NEEDS VERIFICATION: Which KFM panels require window-query receipts or stored analytical view definitions?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 3, 23; basis: window functions and analytical query concepts; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.8 KFM-P18-INV-027 — Recursive CTEs for hierarchies, networks, and calendar scaffolds

Idea card metadata - Stable ID: KFM-P18-INV-027 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 27 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM should use recursive-query patterns for explicit traversals of hierarchies, networks, lineage chains, and generated time scaffolds.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should use recursive-query patterns for explicit traversals of hierarchies, networks, lineage chains, and generated time scaffolds. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired

to a public KFM surface. - **NEEDS VERIFICATION**: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - **PROPOSED**: It gives temporal and hierarchical analyses a deterministic query form rather than hidden application logic.

Related ideas - **CONFIRMED** baseline link: KFM-IDX-PIP-009. - **CONFIRMED** baseline link: KFM-IDX-DAT-006.

Dependencies - **PROPOSED**: Recursive-query guardrails - **PROPOSED**: maximum-depth controls - **PROPOSED**: cycle detection - **PROPOSED**: time scaffold contract

Tensions - **PROPOSED**: Recursive queries can be expensive or misleading if cycles and missing parents are not handled.

Expansion directions - **PROPOSED**: Add recursion-depth and cycle-result fields to validation reports.

Open questions - **NEEDS VERIFICATION**: Should network traversal live in SQL views, graph projections, or both?

Future work - **PROPOSED**: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - **CONFIRMED** source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 10; basis: recursive CTE hierarchy and generated-date examples; Phase 1 role: reference_source_candidate.

Carry-forward note - **CONFIRMED**: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - **CONFIRMED**: Required schema fields are present in this Markdown card. - **CONFIRMED**: No prior-pass body text is inlined. - **CONFIRMED**: Narrative claims use explicit truth labels. - **UNKNOWN**: Repository implementation status remains unverified.

8.8.9 KFM-P18-INV-028 — CTEs as transparent transformation stages

Idea card metadata - Stable ID: KFM-P18-INV-028 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 28 - Status label: **PROPOSED** - Carry-forward state: **EXPANDED** - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: **PROPOSED** assignment derived from **CONFIRMED** Phase 3 inventory; subject to Phase 5 card-level review - Source support label: **CONFIRMED** - Implementation maturity: **UNKNOWN** — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-003

Normalized statement - **PROPOSED**: KFM should express complex analytical transformations as named CTE stages or equivalent transparent steps before publishing query-derived claims.

Detailed explanation - **CONFIRMED**: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - **CONFIRMED**: Phase 3 inventory recorded this idea with source support label **CONFIRMED** and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - **PROPOSED**: In KFM, this idea should be treated as a governed design candidate: KFM should express complex analytical transformations as named CTE stages or equivalent transparent steps before publishing query-derived claims. - **PROPOSED**: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - **PROPOSED**: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - **UNKNOWN**: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - **NEEDS VERIFICATION**: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - **PROPOSED**: It makes intermediate assumptions inspectable and reduces the risk of opaque, irreproducible analytical claims.

Related ideas - **CONFIRMED** baseline link: KFM-IDX-PIP-001. - **CONFIRMED** baseline link: KFM-IDX-EVD-006.

Dependencies - PROPOSED: Named transformation stages - PROPOSED: query receipt - PROPOSED: intermediate row-count checks

Tensions - PROPOSED: Too many CTEs can become pseudo-pipeline code without lifecycle receipts.

Expansion directions - PROPOSED: Create a query_stage_report shape for derived analytical outputs.

Open questions - NEEDS VERIFICATION: When does a query-stage become a pipeline transform that needs its own RunReceipt?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 22, 30, 32; basis: subqueries and CTEs for decomposing complex logic; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-001, KFM-IDX-EVD-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.10 KFM-P18-INV-029 — Set and join operations as explicit crosswalk semantics

Idea card metadata - Stable ID: KFM-P18-INV-029 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 29 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM should label join, intersection, exception, and union operations according to their semantic purpose in crosswalks and source reconciliation.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should label join, intersection, exception, and union operations according to their semantic purpose in crosswalks and source reconciliation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents technical joins from silently becoming evidentiary equivalence claims.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-005. - CONFIRMED baseline link: KFM-IDX-DAT-004.

Dependencies - PROPOSED: Crosswalk relation type - PROPOSED: join predicate registry - PROPOSED: anti-join/missingness report

Tensions - PROPOSED: A join on shared identifiers can imply a stronger relationship than the evidence supports.

Expansion directions - PROPOSED: Map query operators to KFM relation labels such as same_as, overlaps, excludes, and unresolved.

Open questions - NEEDS VERIFICATION: Which join types require source-role approval before promotion?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 20, 22; basis: joins, subqueries, and set-operator discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.11 KFM-P18-INV-030 — Null, aggregation, and COALESCE handling as evidence caveats

Idea card metadata - Stable ID: KFM-P18-INV-030 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 30 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM should make missingness, NULL handling, aggregation choice, and fallback labels visible in derived analytical outputs.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should make missingness, NULL handling, aggregation choice, and fallback labels visible in derived analytical outputs. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It protects users from interpreting missing data, fallback text, or aggregate summaries as stronger evidence than they are.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-OBS-001.

Dependencies - PROPOSED: Missingness policy - PROPOSED: aggregation metadata - PROPOSED: fallback label registry

Tensions - PROPOSED: Clean labels can hide source gaps and null-driven exclusions.

Expansion directions - PROPOSED: Add denominator and missing_count fields to derived metric manifests.

Open questions - NEEDS VERIFICATION: Should derived indicators expose NULL counts and denominator definitions by default?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 22, 30, 50; basis: aggregate, GROUP BY, and COALESCE/fallback discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.12 KFM-P18-INV-031 — AI data lifecycle as governed model input boundary

Idea card metadata - Stable ID: KFM-P18-INV-031 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 31 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should treat AI-ready data preparation as a governed lifecycle with collection, cleaning, preparation, training, validation, and testing states.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat AI-ready data preparation as a governed lifecycle with collection, cleaning, preparation, training, validation, and testing states. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps AI outputs from outranking the source data and exposes where model inputs came from.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-004. - CONFIRMED baseline link: KFM-IDX-DOC-007.

Dependencies - PROPOSED: EvidenceRef-to-EvidenceBundle resolution - PROPOSED: AI input manifest - PROPOSED: training/validation/test split records

Tensions - PROPOSED: AI pipelines often compress evidence into features that are hard to trace back to source claims.

Expansion directions - PROPOSED: Define `admissible_ai_context` and `model_input_receipt` objects.

Open questions - NEEDS VERIFICATION: What AI input manifests are required before any Focus Mode or model-assisted workflow is allowed?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 79, 80, 83; basis: AI data importance, veracity, preparation, and use in training/validation/testing; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-004, KFM-IDX-DOC-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.13 KFM-P18-INV-032 — Model-type labels for AI/ML outputs

Idea card metadata - Stable ID: KFM-P18-INV-032 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 32 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should label AI and ML outputs by task type, training mode, and evidentiary role rather than presenting them as generic intelligence.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should label AI and ML outputs by task type, training mode, and evidentiary role rather than presenting them as generic intelligence. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It helps reviewers and users distinguish classification, clustering, prediction, and policy-constrained synthesis.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-007. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: Model card - PROPOSED: task taxonomy - PROPOSED: output role label - PROPOSED: review state

Tensions - PROPOSED: A model-type label does not by itself prove a model is valid or fair.

Expansion directions - PROPOSED: Add task_type and model_output_role fields to governed AI envelopes.

Open questions - NEEDS VERIFICATION: Which model families are allowed on public KFM surfaces, and which must remain internal review aids?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 104, 106, 109, 113; basis: machine-learning task types and model-selection discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.14 KFM-P18-INV-033 — Train/test/evaluation discipline for model-assisted claims

Idea card metadata - Stable ID: KFM-P18-INV-033 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 33 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should require train/test/evaluation records before model-assisted classifications, predictions, or rankings can support consequential claims.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should require train/test/evaluation records before model-assisted classifications, predictions, or rankings can support consequential claims. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents untested model outputs from being treated as evidence-ready claims.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-004. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Evaluation report - PROPOSED: validation split receipt - PROPOSED: metric definitions - PROPOSED: review gate

Tensions - PROPOSED: Evaluation metrics may not capture public-risk or sensitivity harms.

Expansion directions - PROPOSED: Create a ModelEvaluationBundle linked to EvidenceBundle and PolicyDecision.

Open questions - NEEDS VERIFICATION: Which evaluation thresholds trigger ABSTAIN rather than ANSWER or publish?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 83, 106, 155; basis: model training/validation/testing and data-splitting concepts; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-004, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.15 KFM-P18-INV-034 — Bias, transparency, and explainability review for AI-supported surfaces

Idea card metadata - Stable ID: KFM-P18-INV-034 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 34 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should treat bias detection, transparency, and explainability as release gates for AI-assisted products, not optional documentation.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat bias detection, transparency, and explainability as release gates for AI-assisted products, not

optional documentation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It protects sensitive public decisions and prevents plausible model language from masking inequity or unsupported inference.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-001. - CONFIRMED baseline link: KFM-IDX-POL-002.

Dependencies - PROPOSED: Bias review checklist - PROPOSED: explainability note - PROPOSED: sensitivity and rights policy - PROPOSED: finite outcome envelope

Tensions - PROPOSED: Explainability artifacts can be superficial if not connected to evidence and review state.

Expansion directions - PROPOSED: Add AI risk gates to release self-check for any AI-generated or AI-assisted text.

Open questions - NEEDS VERIFICATION: What bias-review evidence is sufficient for each KFM domain lane?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 349, 350; basis: bias detection, mitigation, and transparency concerns; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-001, KFM-IDX-POL-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.16 KFM-P18-INV-035 — Data quality dimensions as feature-source gates

Idea card metadata - Stable ID: KFM-P18-INV-035 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 35 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should treat AI data volume, variety, velocity, veracity, and value as source-admission checks when model features are derived from evidence.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat AI data volume, variety, velocity, veracity, and value as source-admission checks when model features are derived from evidence. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It gives model feature engineering the same fail-closed posture as source ingestion.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-OBS-005.

Dependencies - PROPOSED: FeatureSourceDescriptor - PROPOSED: data-quality thresholds - PROPOSED: staleness policy - PROPOSED: noise caveat

Tensions - PROPOSED: High-volume data may still be low authority or unsafe to publish.

Expansion directions - PROPOSED: Add feature_source_quality fields to source admission records.

Open questions - NEEDS VERIFICATION: Should AI feature sources use the same SourceDescriptor schema as non-AI sources or an extension?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 79, 80; basis: AI data characteristics, veracity, and value discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.17 KFM-P18-INV-054 — Positions, networks, fields, and transformations as distinct model families

Idea card metadata - Stable ID: KFM-P18-INV-054 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 54 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should distinguish positional features, network relationships, continuous fields, and transformations as different spatial model families with different validation rules.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should distinguish positional features, network relationships, continuous fields, and transformations as different spatial model families with different validation rules. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A route graph, soil-moisture surface, point occurrence, and projected geometry cannot share one generic validation rubric.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-002. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: domain object family registry - PROPOSED: validator dispatch by spatial_model_type

Tensions - PROPOSED: Cross-domain products often combine model families, which can obscure where uncertainty enters.

Expansion directions - PROPOSED: Add `spatial_model_type` to layer, feature, and analysis manifests.

Open questions - NEEDS VERIFICATION: Can the same EvidenceBundle safely cover multiple spatial model families?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 185; basis: positions, networks, fields, and transformations chapter framing; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.18 KFM-P18-INV-061 — TOD indices as composite indicators with mode-shift assumptions

Idea card metadata - Stable ID: KFM-P18-INV-061 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 61 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM should treat transit-oriented-development indices as composite indicators whose inputs, weights, travel-mode assumptions, and geography must be inspectable.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat transit-oriented-development indices as composite indicators whose inputs, weights, travel-mode assumptions, and geography must be inspectable. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A TOD score can influence investment and settlement interpretation; its assumptions need to be visible and revisable.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-013. - CONFIRMED baseline link: KFM-IDX-CAT-002.

Dependencies - PROPOSED: CompositeIndicator schema - PROPOSED: transport geography version

Tensions - PROPOSED: One score can conceal conflicting accessibility, equity, and land-use dimensions.

Expansion directions - PROPOSED: Add `TOD_indicator_profile` with inputs, weights, mode assumptions, and update cadence.

Open questions - NEEDS VERIFICATION: Can TOD-style indicators be reused for Kansas frontier settlement without importing urban-only assumptions?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 7, 282; basis: Transit Oriented Development index and modal-shift planning context; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.19 KFM-P18-INV-064 — Temporal query class labels for public explanations

Idea card metadata - Stable ID: KFM-P18-INV-064 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 64 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should label whether a query is current-state, prior-state, sequenced, nonsequenced, or tracking-log based when the distinction affects a public claim.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should label whether a query is current-state, prior-state, sequenced, nonsequenced, or tracking-log based when the distinction affects a public claim. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: The same dataset can answer different questions depending on whether the user asks what was true then, what KFM recorded then, or what is released now.

Related ideas - CONFIRMED baseline link: KFM-IDX-OBS-003. - CONFIRMED baseline link: KFM-IDX-CAT-003.

Dependencies - PROPOSED: QueryReceipt - PROPOSED: temporal_query_type enum

Tensions - PROPOSED: Most users do not expect temporal query semantics, but hiding them can mislead.

Expansion directions - PROPOSED: Add temporal_query_type and time_basis to query-derived outputs.

Open questions - NEEDS VERIFICATION: Which map popups should expose temporal query class versus only a simplified date label?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 143, 145, 219, 253, 277; basis: current/prior/sequenced queries, tracking logs, transaction-time and bitemporal topics; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.20 KFM-P18-INV-065 — Aggregates, repositories, and domain events for lane boundaries

Idea card metadata - Stable ID: KFM-P18-INV-065 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 65 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should use aggregates, repositories, domain events, and layered architecture as design checks for keeping domain lanes coherent and integration surfaces explicit.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should use aggregates, repositories, domain events, and layered architecture as design checks for keeping domain lanes coherent and integration surfaces explicit. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reduces the risk that domain-specific source quirks leak directly into shared schemas, APIs, or public maps.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-008. - CONFIRMED baseline link: KFM-IDX-CAT-002.

Dependencies - PROPOSED: DomainLane README - PROPOSED: context map registry

Tensions - PROPOSED: DDD terms should guide boundaries without creating unnecessary implementation ceremony.

Expansion directions - PROPOSED: Add a bounded_context and aggregate_boundary section to domain-lane architecture docs.

Open questions - NEEDS VERIFICATION: Which KFM object families are true aggregates versus cross-domain projections?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 2, 10, 13, 16, 17, 29, 36; basis: DDD pattern catalog: layered architecture, domain events, aggregates, repositories, context map, published language; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-008, KFM-IDX-CAT-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.21 KFM-P18-INV-069 — CASE and conditional aggregation as auditable classification rules

Idea card metadata - Stable ID: KFM-P18-INV-069 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 69 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM should persist conditional classification logic such as CASE expressions and conditional aggregates as named rules, not hidden query code.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should persist conditional classification logic such as CASE expressions and conditional aggregates as named rules, not hidden query code. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A public category such as high risk, low flow, eligible, or missing support may depend on thresholds that need review and correction.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-EVD-004.

Dependencies - PROPOSED: RuleDefinition - PROPOSED: ThresholdReceipt

Tensions - PROPOSED: Readable categories often simplify continuous values and can hide edge cases.

Expansion directions - PROPOSED: Add rule_id and rule_version to query-derived class labels.

Open questions - NEEDS VERIFICATION: Which classifications need steward approval before public release?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 25; basis: CASE WHEN examples and aggregate/conditional analytical query patterns; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.22 KFM-P18-INV-070 — Missing-data and feature-preprocessing decisions as evidence transforms

Idea card metadata - Stable ID: KFM-P18-INV-070 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 70 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should treat missing-data handling, data splitting, feature engineering, scaling, and preparation for AI as governed evidence transformations.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat missing-data handling, data splitting, feature engineering, scaling, and preparation for AI as governed evidence transformations. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream

of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Model outputs depend strongly on how evidence was cleaned, imputed, partitioned, and transformed before training or inference.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-002. - CONFIRMED baseline link: KFM-IDX-AIG-005.

Dependencies - PROPOSED: FeatureReceipt - PROPOSED: ModelEvaluationRecord

Tensions - PROPOSED: Preprocessing may improve model performance while reducing traceability if not recorded.

Expansion directions - PROPOSED: Add preprocessing_steps, missingness_policy, split_strategy, and feature_lineage fields to model receipts.

Open questions - NEEDS VERIFICATION: Should imputed features be eligible for consequential public claims without separate caveats?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 80, 83, 104, 146, 155; basis: AI data lifecycle, ethics, workflow, missing data, and data splitting sections; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-002, KFM-IDX-AIG-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.23 KFM-P18-INV-071 — Algorithm limitation labels for classifications and clusters

Idea card metadata - Stable ID: KFM-P18-INV-071 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 71 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should label model-assisted classifications, neighbor-based predictions, and clusters with the algorithm's assumptions, limitations, validation status, and intended use.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should label model-assisted classifications, neighbor-based predictions, and clusters with the algorithm's assumptions, limitations, validation status, and intended use. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A cluster or nearest-neighbor label can look authoritative on a map even when it is sensitive to scaling, distance metrics, initialization, or class imbalance.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-004. - CONFIRMED baseline link: KFM-IDX-POL-005.

Dependencies - PROPOSED: ModelCard - PROPOSED: algorithm_assumption fields - PROPOSED: evaluation thresholds

Tensions - PROPOSED: Simple algorithms are explainable but may still be unsuitable for high-consequence claims.

Expansion directions - PROPOSED: Create algorithm_limitations and intended_use labels in model-derived layer metadata.

Open questions - NEEDS VERIFICATION: What model classes should default to ABSTAIN for insufficient evaluation?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 100, 123, 126, 131, 134; basis: machine-learning workflow, KNN and K-Means strengths, limitations, and optimization topics; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.24 KFM-P18-INV-082 — Global water-resource layers need classification and denominator transparency

Idea card metadata - Stable ID: KFM-P18-INV-082 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 82 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should expose classification denominators and physical-state distinctions when publishing water-resource or resource-budget layers.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should expose classification denominators and physical-state distinctions when publishing water-resource or resource-budget layers. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents resource maps from mixing incompatible denominators or hiding the scarcity implied by classification choices.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-013.

Dependencies - PROPOSED: Indicator denominator metadata - PROPOSED: Resource category controlled vocabulary

Tensions - PROPOSED: Percent of freshwater and percent of total water can tell different stories.

Expansion directions - PROPOSED: Add denominator_basis and resource_state fields to resource-budget indicators.

Open questions - NEEDS VERIFICATION: Should KFM publish paired denominators by default for scarce-resource indicators?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 109, 110, 111, 115; basis: global water resource categories, state/phase distinctions, and percentage tables; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.25 KFM-P18-INV-083 — Raster suitability workflows need criteria weights and exclusion receipts

Idea card metadata - Stable ID: KFM-P18-INV-083 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 83 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should record suitability criteria, raster reclassification rules, exclusions, and area calculations for multi-criteria site-selection workflows.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record suitability criteria, raster reclassification rules, exclusions, and area calculations for multi-criteria site-selection workflows. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It turns suitability maps into inspectable derived products rather than opaque recommendation surfaces.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-005. - CONFIRMED baseline link: KFM-IDX-DAT-013.

Dependencies - PROPOSED: SuitabilityModelReceipt - PROPOSED: RasterTransformReceipt - PROPOSED: Criteria registry

Tensions - PROPOSED: A suitability output can be misread as recommendation rather than criteria-constrained candidate set.

Expansion directions - PROPOSED: Define criteria_weight, exclusion_rule, reclass_table_hash, and output_area_method fields.

Open questions - NEEDS VERIFICATION: How should KFM express suitability uncertainty when criteria are qualitative?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 132, 133, 134, 135, 136, 138; basis: Mars landing raster reclassification, slope/elevation criteria, raster calculation, vector conversion, and zonal/area analysis; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-005, KFM-IDX-DAT-013 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.26 KFM-P18-INV-084 — Age-sensitive built-environment walking indicators

Idea card metadata - Stable ID: KFM-P18-INV-084 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 84 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM should support age-sensitive walking and access indicators that separate macro built-environment attributes from perceived micro attributes.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should support age-sensitive walking and access indicators that separate macro built-environment attributes from perceived micro attributes. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It gives KFM settlement and infrastructure analyses a way to avoid one-size-fits-all accessibility metrics.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-010.

Dependencies - PROPOSED: Accessibility indicator registry - PROPOSED: Population group metadata - PROPOSED: Survey/perception evidence

Tensions - PROPOSED: Perceived safety and aesthetics may not match measured network accessibility.

Expansion directions - PROPOSED: Create access_indicator_profile with target_population, macro_attributes, micro_attributes, and evidence_type fields.

Open questions - NEEDS VERIFICATION: Which perceived attributes can be public without exposing individual survey data?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 76, 77, 78; basis: older-adult walking and macro/micro built-environment attribute discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.27 KFM-P18-INV-087 — Water-supply preference models need stated-choice caveats

Idea card metadata - Stable ID: KFM-P18-INV-087 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 87 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM should label water-supply preference and policy models with stated-choice design, attribute levels, and respondent-understanding caveats.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should label water-supply preference and policy models with stated-choice design, attribute levels, and respondent-understanding caveats. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents preference models from being treated as direct observations of behavior or policy success.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-010. - CONFIRMED baseline link: KFM-IDX-OBS-004.

Dependencies - PROPOSED: Survey design metadata - PROPOSED: Model caveat fields - PROPOSED: Policy indicator registry

Tensions - PROPOSED: Stated preferences can differ from actual behavior under cost, knowledge, or service constraints.

Expansion directions - PROPOSED: Add `choice_set_design`, `attribute_levels`, `opt_out_allowed`, `respondent_understanding_caveat`, and `model_type` fields.

Open questions - NEEDS VERIFICATION: Can KFM publish preference-model outputs without individual respondent data?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 296, 298, 300, 302, 303, 305; basis: Yogyakarta stated choice experiment, water supply attributes, and model limitations; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.28 KFM-P18-INV-088 — Urban-growth scenarios are planning evidence, not predictions

Idea card metadata - Stable ID: KFM-P18-INV-088 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 88 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM should label urban-growth simulations and scenario outputs as planning evidence with drivers, assumptions, and comparison baselines, not as predictions.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should label urban-growth simulations and scenario outputs as planning evidence with drivers, assumptions, and comparison baselines, not as predictions. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reduces the risk that scenario maps are cited as deterministic forecasts.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-013. - CONFIRMED baseline link: KFM-IDX-OBS-002.

Dependencies - PROPOSED: ScenarioManifest - PROPOSED: Driver registry - PROPOSED: Baseline comparison receipt

Tensions - PROPOSED: A scenario is plausible under assumptions, not guaranteed.

Expansion directions - PROPOSED: Add `scenario_assumption_set`, `driver_weights`, `baseline_period`, and `not_forecast` label.

Open questions - NEEDS VERIFICATION: Which KFM UI states should distinguish forecast, scenario, model, and observed trend?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 16, 18, 226, 228; basis: urban-growth models, scenario-based planning, drivers, land-cover change, and baseline comparison; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-013, KFM-IDX-OBS-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.29 KFM-P18-INV-103 — Geostatistical products require uncertainty and model metadata

Idea card metadata - Stable ID: KFM-P18-INV-103 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 103 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should publish geostatistical surfaces only with interpolation method, sampling support, assumptions, and uncertainty metadata.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should publish geostatistical surfaces only with interpolation method, sampling support, assumptions, and uncertainty metadata. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It protects readers from treating interpolated surfaces as direct observations.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-009. - CONFIRMED baseline link: KFM-IDX-OBS-003.

Dependencies - PROPOSED: ModelRunReceipt - PROPOSED: Raster/COG manifest - PROPOSED: Uncertainty schema

Tensions - PROPOSED: Statistically elegant surfaces can appear more authoritative than their sampling support allows.

Expansion directions - PROPOSED: Add interpolation_method, sample_support, model_parameters, and uncertainty_surface references.

Open questions - NEEDS VERIFICATION: Which uncertainty fields are mandatory for interpolated public layers?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 271, 295; basis: geostatistics, spatial processes, and kriging discussion; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.30 KFM-P18-INV-104 — Spatial model family labels for positions, networks, fields, and transformations

Idea card metadata - Stable ID: KFM-P18-INV-104 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 104 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should tag spatial products by model family—position, network, field, transformation, or administrative space—before applying analytics or visualization defaults.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be

treated as a governed design candidate: KFM should tag spatial products by model family—position, network, field, transformation, or administrative space—before applying analytics or visualization defaults. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents KFM from applying point-feature assumptions to field surfaces or network logic to administrative polygons.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: Domain model registry - PROPOSED: Layer profile registry - PROPOSED: Validator dispatch

Tensions - PROPOSED: Cross-domain products often combine model families and need explicit relation labels.

Expansion directions - PROPOSED: Create model_family enum and validator-specific requirements.

Open questions - NEEDS VERIFICATION: Should model_family be a required field on every LayerManifest?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 174, 253, 290; basis: positions, networks, fields, transformations, and GI analysis chapter structure; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.31 KFM-P18-INV-105 — Window frame semantics for analytical panels

Idea card metadata - Stable ID: KFM-P18-INV-105 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 105 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM analytical SQL should record window partitioning, ordering, and frame rules when computing rolling, ranked, or panel-style metrics.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM analytical SQL should record window partitioning, ordering, and frame rules when computing rolling, ranked, or panel-style metrics. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes derived rankings, rolling averages, and release-panel metrics reproducible.

Related ideas - CONFIRMED baseline link: KFM-IDX-OBS-004. - CONFIRMED baseline link: KFM-IDX-PIP-008.

Dependencies - PROPOSED: MetricCalculationReceipt - PROPOSED: SQL transformation registry

Tensions - PROPOSED: Different frame definitions can produce different public indicators from the same table.

Expansion directions - PROPOSED: Add window_partition, window_order, frame_clause, and metric_basis fields.

Open questions - NEEDS VERIFICATION: Which rolling metrics require frame metadata in public explanations?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 23, 24; basis: OVER clause, partitions, window frames, and ordering discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-OBS-004, KFM-IDX-PIP-008 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.32 KFM-P18-INV-107 — Database views as documented analytical contract boundaries

Idea card metadata - Stable ID: KFM-P18-INV-107 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 107 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM may use database views or equivalent materialized views only when their definition, purpose, and validation status are documented as contract surfaces.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM may use database views or equivalent materialized views only when their definition, purpose, and validation status are documented as contract surfaces. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps SQL convenience layers from becoming undocumented shadow truth.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-004. - CONFIRMED baseline link: KFM-IDX-MDP-001.

Dependencies - PROPOSED: SQL contract registry - PROPOSED: View validation fixtures

Tensions - PROPOSED: Views can hide joins, filters, and caveats if not paired with receipts.

Expansion directions - PROPOSED: Create view_manifest records with SQL digest, upstream tables, and validation status.

Open questions - NEEDS VERIFICATION: Should every public-facing SQL view have a stable spec_hash?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 18, 103; basis: database views, maintainability, and CREATE VIEW discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.33 KFM-P18-INV-108 — Training/testing split reproducibility

Idea card metadata - Stable ID: KFM-P18-INV-108 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 108 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should record training, validation, and testing split methods for any model-assisted product that influences public claims.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record training, validation, and testing split methods for any model-assisted product that influences public claims. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents model performance claims from being detached from the data partitioning that produced them.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-005. - CONFIRMED baseline link: KFM-IDX-OBS-005.

Dependencies - PROPOSED: ModelRunReceipt - PROPOSED: Training dataset manifest - PROPOSED: Evaluation report

Tensions - PROPOSED: Random or leakage-prone splits can make model performance look stronger than it is.

Expansion directions - PROPOSED: Add split_method, random_seed, leakage_check, and holdout_scope fields.

Open questions - NEEDS VERIFICATION: Which model outputs are consequential enough to require public split metadata?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 7, 100, 155; basis: training/testing split and model performance context; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-005, KFM-IDX-OBS-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.34 KFM-P18-INV-109 — Confusion-matrix and precision/recall evidence for classification layers

Idea card metadata - Stable ID: KFM-P18-INV-109 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 109 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM classification outputs should carry confusion-matrix, precision, recall, F1, and class-balance caveats where classification accuracy matters.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM classification outputs should carry confusion-matrix, precision, recall, F1, and class-balance caveats where classification accuracy matters. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It lets users distinguish model-supported classifications from ground-truth observations and understand class-specific risk.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-006. - CONFIRMED baseline link: KFM-IDX-POL-005.

Dependencies - PROPOSED: EvaluationReport schema - PROPOSED: Classified layer manifest

Tensions - PROPOSED: High overall accuracy can mask poor performance on rare or high-consequence classes.

Expansion directions - PROPOSED: Add evaluation_metric_bundle with class-level scores and support counts.

Open questions - NEEDS VERIFICATION: Should rare-species, hazard, or archaeology classifiers have stricter per-class thresholds?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 107, 166, 219; basis: precision, recall, F1-score, and confusion matrix discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-006, KFM-IDX-POL-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.35 KFM-P18-INV-111 — AI data lifecycle as a governed intake-to-use lane

Idea card metadata - Stable ID: KFM-P18-INV-111 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 111 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should align AI training and inference inputs with a lifecycle that covers collection, storage, cleaning, preparation, utilization, and ethical review.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should align AI training and inference inputs with a lifecycle that covers collection, storage, cleaning, preparation, utilization, and ethical review. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes model-supporting data subject to the same KFM evidence, rights, and sensitivity gates as spatial data.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-001. - CONFIRMED baseline link: KFM-IDX-POL-006.

Dependencies - PROPOSED: AI source registry - PROPOSED: Policy precheck - PROPOSED: Data-use review

Tensions - PROPOSED: Model data can inherit sensitive or biased properties from upstream sources.

Expansion directions - PROPOSED: Add ai_data_use_scope and ethical_review_status to model-related manifests.

Open questions - NEEDS VERIFICATION: Where should KFM record data-use consent for model training versus inference?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 80, 81, 82, 83; basis: AI data lifecycle and ethical considerations discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-001, KFM-IDX-POL-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.36 KFM-P18-INV-121 — Aggregates and repositories as consistency boundaries

Idea card metadata - Stable ID: KFM-P18-INV-121 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 121 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators,

Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should use aggregate and repository concepts to define where consistency is enforced and how queries express domain concepts.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should use aggregate and repository concepts to define where consistency is enforced and how queries express domain concepts. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents scattered updates from breaking EvidenceBundle, DecisionEnvelope, ReleaseManifest, or correction consistency.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-006. - CONFIRMED baseline link: KFM-IDX-CAT-003.

Dependencies - PROPOSED: Bounded-context model - PROPOSED: Repository interface contract

Tensions - PROPOSED: Large aggregates reduce independence; tiny aggregates can fail to enforce invariants.

Expansion directions - PROPOSED: Add aggregate_root and invariant list to object-family register.

Open questions - NEEDS VERIFICATION: Which KFM object families need aggregate roots?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 20, 24; basis: aggregates, consistency, and repositories expressed in ubiquitous language; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-006, KFM-IDX-CAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.37 KFM-P18-INV-132 — Window ordering and tie-breakers as reproducibility requirements

Idea card metadata - Stable ID: KFM-P18-INV-132 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 132 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM should require deterministic ORDER BY clauses and tie-breaker fields for window-function based rankings, deltas, moving averages, and panels.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM

should require deterministic ORDER BY clauses and tie-breaker fields for window-function based rankings, deltas, moving averages, and panels. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents release-to-release drift caused by unspecified row ordering.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-008.

Dependencies - PROPOSED: Analytic SQL linting - PROPOSED: IndicatorManifest

Tensions - PROPOSED: Natural ordering may appear obvious in prose but still be nondeterministic in SQL.

Expansion directions - PROPOSED: Add SQL fixture tests that fail on ambiguous window ordering.

Open questions - NEEDS VERIFICATION: Should KFM enforce query-lint rules for all release analytics?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 2, 23, 26; basis: window function, partition, ordering, and analytical calculation discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.38 KFM-P18-INV-133 — Recursive CTE termination and depth guards

Idea card metadata - Stable ID: KFM-P18-INV-133 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 133 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM should require termination criteria, maximum depth, and cycle handling for recursive CTEs used in hierarchies, networks, route graphs, calendars, or lineage traversal.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should require termination criteria, maximum depth, and cycle handling for recursive CTEs used in hierarchies, networks, route graphs, calendars, or lineage traversal. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps network, lineage, and hierarchy queries from producing runaway, cyclic, or misleading derived results.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-005. - CONFIRMED baseline link: KFM-IDX-OBS-003.

Dependencies - PROPOSED: Graph/query validators - PROPOSED: Lineage traversal policy

Tensions - PROPOSED: Strict depth caps can hide valid deep lineage unless the cap is justified and visible.

Expansion directions - PROPOSED: Emit recursion_depth and cycle_detected fields in query receipts.

Open questions - NEEDS VERIFICATION: What depth limits are appropriate per domain lane?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 6, 10, 15; basis: recursive CTE examples for hierarchies, calendars, and generated sequences; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-005, KFM-IDX-OBS-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.39 KFM-P18-INV-134 — Machine-learning workflow failures as abstain triggers

Idea card metadata - Stable ID: KFM-P18-INV-134 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 134 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should convert missing training/test records, weak evaluation, unhandled imbalance, leakage risk, or unexplained preprocessing into ABSTAIN or DENY outcomes for model-assisted claims.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should convert missing training/test records, weak evaluation, unhandled imbalance, leakage risk, or unexplained preprocessing into ABSTAIN or DENY outcomes for model-assisted claims. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents model output from appearing as evidence when the workflow does not support the claim.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-001. - CONFIRMED baseline link: KFM-IDX-DOC-003.

Dependencies - PROPOSED: ModelCard - PROPOSED: EvaluationReport - PROPOSED: RuntimeResponseEnvelope

Tensions - PROPOSED: Domain experts may need override paths, but overrides still need review and receipts.

Expansion directions - PROPOSED: Add model_evidence_closure checks to Focus Mode and model-derived layer gates.

Open questions - NEEDS VERIFICATION: Which ML failure modes produce ABSTAIN versus DENY?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 104, 106, 107; basis: machine-learning workflow and challenges discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-001, KFM-IDX-DOC-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.40 KFM-P18-INV-135 — Ethical data-use review as AI source-admission gate

Idea card metadata - Stable ID: KFM-P18-INV-135 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 135 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should require privacy, bias, transparency, and consent checks before data can be admitted as AI training, evaluation, embedding, or retrieval material.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should require privacy, bias, transparency, and consent checks before data can be admitted as AI training, evaluation, embedding, or retrieval material. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It protects living-person, sensitive-location, and steward-controlled data from becoming hidden model substrate.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-005. - CONFIRMED baseline link: KFM-IDX-POL-003.

Dependencies - PROPOSED: SourceDescriptor - PROPOSED: PolicyDecision - PROPOSED: Consent/rights registry

Tensions - PROPOSED: Open metadata can still leak sensitive patterns when used for embeddings or model training.

Expansion directions - PROPOSED: Add ai_use_allowed, embedding_allowed, and training_allowed fields to source descriptors.

Open questions - NEEDS VERIFICATION: Who reviews AI-use permissions for public but sensitive sources?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 80, 83, 84, 350, 352; basis: data lifecycle and ethical considerations in AI data usage; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-005, KFM-IDX-POL-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.41 KFM-P18-INV-136 — Learning-mode labels for model-assisted surfaces

Idea card metadata - Stable ID: KFM-P18-INV-136 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 136 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should label model-assisted outputs by learning mode so supervised classifications, unsupervised clusters, semi-supervised labels, and reinforcement policies carry distinct evidence expectations.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should label model-assisted outputs by learning mode so supervised classifications, unsupervised clusters, semi-supervised labels, and reinforcement policies carry distinct evidence expectations. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents a cluster, classifier, or policy recommendation from being interpreted with the wrong confidence and validation frame.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-002.

Dependencies - PROPOSED: ModelCard - PROPOSED: LayerManifest - PROPOSED: Focus Mode response schema

Tensions - PROPOSED: Hybrid methods may require multiple labels and a plain-language explanation.

Expansion directions - PROPOSED: Extend model limitation labels with learning_mode and validation_basis.

Open questions - NEEDS VERIFICATION: How should KFM explain unsupervised clusters to nontechnical users?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 101, 102, 103, 114, 115, 116; basis: supervised, unsupervised, semi-supervised, and reinforcement learning discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.42 KFM-P18-INV-143 — Sequenced and nonsequenced temporal query labels

Idea card metadata - Stable ID: KFM-P18-INV-143 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 143 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should label temporal queries as current, prior-state, sequenced, nonsequenced, valid-time, transaction-time, or bitemporal where that distinction changes claim meaning.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should label temporal queries as current, prior-state, sequenced, non-sequenced, valid-time, transaction-time, or bitemporal where that distinction changes claim meaning. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps users from confusing what was true in the world with what was recorded or released at a later time.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-DAT-009.

Dependencies - PROPOSED: Temporal query registry - PROPOSED: EvidenceDrawerPayload - PROPOSED: Focus Mode envelope

Tensions - PROPOSED: Plain-language answers must remain readable while preserving temporal exactness.

Expansion directions - PROPOSED: Add `temporal_query_class` to query receipts.

Open questions - NEEDS VERIFICATION: Which UI surfaces need visible temporal-query labels?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 143, 145, 156, 173; basis: temporal database discussion of current, prior, sequenced, and nonsequenced queries; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.43 KFM-P18-INV-155 — DISTINCT and NULL semantics as deduplication risk

Idea card metadata - Stable ID: KFM-P18-INV-155 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 155 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM should document deduplication semantics for DISTINCT-like operations, especially NULL handling and multi-column uniqueness, before counts or entity matches are promoted.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should document deduplication semantics for DISTINCT-like operations, especially NULL handling and multi-column uniqueness, before counts or entity matches are promoted. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reduces the risk that source counts, occurrence counts, or crosswalk matches change because a query silently treats null or paired values differently than reviewers expect.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-008. - CONFIRMED baseline link: KFM-IDX-DAT-004.

Dependencies - PROPOSED: Metric definition - PROPOSED: SQL dialect profile - PROPOSED: Reconciliation tests

Tensions - PROPOSED: Different database engines and query planners may differ in edge behavior or performance.

Expansion directions - PROPOSED: Add fixture datasets with nulls, duplicates, and multi-column uniqueness cases.

Open questions - NEEDS VERIFICATION: Should KFM use SQL DISTINCT for identity reconciliation or require explicit deterministic grouping?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 21; basis: Advanced SQL discussion of DISTINCT, unique pairs, and NULL behavior; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.44 KFM-P18-INV-156 — SQL dialect verification for set-operator compatibility

Idea card metadata - Stable ID: KFM-P18-INV-156 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 156 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM should treat set-operator syntax and duplicate behavior as dialect-sensitive and verify the target engine before using it in validation, reconciliation, or release SQL.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before

Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat set-operator syntax and duplicate behavior as dialect-sensitive and verify the target engine before using it in validation, reconciliation, or release SQL. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps cross-source gap analysis from depending on unverified SQL dialect behavior or source-text claims about database compatibility.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-004. - CONFIRMED baseline link: KFM-IDX-PIP-006.

Dependencies - PROPOSED: SQL engine profile - PROPOSED: Validation query library - PROPOSED: No-network fixtures

Tensions - PROPOSED: Source references to dialect support may be stale or inaccurate and need engine documentation or tests.

Expansion directions - PROPOSED: Create per-engine fixtures for UNION, UNION ALL, INTERSECT, EXCEPT/MINUS, duplicate retention, and null handling.

Open questions - NEEDS VERIFICATION: Which SQL engines are allowed in KFM processing and validation lanes?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 21; basis: set-operator discussion naming UNION, UNION ALL, INTERSECT, MINUS, and EXCEPT across database dialects; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.45 KFM-P18-INV-157 — COUNT denominator semantics for public metrics

Idea card metadata - Stable ID: KFM-P18-INV-157 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 157 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM public metrics should state whether counts include all rows, only non-null column values, distinct values, grouped rows, or filtered subsets.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM public metrics should state whether counts include all rows, only non-null column values, distinct values, grouped rows, or filtered subsets. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested,

released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents maps, dashboards, and AI explanations from comparing denominators that were calculated under different SQL semantics.

Related ideas - CONFIRMED baseline link: KFM-IDX-OBS-002. - CONFIRMED baseline link: KFM-IDX-CAT-004.

Dependencies - PROPOSED: Metric registry - PROPOSED: SQL query receipt - PROPOSED: Dashboard/Evidence Drawer labels

Tensions - PROPOSED: Too much denominator detail can clutter public UI unless summarized well.

Expansion directions - PROPOSED: Add a metric-denominator field to analytical receipt fragments.

Open questions - NEEDS VERIFICATION: Which public metrics require denominator notes in popups versus only in the Evidence Drawer?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 22; basis: aggregate-function discussion distinguishing COUNT(*) from COUNT(column) and grouping behavior; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.46 KFM-P18-INV-158 — Correlation aggregates as relationship evidence, not causality

Idea card metadata - Stable ID: KFM-P18-INV-158 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 158 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM should label correlation aggregates as statistical relationship evidence and deny or abstain on causal wording unless a stronger causal design is present.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should label correlation aggregates as statistical relationship evidence and deny or abstain on causal wording unless a stronger causal design is present. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It protects planning, hazards, social, and environmental layers from overclaiming cause from a convenient SQL-derived statistic.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-007. - CONFIRMED baseline link: KFM-IDX-POL-007.

Dependencies - PROPOSED: Analytical method registry - PROPOSED: Claim wording validator - PROPOSED: Evidence Drawer caveat fields

Tensions - PROPOSED: Some policy audiences may expect causal language from correlational dashboards.

Expansion directions - PROPOSED: Define language checks for correlation, association, contribution, exposure, and cause.

Open questions - NEEDS VERIFICATION: What causal-design evidence is sufficient to upgrade from correlation to causal claim?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 22; basis: aggregate-function discussion mentioning CORR() as a lesser-known aggregate for calculating correlation; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.47 KFM-P18-INV-159 — Model-selection rationale for model-assisted layers

Idea card metadata - Stable ID: KFM-P18-INV-159 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 159 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should require a model-selection rationale that links the problem type, data characteristics, interpretability need, and policy consequence to the chosen model family.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should require a model-selection rationale that links the problem type, data characteristics, interpretability need, and policy consequence to the chosen model family. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps model-assisted map layers from becoming opaque algorithm choices with no evidence of fitness for the domain question.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-007. - CONFIRMED baseline link: KFM-IDX-POL-003.

Dependencies - PROPOSED: Model card - PROPOSED: Training receipt - PROPOSED: Policy consequence rating

Tensions - PROPOSED: Interpretable models may underperform, while high-performing models may be harder to explain.

Expansion directions - PROPOSED: Add a model-family rationale block to model-assisted layer metadata.

Open questions - NEEDS VERIFICATION: When does model complexity require reviewer escalation or public caveat wording?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 106; basis: machine-learning workflow section listing model-selection choices such as regression, trees, forests, SVMs, and neural networks; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-007, KFM-IDX-POL-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.48 KFM-P18-INV-160 — Hyperparameter tuning receipts for model claims

Idea card metadata - Stable ID: KFM-P18-INV-160 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 160 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should record hyperparameter search space, search method, selected values, validation metric, and random seed for any model whose output supports a public claim.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record hyperparameter search space, search method, selected values, validation metric, and random seed for any model whose output supports a public claim. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, run-time routing, or release state.

Why it matters - PROPOSED: It makes model-assisted surfaces reproducible and reviewable rather than dependent on an undocumented tuning run.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-AIG-008. - CONFIRMED baseline link: KFM-IDX-PIP-010.

Dependencies - PROPOSED: Model training receipt - PROPOSED: Evaluation report - PROPOSED: Random seed policy

Tensions - PROPOSED: Search spaces can be large; receipts need useful summaries without hiding important choices.

Expansion directions - PROPOSED: Define minimum hyperparameter receipt fields for classical ML and neural models.

Open questions - NEEDS VERIFICATION: Which tuning changes require a new release, reviewer approval, or only a model-card update?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 107; basis: machine-learning workflow section describing hyperparameters and Grid Search / Random Search tuning; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-001, KFM-IDX-AIG-008, KFM-IDX-PIP-010 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.49 KFM-P18-INV-163 — Overfitting and underfitting as public model limitation labels

Idea card metadata - Stable ID: KFM-P18-INV-163 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 163 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM model-assisted outputs should carry limitation labels for overfitting, underfitting, gradient instability, and computational constraint where those risks were detected or not assessed.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM model-assisted outputs should carry limitation labels for overfitting, underfitting, gradient instability, and computational constraint where those risks were detected or not assessed. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents model outputs from being presented as equally reliable when training diagnostics indicate weak generalization or unstable learning.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-007. - CONFIRMED baseline link: KFM-IDX-POL-007.

Dependencies - PROPOSED: Evaluation report - PROPOSED: Model card - PROPOSED: Focus Mode caveat rules

Tensions - PROPOSED: Not all model families expose the same diagnostics.

Expansion directions - PROPOSED: Define model-family-specific limitation labels and abstain triggers.

Open questions - NEEDS VERIFICATION: What evidence is enough to say overfitting risk is controlled?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 184; basis: neural-network training section listing overfitting, underfitting, gradient issues, and computational costs as challenges; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.50 KFM-P18-INV-165 — Hands-on modelers as governance review discipline

Idea card metadata - Stable ID: KFM-P18-INV-165 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 165 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM domain model changes should include reviewers who understand both the domain vocabulary and the implementation expression of that model.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM domain model changes should include reviewers who understand both the domain vocabulary and the implementation expression of that model. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reduces the risk that schemas, validators, docs, and code drift apart while each appears locally reasonable.

Related ideas - CONFIRMED baseline link: KFM-IDX-MDP-003. - CONFIRMED baseline link: KFM-IDX-DOC-004.

Dependencies - PROPOSED: Reviewer matrix - PROPOSED: ADR process - PROPOSED: Domain steward role

Tensions - PROPOSED: Small teams may not have distinct domain and implementation experts for every lane.

Expansion directions - PROPOSED: Add a reviewer-role field to domain model ADRs and schema changes.

Open questions - NEEDS VERIFICATION: Which KFM changes require domain-steward plus implementer co-review?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 14; basis: DDD Hands-on Modelers pattern requiring technical contributors to touch code and code-changers to understand the model; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.51 KFM-P18-INV-166 — Refactoring-to-insight requires ADR and evidence trace

Idea card metadata - Stable ID: KFM-P18-INV-166 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 166 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators,

Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should require model refactors driven by domain insight to cite the evidence, expert review, ADR, migration note, and rollback path that justify the deeper model.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should require model refactors driven by domain insight to cite the evidence, expert review, ADR, migration note, and rollback path that justify the deeper model. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It separates useful domain-learning refactors from style-only rewrites that may break provenance, compatibility, or public claim semantics.

Related ideas - CONFIRMED baseline link: KFM-IDX-MDP-005. - CONFIRMED baseline link: KFM-IDX-DOC-008.

Dependencies - PROPOSED: ADR - PROPOSED: Migration note - PROPOSED: Regression tests

Tensions - PROPOSED: Over-documenting small naming improvements may slow iteration.

Expansion directions - PROPOSED: Define thresholds for insight-driven refactor versus minor cleanup.

Open questions - NEEDS VERIFICATION: What evidence qualifies as domain insight for a schema-breaking refactor?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 15; basis: DDD Refactoring Toward Deeper Insight pattern linking model refinement to domain insight and close expert involvement; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-MDP-005, KFM-IDX-DOC-008 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.52 KFM-P18-INV-219 — Training/test split and time-leakage guards for model claims

Idea card metadata - Stable ID: KFM-P18-INV-219 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 219 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM model-assisted layers should record training/test split strategy and prevent future or validation data from leaking into model fitting, especially for time-aware analyses.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded

this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM model-assisted layers should record training/test split strategy and prevent future or validation data from leaking into model fitting, especially for time-aware analyses. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It protects model performance claims and temporal predictions from false validation.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-001. - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Model card - PROPOSED: Training split receipt - PROPOSED: Temporal split policy - PROPOSED: Evaluation evidence

Tensions - PROPOSED: Random splits can inflate performance for spatial or temporal autocorrelated data.

Expansion directions - PROPOSED: Add split_strategy, leakage_checks, time_cutoff, and spatial_holdout fields to model evidence receipts.

Open questions - NEEDS VERIFICATION: What split strategies are mandatory for KFM environmental, hazard, and historical prediction tasks?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 156, 160, 163; basis: training/testing split, overfitting prevention, time-based split, and data leakage avoidance for time-dependent tasks; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-001, KFM-IDX-DAT-003, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.53 KFM-P18-INV-220 — Imbalanced classification metrics and confusion-matrix disclosure

Idea card metadata - Stable ID: KFM-P18-INV-220 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 220 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM classification-derived layers should disclose confusion-matrix counts and task-appropriate metrics such as precision, recall, F1, ROC/AUC, or class-specific errors instead of relying on accuracy alone.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM classification-derived layers should disclose confusion-matrix counts and task-appropriate metrics such as precision, recall, F1, ROC/AUC, or class-specific errors instead of relying on accuracy alone. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5.

- PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents a high headline accuracy from hiding poor performance on rare or sensitive classes.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-003. - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: Model card - PROPOSED: Class distribution - PROPOSED: Confusion matrix - PROPOSED: Threshold policy

Tensions - PROPOSED: For rare-species, hazard, or sensitive-location classes, false positives and false negatives have asymmetric consequences.

Expansion directions - PROPOSED: Require class_distribution and confusion_matrix fields for classification-derived public layers.

Open questions - NEEDS VERIFICATION: Which metric thresholds should be policy gates rather than reviewer guidance?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 165, 166, 219; basis: classification metrics for imbalanced datasets, precision, recall, F1-score, and confusion-matrix counts; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-003, KFM-IDX-DAT-003, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.54 KFM-P18-INV-221 — Feature-engineering and preprocessing receipts for AI-derived layers

Idea card metadata - Stable ID: KFM-P18-INV-221 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 221 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM AI-derived evidence should record preprocessing, missing-data handling, encoding, normalization, feature engineering, and data-cleaning steps before model outputs support claims.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM AI-derived evidence should record preprocessing, missing-data handling, encoding, normalization, feature engineering, and data-cleaning steps before model outputs support claims. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION:

Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes model-derived layers reproducible and exposes transforms that can bias outputs.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-001. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: Preprocessing pipeline - PROPOSED: Feature registry - PROPOSED: Missing-data method - PROPOSED: Transform receipt

Tensions - PROPOSED: Feature engineering can embed subjective assumptions or leak target information.

Expansion directions - PROPOSED: Add preprocessing_receipt and feature_registry records for AI-derived map/model products.

Open questions - NEEDS VERIFICATION: How should KFM label features derived from protected, sensitive, or weakly sourced variables?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 34, 67, 83; basis: data preprocessing, missing values, encoding, normalizing, data cleaning, and feature engineering as preparation for AI models; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-001, KFM-IDX-EVD-001, KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.55 KFM-P18-INV-226 — Window-function analytics preserve row-level evidence

Idea card metadata - Stable ID: KFM-P18-INV-226 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 226 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM analytical queries should use window-function style outputs when a reviewer needs aggregate context without collapsing individual evidence rows.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM analytical queries should use window-function style outputs when a reviewer needs aggregate context without collapsing individual evidence rows. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It lets KFM show summary context while preserving the row-level trace needed for EvidenceBundle resolution.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-OBS-003.

Dependencies - PROPOSED: SQL view or query contract - PROPOSED: Evidence row identifiers - PROPOSED: Analytic output schema

Tensions - PROPOSED: Analytic convenience can obscure which records supported each summary.

Expansion directions - PROPOSED: Add windowed-summary fixtures for county-year panels, hydrology stations, occurrence counts, and release QA reports.

Open questions - NEEDS VERIFICATION: Which analytic outputs should be exposed as public claims versus internal review context?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 23, 24; basis: window functions and aggregate window functions compute analytics over partitions while rows remain intact; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-EVD-001, KFM-IDX-OBS-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.56 KFM-P18-INV-227 — Recursive CTE calendars and hierarchy fixtures require bounds

Idea card metadata - Stable ID: KFM-P18-INV-227 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 227 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM should allow recursive SQL patterns for hierarchy traversal, generated date sequences, and gap analysis only with explicit recursion bounds and validation fixtures.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should allow recursive SQL patterns for hierarchy traversal, generated date sequences, and gap analysis only with explicit recursion bounds and validation fixtures. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It supports time-aware and hierarchy-aware analysis while preventing run-away recursion or unreviewed synthetic records.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-OBS-004.

Dependencies - PROPOSED: Recursion bound - PROPOSED: Cycle detection - PROPOSED: Calendar/version fixture

Tensions - PROPOSED: Generated rows can look as authoritative as sourced observations if not labeled.

Expansion directions - PROPOSED: Create recursive-query fixtures for administrative hierarchies, graph traversals, date spans, and missing-period detection.

Open questions - NEEDS VERIFICATION: How should generated calendar rows be labeled in evidence tables and release manifests?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 5, 10, 15; basis: recursive CTEs traverse hierarchies and generate sequences or dates such as calendars; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-PIP-004, KFM-IDX-OBS-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.57 KFM-P18-INV-229 — Pre-filtered aggregation as a performance and correctness gate

Idea card metadata - Stable ID: KFM-P18-INV-229 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 229 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM aggregate queries should filter inadmissible or out-of-scope records before grouping and aggregation, then record the filter conditions used.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM aggregate queries should filter inadmissible or out-of-scope records before grouping and aggregation, then record the filter conditions used. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reduces compute waste and prevents invalid records from entering summary claims.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-OBS-004. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: Query plan review - PROPOSED: Policy filter contract - PROPOSED: Aggregate receipt

Tensions - PROPOSED: Filtering for performance must not hide policy exclusions from reviewers.

Expansion directions - PROPOSED: Add aggregate_receipt fields for where_clause_hash, excluded_count, policy_exclusion_count, and input_row_count.

Open questions - NEEDS VERIFICATION: Which aggregates require human review when records are excluded by policy?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 55; basis: aggregation is expensive and filtering rows before grouping reduces input and improves efficiency; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.58 KFM-P18-INV-230 — Missing-data mechanism and imputation receipts

Idea card metadata - Stable ID: KFM-P18-INV-230 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 230 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM model, index, and sensor-derived outputs should record whether missingness is random, systematic, time-related, or potentially biased before any imputation or deletion is accepted.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM model, index, and sensor-derived outputs should record whether missingness is random, systematic, time-related, or potentially biased before any imputation or deletion is accepted. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes uncertainty and bias visible before a polished layer or model output reaches public use.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-001. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: Missingness audit - PROPOSED: Imputation method receipt - PROPOSED: Bias review - PROPOSED: Model card

Tensions - PROPOSED: Imputation can reduce gaps while adding unstated assumptions.

Expansion directions - PROPOSED: Add missing_data_receipt fields for mechanism, affected fields, method, rows_removed, values_imputed, and bias caveat.

Open questions - NEEDS VERIFICATION: When should missingness force ABSTAIN instead of imputation?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 147, 149, 150, 151, 152; basis: missing data causes, time-related gaps, not-missing-at-random bias, deletion, and imputation strategies; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-001, KFM-IDX-EVD-001, KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.59 KFM-P18-INV-231 — Model-evaluation receipts must prove generalization, not just fit

Idea card metadata - Stable ID: KFM-P18-INV-231 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 231 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM model-derived claims should carry evaluation evidence for generalization to unseen data and disclose overfitting, underfitting, and interpretability risks.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM model-derived claims should carry evaluation evidence for generalization to unseen data and disclose overfitting, underfitting, and interpretability risks. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents model-backed map layers from claiming reliability based only on training performance.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-003. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: Evaluation dataset - PROPOSED: Metric receipt - PROPOSED: Generalization statement - PROPOSED: Reviewer threshold

Tensions - PROPOSED: A high metric can still be inappropriate for high-consequence or imbalanced public claims.

Expansion directions - PROPOSED: Add evaluation_receipt fields for holdout type, metric set, failure classes, and reviewer acceptance threshold.

Open questions - NEEDS VERIFICATION: Which KFM model outputs require independent steward review before release?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 108, 164, 165, 166; basis: model evaluation, generalization to unseen data, overfitting/underfitting, and task-specific metrics; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-003, KFM-IDX-EVD-001, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.60 KFM-P18-INV-260 — K-means cluster choices require parameter and stability receipts

Idea card metadata - Stable ID: KFM-P18-INV-260 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 260 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM clustering-derived segments should record k, initialization method, distance metric, feature scaling, convergence, and stability checks before use as evidence.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM clustering-derived segments should record k, initialization method, distance metric, feature scaling, convergence, and stability checks before use as evidence. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, run-time routing, or release state.

Why it matters - PROPOSED: It makes exploratory segmentation auditable and prevents cluster labels from becoming unexamined categories.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-001. - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Model run receipt - PROPOSED: feature set - PROPOSED: normalization receipt - PROPOSED: stability report

Tensions - PROPOSED: Clusters can look authoritative even when k and initialization are arbitrary.

Expansion directions - PROPOSED: Add clustering_receipt fields for k, initialization, distance_metric, features, scaling, inertia, silhouette, and random_seed.

Open questions - NEEDS VERIFICATION: Which cluster-stability threshold is enough for public exploratory layers?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 131, 132, 133; basis: K-Means, user-defined k, centroids, Euclidean distance, random initialization, iterative refinement, and K-Means++; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-001, KFM-IDX-DAT-003, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.61 KFM-P18-INV-261 — KNN distance metrics and scaling are evidence-critical parameters

Idea card metadata - Stable ID: KFM-P18-INV-261 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 261 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM KNN-assisted classification should record distance metric, k, scaling/preprocessing, feature space, and noise sensitivity before any class label is exposed.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM KNN-assisted classification should record distance metric, k, scaling/preprocessing, feature space, and noise sensitivity before any class label is exposed. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents nearest-neighbor classifications from hiding how similarity was defined.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-001. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: Classification model receipt - PROPOSED: feature preprocessing - PROPOSED: evaluation report

Tensions - PROPOSED: Changing distance metric or scale can change the class assignment.

Expansion directions - PROPOSED: Add knn_receipt with k, distance_metric, scaling_method, feature_set_hash, neighbor_ids, and confidence/abstain rule.

Open questions - NEEDS VERIFICATION: Should KNN neighbor IDs be shown in Evidence Drawer for model-assisted labels?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 123, 124, 130; basis: KNN classification, feature space, distance metrics, data quality, scalability, and sensitivity to noise; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-001, KFM-IDX-EVD-001, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.62 KFM-P18-INV-262 — Linear regression assumptions become model limitation labels

Idea card metadata - Stable ID: KFM-P18-INV-262 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 262 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label:

CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM regression-derived claims should expose assumption checks and known violations as limitation labels, not just coefficients or predicted values.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM regression-derived claims should expose assumption checks and known violations as limitation labels, not just coefficients or predicted values. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reduces the risk of treating a simple predictive model as causal or universally valid.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-001. - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Model card - PROPOSED: assumption checks - PROPOSED: residual diagnostics - PROPOSED: scope label

Tensions - PROPOSED: Regression outputs can be overread as causal relationships.

Expansion directions - PROPOSED: Add regression_model_receipt fields for assumptions_checked, diagnostics, coefficient_table, residual_summary, and causal_claim_denied.

Open questions - NEEDS VERIFICATION: Which assumption violations should force ABSTAIN versus visible caveat?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 116, 117, 118, 120, 121; basis: linear regression objective, assumptions, advantages, disadvantages, and applications; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-001, KFM-IDX-DAT-003, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.63 KFM-P18-INV-263 — Data visualization choices in AI analysis are interpretive outputs

Idea card metadata - Stable ID: KFM-P18-INV-263 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 263 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM AI and data-analysis visualizations should carry chart intent, selected variables, transformation, and audience caveats because visualizations are interpretive outputs.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM AI and data-analysis visualizations should carry chart intent, selected variables, transformation, and audience caveats because visualizations are interpretive outputs. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps charts, dashboards, and model visuals from being treated as raw evidence.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-001. - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-MDP-003.

Dependencies - PROPOSED: Visualization manifest - PROPOSED: transformation receipt - PROPOSED: evidence source links

Tensions - PROPOSED: Charts can persuade even when they omit uncertainty or excluded data.

Expansion directions - PROPOSED: Add visualization_receipt fields for variables, transform, aggregation, chart_type, omitted_data, and intended_message.

Open questions - NEEDS VERIFICATION: Should visualization receipts be mandatory for all public dashboards?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 67, 74, 75; basis: Matplotlib visualizes data for insights and practical data-analysis workflows include visualization and advanced analysis; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.64 KFM-P18-INV-265 — Correlated subqueries expose outer-query dependency in analytic receipts

Idea card metadata - Stable ID: KFM-P18-INV-265 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 265 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM query receipts should record when a subquery depends on the outer query because result semantics cannot be understood from the subquery alone.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM query receipts should record when a subquery depends on the outer query because result semantics cannot be understood from the subquery alone. - PROPOSED: Phase 3 scope note: Inventory-level candidate record;

full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It helps reviewers identify where row-by-row dependency may affect performance, reproducibility, or interpretation.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-OBS-004. - CONFIRMED baseline link: KFM-IDX-MDP-003.

Dependencies - PROPOSED: Query receipt - PROPOSED: explain-plan check - PROPOSED: outer dependency list

Tensions - PROPOSED: Correlated subqueries may be correct but opaque and expensive.

Expansion directions - PROPOSED: Add `correlated_subquery_receipt` fields for `outer_columns_referenced`, `predicate`, `expected_cardinality`, and `performance_budget`.

Open questions - NEEDS VERIFICATION: When should correlated logic be rewritten as joins or CTEs for inspectability?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 2, 3; basis: correlated subqueries reference outer-query columns for complex filtering or calculations; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.65 KFM-P18-INV-284 — Connector unit tests are source-shape regression controls

Idea card metadata - Stable ID: KFM-P18-INV-284 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 284 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM web connectors should include tests that verify expected extraction behavior against fixtures so source-shape drift is detected before promotion.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM web connectors should include tests that verify expected extraction behavior against fixtures so source-shape drift is detected before promotion. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It converts fragile scraping assumptions into observable, failing tests rather than hidden runtime drift.

Related ideas - CONFIRMED baseline link: KFM-IDX-SEC-010. - CONFIRMED baseline link: KFM-IDX-PIP-003.

Dependencies - PROPOSED: HTML fixture - PROPOSED: connector test harness - PROPOSED: selector drift report

Tensions - PROPOSED: Passing fixture tests cannot prove the live site has not changed after the fixture was captured.

Expansion directions - PROPOSED: Add live-probe quarantine when fixture and live selector behavior diverge.

Open questions - NEEDS VERIFICATION: How often should source-shape fixtures be refreshed and who reviews refresh diffs?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 45, 51; basis: The scraping source introduces unit tests as written requirements and shows selector-heavy extraction that can be brittle when page structure lacks helpful IDs/classes.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-SEC-010, KFM-IDX-PIP-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.66 KFM-P18-INV-306 — Published languages for interchange without freezing internal models

Idea card metadata - Stable ID: KFM-P18-INV-306 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 306 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM cross-lane interchange should use documented published languages while translating into and out of lane models so internal models are not frozen by interchange needs.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM cross-lane interchange should use documented published languages while translating into and out of lane models so internal models are not frozen by interchange needs. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It supports stable public contracts without blocking internal model improvement.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-001. - CONFIRMED baseline link: KFM-IDX-DOC-002.

Dependencies - PROPOSED: PublishedLanguage register - PROPOSED: crosswalks - PROPOSED: translation tests

Tensions - PROPOSED: Too much translation can hide semantic loss; too little can freeze or corrupt lane models.

Expansion directions - PROPOSED: Create published-language entries for EvidenceBundle, DecisionEnvelope, SourceDescriptor, and LayerManifest candidates.

Open questions - NEEDS VERIFICATION: Which KFM interchange languages should be versioned independently from schemas?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 43; basis: DDD reference says a published language provides a common medium for communication and warns that using existing domain models directly as interchange languages can freeze them.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-001, KFM-IDX-DOC-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.67 KFM-P18-INV-310 — ROLLUP and CUBE subtotal semantics require hierarchy receipts

Idea card metadata - Stable ID: KFM-P18-INV-310 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 310 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM SQL-derived public metrics should record hierarchy and subtotal semantics when GROUP BY ROLLUP or CUBE is used.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM SQL-derived public metrics should record hierarchy and subtotal semantics when GROUP BY ROLLUP or CUBE is used. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents subtotal rows from being read as ordinary observations or double-counted in public panels.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-010. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: metric query receipt - PROPOSED: grouping hierarchy - PROPOSED: row_type label

Tensions - PROPOSED: Subtotals improve readability but can confuse downstream joins and charts.

Expansion directions - PROPOSED: Add row_type = detail | subtotal | total and grouping_level fields to metric outputs.

Open questions - NEEDS VERIFICATION: How should Evidence Drawer explain subtotal rows in tables and charts?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 26; basis: Advanced SQL source states GROUP BY has advanced features like CUBE and ROLLUP that return totals and subtotals for each group.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.68 KFM-P18-INV-311 — WITH CHECK OPTION as a view-level data-quality guardrail

Idea card metadata - Stable ID: KFM-P18-INV-311 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 311 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM SQL views used as contract surfaces should consider WITH CHECK OPTION-style constraints when inserts or updates through the view must preserve the view predicate.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM SQL views used as contract surfaces should consider WITH CHECK OPTION-style constraints when inserts or updates through the view must preserve the view predicate. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It helps distinguish read-only analytic views from updateable governed contract views with invariant enforcement.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-007. - CONFIRMED baseline link: KFM-IDX-DAT-010.

Dependencies - PROPOSED: view contract - PROPOSED: predicate assertion - PROPOSED: update policy

Tensions - PROPOSED: Updateable views can simplify workflows but blur canonical-store boundaries if not governed.

Expansion directions - PROPOSED: Document which KFM views are read-only, updateable, or prohibited from direct mutation.

Open questions - NEEDS VERIFICATION: Should public-facing KFM views ever be updateable, or only internal governed tools?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 104; basis: Advanced SQL source says WITH CHECK OPTION ensures UPDATE and INSERT operations satisfy view-definition conditions, otherwise returning an error; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.69 KFM-P18-INV-312 — Timestamp-difference metrics require explicit unit conversion

Idea card metadata - Stable ID: KFM-P18-INV-312 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 312 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM SQL metrics that compare timestamps should record the extraction unit and conversion rule used to compute elapsed time.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM SQL metrics that compare timestamps should record the extraction unit and conversion rule used to compute elapsed time. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It avoids hidden unit conversions in temporal KPIs, eligibility windows, and event-response metrics.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-004. - CONFIRMED baseline link: KFM-IDX-OBS-003.

Dependencies - PROPOSED: metric unit registry - PROPOSED: temporal query receipt - PROPOSED: timezone policy

Tensions - PROPOSED: Different databases and timezones can produce subtly different elapsed-time results.

Expansion directions - PROPOSED: Add elapsed_time_unit and timezone_assumption fields to temporal metric receipts.

Open questions - NEEDS VERIFICATION: Should all KFM elapsed-time metrics normalize to UTC seconds before display?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 36, 38; basis: Advanced SQL source demonstrates using EXTRACT(EPOCH FROM timestamp difference), dividing seconds by 3600, and filtering records within 168 hours.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN:

Repository implementation status remains unverified.

8.8.70 KFM-P18-INV-313 — Missing-data method receipts distinguish deletion, imputation, and advanced repair

Idea card metadata - Stable ID: KFM-P18-INV-313 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 313 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM model and analytic outputs should record the missing-data method used, why it was chosen, and the bias or information-loss caveat attached to it.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM model and analytic outputs should record the missing-data method used, why it was chosen, and the bias or information-loss caveat attached to it. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps incomplete observations from being silently transformed into clean-looking outputs.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-004. - CONFIRMED baseline link: KFM-IDX-DAT-010.

Dependencies - PROPOSED: missingness classification - PROPOSED: imputation parameters - PROPOSED: bias caveat

Tensions - PROPOSED: Simple methods are transparent but may bias results; advanced methods can be harder to explain.

Expansion directions - PROPOSED: Add missing_data_method and missingness_assumption fields to analytic receipts.

Open questions - NEEDS VERIFICATION: Should KFM deny public model release when missingness is non-random and unmodeled?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 147, 151, 152, 155; basis: AI Concepts source states missing data can distort analyses and models, describes dropping rows and imputation, and notes bias, information loss, and method selection tradeoffs.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-004, KFM-IDX-DAT-010 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.71 KFM-P18-INV-315 — Model split choice as an evaluation-fairness record

Idea card metadata - Stable ID: KFM-P18-INV-315 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 315 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM model receipts should record random, stratified, time-based, or other split choices and explain how the split supports fair evaluation.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM model receipts should record random, stratified, time-based, or other split choices and explain how the split supports fair evaluation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes model validation posture inspectable instead of reducing performance claims to a single metric.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-004. - CONFIRMED baseline link: KFM-IDX-OBS-003.

Dependencies - PROPOSED: split method - PROPOSED: train/test ranges - PROPOSED: data type label - PROPOSED: leakage check

Tensions - PROPOSED: Different split methods can produce different performance estimates.

Expansion directions - PROPOSED: Add split_method and split_rationale fields to ModelEvaluationReceipt.

Open questions - NEEDS VERIFICATION: Should spatial, temporal, and stratified splits be combined for some KFM models?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 160, 163; basis: AI Concepts source states train/test splitting is critical, that time-based splits preserve temporal order and avoid leakage, and that split choice should depend on data type.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-004, KFM-IDX-OBS-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.72 KFM-P18-INV-340 — AI data lifecycle as a governed transform chain

Idea card metadata - Stable ID: KFM-P18-INV-340 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 340 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM AI-assisted outputs should record the data lifecycle from collection through cleaning, preparation, training or inference, evaluation, and deployment-facing use as evidence transforms.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM AI-assisted outputs should record the data lifecycle from collection through cleaning, preparation, training or inference, evaluation, and deployment-facing use as evidence transforms. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps AI outputs subordinate to traceable data operations.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-004. - CONFIRMED baseline link: KFM-IDX-DOC-007.

Dependencies - PROPOSED: data source - PROPOSED: cleaning transform - PROPOSED: feature preparation - PROPOSED: evaluation report - PROPOSED: inference receipt

Tensions - PROPOSED: A generic AI lifecycle can obscure domain-specific evidence and policy needs unless specialized by lane.

Expansion directions - PROPOSED: Map AI lifecycle fields to EvidenceBundle and AIReceipt fields during Phase 5.

Open questions - NEEDS VERIFICATION: Which lifecycle fields are mandatory for KFM Focus Mode versus offline model experiments?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 77, 84, 104, 106; basis: AI Concepts Using Python describes data as central to AI, names a data lifecycle, and outlines machine-learning workflow and challenges.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-004, KFM-IDX-DOC-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.73 KFM-P18-INV-342 — Train-test split and evaluation records for model-assisted map claims

Idea card metadata - Stable ID: KFM-P18-INV-342 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 342 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM model-assisted classifications, risk scores, or suitability layers should retain train/test split, evaluation metrics, leakage checks, and validation-set provenance before any public map claim.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category

before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM model-assisted classifications, risk scores, or suitability layers should retain train/test split, evaluation metrics, leakage checks, and validation-set provenance before any public map claim. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It helps reviewers distinguish validated model outputs from untested exploratory classifications.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-004. - CONFIRMED baseline link: KFM-IDX-POL-007.

Dependencies - PROPOSED: training set - PROPOSED: test set - PROPOSED: split method - PROPOSED: evaluation metric - PROPOSED: leakage check

Tensions - PROPOSED: Small or spatially autocorrelated datasets can make conventional splits misleading.

Expansion directions - PROPOSED: Add `spatial_split_warning` to model-derived layer reviews.

Open questions - NEEDS VERIFICATION: When should KFM require spatial or temporal cross-validation rather than random splits?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — `AI_Concepts_Using_Python.pdf`; pages 155; basis: AI Concepts Using Python includes splitting data into training and testing sets as a practical data-analysis topic.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-004, KFM-IDX-POL-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.74 KFM-P18-INV-359 — Mars landing suitability as a transferable site-selection exercise

Idea card metadata - Stable ID: KFM-P18-INV-359 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 359 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM can adapt raster reclassification, terrain analysis, and suitability-screening patterns from site-selection exercises, but must keep suitability models distinct from authoritative site decisions.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM can adapt raster reclassification, terrain analysis, and suitability-screening patterns from site-selection exercises, but must keep suitability models distinct from authoritative site decisions. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired

to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It gives KFM a reusable pattern for suitability maps without turning model outputs into approval decisions.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: criteria list - PROPOSED: raster transform - PROPOSED: terrain metric - PROPOSED: decision role label

Tensions - PROPOSED: A suitability map can be mistaken for a recommendation unless its non-authoritative role is visible.

Expansion directions - PROPOSED: Create a site_selection_model evidence profile for terrestrial KFM use.

Open questions - NEEDS VERIFICATION: Which site-selection contexts require review before publication?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 3, 132; basis: Earth, Space, and Environmental Science Explorations with ArcGIS Pro includes “Plan your Mars landing” with raster reclassification, attribute extraction, and terrain analysis objectives.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.75 KFM-P18-INV-368 — AI missing-data handling as an evidence-quality decision

Idea card metadata - Stable ID: KFM-P18-INV-368 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 368 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM model-assisted analyses should record missing-data handling choices as evidence-quality decisions rather than hidden preprocessing defaults.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM model-assisted analyses should record missing-data handling choices as evidence-quality decisions rather than hidden preprocessing defaults. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Missing-data choices can change model outputs, spatial patterns, and downstream claims.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-006. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: preprocessing pipeline - PROPOSED: missingness profile - PROPOSED: imputation rule - PROPOSED: model receipt

Tensions - PROPOSED: Imputation can fabricate apparent precision if not labeled.

Expansion directions - PROPOSED: Define missing_data_policy fields for model-assisted evidence.

Open questions - NEEDS VERIFICATION: When does missingness force ABSTAIN instead of model output?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 67, 100, 146; basis: AI Concepts Using Python covers handling missing data in practical data analysis and notes Pandas support for missing data and data cleaning.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-006, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.76 KFM-P18-INV-369 — Mathematical foundation notes as model explanation metadata

Idea card metadata - Stable ID: KFM-P18-INV-369 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 369 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM model cards should record the mathematical assumptions needed to interpret model outputs, including vector/matrix representation and probability-based uncertainty treatment where applicable.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM model cards should record the mathematical assumptions needed to interpret model outputs, including vector/matrix representation and probability-based uncertainty treatment where applicable. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Users cannot evaluate model-assisted map claims when the representation and uncertainty assumptions are invisible.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-006. - CONFIRMED baseline link: KFM-IDX-MDP-003.

Dependencies - PROPOSED: model card - PROPOSED: feature representation - PROPOSED: uncertainty method - PROPOSED: evaluation report

Tensions - PROPOSED: Too much mathematical detail can reduce public readability.

Expansion directions - PROPOSED: Add public and technical layers to model explanation metadata.

Open questions - NEEDS VERIFICATION: Which mathematical assumptions are material for each KFM model family?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 92, 94, 97; basis: AI Concepts Using Python describes linear algebra as the language of data and probability theory as a way to model uncertainty in AI systems.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.77 KFM-P18-INV-379 — Variogram and sampling metadata for interpolated surfaces

Idea card metadata - Stable ID: KFM-P18-INV-379 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 379 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM geostatistical or interpolated layers should carry sampling design, model family, variogram or covariance assumptions, and uncertainty surfaces as source-linked metadata.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM geostatistical or interpolated layers should carry sampling design, model family, variogram or covariance assumptions, and uncertainty surfaces as source-linked metadata. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Interpolated maps can be visually smooth while masking sparse sampling or model assumptions.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: sample locations - PROPOSED: interpolation method - PROPOSED: uncertainty layer - PROPOSED: model receipt

Tensions - PROPOSED: Detailed geostatistical metadata may exceed what casual map users can interpret.

Expansion directions - PROPOSED: Define a geostatistical_surface_profile for KFM raster and contour products.

Open questions - NEEDS VERIFICATION: What uncertainty visualization should accompany interpolated public layers?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 271, 290, 301; basis: A Primer of GIS includes geostatistics in the GI analysis section and index/table-of-contents references to geostatistical concepts.; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-012, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.78 KFM-P18-INV-380 — FIRST_VALUE and LAST_VALUE anchors for released observation panels

Idea card metadata - Stable ID: KFM-P18-INV-380 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 380 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM SQL-derived observation panels should record FIRST_VALUE, LAST_VALUE, and ordering semantics when analytics compare a current value to a baseline or endpoint.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM SQL-derived observation panels should record FIRST_VALUE, LAST_VALUE, and ordering semantics when analytics compare a current value to a baseline or endpoint. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Endpoint comparisons can change when ordering, partitioning, or tie-breaking changes, so public metrics need reproducible analytic semantics.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-010. - CONFIRMED baseline link: KFM-IDX-EVD-006.

Dependencies - PROPOSED: window function - PROPOSED: partition key - PROPOSED: order key - PROPOSED: tie-breaker - PROPOSED: analytic receipt

Tensions - PROPOSED: Endpoint functions can be misread as temporal truth if the ordering column is not time.

Expansion directions - PROPOSED: Add endpoint_function_semantics to SQL analytics receipts.

Open questions - NEEDS VERIFICATION: Which public panels need baseline/current endpoint disclosure?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 13; basis: Advanced SQL Concepts describes LEAD, LAG, FIRST_VALUE, LAST_VALUE, and NTH_VALUE as analytic functions based on ordering and partitioning.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-010, KFM-IDX-EVD-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.79 KFM-P18-INV-382 — Subquery-in-FROM outputs as named analytical intermediates

Idea card metadata - Stable ID: KFM-P18-INV-382 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 382 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM SQL workflows should name, hash, and describe subqueries used in FROM clauses when they produce intermediate datasets used for public metrics or map layers.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM SQL workflows should name, hash, and describe subqueries used in FROM clauses when they produce intermediate datasets used for public metrics or map layers. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Complex intermediate tables can determine final claims even when only the final query result is published.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-010. - CONFIRMED baseline link: KFM-IDX-EVD-006.

Dependencies - PROPOSED: intermediate query - PROPOSED: query hash - PROPOSED: metric definition - PROPOSED: run receipt

Tensions - PROPOSED: Not every subquery warrants a named artifact; thresholding is needed.

Expansion directions - PROPOSED: Define when derived SQL intermediates become receipt-bearing artifacts.

Open questions - NEEDS VERIFICATION: What complexity threshold triggers a named intermediate receipt?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 12; basis: Advanced SQL Concepts describes subqueries in the FROM clause as a way to derive product/category/sales data and aggregate results.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-010, KFM-IDX-EVD-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.80 KFM-P18-INV-383 — SQL performance optimization as reproducibility evidence for large panels

Idea card metadata - Stable ID: KFM-P18-INV-383 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 383 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM should record performance-relevant SQL choices for large public panels when query optimization, views, or stored procedures materially affect reproducibility, cost, or completeness.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record performance-relevant SQL choices for large public panels when query optimization, views, or stored procedures materially affect reproducibility, cost, or completeness. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Performance choices can change whether a panel is feasible, current, or complete; they are not only engineering trivia.

Related ideas - CONFIRMED baseline link: KFM-IDX-OBS-002. - CONFIRMED baseline link: KFM-IDX-MDP-003.

Dependencies - PROPOSED: query plan - PROPOSED: view or procedure - PROPOSED: test result - PROPOSED: cost budget

Tensions - PROPOSED: Capturing query-plan details may be database-specific and noisy.

Expansion directions - PROPOSED: Add a lightweight `sql_runtime_profile` for published analytical panels.

Open questions - NEEDS VERIFICATION: Which SQL runtime details belong in receipts versus operational logs?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 18; basis: Advanced SQL Concepts lists query optimization, data integrity, thorough testing, views or stored procedures, and documentation as best practices for complex SQL.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.81 KFM-P18-INV-384 — Scikit-learn workflow artifacts as model-evidence chain

Idea card metadata - Stable ID: KFM-P18-INV-384 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 384 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM model-assisted layers should preserve train/test split, estimator configuration, fitted artifact digest, evaluation report, and prediction output as a linked model-evidence chain.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category

before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM model-assisted layers should preserve train/test split, estimator configuration, fitted artifact digest, evaluation report, and prediction output as a linked model-evidence chain. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A model output cannot be audited without knowing how the estimator was configured, trained, evaluated, and applied.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-006. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: model card - PROPOSED: training data - PROPOSED: estimator config - PROPOSED: evaluation report - PROPOSED: prediction layer

Tensions - PROPOSED: Model libraries abstract implementation details, which can obscure defaults.

Expansion directions - PROPOSED: Create a model_workflow_receipt template for scikit-style estimators.

Open questions - NEEDS VERIFICATION: Which estimator defaults must be captured to make a model reproducible?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 99, 120, 137, 140; basis: AI Concepts Using Python discusses Scikit-learn and practical machine-learning workflows including model implementation and examples.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-006, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.82 KFM-P18-INV-385 — Hybrid learning-mode labels for semi-supervised and reinforcement outputs

Idea card metadata - Stable ID: KFM-P18-INV-385 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 385 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM AI-supported surfaces should label semi-supervised and reinforcement-learning outputs separately from supervised, unsupervised, or rule-based products.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM AI-supported surfaces should label semi-supervised and reinforcement-learning outputs separately from supervised, unsupervised, or rule-based products. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the

planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Learning mode affects evidence interpretation, evaluation strategy, and release risk.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-006. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: learning mode - PROPOSED: training signal - PROPOSED: reward or label source - PROPOSED: evaluation method

Tensions - PROPOSED: A single workflow can combine modes, making simple labels incomplete.

Expansion directions - PROPOSED: Add learning_mode enum support for hybrid model products.

Open questions - NEEDS VERIFICATION: How should KFM disclose reward functions for reinforcement-learning-derived outputs?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 102, 115, 116; basis: AI Concepts Using Python discusses supervised, unsupervised, semi-supervised, and reinforcement learning as distinct machine-learning paradigms.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-006, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.83 KFM-P18-INV-386 — Visualization artifacts for non-map AI analysis

Idea card metadata - Stable ID: KFM-P18-INV-386 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 386 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should treat charts, plots, and diagnostic visualizations used in AI/data-analysis review as receipt-bearing artifacts when they support model selection, evaluation, or public explanation.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat charts, plots, and diagnostic visualizations used in AI/data-analysis review as receipt-bearing artifacts when they support model selection, evaluation, or public explanation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Model-review charts can influence release decisions even when they are not map layers.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-006. - CONFIRMED baseline link: KFM-IDX-MDP-003.

Dependencies - PROPOSED: diagnostic chart - PROPOSED: plot specification - PROPOSED: input data hash - PROPOSED: model evaluation receipt

Tensions - PROPOSED: Diagnostic plots are easy to overinterpret without statistical context.

Expansion directions - PROPOSED: Add chart_artifact records to model evaluation packages.

Open questions - NEEDS VERIFICATION: Which diagnostic visualizations are mandatory for KFM model reviews?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 69, 74, 75; basis: AI Concepts Using Python covers Matplotlib and data visualization for insights in practical data analysis.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.84 KFM-P18-INV-393 — Tracking-log report reconstruction windows

Idea card metadata - Stable ID: KFM-P18-INV-393 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 393 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM report receipts should state the transaction-time or tracking-log window used to reconstruct the data seen by a published report or dashboard.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM report receipts should state the transaction-time or tracking-log window used to reconstruct the data seen by a published report or dashboard. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A user must know whether a report reflects current data, a historical database state, or a reconstructed after-image sequence.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-006. - CONFIRMED baseline link: KFM-IDX-OBS-001.

Dependencies - PROPOSED: tracking log - PROPOSED: transaction time - PROPOSED: report receipt - PROPOSED: after-image policy

Tensions - PROPOSED: Tracking-log retention may conflict with storage and privacy policies.

Expansion directions - PROPOSED: Add report_reconstruction_time_window to dashboard receipts.

Open questions - NEEDS VERIFICATION: How long should KFM retain transaction histories for public reports?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 219, 222, 235; basis: Developing Time-Oriented Database Applications in SQL discusses retaining a tracking log, queries, modifications, and using after-images consistently; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-006, KFM-IDX-OBS-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.85 KFM-P18-INV-399 — Scraper unit tests as source-adapter regression evidence

Idea card metadata - Stable ID: KFM-P18-INV-399 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 399 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM web-source connectors should include parser and retrieval tests that prove the adapter handles expected HTML, broken links, missing fields, and persistence behavior without live-network dependence.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM web-source connectors should include parser and retrieval tests that prove the adapter handles expected HTML, broken links, missing fields, and persistence behavior without live-network dependence. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A source adapter can silently break when website structure changes; no-network tests provide a controlled regression floor.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-006. - CONFIRMED baseline link: KFM-IDX-SEC-011.

Dependencies - PROPOSED: fixture HTML - PROPOSED: parser test - PROPOSED: error case - PROPOSED: persistence assertion

Tensions - PROPOSED: Tests based on old pages can become stale if they are not refreshed under governance.

Expansion directions - PROPOSED: Create scraped_source_fixture_refresh receipts.

Open questions - NEEDS VERIFICATION: How often should web-source fixtures be refreshed and re-approved?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 31, 45; basis: Instant Web Scraping with Java includes sections on handling errors and writing tests for scraping workflows; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-006, KFM-IDX-SEC-011 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.86 KFM-P18-INV-404 — Transaction boundary receipts for data-changing SQL operations

Idea card metadata - Stable ID: KFM-P18-INV-404 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 404 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM database mutation batches should emit receipts that state transaction boundaries, commit/rollback outcome, affected row counts, and recovery target.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM database mutation batches should emit receipts that state transaction boundaries, commit/rollback outcome, affected row counts, and recovery target. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A published claim depends on knowing whether a database change committed, rolled back, or partially failed under a controlled transaction policy.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-EVD-002.

Dependencies - PROPOSED: RunReceipt - PROPOSED: database migration policy - PROPOSED: mutation validator - PROPOSED: rollback target

Tensions - PROPOSED: Read-only analytical queries do not need the same receipt burden as write operations.

Expansion directions - PROPOSED: Add transaction_outcome fields to database run receipts for mutating jobs.

Open questions - NEEDS VERIFICATION: What minimum mutation metadata should be exposed to public release auditors?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 59, 61; basis: Advanced SQL Concepts distinguishes DML operations, COMMIT, and ROLLBACK and shows rollback behavior as part of SQL operations.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-004, KFM-IDX-EVD-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.87 KFM-P18-INV-405 — HAVING filters as post-aggregation policy decisions

Idea card metadata - Stable ID: KFM-P18-INV-405 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 405 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM analytical panels should disclose filters applied after aggregation separately from row-level WHERE filters, especially when thresholds shape public conclusions.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM analytical panels should disclose filters applied after aggregation separately from row-level WHERE filters, especially when thresholds shape public conclusions. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Post-aggregation filters can hide groups or suppress small denominators; disclosure prevents threshold logic from masquerading as source fact.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: analysis manifest - PROPOSED: denominator policy - PROPOSED: threshold disclosure - PROPOSED: SQL query receipt

Tensions - PROPOSED: Some privacy-preserving suppression thresholds cannot be fully exposed to all audiences.

Expansion directions - PROPOSED: Record `aggregation_filter_stage` in analytical query manifests.

Open questions - NEEDS VERIFICATION: How should privacy thresholds be disclosed without weakening privacy protections?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 54, 55; basis: Advanced SQL Concepts discusses HAVING and efficient query strategy around aggregation and filtering.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.88 KFM-P18-INV-406 — Query-cost and performance budget as reproducibility metadata

Idea card metadata - Stable ID: KFM-P18-INV-406 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 406 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM large analytical queries should record performance-relevant assumptions, indexes, and input-size budgets when runtime affects repeatability or operational cost.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM large analytical queries should record performance-relevant assumptions, indexes, and input-size budgets when runtime affects repeatability or operational cost. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A reproducible analytical result includes not only SQL semantics but also whether the query can be run within governed cost, time, and infrastructure constraints.

Related ideas - CONFIRMED baseline link: KFM-IDX-OBS-003.

Dependencies - PROPOSED: query plan snapshot - PROPOSED: data size receipt - PROPOSED: cost budget - PROPOSED: observability event

Tensions - PROPOSED: Query-plan capture can be database-specific and unstable across engines.

Expansion directions - PROPOSED: Add optional query_cost_profile to heavy analytical receipts.

Open questions - NEEDS VERIFICATION: Which query-cost signals are portable enough for KFM receipts?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 54, 55; basis: Advanced SQL Concepts emphasizes query optimization and performance when working with large datasets or complex operations.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-OBS-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.89 KFM-P18-INV-407 — Cross-validation and overfitting labels for predictive layers

Idea card metadata - Stable ID: KFM-P18-INV-407 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 407 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM predictive or model-assisted layers should carry cross-validation, overfitting, and generalization labels before their outputs can support public interpretation.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM predictive or model-assisted layers should carry cross-validation, overfitting, and generalization labels before their outputs can support public interpretation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback

target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Predictive layers can look authoritative on a map even when model generalization is weak; the validation posture must travel with the rendered output.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-005. - CONFIRMED baseline link: KFM-IDX-DAT-011.

Dependencies - PROPOSED: model card - PROPOSED: validation split receipt - PROPOSED: metrics registry - PROPOSED: Evidence Drawer model section

Tensions - PROPOSED: Simple models and complex neural networks need different but comparable validation summaries.

Expansion directions - PROPOSED: Create model_validation_summary fields for predictive map outputs.

Open questions - NEEDS VERIFICATION: What thresholds should trigger ABSTAIN for model-assisted claims?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 106, 155, 184; basis: AI Concepts Using Python introduces machine-learning workflow, challenges, training/testing splits, and tools for model training.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-005, KFM-IDX-DAT-011 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.90 KFM-P18-INV-408 — Feature scaling and normalization transform receipts

Idea card metadata - Stable ID: KFM-P18-INV-408 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 408 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM ML and analytical feature pipelines should record scaling, normalization, encoding, and feature-selection transforms as evidence-bearing processing steps.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM ML and analytical feature pipelines should record scaling, normalization, encoding, and feature-selection transforms as evidence-bearing processing steps. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Feature transforms can change model behavior and interpretability; the public claim must know whether a value is raw, standardized, encoded, or engineered.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-AIG-005.

Dependencies - PROPOSED: feature registry - PROPOSED: transform manifest - PROPOSED: model card - PROPOSED: run receipt

Tensions - PROPOSED: Some transformations are obvious to engineers but invisible to public reviewers unless named.

Expansion directions - PROPOSED: Add feature_transform_receipt as a reusable object in model-assisted lanes.

Open questions - NEEDS VERIFICATION: Should feature transforms be versioned independently from model versions?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 70, 73, 139; basis: AI Concepts Using Python covers data cleaning/preparation, data analysis, and machine-learning workflows with preprocessing steps.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-004, KFM-IDX-AIG-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.91 KFM-P18-INV-409 — AI visualization outputs remain explanatory derivatives

Idea card metadata - Stable ID: KFM-P18-INV-409 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 409 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: Charts, plots, and non-map visualizations created during AI/data analysis should be treated as explanatory derivatives that link back to evidence, not root truth.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: Charts, plots, and non-map visualizations created during AI/data analysis should be treated as explanatory derivatives that link back to evidence, not root truth. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A plot can strongly shape interpretation while being only one rendering of processed data; KFM needs the same trust membrane for charts as for maps.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-004. - CONFIRMED baseline link: KFM-IDX-DAT-013.

Dependencies - PROPOSED: visualization manifest - PROPOSED: input data digest - PROPOSED: style/render config - PROPOSED: citation validation

Tensions - PROPOSED: Exploratory visualizations may be useful internally but unsuitable for publication.

Expansion directions - PROPOSED: Extend release manifests to include chart artifacts with input digests.

Open questions - NEEDS VERIFICATION: Which chart artifacts need Evidence Drawer equivalents?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 67, 69, 74; basis: AI Concepts Using Python describes Matplotlib and data visualization as tools for insight and analysis.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-004, KFM-IDX-DAT-013 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.92 KFM-P18-INV-415 — Sequenced versus nonsequenced query label in public explanations

Idea card metadata - Stable ID: KFM-P18-INV-415 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 415 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM public explanations over temporal data should disclose whether an answer used current-state, sequenced, or nonsequenced temporal query semantics.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM public explanations over temporal data should disclose whether an answer used current-state, sequenced, or nonsequenced temporal query semantics. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Different temporal query modes answer different questions; without the mode label, users can mistake historical summaries for current facts or vice versa.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-010. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: query receipt - PROPOSED: time scope field - PROPOSED: Evidence Drawer text - PROPOSED: API response envelope

Tensions - PROPOSED: Temporal query terminology may need translation for nontechnical audiences.

Expansion directions - PROPOSED: Add temporal_query_mode to runtime response envelopes for time-aware panels.

Open questions - NEEDS VERIFICATION: What simple language best explains sequenced query semantics to public users?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 145, 156; basis: Developing Time-Oriented Database Applications in SQL dis-

tinguishes current-state extraction, prior-state extraction, sequenced queries, and nonsequenced variants.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-010, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.93 KFM-P18-INV-417 — Knowledge-level models for policy and source-role rules

Idea card metadata - Stable ID: KFM-P18-INV-417 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 417 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM can model policy, source roles, and admissibility rules at a knowledge level so rules can evolve without rewriting every domain object.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM can model policy, source roles, and admissibility rules at a knowledge level so rules can evolve without rewriting every domain object. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Source authority, sensitivity, and admissibility rules cross many lanes; a knowledge-level pattern can reduce duplicated policy logic.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-CAT-005.

Dependencies - PROPOSED: policy registry - PROPOSED: source-role registry - PROPOSED: rule versioning - PROPOSED: DecisionEnvelope

Tensions - PROPOSED: Over-abstract policy models can become too opaque for reviewers.

Expansion directions - PROPOSED: Explore a knowledge-level rule registry for source roles and sensitivity classes.

Open questions - NEEDS VERIFICATION: Which policy decisions should stay declarative and which should remain explicit in domain schemas?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 51, 52; basis: Domain-Driven Design Reference includes Knowledge Level and Pluggable Component Framework in large-scale structure patterns.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN:

Repository implementation status remains unverified.

8.8.94 KFM-P18-INV-429 — Set-operator merge semantics for multi-source evidence panels

Idea card metadata - Stable ID: KFM-P18-INV-429 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 429 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM multi-source analytical panels should record whether rows were merged with UNION, UNION ALL, INTERSECT, or EXCEPT-style semantics.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM multi-source analytical panels should record whether rows were merged with UNION, UNION ALL, INTERSECT, or EXCEPT-style semantics. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Duplicate removal or retention can materially change counts, evidence coverage, and disagreement analysis.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-010. - CONFIRMED baseline link: KFM-IDX-EVD-002.

Dependencies - PROPOSED: transform receipt - PROPOSED: SQL lineage - PROPOSED: data-quality report - PROPOSED: analytical panel schema

Tensions - PROPOSED: Some engines differ in supported set operators.

Expansion directions - PROPOSED: Add set_operator_semantics to analytical transform receipts.

Open questions - NEEDS VERIFICATION: How should engine-specific operator differences be represented in portable receipts?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 21, 55, 82; basis: Advanced SQL Concepts distinguishes UNION, UNION ALL, INTERSECT, MINUS/EXCEPT, and their duplicate-handling semantics.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-010, KFM-IDX-EVD-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.95 KFM-P18-INV-430 — NULL-sensitive aggregate metrics for data-quality reports

Idea card metadata - Stable ID: KFM-P18-INV-430 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 430 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label:

CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM data-quality metrics should disclose NULL treatment for counts, distinct counts, averages, and grouped summaries.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM data-quality metrics should disclose NULL treatment for counts, distinct counts, averages, and grouped summaries. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Data-quality conclusions can be wrong when missingness is hidden inside aggregate behavior.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-005. - CONFIRMED baseline link: KFM-IDX-DAT-012.

Dependencies - PROPOSED: DQ report - PROPOSED: SQL transform receipt - PROPOSED: missingness profile - PROPOSED: validator

Tensions - PROPOSED: Null treatment differs between human expectations and SQL defaults.

Expansion directions - PROPOSED: Add null_semantics fields to aggregate metric reports.

Open questions - NEEDS VERIFICATION: Which aggregate functions require mandatory NULL semantics disclosure?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 22, 36, 45; basis: Advanced SQL Concepts notes NULL handling in DISTINCT and COUNT(column), including that COUNT(column) only counts non-NULL values.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-005, KFM-IDX-DAT-012 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.96 KFM-P18-INV-431 — Feature-scaling receipts for model-assisted spatial analysis

Idea card metadata - Stable ID: KFM-P18-INV-431 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 431 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM model-assisted spatial analyses should emit receipts for feature scaling, normalization, and preprocessing choices.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate:

KFM model-assisted spatial analyses should emit receipts for feature scaling, normalization, and preprocessing choices. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Scaling decisions can change model behavior and should be auditable before model outputs influence map claims.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-004. - CONFIRMED baseline link: KFM-IDX-EVD-002.

Dependencies - PROPOSED: model run receipt - PROPOSED: preprocessing manifest - PROPOSED: feature schema - PROPOSED: evaluation report

Tensions - PROPOSED: Some preprocessing is embedded inside pipelines and easy to overlook.

Expansion directions - PROPOSED: Create a preprocessing_receipt object for model-assisted lanes.

Open questions - NEEDS VERIFICATION: How much preprocessing detail should public Evidence Drawer expose?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 128, 139, 144; basis: AI Concepts Using Python discusses feature scaling and machine-learning workflows using Python libraries.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-004, KFM-IDX-EVD-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.97 KFM-P18-INV-432 — Confusion-matrix evidence for classification layers

Idea card metadata - Stable ID: KFM-P18-INV-432 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 432 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM classification layers should attach confusion-matrix or comparable evaluation evidence before being used for consequential public interpretation.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM classification layers should attach confusion-matrix or comparable evaluation evidence before being used for consequential public interpretation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Classification maps can look definitive even when class-level errors are uneven; evaluation evidence keeps uncertainty visible.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-005. - CONFIRMED baseline link: KFM-IDX-DAT-008.

Dependencies - PROPOSED: classification model card - PROPOSED: evaluation receipt - PROPOSED: Layer-Manifest - PROPOSED: EvidenceBundle

Tensions - PROPOSED: Not every classified layer has labeled ground-truth data.

Expansion directions - PROPOSED: Add class-level accuracy fields to model-assisted LayerManifest entries.

Open questions - NEEDS VERIFICATION: What minimum evaluation evidence is acceptable for unsupervised or weakly supervised maps?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 166, 219; basis: AI Concepts Using Python includes confusion matrix material as part of classification evaluation.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-005, KFM-IDX-DAT-008 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.98 KFM-P18-INV-433 — Deployment-monitoring records for model-assisted services

Idea card metadata - Stable ID: KFM-P18-INV-433 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 433 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM model-assisted services should preserve deployment-monitoring records for drift, stale inputs, and degraded performance.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM model-assisted services should preserve deployment-monitoring records for drift, stale inputs, and degraded performance. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A validated model can become unsafe when source conditions or input distributions change after release.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-006. - CONFIRMED baseline link: KFM-IDX-OBS-001.

Dependencies - PROPOSED: monitoring dashboard - PROPOSED: drift report - PROPOSED: runtime envelope - PROPOSED: rollback target

Tensions - PROPOSED: Monitoring may require metrics that are unavailable for low-volume lanes.

Expansion directions - PROPOSED: Define `model_runtime_monitoring_receipt` for Focus Mode or model-assisted maps.

Open questions - NEEDS VERIFICATION: Which model drift indicators should trigger ABSTAIN or DENY?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — `AI_Concepts_Using_Python.pdf`; pages 186, 191, 205; basis: AI Concepts Using Python discusses deployment and monitoring concepts in machine-learning workflows.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-006, KFM-IDX-OBS-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.99 KFM-P18-INV-438 — Bitemporal correction model for released claim history

Idea card metadata - Stable ID: KFM-P18-INV-438 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 438 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM released claims should distinguish when a fact was valid from when KFM knew, recorded, corrected, or published it.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (`developing-time-oriented-database-applications-in-sql.pdf`) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM released claims should distinguish when a fact was valid from when KFM knew, recorded, corrected, or published it. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Corrections and historical truth are different temporal dimensions; bitemporal modeling preserves both.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-002. - CONFIRMED baseline link: KFM-IDX-EVD-004.

Dependencies - PROPOSED: claim table - PROPOSED: correction notice - PROPOSED: release manifest - PROPOSED: temporal validator

Tensions - PROPOSED: Bitemporal storage can be heavier than early thin-slice needs.

Expansion directions - PROPOSED: Define bitemporal fields for correction lineage and public claim history.

Open questions - NEEDS VERIFICATION: Which early object families require full bitemporal support?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — `developing-time-oriented-database-applications-in-sql.pdf`; pages 11, 14, 22, 278; basis: Developing Time-Oriented Database Applications in SQL distinguishes valid time, transaction time, and bitemporal tables.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-002, KFM-IDX-EVD-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.100 KFM-P18-INV-454 — ROLLUP and CUBE subtotal semantics for aggregate releases

Idea card metadata - Stable ID: KFM-P18-INV-454 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 454 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM aggregate tables should label whether subtotals come from GROUP BY, ROLLUP, CUBE, or equivalent logic so readers can distinguish detail rows from roll-up totals.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM aggregate tables should label whether subtotals come from GROUP BY, ROLLUP, CUBE, or equivalent logic so readers can distinguish detail rows from roll-up totals. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Composite public indicators can mislead when subtotals and leaf-level observations are not explicitly distinguished.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-004. - CONFIRMED baseline link: KFM-IDX-OBS-002.

Dependencies - PROPOSED: SQL query plan - PROPOSED: aggregate table - PROPOSED: metric documentation - PROPOSED: review fixture

Tensions - PROPOSED: SQL dialects vary in grouping extensions and grouping-id syntax.

Expansion directions - PROPOSED: Add aggregate_semantics metadata to released analytical panels.

Open questions - NEEDS VERIFICATION: Should roll-up rows receive different evidence IDs than detail rows?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 26; basis: The Advanced SQL source explains that CUBE and ROLLUP return totals and subtotals for grouped values while HAVING filters aggregate results.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-004, KFM-IDX-OBS-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.101 KFM-P18-INV-455 — Database views as governed analytical interfaces

Idea card metadata - Stable ID: KFM-P18-INV-455 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 455 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM may use SQL views or equivalent named query interfaces as governed analytical surfaces, provided each view has owner, purpose, source scope, and validation metadata.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM may use SQL views or equivalent named query interfaces as governed analytical surfaces, provided each view has owner, purpose, source scope, and validation metadata. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Named views can reduce repeated ad hoc logic, but they must not become invisible authorities outside source and policy review.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-010. - CONFIRMED baseline link: KFM-IDX-DAT-001.

Dependencies - PROPOSED: query registry - PROPOSED: view definition - PROPOSED: validation tests - PROPOSED: source lineage

Tensions - PROPOSED: A view can hide complex joins and filters from reviewers if not ledgered.

Expansion directions - PROPOSED: Inventory analytical views with spec_hash and source lineage.

Open questions - NEEDS VERIFICATION: Which views are public-safe interfaces and which are internal-only?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 18, 102, 103; basis: The Advanced SQL source recommends database views or stored procedures for code organization and maintainability, and later discusses views as reusable query constructs.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-010, KFM-IDX-DAT-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.102 KFM-P18-INV-456 — Missing-data treatment receipts for AI-derived analyses

Idea card metadata - Stable ID: KFM-P18-INV-456 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 456 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM AI-assisted or model-derived analyses should emit a receipt describing missing-data treatment, including dropped records, imputations, method choice, and reason.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM AI-assisted or model-derived analyses should emit a receipt describing missing-data treatment, including dropped records, imputations, method choice, and reason. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, run-time routing, or release state.

Why it matters - PROPOSED: Missing-data choices can change a model output as materially as the model itself.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-AIG-004.

Dependencies - PROPOSED: model input dataset - PROPOSED: transform receipt - PROPOSED: validation report - PROPOSED: EvidenceBundle

Tensions - PROPOSED: Imputation can look like observed evidence if not clearly labeled.

Expansion directions - PROPOSED: Add missing_data_treatment blocks to model-run receipts.

Open questions - NEEDS VERIFICATION: Which imputation methods require reviewer approval before release?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 146, 155; basis: The AI/Python source treats handling missing data as a critical data-analysis step and notes that method choice affects the reliability and accuracy of insights.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-001, KFM-IDX-AIG-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.103 KFM-P18-INV-465 — Aggregate boundaries as validation units

Idea card metadata - Stable ID: KFM-P18-INV-465 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 465 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM domain lanes should define aggregate boundaries whose root records carry invariants, validation scope, and external-reference rules.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM domain lanes should define aggregate boundaries whose root records carry invariants, validation scope, and external-reference rules. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full

idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Validation and promotion are easier to audit when each domain object family has an explicit consistency boundary.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-DAT-001.

Dependencies - PROPOSED: bounded context - PROPOSED: aggregate root - PROPOSED: repository/query surface - PROPOSED: fixture suite

Tensions - PROPOSED: Aggregate boundaries may conflict with cross-domain graph relationships.

Expansion directions - PROPOSED: Identify aggregate roots for high-risk KFM domains before API design.

Open questions - NEEDS VERIFICATION: Can cross-domain EvidenceBundle references remain outside aggregate consistency checks?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 23, 24, 25; basis: The DDD reference describes aggregates with root entities, repositories for query access to aggregates, and factories for consistent aggregate creation.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-004, KFM-IDX-DAT-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.104 KFM-P18-INV-477 — Choropleth classification scheme disclosure

Idea card metadata - Stable ID: KFM-P18-INV-477 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 477 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM choropleth or thematic aggregate layers should name their classification method, class count, geography unit, and value range assumptions in the released layer metadata.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM choropleth or thematic aggregate layers should name their classification method, class count, geography unit, and value range assumptions in the released layer metadata. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Classification choices can materially change the public visual interpretation of socioeconomic, environmental, or hazard indicators.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-001. - CONFIRMED baseline link: KFM-IDX-CAT-007.

Dependencies - PROPOSED: classification scheme - PROPOSED: legend - PROPOSED: aggregate geography - PROPOSED: catalog record

Tensions - PROPOSED: Human-readable legends may hide the statistical assumptions behind classes.

Expansion directions - PROPOSED: Define a `classification_profile` object for choropleth layers and dashboards.

Open questions - NEEDS VERIFICATION: Should classification profiles be required for all binned colors or only for analytical choropleths?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — `a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf`; pages 214, 215, 228; basis: The GIS primer treats choropleth mapping and classification choices as central cartographic representation techniques and notes that equal-interval classification is suited only to some value ranges.; Phase 1 role: `candidate_synthesis_source`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.105 KFM-P18-INV-478 — Recursive CTE traversal for evidence-graph hierarchies

Idea card metadata - Stable ID: KFM-P18-INV-478 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 478 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM graph and catalog QA can use recursive CTE-style traversal to inspect source, geography, administrative, and evidence hierarchies without treating the query view as canonical truth.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (`Advanced-SQL-Concepts.pdf`) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM graph and catalog QA can use recursive CTE-style traversal to inspect source, geography, administrative, and evidence hierarchies without treating the query view as canonical truth. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Hierarchy traversal is useful for provenance, taxonomy, watershed, route, and category checks, but it remains a derived inspection surface.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-001. - CONFIRMED baseline link: KFM-IDX-CAT-004.

Dependencies - PROPOSED: recursive CTE - PROPOSED: hierarchy - PROPOSED: graph projection - PROPOSED: EvidenceRef

Tensions - PROPOSED: Recursive traversal can mask cycles or ambiguous parentage if not constrained.

Expansion directions - PROPOSED: Create recursive-query fixtures that detect loops in source and category hierarchies.

Open questions - NEEDS VERIFICATION: Which hierarchy families should have cycle-detection fixtures first?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 6, 10, 15; basis: The advanced SQL source presents recursive CTEs for traversing hierarchies and generating hierarchical data.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-001, KFM-IDX-CAT-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.106 KFM-P18-INV-479 — Window-frame definitions for rolling public metrics

Idea card metadata - Stable ID: KFM-P18-INV-479 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 479 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM rolling metrics should record partition, ordering, and frame definitions so moving averages, ranks, and trend indicators can be reproduced and audited.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM rolling metrics should record partition, ordering, and frame definitions so moving averages, ranks, and trend indicators can be reproduced and audited. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A rolling average or ranking can change meaning when the frame changes; public metrics need deterministic analytical metadata.

Related ideas - CONFIRMED baseline link: KFM-IDX-OBS-001. - CONFIRMED baseline link: KFM-IDX-PIP-010.

Dependencies - PROPOSED: window function - PROPOSED: frame definition - PROPOSED: metric label - PROPOSED: ReleaseManifest

Tensions - PROPOSED: Different databases express frames differently, so cross-engine parity needs tests.

Expansion directions - PROPOSED: Add metric_window_profile to aggregate-release fixtures.

Open questions - NEEDS VERIFICATION: Which indicators in later chapters depend on rolling windows or ranks?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 23; basis: The advanced SQL source explains that window functions use OVER clauses, partitions, ordering, and window frames to specify analytical input.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-OBS-001, KFM-IDX-PIP-010 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.8.107 KFM-P18-INV-480 — Subtotal semantics for ROLLUP and CUBE outputs

Idea card metadata - Stable ID: KFM-P18-INV-480 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 480 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM aggregate releases should distinguish raw group rows, subtotal rows, and grand-total rows when ROLLUP or CUBE-like semantics are used.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM aggregate releases should distinguish raw group rows, subtotal rows, and grand-total rows when ROLLUP or CUBE-like semantics are used. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Subtotal rows can be mistaken for observations unless their aggregation level is explicit.

Related ideas - CONFIRMED baseline link: KFM-IDX-OBS-001. - CONFIRMED baseline link: KFM-IDX-CAT-003.

Dependencies - PROPOSED: GROUP BY - PROPOSED: ROLLUP - PROPOSED: CUBE - PROPOSED: aggregate release

Tensions - PROPOSED: Additional aggregation-level fields may complicate downstream charts.

Expansion directions - PROPOSED: Define aggregate_level and grouping_keys in analytical panel schemas.

Open questions - NEEDS VERIFICATION: Should subtotal semantics be normalized across SQL and non-SQL aggregators?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 26; basis: The advanced SQL source describes GROUP BY advanced features such as CUBE and ROLLUP for totals and subtotals.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-OBS-001, KFM-IDX-CAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN:

Repository implementation status remains unverified.

8.8.108 KFM-P18-INV-482 — Unsupervised cluster labels as provisional hypotheses

Idea card metadata - Stable ID: KFM-P18-INV-482 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 482 - Status label: PROPOSED - Carry-forward state: NEW - Category: ANA — Analysis, Indicators, Statistics, Machine Learning, and Model Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM outputs from unsupervised clustering should be labeled as provisional pattern hypotheses until domain review, evidence linkage, and policy checks support stronger claims.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM outputs from unsupervised clustering should be labeled as provisional pattern hypotheses until domain review, evidence linkage, and policy checks support stronger claims. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Cluster IDs can look authoritative even when they are algorithmic groupings without domain meaning.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-004. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: K-Means - PROPOSED: cluster label - PROPOSED: review note - PROPOSED: EvidenceBundle

Tensions - PROPOSED: Clusters may be useful for exploration but unsafe as public labels.

Expansion directions - PROPOSED: Create cluster_interpretation_status values: exploratory, reviewed, rejected, superseded.

Open questions - NEEDS VERIFICATION: Which KFM domains may use clustering only behind review surfaces?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 132, 138; basis: The AI/Python source describes clustering and K-Means as unsupervised techniques for organizing data into groups or clusters.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9 FIE - Field Capture, Remote Sensing, 3D, and Archaeological Interpretation

Status: CONFIRMED source support for grouped Phase 3 ideas; PROPOSED category synthesis

Definition: Field capture, remote sensing, 3D/2.5D representation, archaeological interpretation, visibility, volumes, and acquisition paradata.

CONFIRMED: This chapter contains 59 Pass 18 idea cards assigned to FIE.

Idea ID	Title	Carry state	Status	Source IDs
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KFM-P18-INV-006 | Flood-hazard exposure enrichment workflow | NEW | PROPOSED | SRC-P18-010 |

KFM-P18-INV-007 | Mobile field collection as evidence intake | NEW | PROPOSED | SRC-P18-010 |

KFM-P18-INV-016 | 3D archaeology as evidence-rich spatial analysis | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-017 | 2.5D versus true 3D distinction | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-019 | Trowel-edge anomaly detection as reviewed field inference | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-020 | Subsurface, volume, and visibility analyses as derived evidence | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-053 | Remote sensing as sensor-mediated evidence | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-057 | Hazard extent, response, and climate context as separate evidence roles | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-059 | Climate scenario maps require model and metadata compliance labels | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-079 | Hurricane tracks as temporal event-impact bundles | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-080 | Wildfire extent and response are separate claim roles | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-081 | Volcanic hazard change mapping requires georeference and workflow receipts | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-089 | 3D GIS should support multi-model archaeological interpretation | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-090 | Volumetric and boundary models need acquisition and analysis lineage | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-091 | 3D visualization settings are analytical choices | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-113 | 3D visibility analysis as bounded interpretive inference | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-114 | Trowel-edge 3D capture as field review loop | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-124 | Mobile field collection requires device, user, time, and attachment receipts | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-137 | Point clouds, meshes, textures, and volumes as separate artifact roles | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-138 | Scale and georeferencing receipts for 3D models | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-139 | 3D field query logs as review evidence | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-144 | Date columns are not ordinary fields in governed temporal data | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-197 | Geostatistical surfaces as model-backed evidence | NEW | PROPOSED | SRC-P18-002 |

KFM-P18-INV-198 | Raster field cells do not imply hard natural boundaries | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-207 | Scenario-based urban growth and flood risk as modeled evidence | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-209 | 2.5D surfaces are not true 3D evidence for complex geometry | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-211 | Interpretive versus reality-based 3D model labels | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-212 | 3D boundary and density maps are query-derived derivatives | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-213 | 3D visibility analysis needs assumption receipts | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-214 | Volumetric analysis needs legacy-data and interpolation caveats | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-248 | Offline field-capture packages need redaction and sync receipts | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-264 | SQL views and stored procedures as documented analytic contract surfaces | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-271 | 2D orthophotos derived from 3D models need model-to-map lineage | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-292 | Realistic and interpretive 3D models need separate labels | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-293 | 3D GIS dimensional capacity must be recorded explicitly | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-294 | Trowel-edge 3D GIS becomes field-recording evidence | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-295 | Surface and subsurface analyses require preservation-state context | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-296 | 3D visibility analysis is simulation evidence | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-316 | Hurricane translation speed as a hazard-impact variable | NEW | PROPOSED | SRC-P18-010 |

KFM-P18-INV-317 | GCP distribution and visibility as photogrammetry evidence quality | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-318 | Manual versus GNSS-collected point distinction in field maps | NEW | PROPOSED | SRC-P18-010 |

KFM-P18-INV-319 | Climate scenario assumptions and ensembles as layer metadata | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-323 | Interactive 3D web GIS needs collaboration and reuse metadata | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-324 | 3D visibility metrics require observer and obstacle assumption receipts | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-347 | Regulatory floodplain labels are not observed inundation evidence | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-361 | Field Maps collection as attachment-aware evidence capture | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-362 | Interpretative and realistic 3D models require separate knowledge-character labels | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-363 | Surface and subsurface 3D analysis as layered evidence rather than a single scene | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-387 | Georeferenced 3D imports need scale and orientation QA | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-388 | 2.5D comparison views disclose vertical-surface loss | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-397 | Emission-scenario horizons for climate-change visualizations | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-410 | 3D GIS paradata for acquisition and interpretation choices | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-434 | Extended-matrix links for 3D archaeological interpretation | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-452 | Remote-sensing resolution metadata for raster and imagery layers | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-458 | 3D visibility analysis assumption disclosure | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-459 | Surface and subsurface damage-assessment evidence objects | NEW | PROPOSED | SRC-P18-005 |

KFM-P18-INV-467 | Field Maps GCP observations as governed field evidence | NEW | PROPOSED | SRC-P18-010 |

KFM-P18-INV-484 | 3D model georeferencing and scale receipts | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-485 | 2.5D versus true 3D representation labels | NEW | PROPOSED | SRC-P18-005 |

8.9.1 KFM-P18-INV-006 — Flood-hazard exposure enrichment workflow

Idea card metadata - Stable ID: KFM-P18-INV-006 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 6 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM can model flood exposure as a governed overlay between hazard geometry and demographic or socioeconomic context, with clear source and use limitations.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM can model flood exposure as a governed overlay between hazard geometry and demographic or socioeconomic context, with clear source and use limitations. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It supports public-interest exposure analysis without treating a tutorial workflow as regulatory truth.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-005. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: Hazard source descriptor; demographic source descriptor; overlay receipt

Tensions - PROPOSED: Exposure analysis may imply individual risk if spatial aggregation is too precise.

Expansion directions - PROPOSED: Define a FloodExposureClaim fixture using public-safe generalized geometry.

Open questions - NEEDS VERIFICATION: What aggregation threshold is sufficient for public demographic overlays?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 7, 8, 11; basis: flood hazard and enrichment exercise; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.2 KFM-P18-INV-007 — Mobile field collection as evidence intake

Idea card metadata - Stable ID: KFM-P18-INV-007 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 7 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should treat field-collected points, attachments, popups, and ground-control metadata as governed observation intake, not informal map edits.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat field-collected points, attachments, popups, and ground-control metadata as governed observation intake, not informal map edits. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Field evidence can be high value but is easy to over-publish without identity, time, rights, and precision controls.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-POL-002.

Dependencies - PROPOSED: Observation schema; attachment manifest; field-run receipt

Tensions - PROPOSED: Mobile convenience conflicts with strict evidence and privacy controls.

Expansion directions - PROPOSED: Add a field_observation_receipt inventory family for future cards.

Open questions - NEEDS VERIFICATION: How should KFM verify collector identity and device time for field observations?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 85, 86; basis: Field Maps data collection and GCP workflow; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.3 KFM-P18-INV-016 — 3D archaeology as evidence-rich spatial analysis

Idea card metadata - Stable ID: KFM-P18-INV-016 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 16 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM archaeology should treat 3D GIS as an evidentiary analysis environment, not only a visualization upgrade.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM archaeology should treat 3D GIS as an evidentiary analysis environment, not only a visualization upgrade. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: 3D data can carry stratigraphic, volumetric, and interpretive evidence that 2D maps flatten.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-POL-002.

Dependencies - PROPOSED: Archaeology sensitivity policy; 3D artifact provenance

Tensions - PROPOSED: 3D models can reveal sensitive archaeological locations or details.

Expansion directions - PROPOSED: Add a 3D archaeological artifact provenance and sensitivity pattern.

Open questions - NEEDS VERIFICATION: Which 3D archaeology outputs are public, generalized, steward-only, or denied?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 1, 29, 55; basis: 3D GIS and archaeological knowledge production framing; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.4 KFM-P18-INV-017 — 2.5D versus true 3D distinction

Idea card metadata - Stable ID: KFM-P18-INV-017 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 17 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should distinguish draped/2.5D terrain representations from true 3D geometries in evidence and UI labels.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should distinguish draped/2.5D terrain representations from true 3D geometries in evidence and UI labels. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A scene can look three-dimensional while losing evidence relevant to vertical or subsurface features.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012.

Dependencies - PROPOSED: 3D/2.5D layer type labels; scene metadata

Tensions - PROPOSED: Public UI may prefer simple 2.5D renderings for performance.

Expansion directions - PROPOSED: Add geometry_dimensionality and vertical_validity fields to 3D scene manifests.

Open questions - NEEDS VERIFICATION: Should 2.5D scenes be prohibited from supporting subsurface or vertical claims?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 11, 12, 109; basis: 2.5D/3D representation and 3D surface model examples; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.5 KFM-P18-INV-019 — Trowel-edge anomaly detection as reviewed field inference

Idea card metadata - Stable ID: KFM-P18-INV-019 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 19 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED -

Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should treat field anomaly detection from 3D/terrain analysis as a reviewed inference that requires source method, analyst, and uncertainty metadata.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat field anomaly detection from 3D/terrain analysis as a reviewed inference that requires source method, analyst, and uncertainty metadata. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Anomaly detection is a candidate-finding technique, not proof of an archaeological site or hazard by itself.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-002. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: InferenceClaim schema; reviewer state; sensitivity rules

Tensions - PROPOSED: Fast field workflows may pressure premature publication.

Expansion directions - PROPOSED: Create candidate_feature_not_confirmed status guidance.

Open questions - NEEDS VERIFICATION: Which field analyses require steward review before leaving WORK/QUARANTINE?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 83, 96; basis: trowel-edge anomaly detection and field GIS recording; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.6 KFM-P18-INV-020 — Subsurface, volume, and visibility analyses as derived evidence

Idea card metadata - Stable ID: KFM-P18-INV-020 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 20 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should catalog visibility, volume, and subsurface analyses as derived evidence objects with inputs, method, and uncertainty recorded.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should catalog visibility, volume, and subsurface analyses as derived evidence objects with inputs, method,

and uncertainty recorded. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Derived spatial analyses can materially change interpretation and need reproducibility and caveats.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-004. - CONFIRMED baseline link: KFM-IDX-DAT-012.

Dependencies - PROPOSED: DerivedAnalysisReceipt; input artifact digests

Tensions - PROPOSED: Methods vary by software, data resolution, and analyst choices.

Expansion directions - PROPOSED: Add DerivedSpatialAnalysis as a cross-domain candidate object.

Open questions - NEEDS VERIFICATION: What method metadata is enough to make a derived spatial analysis reproducible?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 8, 109; basis: surface/subsurface, visibility, and volume analysis coverage; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.7 KFM-P18-INV-053 — Remote sensing as sensor-mediated evidence

Idea card metadata - Stable ID: KFM-P18-INV-053 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 53 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should label remote-sensing layers by sensor, electromagnetic signal type, processing level, and evidentiary role rather than treating imagery as direct observation.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should label remote-sensing layers by sensor, electromagnetic signal type, processing level, and evidentiary role rather than treating imagery as direct observation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps imagery, classifications, anomaly surfaces, and derived indices from being mistaken for field-confirmed truth.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-005. - CONFIRMED baseline link: KFM-IDX-POL-006.

Dependencies - PROPOSED: RemoteSensingSourceDescriptor - PROPOSED: processing_level metadata

Tensions - PROPOSED: Imagery can be highly valuable while still being indirect and method-dependent.

Expansion directions - PROPOSED: Add sensor_platform, signal_type, acquisition_time, processing_level, and validation_status fields.

Open questions - NEEDS VERIFICATION: Which remote-sensing outputs require human review before public release?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 171; basis: remote sensing definition and electromagnetic-radiation acquisition model; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-005, KFM-IDX-POL-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.8 KFM-P18-INV-057 — Hazard extent, response, and climate context as separate evidence roles

Idea card metadata - Stable ID: KFM-P18-INV-057 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 57 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should separate hazard extent, response activity, exposure, and climate-context layers when explaining hurricane or wildfire events.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should separate hazard extent, response activity, exposure, and climate-context layers when explaining hurricane or wildfire events. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A burn perimeter, response footprint, damage estimate, and climate projection answer different questions and should not collapse into one hazard truth layer.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-003. - CONFIRMED baseline link: KFM-IDX-DAT-013.

Dependencies - PROPOSED: HazardLayerRole enum - PROPOSED: event_time and source_time fields

Tensions - PROPOSED: Public users may conflate contextual climate layers with event causation claims.

Expansion directions - PROPOSED: Add hazard_layer_role values for extent, exposure, response, projection, and context.

Open questions - NEEDS VERIFICATION: Which hazard contexts are safe for public use without life-safety implications?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 18, 44, 45; basis: historical hurricane impact, wildfire extent/response, and climate-context discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-003, KFM-IDX-DAT-013 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.9 KFM-P18-INV-059 — Climate scenario maps require model and metadata compliance labels

Idea card metadata - Stable ID: KFM-P18-INV-059 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 59 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should require scenario, emissions, model, metadata-compliance, and uncertainty labels for climate-change visualization layers.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should require scenario, emissions, model, metadata-compliance, and uncertainty labels for climate-change visualization layers. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Scenario maps are not observations; without scenario and metadata labels they can be misread as current fact or deterministic forecast.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-002. - CONFIRMED baseline link: KFM-IDX-DAT-006.

Dependencies - PROPOSED: ScenarioLayerProfile - PROPOSED: metadata compliance validator

Tensions - PROPOSED: Simplifying scenarios improves usability but can hide model assumptions.

Expansion directions - PROPOSED: Add scenario_id, model_family, emissions_pathway, temporal_horizon, and compliance_status.

Open questions - NEEDS VERIFICATION: Should climate scenario layers be excluded from ordinary factual search results unless labeled as scenario?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 121; basis: climate-change emission scenario exercise with metadata viewing/editing and compliance verification; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-002, KFM-IDX-DAT-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.10 KFM-P18-INV-079 — Hurricane tracks as temporal event-impact bundles

Idea card metadata - Stable ID: KFM-P18-INV-079 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 79 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should model hurricane or storm-track examples as event-impact bundles that separate track geometry, intensity attributes, damages, and exposed populations.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should model hurricane or storm-track examples as event-impact bundles that separate track geometry, intensity attributes, damages, and exposed populations. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It avoids collapsing a hazard path into a complete explanation of impact or vulnerability.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-009. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: HazardEvent schema - PROPOSED: ImpactObservation schema - PROPOSED: Exposure join policy

Tensions - PROPOSED: Past impacts may be used to imply future risk beyond evidence support.

Expansion directions - PROPOSED: Add event_track, event_intensity, impact_record, and exposure_context roles to hazard layers.

Open questions - NEEDS VERIFICATION: Which damage estimates are admissible as evidence versus contextual summaries?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 18, 19, 20, 21, 22; basis: historical hurricane track, Saffir-Simpson, impact, definition query, and statistics exercise; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-009, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.11 KFM-P18-INV-080 — Wildfire extent and response are separate claim roles

Idea card metadata - Stable ID: KFM-P18-INV-080 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 80 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should keep wildfire extent, alarm/containment timing, responsible agency, and modeled future burned-area context as separate claim roles.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should keep wildfire extent, alarm/containment timing, responsible agency, and modeled future burned-area context as separate claim roles. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents a map of burned polygons from being mistaken for a complete response, causation, or future-risk claim.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-009. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: Hazard layer roles - PROPOSED: Agency/source authority registry - PROPOSED: Temporal event metadata

Tensions - PROPOSED: Response agency symbology may be read as responsibility or legal authority without context.

Expansion directions - PROPOSED: Define extent_observation, response_assignment, modeled_context, and event_timing roles.

Open questions - NEEDS VERIFICATION: How should KFM label modeled future fire context versus observed fire history?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 44, 45, 46, 47, 50; basis: wildfire extent, agency response, alarm/containment dates, and modeled burned-area context; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-009, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.12 KFM-P18-INV-081 — Volcanic hazard change mapping requires georeference and workflow receipts

Idea card metadata - Stable ID: KFM-P18-INV-081 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 81 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from

CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should require georeference control-point evidence and workflow receipts for change maps derived from historical or event images.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should require georeference control-point evidence and workflow receipts for change maps derived from historical or event images. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes derived change polygons auditable back to image dates, control points, digitization choices, and geoprocessing steps.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-004. - CONFIRMED baseline link: KFM-IDX-PIP-004.

Dependencies - PROPOSED: GeoreferenceReceipt - PROPOSED: DigitizationReceipt - PROPOSED: ChangeAnalysisReceipt

Tensions - PROPOSED: Digitized polygons may look authoritative even when control points or source images are weak.

Expansion directions - PROPOSED: Create image-derived feature lineage fields for control_point_count, residual_error, source_image_date, and digitizer_review.

Open questions - NEEDS VERIFICATION: What residual-error threshold blocks publication of image-derived boundaries?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 69, 70, 71, 72, 73, 75; basis: Kilauea image georeferencing, control points, digitized lava polygons, and task workflow for change calculation; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-004, KFM-IDX-PIP-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.13 KFM-P18-INV-089 — 3D GIS should support multi-model archaeological interpretation

Idea card metadata - Stable ID: KFM-P18-INV-089 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 89 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should support multiple archaeological 3D scenarios or models linked to evidence rather than forcing one static spatial interpretation.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should support multiple archaeological 3D scenarios or models linked to evidence rather than forcing one static spatial interpretation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It preserves interpretive plurality while keeping evidence, model status, and review state inspectable.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-006. - CONFIRMED baseline link: KFM-IDX-DAT-013.

Dependencies - PROPOSED: InterpretationManifest - PROPOSED: EvidenceBundle links - PROPOSED: Review state

Tensions - PROPOSED: Multiple scenarios can confuse users if not clearly labeled.

Expansion directions - PROPOSED: Add interpretation_status, scenario_id, evidence_support_level, and rejected_alternative links.

Open questions - NEEDS VERIFICATION: How should public UI show competing archaeological interpretations safely?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 24, 30, 31; basis: multi-model/scenario interpretation and critical GIS discussion in archaeological 3D GIS; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-006, KFM-IDX-DAT-013 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.14 KFM-P18-INV-090 — Volumetric and boundary models need acquisition and analysis lineage

Idea card metadata - Stable ID: KFM-P18-INV-090 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 90 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should treat volumetric, boundary, and solid 3D models as derived artifacts with acquisition, processing, and analysis lineage.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat volumetric, boundary, and solid 3D models as derived artifacts with acquisition, processing, and analysis lineage. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source

role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes 3D evidence auditable as a sequence of capture, georeferencing, modeling, and interpretation steps.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-004. - CONFIRMED baseline link: KFM-IDX-DAT-013.

Dependencies - PROPOSED: 3DArtifactManifest - PROPOSED: Processing receipt - PROPOSED: Source capture metadata

Tensions - PROPOSED: 3D scenes can appear more realistic than their evidence support warrants.

Expansion directions - PROPOSED: Define capture_method, processing_tool, georeference_basis, mesh_resolution, and volume_method fields.

Open questions - NEEDS VERIFICATION: What 3D model classes should be withheld or generalized for archaeology?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 10, 11, 14, 26, 27; basis: 3D volumes, volumetric visualisation, boundary models, and volumetric workflow descriptions; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-004, KFM-IDX-DAT-013 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.15 KFM-P18-INV-091 — 3D visualization settings are analytical choices

Idea card metadata - Stable ID: KFM-P18-INV-091 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 91 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should record important 3D visualization settings such as shaders, orthophoto draping, lighting, and viewpoint when they affect interpretation.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record important 3D visualization settings such as shaders, orthophoto draping, lighting, and viewpoint when they affect interpretation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents a rendered scene, screenshot, or view from acting as unrecorded evidence transformation.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-013. - CONFIRMED baseline link: KFM-IDX-OBS-003.

Dependencies - PROPOSED: SceneManifest - PROPOSED: PreviewRendererReceipt - PROPOSED: Style/lighting metadata

Tensions - PROPOSED: Changing viewpoint or lighting can materially change what users infer.

Expansion directions - PROPOSED: Add camera_pose, shader_profile, lighting_profile, draped_source, and visibility_method fields.

Open questions - NEEDS VERIFICATION: Which 3D visualization changes require a new preview hash?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 11, 12, 13, 15; basis: shader, orthophoto, visibility, viewpoint, light-source, and 3D visualization references; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.16 KFM-P18-INV-113 — 3D visibility analysis as bounded interpretive inference

Idea card metadata - Stable ID: KFM-P18-INV-113 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 113 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should treat 3D visibility and sensory analyses as bounded interpretations with parameters, viewpoint assumptions, and review status.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat 3D visibility and sensory analyses as bounded interpretations with parameters, viewpoint assumptions, and review status. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It supports powerful spatial interpretation without overclaiming certainty about past experience.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012.

Dependencies - PROPOSED: VisibilityAnalysisReceipt - PROPOSED: 3D scene manifest

Tensions - PROPOSED: Visibility results depend on model completeness, observer assumptions, and reconstruction choices.

Expansion directions - PROPOSED: Add visibility_parameters, observer_profile, and uncertainty_note fields.

Open questions - NEEDS VERIFICATION: Which viewpoint and obstruction parameters must be recorded?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 8, 14, 96; basis: 3D visibility and volumetric visibility analysis discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.17 KFM-P18-INV-114 — Trowel-edge 3D capture as field review loop

Idea card metadata - Stable ID: KFM-P18-INV-114 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 114 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should support 3D field capture as a reviewable loop linking acquisition, georeferencing, field interpretation, and later laboratory analysis.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should support 3D field capture as a reviewable loop linking acquisition, georeferencing, field interpretation, and later laboratory analysis. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes field interpretation reversible and reviewable instead of isolating it from later evidence review.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-004. - CONFIRMED baseline link: KFM-IDX-DAT-010.

Dependencies - PROPOSED: FieldRunReceipt - PROPOSED: 3D capture manifest - PROPOSED: Review record

Tensions - PROPOSED: Real-time field interpretation can become sticky if later lab evidence contradicts it.

Expansion directions - PROPOSED: Add field_capture_stage and interpretation_revision links.

Open questions - NEEDS VERIFICATION: How should field interpretation be superseded without losing the original observation?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 26, 57, 62; basis: trowel-edge 3D GIS, field investigations, and real-time interpretation discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-004, KFM-IDX-DAT-010 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.18 KFM-P18-INV-124 — Mobile field collection requires device, user, time, and attachment receipts

Idea card metadata - Stable ID: KFM-P18-INV-124 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 124 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM mobile field intake should record who/what collected the observation, collection time, attachment provenance, and feature-layer publication state.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM mobile field intake should record who/what collected the observation, collection time, attachment provenance, and feature-layer publication state. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps field observations inspectable as evidence rather than just points that appear on a map.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-002. - CONFIRMED baseline link: KFM-IDX-POL-005.

Dependencies - PROPOSED: Field observation schema - PROPOSED: Attachment manifest - PROPOSED: EvidenceBundle

Tensions - PROPOSED: Mobile collection can expose sensitive locations or personal metadata if not redacted.

Expansion directions - PROPOSED: Add field_device_receipt, attachment_rights, and redaction profile fields.

Open questions - NEEDS VERIFICATION: Which EXIF/device fields should be retained, redacted, or generalized?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 85, 86; basis: Field Maps feature layer, attachments, web map, and field data collection workflow; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-002, KFM-IDX-POL-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.19 KFM-P18-INV-137 — Point clouds, meshes, textures, and volumes as separate artifact roles

Idea card metadata - Stable ID: KFM-P18-INV-137 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 137 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should assign separate artifact roles to point clouds, meshes, textures, volumes, orthoimages, and interpretive 3D reconstructions.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should assign separate artifact roles to point clouds, meshes, textures, volumes, orthoimages, and interpretive 3D reconstructions. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It allows provenance, rights, precision, and interpretation status to travel with each 3D artifact type.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-011. - CONFIRMED baseline link: KFM-IDX-EVD-002.

Dependencies - PROPOSED: 3DArtifactManifest - PROPOSED: EvidenceBundle - PROPOSED: SceneManifest

Tensions - PROPOSED: Users may experience a single 3D scene even when it is composed of evidence with different authority.

Expansion directions - PROPOSED: Define role enums for captured_3d, derived_3d, interpretive_3d, volume, texture, and orthoprojection.

Open questions - NEEDS VERIFICATION: Which 3D artifact roles can be public by default?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 10, 12, 24, 47, 48; basis: 3D GIS discussion of point clouds, 3D models, textures, and volumetric/visual artifacts; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-011, KFM-IDX-EVD-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.20 KFM-P18-INV-138 — Scale and georeferencing receipts for 3D models

Idea card metadata - Stable ID: KFM-P18-INV-138 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 138 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should require scale, georeferencing, coordinate basis, and registration receipts before 3D models support spatial claims.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should require scale, georeferencing, coordinate basis, and registration receipts before 3D models support spatial claims. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents visually convincing 3D models from implying spatial accuracy they do not have.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-CAT-002.

Dependencies - PROPOSED: 3DArtifactManifest - PROPOSED: TransformReceipt - PROPOSED: Coordinate reference registry

Tensions - PROPOSED: Some 3D models support interpretation but not precise measurement.

Expansion directions - PROPOSED: Add 3d_registration_method and scale_control fields.

Open questions - NEEDS VERIFICATION: What tolerance levels are acceptable for public 3D spatial claims?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 24, 27, 32, 35, 42, 43; basis: georeferenced and scaled 3D model discussion in archaeological 3D GIS; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-012, KFM-IDX-CAT-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.21 KFM-P18-INV-139 — 3D field query logs as review evidence

Idea card metadata - Stable ID: KFM-P18-INV-139 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 139 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should preserve important 3D field query sessions, annotations, and review decisions as evidence-adjacent records when they shape archaeological interpretation.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should preserve important 3D field query sessions, annotations, and review decisions as evidence-adjacent records when they shape archaeological interpretation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path,

and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It lets later reviewers understand how an interpretation emerged from 3D inspection and specialist input.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-007. - CONFIRMED baseline link: KFM-IDX-DOC-007.

Dependencies - PROPOSED: ReviewLog - PROPOSED: SceneManifest - PROPOSED: EvidenceDrawerPayload

Tensions - PROPOSED: Not every exploratory camera movement should become permanent evidence.

Expansion directions - PROPOSED: Define which 3D queries or annotations become review records.

Open questions - NEEDS VERIFICATION: What is the retention rule for field-review annotations?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 2, 13, 52, 55, 67, 79; basis: 3D GIS platforms linking, querying, and analyzing archaeological data in field and laboratory contexts; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.22 KFM-P18-INV-144 — Date columns are not ordinary fields in governed temporal data

Idea card metadata - Stable ID: KFM-P18-INV-144 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 144 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should treat valid-time, transaction-time, observed-time, source-time, retrieval-time, release-time, and correction-time fields as typed temporal semantics, not generic date columns.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat valid-time, transaction-time, observed-time, source-time, retrieval-time, release-time, and correction-time fields as typed temporal semantics, not generic date columns. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents silent temporal ambiguity in claims, releases, corrections, and observations.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-008. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Temporal field registry - PROPOSED: Schema validators - PROPOSED: Release-Manifest

Tensions - PROPOSED: Adding temporal semantics increases schema complexity but reduces false certainty.

Expansion directions - PROPOSED: Define required temporal role enums for domain observations and release objects.

Open questions - NEEDS VERIFICATION: Which temporal roles are mandatory for every EvidenceBundle?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 17, 18, 111, 113, 117, 119; basis: temporal database preface and state-table chapters warning about date columns, valid time, keys, and constraints; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-008, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.23 KFM-P18-INV-197 — Geostatistical surfaces as model-backed evidence

Idea card metadata - Stable ID: KFM-P18-INV-197 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 197 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should label geostatistical and interpolated surfaces as model-backed evidence with sample, method, uncertainty, and aggregation metadata rather than as observed everywhere.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should label geostatistical and interpolated surfaces as model-backed evidence with sample, method, uncertainty, and aggregation metadata rather than as observed everywhere. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It guards against reading a continuous surface as if every cell were directly measured.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Model metadata - PROPOSED: Sample design - PROPOSED: Uncertainty profile - PROPOSED: Layer disclaimer

Tensions - PROPOSED: A smooth surface can overstate certainty and hide sparse or biased sampling.

Expansion directions - PROPOSED: Add model-surface labels for interpolated rasters and geostatistical products.

Open questions - NEEDS VERIFICATION: What uncertainty fields should be mandatory for public model surfaces?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 272, 273; basis: geostatistical GIS operations, Monte Carlo simulations, fuzzy-set theory, variance, and complex analysis of continuous characteristics; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.24 KFM-P18-INV-198 — Raster field cells do not imply hard natural boundaries

Idea card metadata - Stable ID: KFM-P18-INV-198 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 198 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM raster and field layers should disclose when cell boundaries are storage artifacts rather than sharp boundaries in the underlying phenomenon.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM raster and field layers should disclose when cell boundaries are storage artifacts rather than sharp boundaries in the underlying phenomenon. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents users from treating soil moisture, air quality, elevation, climate, or suitability cells as exact jurisdiction-like polygons.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-DOC-007.

Dependencies - PROPOSED: Raster layer metadata - PROPOSED: Field/model type label - PROPOSED: Map popup caveat

Tensions - PROPOSED: Some public users need simplified maps; caveats must be visible but not overwhelming.

Expansion directions - PROPOSED: Add field_continuity and cell_boundary_caveat metadata to raster/model layers.

Open questions - NEEDS VERIFICATION: How should MapLibre popups explain cell-boundary caveats without dense technical prose?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 190, 191; basis: field representation, TINs, transformations, and warning that raster cell boundaries should not be interpreted as sharp boundaries of nondiscrete phenomena; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-POL-001, KFM-IDX-DOC-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.25 KFM-P18-INV-207 — Scenario-based urban growth and flood risk as modeled evidence

Idea card metadata - Stable ID: KFM-P18-INV-207 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 207 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM scenario outputs for urban growth, flood risk, or resilience should be labeled as modeled scenario evidence with assumptions, input models, and comparison scope preserved.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM scenario outputs for urban growth, flood risk, or resilience should be labeled as modeled scenario evidence with assumptions, input models, and comparison scope preserved. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents scenario maps from being mistaken for observed future outcomes.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Scenario manifest - PROPOSED: Model inputs - PROPOSED: Assumption register - PROPOSED: Risk-layer caveat

Tensions - PROPOSED: Scenarios are useful for planning but can overstate certainty if rendered like observations.

Expansion directions - PROPOSED: Add ScenarioEvidenceBundle profile for assumptions, input releases, model version, and comparison purpose.

Open questions - NEEDS VERIFICATION: How should KFM distinguish regulatory flood hazard, observed flood event, and scenario flood-risk outputs in the UI?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 308, 317; basis: urban-growth scenarios, land-cover projections, flooding model inputs, and scenario comparison for flood impacts; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-POL-001, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.26 KFM-P18-INV-209 — 2.5D surfaces are not true 3D evidence for complex geometry

Idea card metadata - Stable ID: KFM-P18-INV-209 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 209 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should distinguish true 3D records from 2.5D surfaces when vertical surfaces, overhangs, interiors, or complex archaeological features matter.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should distinguish true 3D records from 2.5D surfaces when vertical surfaces, overhangs, interiors, or complex archaeological features matter. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents a terrain-like surface from being used as evidence for geometry it cannot represent.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Geometry representation label - PROPOSED: 3D scene manifest - PROPOSED: Vertical-feature caveat

Tensions - PROPOSED: 2.5D may be sufficient for broad terrain but misleading for buildings, trenches, caves, or strata.

Expansion directions - PROPOSED: Add `geometry_dimensionality` and `vertical_geometry_supported` fields to scene/layer manifests.

Open questions - NEEDS VERIFICATION: Which KFM lanes require true 3D support versus 2.5D terrain context only?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 52, 61; basis: 2.5D representations cannot display more than one elevation value per x/y coordinate and are problematic for complex archaeological features; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.27 KFM-P18-INV-211 — Interpretive versus reality-based 3D model labels

Idea card metadata - Stable ID: KFM-P18-INV-211 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 211 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM 3D scenes should label whether a model is reality-based, interpretive, reconstructed, or a hybrid, and should connect interpretive elements to evidence and uncertainty.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM 3D scenes should label whether a model is reality-based, interpretive, reconstructed, or a hybrid, and should connect interpretive elements to evidence and uncertainty. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents compelling 3D reconstructions from being mistaken for directly observed reality.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: SceneManifest - PROPOSED: Interpretation label - PROPOSED: Evidence links - PROPOSED: Uncertainty annotation

Tensions - PROPOSED: Highly realistic renderings can overstate certainty more than text descriptions.

Expansion directions - PROPOSED: Add interpretive_status enum and per-component evidence links for 3D scenes.

Open questions - NEEDS VERIFICATION: How should KFM visually differentiate observed geometry from reconstructed or inferred 3D content?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 24, 43; basis: 3D models in knowledge production and examples contrasting digital interpretative and realistic/reality-based models; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-001, KFM-IDX-DAT-003, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.28 KFM-P18-INV-212 — 3D boundary and density maps are query-derived derivatives

Idea card metadata - Stable ID: KFM-P18-INV-212 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 212 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED -

Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM 3D distribution, density, or boundary maps should be treated as query-derived derivatives with inputs, interpolation choices, field documentation quality, and specialist data categories recorded.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM 3D distribution, density, or boundary maps should be treated as query-derived derivatives with inputs, interpolation choices, field documentation quality, and specialist data categories recorded. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It preserves the difference between original field observations and analytical 3D products.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-CAT-003.

Dependencies - PROPOSED: 3D boundary records - PROPOSED: Query receipt - PROPOSED: Specialist-data registry - PROPOSED: Derivative layer manifest

Tensions - PROPOSED: Derivative maps depend on field documentation quality and interpolation boundaries.

Expansion directions - PROPOSED: Add derivative_3d_query_receipt with source categories, query, interpolation, and quality caveats.

Open questions - NEEDS VERIFICATION: Which 3D derivatives should be citable in public narratives versus kept as review artifacts?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 100; basis: 3D boundary representations connected to database entries and query-derived 3D distribution or density maps from specialist excavation data; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-EVD-001, KFM-IDX-CAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.29 KFM-P18-INV-213 — 3D visibility analysis needs assumption receipts

Idea card metadata - Stable ID: KFM-P18-INV-213 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 213 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM visibility, viewshed, and line-of-sight analyses should record observation points, target features, 2.5D/3D data choice, obstruction assumptions, and purpose before supporting interpretive claims.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM visibility, viewshed, and line-of-sight analyses should record observation points, target features, 2.5D/3D data choice, obstruction assumptions, and purpose before supporting interpretive claims. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes perceptual or visibility claims reproducible and properly caveated.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: Visibility model inputs - PROPOSED: Observer/target definitions - PROPOSED: Obstruction model - PROPOSED: 3D/2.5D label

Tensions - PROPOSED: Visibility is interpretive and assumption-sensitive, especially in historical or reconstructed scenes.

Expansion directions - PROPOSED: Add VisibilityAnalysisReceipt for observer positions, targets, occlusion assumptions, data dimensionality, and uncertainty.

Open questions - NEEDS VERIFICATION: Should KFM expose visibility analyses as map layers, story evidence, or reviewer-only products?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 63, 119; basis: viewshed and line-of-sight analysis, target features, observing points, and limitations of 2.5D visibility analysis compared with fully 3D approaches; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.30 KFM-P18-INV-214 — Volumetric analysis needs legacy-data and interpolation caveats

Idea card metadata - Stable ID: KFM-P18-INV-214 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 214 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM volumetric or subsurface analyses should disclose legacy data integration, reconstructed context, unit boundaries, density/frequency methods, and uncertainty in original stratigraphic relationships.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before

Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM volumetric or subsurface analyses should disclose legacy data integration, reconstructed context, unit boundaries, density/frequency methods, and uncertainty in original stratigraphic relationships. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents volumes and 3D density maps from appearing more certain than the underlying documentation supports.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: Legacy source register - PROPOSED: Volume model - PROPOSED: Density method - PROPOSED: Uncertainty note

Tensions - PROPOSED: Legacy excavation records may not support the same precision as modern 3D capture.

Expansion directions - PROPOSED: Create `volumetric_derivative_receipt` fields for legacy inputs, reconstruction method, units, and density calculations.

Open questions - NEEDS VERIFICATION: What minimum evidence quality is required before legacy data can support 3D volume claims?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 142, 147; basis: legacy data integration in a digitally reconstructed cave, volumetric information, 3D density maps, and distribution patterns of artefacts and ecofacts; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.31 KFM-P18-INV-248 — Offline field-capture packages need redaction and sync receipts

Idea card metadata - Stable ID: KFM-P18-INV-248 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 248 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM offline mobile map packages should carry release digests, expiration state, collection permissions, redaction rules, and sync receipts before field-collected records reenter the pipeline.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM offline mobile map packages should carry release digests, expiration state, collection permissions, redaction rules, and sync receipts before field-collected records reenter the pipeline. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session

repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents offline collection from becoming an ungoverned side channel around source, sensitivity, and release controls.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Offline package manifest - PROPOSED: Field capture receipt - PROPOSED: Redaction policy - PROPOSED: Sync validator

Tensions - PROPOSED: Offline work can outlive release state or collect sensitive observations before policy review.

Expansion directions - PROPOSED: Add `offline_capture_receipt` fields for `package_digest`, `release_id`, `device_scope`, `sync_time`, `redaction_applied`, and `denied_fields`.

Open questions - NEEDS VERIFICATION: How should KFM expire or revoke offline map packages after policy changes?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 16, 28; basis: offline PMTiles/MBTiles basemaps and offline-first field capture packages with digest, expiry, and redaction implications; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-POL-001, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.32 KFM-P18-INV-264 — SQL views and stored procedures as documented analytic contract surfaces

Idea card metadata - Stable ID: KFM-P18-INV-264 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 264 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM SQL-derived products should treat views and stored procedures as versioned analytic contracts with documented purpose, inputs, outputs, and validation tests.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM SQL-derived products should treat views and stored procedures as versioned analytic contracts with documented purpose, inputs, outputs, and validation tests. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps database logic from becoming invisible infrastructure that shapes public claims without review.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-MDP-003. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: SQL contract registry - PROPOSED: test fixture - PROPOSED: query hash - PROPOSED: change log

Tensions - PROPOSED: Stored database logic may be harder to audit than plain declarative source files.

Expansion directions - PROPOSED: Add `analytic_sql_contract` fields for `object_name`, `version`, `inputs`, `outputs`, `query_hash`, and `tests`.

Open questions - NEEDS VERIFICATION: Which SQL objects should require ADR review before changing released metrics?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 3, 4; basis: best practices: optimize, ensure integrity, test complex queries, consider views/stored procedures, and document SQL code; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-MDP-003, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.33 KFM-P18-INV-271 — 2D orthophotos derived from 3D models need model-to-map lineage

Idea card metadata - Stable ID: KFM-P18-INV-271 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 271 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should record when a 2D orthophoto or plan layer is derived from a 3D model, including georectification and source-model provenance.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record when a 2D orthophoto or plan layer is derived from a 3D model, including georectification and source-model provenance. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents derived 2D map products from losing the 3D acquisition and interpretation context that produced them.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-CAT-002.

Dependencies - PROPOSED: 3D source model - PROPOSED: orthophoto artifact - PROPOSED: georectification receipt - PROPOSED: planning-use caveat

Tensions - PROPOSED: A clean 2D orthophoto may hide 3D uncertainty or model-processing choices.

Expansion directions - PROPOSED: Add derived_orthophoto_receipt fields for source_model_id, georectification_method, GCPs, projection, and derived_scale.

Open questions - NEEDS VERIFICATION: Should model-derived orthophotos require both 2D and 3D evidence drawer entries?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 57; basis: 2D orthophoto generated from a 3D model, georectified, and imported into project intrasite GIS for planning; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-EVD-001, KFM-IDX-CAT-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.34 KFM-P18-INV-292 — Realistic and interpretive 3D models need separate labels

Idea card metadata - Stable ID: KFM-P18-INV-292 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 292 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM 3D archaeological and historical models should distinguish reality-based capture from interpretive reconstruction before publication or Story Node use.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM 3D archaeological and historical models should distinguish reality-based capture from interpretive reconstruction before publication or Story Node use. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents visually persuasive 3D scenes from being mistaken for direct evidence.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-POL-004.

Dependencies - PROPOSED: 3D asset manifest - PROPOSED: model_type label - PROPOSED: EvidenceBundle link

Tensions - PROPOSED: Interpretive models can communicate well while overstating certainty if unlabeled.

Expansion directions - PROPOSED: Add model_type enum values for reality_based_capture, interpretive_reconstruction, hybrid, and simulation.

Open questions - NEEDS VERIFICATION: What visual badge should distinguish captured from interpretive 3D geometry in public scenes?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 42, 43, 44; basis: The archaeological 3D GIS source discusses 3D models and knowledge production, including realistic and interpretive models and their role in archaeological understanding.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.35 KFM-P18-INV-293 — 3D GIS dimensional capacity must be recorded explicitly

Idea card metadata - Stable ID: KFM-P18-INV-293 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 293 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM 3D GIS artifacts should record whether they are full 3D, 2.5D, surface-only, boundary, or volumetric representations, with associated limits.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM 3D GIS artifacts should record whether they are full 3D, 2.5D, surface-only, boundary, or volumetric representations, with associated limits. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps vertical, volumetric, and surface claims from being inferred from artifacts that cannot support them.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-DOC-007.

Dependencies - PROPOSED: 3D asset schema - PROPOSED: dimensionality label - PROPOSED: analysis limitation note

Tensions - PROPOSED: A visually detailed model may lack volumetric capability.

Expansion directions - PROPOSED: Add dimensionality and analysis_supported fields to 3D asset manifests.

Open questions - NEEDS VERIFICATION: Which KFM 3D analyses require volumetric data rather than surface meshes?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 52, 61, 62; basis: The source discusses 3D GIS and contrasts 2.5D orthophotos with 3D surface models and boundary representations.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-012, KFM-IDX-DOC-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.36 KFM-P18-INV-294 — Trowel-edge 3D GIS becomes field-recording evidence

Idea card metadata - Stable ID: KFM-P18-INV-294 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 294 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM field archaeology workflows should treat on-site 3D GIS as evidence-producing documentation with acquisition, scale, georeference, revision, and analyst records.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM field archaeology workflows should treat on-site 3D GIS as evidence-producing documentation with acquisition, scale, georeference, revision, and analyst records. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It connects field capture, interpretation, and later review without pretending that the map scene itself is root truth.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-DAT-012.

Dependencies - PROPOSED: field acquisition receipt - PROPOSED: georeference metadata - PROPOSED: review record

Tensions - PROPOSED: Field 3D data can aid interpretation but still requires rights, sensitivity, and review controls before public release.

Expansion directions - PROPOSED: Create a field_3d_capture_receipt with acquisition method, operator, scale, GCPs, and revision history.

Open questions - NEEDS VERIFICATION: How should KFM preserve interim excavation interpretations that are later revised?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 78, 79, 80, 81; basis: The archaeological 3D GIS source includes a chapter on deploying 3D GIS at the trowel's edge and examples of field investigation documentation.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.37 KFM-P18-INV-295 — Surface and subsurface analyses require preservation-state context

Idea card metadata - Stable ID: KFM-P18-INV-295 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 295 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM surface/subsurface 3D analyses should record preservation state, threat context, method, and morphology-change evidence before supporting cultural-heritage claims.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM surface/subsurface 3D analyses should record preservation state, threat context, method, and morphology-change evidence before supporting cultural-heritage claims. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It ensures at-risk cultural-heritage layers are evidence-bounded and sensitivity-aware.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-004. - CONFIRMED baseline link: KFM-IDX-DAT-012.

Dependencies - PROPOSED: preservation-state record - PROPOSED: method metadata - PROPOSED: sensitivity review

Tensions - PROPOSED: High-detail preservation evidence can increase exposure risk for vulnerable sites.

Expansion directions - PROPOSED: Add preservation_state and threat_context fields to archaeology 3D analysis records.

Open questions - NEEDS VERIFICATION: Which preservation-threat fields should be public, generalized, or restricted?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 106; basis: The source describes surface and subsurface assessment of threatened archaeological or cultural heritage sites and mapping morphology changes with 3D methods.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.38 KFM-P18-INV-296 — 3D visibility analysis is simulation evidence

Idea card metadata - Stable ID: KFM-P18-INV-296 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 296 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM visibility studies should be labeled as simulations with observer assumptions, model inputs, terrain/building data, and uncertainty, not as direct observation evidence.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM visibility studies should be labeled as simulations with observer assumptions, model inputs, terrain/building data, and uncertainty, not as direct observation evidence. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It guards against overstating modeled sightlines as historical fact.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-EVD-004.

Dependencies - PROPOSED: simulation manifest - PROPOSED: observer assumption record - PROPOSED: input terrain/scene evidence

Tensions - PROPOSED: Visibility models can be compelling while dependent on uncertain assumptions about past terrain and structures.

Expansion directions - PROPOSED: Add observer_height, visibility_algorithm, input_scene_id, and uncertainty fields to simulation evidence.

Open questions - NEEDS VERIFICATION: What confidence label should apply when a visibility model depends on reconstructed terrain or buildings?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 119; basis: The source describes 3D visibility studies as simulations of human sight used to evaluate monument or feature visibility to hypothetical observers.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-012, KFM-IDX-EVD-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.39 KFM-P18-INV-316 — Hurricane translation speed as a hazard-impact variable

Idea card metadata - Stable ID: KFM-P18-INV-316 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 316 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM hurricane or storm-impact records should separate wind category, translation speed, flooding, damage, and exposure variables rather than treating the storm category as sufficient.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea

should be treated as a governed design candidate: KFM hurricane or storm-impact records should separate wind category, translation speed, flooding, damage, and exposure variables rather than treating the storm category as sufficient. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents public hazard narratives from over-weighting a single event metric.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-001. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: event variable schema - PROPOSED: impact evidence - PROPOSED: exposure denominator

Tensions - PROPOSED: Single-label categories are easy to communicate but can hide the actual impact mechanism.

Expansion directions - PROPOSED: Add translation_speed and impact_mechanism fields to storm-event candidate schemas.

Open questions - NEEDS VERIFICATION: Which hazard metrics belong in public summary versus technical Evidence Drawer?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 53; basis: ArcGIS environmental source describes Hurricane Florence as lower wind category but slow-moving, which contributed to widespread flooding, wind damage, and losses.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.40 KFM-P18-INV-317 — GCP distribution and visibility as photogrammetry evidence quality

Idea card metadata - Stable ID: KFM-P18-INV-317 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 317 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM drone-derived layers should record GCP count, distribution, visibility, contrast, RTK/GNSS source, and matching method as evidence-quality fields.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM drone-derived layers should record GCP count, distribution, visibility, contrast, RTK/GNSS source, and matching method as evidence-quality fields. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS

VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes drone-derived positional accuracy auditable rather than assumed.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: GCP record - PROPOSED: drone image set - PROPOSED: RTK/GNSS source - PROPOSED: matching receipt

Tensions - PROPOSED: More GCPs improve quality but increase field burden and may expose sensitive locations.

Expansion directions - PROPOSED: Add GCP quality section to drone-source EvidenceBundle profiles.

Open questions - NEEDS VERIFICATION: What GCP adequacy threshold should block precise public geometry?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 85, 86; basis: ArcGIS environmental source says GCP targets should be large, clear, high-contrast, sufficiently numerous, evenly distributed, and matched to RTK GPS coordinates and drone photos.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.41 KFM-P18-INV-318 — Manual versus GNSS-collected point distinction in field maps

Idea card metadata - Stable ID: KFM-P18-INV-318 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 318 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM field-collection records should distinguish manually placed test points from device-collected GPS/GNSS observations.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM field-collection records should distinguish manually placed test points from device-collected GPS/GNSS observations. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents training, test, or mock points from being promoted as field-measured observations.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-001. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: collection_mode - PROPOSED: device permission record - PROPOSED: GPS/GNSS metadata

Tensions - PROPOSED: Training workflows often create useful test data that must not enter evidence lanes.

Expansion directions - PROPOSED: Add collection_mode = manual_test | device_gnss | imported | corrected to field observation schemas.

Open questions - NEEDS VERIFICATION: How should KFM quarantine field-training data after exercises?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 94; basis: ArcGIS environmental source notes latitude and longitude may be blank when test points are manually placed rather than collected with a mobile device GPS/GNSS.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.42 KFM-P18-INV-319 — Climate scenario assumptions and ensembles as layer metadata

Idea card metadata - Stable ID: KFM-P18-INV-319 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 319 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM climate layers should expose RCP/scenario assumptions, forcing values, model ensemble basis, and time period comparisons as material metadata.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM climate layers should expose RCP/scenario assumptions, forcing values, model ensemble basis, and time period comparisons as material metadata. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps scenario maps from being interpreted as single deterministic forecasts.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: scenario id - PROPOSED: forcing value - PROPOSED: ensemble description - PROPOSED: baseline and future periods

Tensions - PROPOSED: Scenario labels may be familiar to experts but opaque to public users.

Expansion directions - PROPOSED: Add climate_scenario_metadata and not_prediction caveat fields to climate layer manifests.

Open questions - NEEDS VERIFICATION: How should KFM phrase uncertainty around scenario layers in public popups?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 121, 122, 127; basis: ArcGIS environmental source explains IPCC representative concentration pathways, contrasts RCP 2.6 and RCP 8.5 assumptions, and describes a multi-model ensemble basis.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-012, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.43 KFM-P18-INV-323 — Interactive 3D web GIS needs collaboration and reuse metadata

Idea card metadata - Stable ID: KFM-P18-INV-323 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 323 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM 3D web GIS publications should document collaboration mode, data reuse conditions, platform dependency, and investigation-campaign linkage.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM 3D web GIS publications should document collaboration mode, data reuse conditions, platform dependency, and investigation-campaign linkage. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It treats 3D web publication as a governed research surface, not merely a viewer.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: 3D platform manifest - PROPOSED: data reuse terms - PROPOSED: campaign linkage - PROPOSED: collaboration role

Tensions - PROPOSED: Open 3D access can improve reuse but may expose sensitive archaeological context.

Expansion directions - PROPOSED: Add reuse_terms and campaign_context fields to 3D artifact manifests.

Open questions - NEEDS VERIFICATION: Should archaeological 3D web layers default to staged access rather than public viewing?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 57, 60; basis: Archaeological 3D GIS source describes 3D visualization as communication and analysis in 3D space and notes recent 3D web platforms supporting archaeological practice and data reuse across campaigns.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.44 KFM-P18-INV-324 — 3D visibility metrics require observer and obstacle assumption receipts

Idea card metadata - Stable ID: KFM-P18-INV-324 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 324 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM 3D visibility analyses should record observer positions, target surfaces, obstacle assumptions, sightline frequency calculations, and normalization method.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM 3D visibility analyses should record observer positions, target surfaces, obstacle assumptions, sightline frequency calculations, and normalization method. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps visibility outputs from appearing as direct observations rather than modeled inference.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: observer model - PROPOSED: target surface - PROPOSED: obstacle model - PROPOSED: visibility calculation receipt

Tensions - PROPOSED: Visibility models can be persuasive but depend heavily on assumptions about built form and observer behavior.

Expansion directions - PROPOSED: Add visibility_model_receipt fields to 3D analysis outputs.

Open questions - NEEDS VERIFICATION: Which visibility assumptions should be mandatory in public explanations?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 119, 125; basis: Archaeological 3D GIS source defines visibility studies as simulations of human sight and reports undisturbed sightline frequencies normalized to percentage visibility.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.45 KFM-P18-INV-347 — Regulatory floodplain labels are not observed inundation evidence

Idea card metadata - Stable ID: KFM-P18-INV-347 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 347 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM flood layers should distinguish FEMA Special Flood Hazard Area or 100-year floodplain designations from observed inundation, modeled flood depth, insurance obligation, and exposure analysis.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM flood layers should distinguish FEMA Special Flood Hazard Area or 100-year floodplain designations from observed inundation, modeled flood depth, insurance obligation, and exposure analysis. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It avoids turning regulatory flood zones into unsupported claims about a specific flood event.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-005. - CONFIRMED baseline link: KFM-IDX-POL-003.

Dependencies - PROPOSED: source role - PROPOSED: regulatory designation - PROPOSED: observed event - PROPOSED: model output - PROPOSED: exposure analysis

Tensions - PROPOSED: Users often conflate flood-hazard designation, insurance requirement, and observed water extent.

Expansion directions - PROPOSED: Add regulatory_context_source_role to flood/hazard source descriptors.

Open questions - NEEDS VERIFICATION: How should KFM explain insurance obligations without legal overclaiming?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 7, 8; basis: Earth, Space, and Environmental Science Explorations with ArcGIS Pro defines 100-year floodplains/Special Flood Hazard Areas and discusses flood insurance obligations.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-005, KFM-IDX-POL-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.46 KFM-P18-INV-361 — Field Maps collection as attachment-aware evidence capture

Idea card metadata - Stable ID: KFM-P18-INV-361 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 361 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing,

3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM field-data collection workflows should record coordinates, attachments, collector identity, sharing state, and target purpose as evidence-capture metadata before observations enter domain lanes.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM field-data collection workflows should record coordinates, attachments, collector identity, sharing state, and target purpose as evidence-capture metadata before observations enter domain lanes. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Field observations become more auditable when media, collector role, and layer-sharing state are preserved at intake.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-POL-002.

Dependencies - PROPOSED: collector role - PROPOSED: feature layer - PROPOSED: attachment metadata - PROPOSED: coordinate receipt

Tensions - PROPOSED: Precise field coordinates may require public-safe redaction.

Expansion directions - PROPOSED: Define field_capture_receipt fields for attachment-bearing observations.

Open questions - NEEDS VERIFICATION: When should field attachments be public, restricted, or denied?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 85, 86; basis: Earth, Space, and Environmental Science Explorations with ArcGIS Pro includes collecting field data with Field Maps, creating feature layers, enabling attachments, sharing web maps, and recording target coordinates.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-001, KFM-IDX-POL-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.47 KFM-P18-INV-362 — Interpretative and realistic 3D models require separate knowledge-character labels

Idea card metadata - Stable ID: KFM-P18-INV-362 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 362 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM 3D and scene artifacts should distinguish reality-based surface models from interpretative reconstructions and label each with evidence support, modeling method, and uncertainty.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM 3D and scene artifacts should distinguish reality-based surface models from interpretative reconstructions and label each with evidence support, modeling method, and uncertainty. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A reconstruction can be visually persuasive even when it has less evidence than a measured surface model.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-007. - CONFIRMED baseline link: KFM-IDX-DAT-012.

Dependencies - PROPOSED: model type - PROPOSED: capture method - PROPOSED: interpretation method - PROPOSED: EvidenceBundle link

Tensions - PROPOSED: Users may read photorealistic reconstructions as direct observation.

Expansion directions - PROPOSED: Add model_character to 3D Story Node manifests.

Open questions - NEEDS VERIFICATION: What visual cues should separate observation-derived and interpretation-derived 3D assets?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 10, 24; basis: Archaeological 3D GIS discusses digital interpretative and realistic models, 3D surface models, and 3D data as part of interpretation and representation.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-007, KFM-IDX-DAT-012 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.48 KFM-P18-INV-363 — Surface and subsurface 3D analysis as layered evidence rather than a single scene

Idea card metadata - Stable ID: KFM-P18-INV-363 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 363 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM 3D GIS outputs should separate surface models, subsurface models, point-cloud-derived layers, volume interpretations, and analysis results instead of publishing them as one undifferentiated scene.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before

Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM 3D GIS outputs should separate surface models, subsurface models, point-cloud-derived layers, volume interpretations, and analysis results instead of publishing them as one undifferentiated scene. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps 3D outputs inspectable and prevents interpretation layers from replacing source geometry.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-CAT-007.

Dependencies - PROPOSED: surface asset - PROPOSED: subsurface asset - PROPOSED: volume calculation - PROPOSED: analysis layer manifest

Tensions - PROPOSED: Layered 3D evidence is harder to render simply than a single composite scene.

Expansion directions - PROPOSED: Create a 3D evidence-layer manifest with roles and source links.

Open questions - NEEDS VERIFICATION: Which subsurface layers can be public without sensitive-location risk?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 26, 105, 109; basis: Archaeological 3D GIS describes surface and subsurface analysis, 3D point-cloud-derived data, visibility analysis, and volumes as distinct analytical concerns.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-012, KFM-IDX-CAT-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.49 KFM-P18-INV-387 — Georeferenced 3D imports need scale and orientation QA

Idea card metadata - Stable ID: KFM-P18-INV-387 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 387 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM 3D model imports should record georeferencing method, scale, orientation, coordinate reference, and import-platform limitations before any model is used as spatial evidence.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM 3D model imports should record georeferencing method, scale, orientation, coordinate reference, and import-platform limitations before any model is used as spatial evidence. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is

already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A visually detailed 3D model can be spatially misleading if scale, orientation, or CRS handling is weak.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-CAT-007.

Dependencies - PROPOSED: 3D model asset - PROPOSED: georeferencing method - PROPOSED: CRS - PROPOSED: scale check - PROPOSED: orientation check

Tensions - PROPOSED: Different 3D platforms may represent axes and units differently.

Expansion directions - PROPOSED: Define a 3d_import_quality profile for scene assets.

Open questions - NEEDS VERIFICATION: What tolerances should apply to georeferenced 3D model imports?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 42, 43, 54; basis: Archaeological 3D GIS describes 3D models that are scaled, georeferenced, exported, and imported into GIS platforms.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-012, KFM-IDX-CAT-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.50 KFM-P18-INV-388 — 2.5D comparison views disclose vertical-surface loss

Idea card metadata - Stable ID: KFM-P18-INV-388 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 388 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM 3D/terrain interfaces should explicitly disclose when a 2.5D or orthophoto view cannot represent vertical surfaces that a true 3D model contains.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM 3D/terrain interfaces should explicitly disclose when a 2.5D or orthophoto view cannot represent vertical surfaces that a true 3D model contains. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A flattened view can be useful but can erase evidence carried by walls, sections, or vertical relationships.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-DOC-007.

Dependencies - PROPOSED: 2.5D asset - PROPOSED: 3D asset - PROPOSED: vertical-surface flag - PROPOSED: view warning

Tensions - PROPOSED: Public users may prefer simpler 2D/2.5D maps even when evidence needs 3D context.

Expansion directions - PROPOSED: Add vertical_loss_warning to 2.5D derivative manifests.

Open questions - NEEDS VERIFICATION: When is 2.5D insufficient for a KFM claim?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 38, 57, 61; basis: Archaeological 3D GIS contrasts 2.5D orthoimages with 3D surface models and notes limits for vertical surfaces.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-012, KFM-IDX-DOC-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.51 KFM-P18-INV-397 — Emission-scenario horizons for climate-change visualizations

Idea card metadata - Stable ID: KFM-P18-INV-397 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 397 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM climate scenario layers should record emission scenario, model/source family, temporal horizon, uncertainty context, and non-prediction caveats before public visualization.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM climate scenario layers should record emission scenario, model/source family, temporal horizon, uncertainty context, and non-prediction caveats before public visualization. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Scenario maps can look like forecasts unless assumptions and horizons are explicit.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: scenario id - PROPOSED: time horizon - PROPOSED: model/source family - PROPOSED: uncertainty note

Tensions - PROPOSED: Scenario terms can be confused with observed climate change.

Expansion directions - PROPOSED: Add scenario_horizon and scenario_role labels to climate layer manifests.

Open questions - NEEDS VERIFICATION: Which climate scenario labels should KFM standardize?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 121; basis: Earth, Space, and Environmental Science Explorations with ArcGIS Pro includes a chapter on visualizing global climate change under different emission scenarios.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-012, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.52 KFM-P18-INV-410 — 3D GIS paradata for acquisition and interpretation choices

Idea card metadata - Stable ID: KFM-P18-INV-410 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 410 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM 3D GIS artifacts should store paradata about acquisition methods, processing choices, scaling, georeferencing, and interpretive steps.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM 3D GIS artifacts should store paradata about acquisition methods, processing choices, scaling, georeferencing, and interpretive steps. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A 3D scene may combine scans, photogrammetry, CT, and interpretation; paradata makes those choices reviewable instead of hidden in the model.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-003. - CONFIRMED baseline link: KFM-IDX-DAT-013.

Dependencies - PROPOSED: 3D asset manifest - PROPOSED: georeference receipt - PROPOSED: processing history - PROPOSED: interpretation label

Tensions - PROPOSED: Paradata volume can grow quickly for complex scenes.

Expansion directions - PROPOSED: Define 3d_asset_paradata fields for archaeological and terrain-adjacent artifacts.

Open questions - NEEDS VERIFICATION: How much paradata is required before a 3D model can appear in a public Story Node?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 24, 31, 42; basis: Archaeological 3D GIS describes image-based modeling, laser scanning, georeferenced models, 3D GIS platforms, and model integration workflows.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-003, KFM-IDX-DAT-013 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.53 KFM-P18-INV-434 — Extended-matrix links for 3D archaeological interpretation

Idea card metadata - Stable ID: KFM-P18-INV-434 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 434 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM archaeology and 3D story surfaces should preserve explicit links between observed 3D evidence, interpretive reconstruction, and stratigraphic reasoning.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM archaeology and 3D story surfaces should preserve explicit links between observed 3D evidence, interpretive reconstruction, and stratigraphic reasoning. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: 3D scenes can blend observation and interpretation unless their relationships are modeled and inspectable.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-009. - CONFIRMED baseline link: KFM-IDX-POL-002.

Dependencies - PROPOSED: 3D scene manifest - PROPOSED: archaeology sensitivity policy - PROPOSED: EvidenceBundle - PROPOSED: interpretation link schema

Tensions - PROPOSED: Public 3D context can invite false precision or sensitive-site exposure.

Expansion directions - PROPOSED: Create interpretation_relation fields for 3D archaeological assets.

Open questions - NEEDS VERIFICATION: How should sensitive archaeological links be generalized while retaining reviewability?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 11, 45, 56, 159; basis: Archaeological 3D GIS discusses the extended matrix approach and 3D GIS for linking interpretive and reality-based models.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-009, KFM-IDX-POL-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.54 KFM-P18-INV-452 — Remote-sensing resolution metadata for raster and imagery layers

Idea card metadata - Stable ID: KFM-P18-INV-452 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 452 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM raster and imagery layers should disclose sensor bands plus spatial, temporal, and radiometric resolution before derivative observations are promoted or compared.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM raster and imagery layers should disclose sensor bands plus spatial, temporal, and radiometric resolution before derivative observations are promoted or compared. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Resolution governs what an image can and cannot support; without it, derivative map claims can overstate evidence quality.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-002. - CONFIRMED baseline link: KFM-IDX-CAT-007.

Dependencies - PROPOSED: SourceDescriptor - PROPOSED: raster asset manifest - PROPOSED: sensor metadata - PROPOSED: catalog record

Tensions - PROPOSED: Users may assume high-resolution visual appearance equals high-confidence detection.

Expansion directions - PROPOSED: Require raster layer manifests to carry resolution fields and detection-limit caveats.

Open questions - NEEDS VERIFICATION: Which resolution dimensions should be mandatory for each remote-sensing source family?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 174, 176; basis: The GIS primer describes sensor bands and explains that spatial and temporal resolution affect what can be detected and how often a place is revisited.; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-002, KFM-IDX-CAT-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.55 KFM-P18-INV-458 — 3D visibility analysis assumption disclosure

Idea card metadata - Stable ID: KFM-P18-INV-458 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 458 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from

CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM 3D visibility analyses should disclose observer points, target points, obstacles, model resolution, line-of-sight rules, and interpretation scope.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM 3D visibility analyses should disclose observer points, target points, obstacles, model resolution, line-of-sight rules, and interpretation scope. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Visibility output is an interpretation derived from assumptions, not direct archaeological fact.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-EVD-006.

Dependencies - PROPOSED: 3D model - PROPOSED: observer grid - PROPOSED: target points - PROPOSED: analysis receipt

Tensions - PROPOSED: Small changes in model geometry or observer assumptions can change the result.

Expansion directions - PROPOSED: Require visibility-analysis receipts to list observer/target geometry and obstruction rules.

Open questions - NEEDS VERIFICATION: How should KFM publish uncertain or alternative observer scenarios?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 119, 120, 121, 122; basis: Archaeological 3D GIS describes 3D visibility studies, line-of-sight workflows, observers and targets, obstacles, and the interpretive use of visibility in built spaces.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.56 KFM-P18-INV-459 — Surface and subsurface damage-assessment evidence objects

Idea card metadata - Stable ID: KFM-P18-INV-459 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 459 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM archaeology and heritage workflows should treat 3D surface/subsurface assessments as evidence objects with acquisition method, preservation state, damage model, and quantification limits.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM archaeology and heritage workflows should treat 3D surface/subsurface assessments as evidence objects with acquisition method, preservation state, damage model, and quantification limits. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Damage assessment can guide preservation decisions, but only when acquisition and quantification limits are recorded.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-POL-002.

Dependencies - PROPOSED: 3D acquisition record - PROPOSED: damage assessment - PROPOSED: heritage sensitivity policy - PROPOSED: review note

Tensions - PROPOSED: Publicizing precise heritage-damage locations can create sensitivity and looting risk.

Expansion directions - PROPOSED: Create a public-safe generalized damage-assessment payload pattern.

Open questions - NEEDS VERIFICATION: Which damage metrics can be released publicly for sensitive sites?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 105, 106, 107; basis: Archaeological 3D GIS describes surface and subsurface assessment of threatened cultural heritage sites and use of 3D models for quantitative damage/erosion evaluation.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.57 KFM-P18-INV-467 — Field Maps GCP observations as governed field evidence

Idea card metadata - Stable ID: KFM-P18-INV-467 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 467 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM field-data workflows should treat ground-control-point observations as governed evidence records with coordinates, GNSS/RTK method, attachments, web-map source, and collection context.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM field-data workflows should treat ground-control-point observations as governed evidence records with coordinates, GNSS/RTK method, attachments, web-map source, and collection context. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full

idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Ground control points can anchor later imagery products, so their collection method and context must be auditable.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-DAT-001.

Dependencies - PROPOSED: field observation - PROPOSED: attachment manifest - PROPOSED: position method - PROPOSED: EvidenceBundle

Tensions - PROPOSED: Mobile-collected coordinates may vary by device and collection conditions.

Expansion directions - PROPOSED: Create a synthetic GCP evidence fixture with attachment and positional-method metadata.

Open questions - NEEDS VERIFICATION: Which field observations require attachments before promotion?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 85, 86, 89, 94; basis: The ArcGIS exercise describes collecting ground control points with Field Maps, enabling attachments, recording coordinates, sharing a web map, and using GPS/GNSS or RTK context.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.58 KFM-P18-INV-484 — 3D model georeferencing and scale receipts

Idea card metadata - Stable ID: KFM-P18-INV-484 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 484 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM 3D or terrain-context artifacts should record georeferencing, scale, acquisition method, and model-processing steps as receipts before they support spatial claims.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM 3D or terrain-context artifacts should record georeferencing, scale, acquisition method, and model-processing steps as receipts before they support spatial claims. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A visually compelling 3D model can still have positional or scale uncertainty that must be inspectable.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: 3D model - PROPOSED: georeferencing - PROPOSED: scale - PROPOSED: EvidenceBundle

Tensions - PROPOSED: 3D model metadata can be large and discipline-specific.

Expansion directions - PROPOSED: Define a 3d_artifact_receipt profile for scenes and archaeological/terrain models.

Open questions - NEEDS VERIFICATION: Which 3D artifacts need exact versus generalized public coordinates?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 65, 71, 79; basis: The Archaeological 3D GIS source describes 3D models being scaled, georeferenced, exported, and imported into 3D GIS platforms.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-012, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.9.59 KFM-P18-INV-485 — 2.5D versus true 3D representation labels

Idea card metadata - Stable ID: KFM-P18-INV-485 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 485 - Status label: PROPOSED - Carry-forward state: NEW - Category: FIE — Field Capture, Remote Sensing, 3D, and Archaeological Interpretation - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should label elevation and scene artifacts as 2D, 2.5D, or true 3D where that distinction affects interpretation or measurement.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should label elevation and scene artifacts as 2D, 2.5D, or true 3D where that distinction affects interpretation or measurement. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Users may incorrectly assume a draped orthophoto or height surface can answer true 3D visibility, volume, or vertical-face questions.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-DOC-008.

Dependencies - PROPOSED: 2.5D - PROPOSED: 3D surface model - PROPOSED: representation label - PROPOSED: map layer

Tensions - PROPOSED: Renderer capabilities and data geometry categories do not always align cleanly.

Expansion directions - PROPOSED: Add representation_dimensionality to terrain, scene, and 3D layer manifests.

Open questions - NEEDS VERIFICATION: Should 2.5D labels appear in the Evidence Drawer or only in technical metadata?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 61; basis: The Archaeological 3D GIS source distinguishes 2.5D orthoimages from 3D surface models and notes limitations in representing vertical surfaces.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10 REL - Catalog Closure, Publication, Release, Rollback, and Recompile Discipline

Status: CONFIRMED source support for grouped Phase 3 ideas; PROPOSED category synthesis

Definition: Catalog closure, publication, release, rollback, recompile, map artifact integrity, and downstream delivery readiness.

CONFIRMED: This chapter contains 43 Pass 18 idea cards assigned to REL.

Idea ID	Title	Carry state	Status	Source IDs
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KFM-P18-INV-009 | External GIS publishing is not KFM promotion | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-023 | Report-time consistency and release snapshots | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-047 | STAC/DCAT/PROV asset roles for renderable artifacts | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-058 | Social vulnerability overlays need transparent census joins and caveats | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-075 | Offline field capture must pass redaction and catalog gates before publication | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-102 | Surveying, GPS, and digitization as acquisition-method evidence | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-110 | Feature engineering and normalization as evidence transforms | NEW | PROPOSED | SRC-P18-004 |

KFM-P18-INV-118 | Temporal coalescing before public release | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-128 | Map-type taxonomy as release-profile input | NEW | PROPOSED | SRC-P18-002 |

KFM-P18-INV-130 | Spatial analysis outputs are question-scoped evidence | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-152 | Public-payload scrub rule for API promotion | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-161 | Model deployment as governed release boundary | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-162 | Missing-data method provenance | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-164 | CNN feature-map provenance for image-derived evidence | NEW | PROPOSED | SRC-P18-004 |

KFM-P18-INV-176 | OCI digest-pinned geospatial artifact publication | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-177 | OCI referrers as SBOM and PROV evidence links | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-178 | MediaType and asset-role alignment across STAC, DCAT, and PROV | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-210 | 3D acquisition-method provenance for scan, photogrammetry, and CT | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-228 | Set-operator semantics for release deltas and intersections | EXPANDED | PROPOSED | SRC-P18-003 |

KFM-P18-INV-233 | AI data ethics become release-gate evidence | EXPANDED | PROPOSED | SRC-P18-004 |

KFM-P18-INV-240 | Task-recorded geoprocessing workflows as replayable recipes | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-241 | Field Maps observations need attachment and GCP provenance | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-254 | Demographic enrichment in hazard workflows requires provenance and denominator receipts | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-258 | Knowledge co-production and social learning as review outcomes | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-272 | 3D resolution dimensions become release and analysis constraints | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-299 | Stakeholder MCA weights need decision provenance | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-326 | WZDx GeoJSON as a governed transport-event source family | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-327 | Frictionless-style field validation before GeoJSON or TopoJSON tiling | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-328 | Energy and CO2 telemetry as a map-artifact promotion signal | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-330 | Sentinel-1 SAR layers need sovereignty, mask, and lineage controls | NEW | PROPOSED | SRC-P18-039 |

KFM-P18-INV-346 | External ArcGIS web layer publication is separate from KFM promotion | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-355 | Public-space social-interaction indicators need qualitative evidence links | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-357 | Urban quality-of-life surfaces require indicator provenance | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-364 | Trowel-edge 3D GIS as field-review support, not immediate publication | EXPANDED | PROPOSED | SRC-P18-005 |

KFM-P18-INV-378 | Map metadata elements as minimum release disclosure | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-391 | API lifecycle diagrams aligned to promotion states | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-413 | API versioning and contract-change notices as release duties | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-427 | Datum and coordinate-reference provenance for location claims | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-442 | Social vulnerability enrichment as a derived exposure layer | EXPANDED | PROPOSED | SRC-P18-010 |

KFM-P18-INV-448 | OCI registry references for geospatial artifact provenance | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-451 | Positional method provenance for location claims | NEW | PROPOSED | SRC-P18-002 |

KFM-P18-INV-474 | LayerCatalog search points to catalog records, not raw databases | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-494 | Published language for catalog and schema vocabulary | EXPANDED | PROPOSED | SRC-P18-009 |

8.10.1 KFM-P18-INV-009 — External GIS publishing is not KFM promotion

Idea card metadata - Stable ID: KFM-P18-INV-009 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 9 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should explicitly separate external platform sharing or web-layer publication from governed KFM promotion to PUBLISHED.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should explicitly separate external platform sharing or web-layer publication from governed KFM promotion to PUBLISHED. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It blocks a common shortcut where availability on a map platform is mistaken for rights-cleared, evidence-backed publication.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-002. - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: PromotionDecision and ReleaseManifest separation

Tensions - PROPOSED: External platforms are convenient but can bypass KFM trust membrane.

Expansion directions - PROPOSED: Add a platform_publication_not_kfm_release anti-pattern entry.

Open questions - NEEDS VERIFICATION: How should KFM catalog externally hosted layers that are not KFM-published?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 7, 9; basis: ArcGIS web-layer publication workflow; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-002, KFM-IDX-PIP-004, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.2 KFM-P18-INV-023 — Report-time consistency and release snapshots

Idea card metadata - Stable ID: KFM-P18-INV-023 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 23 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should make public reports and exports reproducible against release snapshots, not merely against whatever data is current when queried.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should make public reports and exports reproducible against release snapshots, not merely against whatever data is current when queried. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A report whose numbers shift silently undermines auditability and correction lineage.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-005. - CONFIRMED baseline link: KFM-IDX-EVD-006. - CONFIRMED baseline link: KFM-IDX-PIP-004.

Dependencies - PROPOSED: ReleaseManifest; report manifest; query snapshot identity

Tensions - PROPOSED: Live dashboards and frozen reports serve different needs.

Expansion directions - PROPOSED: Add report_release_manifest and query_snapshot_ref fields to export/report objects.

Open questions - NEEDS VERIFICATION: Should every public report cite a release manifest and query hash?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 18, 343; basis: report inconsistency and temporal query framing; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-005, KFM-IDX-EVD-006, KFM-IDX-PIP-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.3 KFM-P18-INV-047 — STAC/DCAT/PROV asset roles for renderable artifacts

Idea card metadata - Stable ID: KFM-P18-INV-047 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 47 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should map renderable vector, raster, tile, and metadata artifacts into STAC/DCAT/PROV with explicit asset roles and checksums.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should map renderable vector, raster, tile, and metadata artifacts into STAC/DCAT/PROV with explicit asset roles and checksums. - PROPOSED: Phase 3 scope note: Inventory-level

candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It lets map UI, catalog, evidence drawer, and release validation consume the same artifact truth surface.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-001. - CONFIRMED baseline link: KFM-IDX-CAT-003. - CONFIRMED baseline link: KFM-IDX-CAT-004.

Dependencies - PROPOSED: STAC Item/Asset - PROPOSED: DCAT distribution - PROPOSED: PROV activity - PROPOSED: checksum fields - PROPOSED: asset role enum

Tensions - PROPOSED: Catalog metadata is still not sovereign truth without evidence resolution and policy state.

Expansion directions - PROPOSED: Create asset-role validation for renderable, data, quicklook, metadata, and proof assets.

Open questions - NEEDS VERIFICATION: Which asset roles are required for each KFM map layer type?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 10, 11, 14; basis: STAC/DCAT/PROV mapping and asset-role rows; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-001, KFM-IDX-CAT-003, KFM-IDX-CAT-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.4 KFM-P18-INV-058 — Social vulnerability overlays need transparent census joins and caveats

Idea card metadata - Stable ID: KFM-P18-INV-058 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 58 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should make census joins, derived vulnerability fields, and disaster-impact selection criteria visible when social vulnerability is mapped against hazard areas.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should make census joins, derived vulnerability fields, and disaster-impact selection criteria visible when social vulnerability is mapped against hazard areas. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Vulnerability maps are consequential; they should expose their data joins, missingness, geography, and temporal applicability.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-013. - CONFIRMED baseline link: KFM-IDX-POL-004.

Dependencies - PROPOSED: JoinReceipt - PROPOSED: indicator caveat metadata

Tensions - PROPOSED: Aggregated demographics can be useful while also stigmatizing places or populations if overinterpreted.

Expansion directions - PROPOSED: Add `vulnerability_indicator_bundle` with `geography_version` and `join_method`.

Open questions - NEEDS VERIFICATION: How should KFM label vulnerability maps that rely on outdated Census geography or attributes?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 53; basis: social vulnerability exercise with Census preprocessing, calculated fields, joins, and spatial selection; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-013, KFM-IDX-POL-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.5 KFM-P18-INV-075 — Offline field capture must pass redaction and catalog gates before publication

Idea card metadata - Stable ID: KFM-P18-INV-075 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 75 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should allow offline field capture outputs to become PMTiles, COG, STAC, or PROV artifacts only after redaction, classification, rights, and accessibility checks pass.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should allow offline field capture outputs to become PMTiles, COG, STAC, or PROV artifacts only after redaction, classification, rights, and accessibility checks pass. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Field capture often contains precise locations, photographs, EXIF data, notes, and sensitive context that can leak before review.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-004. - CONFIRMED baseline link: KFM-IDX-POL-005.

Dependencies - PROPOSED: FieldCaptureReceipt - PROPOSED: RedactionReceipt - PROPOSED: CatalogClosure validator

Tensions - PROPOSED: Offline workflows are useful during fieldwork but can bypass normal connected validation if not constrained.

Expansion directions - PROPOSED: Create a field_capture_release_profile covering EXIF redaction, public-safe geometry, catalog closure, and alt-text review.

Open questions - NEEDS VERIFICATION: What field-captured materials require quarantine even after redaction?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 28; basis: offline-first field capture to PMTiles/COG with redaction, STAC, PROV, and accessibility tests; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-004, KFM-IDX-POL-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.6 KFM-P18-INV-102 — Surveying, GPS, and digitization as acquisition-method evidence

Idea card metadata - Stable ID: KFM-P18-INV-102 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 102 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should record whether spatial evidence came from surveying, GPS/GNSS, digitization, remote sensing, or transformed legacy maps, including method limits and permissions.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record whether spatial evidence came from surveying, GPS/GNSS, digitization, remote sensing, or transformed legacy maps, including method limits and permissions. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents a digitized legacy map feature from being mistaken for a surveyed boundary or a sensor observation.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-POL-004.

Dependencies - PROPOSED: SourceDescriptor acquisition_method - PROPOSED: Rights review - PROPOSED: Geometry provenance schema

Tensions - PROPOSED: Older maps may be the only available evidence but can carry copyright, scale, and generalization constraints.

Expansion directions - PROPOSED: Add `acquisition_method`, `source_material_rights`, and `digitization_generalization` fields.

Open questions - NEEDS VERIFICATION: How should KFM express positional confidence by acquisition method?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — `a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf`; pages 150, 151, 155, 160; basis: surveying, GPS/GNSS, digitization, copyright, and generalization discussion; Phase 1 role: `candidate_synthesis_source`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-001, KFM-IDX-POL-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.7 KFM-P18-INV-110 — Feature engineering and normalization as evidence transforms

Idea card metadata - Stable ID: KFM-P18-INV-110 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 110 - Status label: PROPOSED - Carry-forward state: NEW - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should record feature engineering, normalization, and preprocessing steps as evidence transforms rather than treating them as invisible model setup.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (`AI_Concepts_Using_Python.pdf`) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record feature engineering, normalization, and preprocessing steps as evidence transforms rather than treating them as invisible model setup. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps derived features auditable and reversible when model outcomes are challenged.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-003. - CONFIRMED baseline link: KFM-IDX-EVD-003.

Dependencies - PROPOSED: Transform receipt - PROPOSED: Feature manifest - PROPOSED: Model input schema

Tensions - PROPOSED: Feature engineering can embed assumptions that users cannot see from the final model output.

Expansion directions - PROPOSED: Add `FeatureTransformReceipt` with input evidence refs and transform parameters.

Open questions - NEEDS VERIFICATION: How should derived features link back to original `EvidenceBundles`?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 81, 83, 194; basis: data lifecycle, feature engineering, normalization, and model utilization discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.8 KFM-P18-INV-118 — Temporal coalescing before public release

Idea card metadata - Stable ID: KFM-P18-INV-118 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 118 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should coalesce value-equivalent and interval-adjacent temporal records when needed for public explanation, while preserving original evidence and transaction history.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should coalesce value-equivalent and interval-adjacent temporal records when needed for public explanation, while preserving original evidence and transaction history. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reduces misleading duplicate fragments without erasing the history needed for audit.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-007. - CONFIRMED baseline link: KFM-IDX-EVD-006.

Dependencies - PROPOSED: Temporal validator - PROPOSED: Release transformation receipt

Tensions - PROPOSED: Over-coalescing can hide meaningful source distinctions or correction events.

Expansion directions - PROPOSED: Add coalescing_policy and original_interval_refs to release receipts.

Open questions - NEEDS VERIFICATION: When should KFM coalesce for display only versus processed truth?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 183, 184; basis: temporal coalescing and duplicate-elimination discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-007, KFM-IDX-EVD-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.9 KFM-P18-INV-128 — Map-type taxonomy as release-profile input

Idea card metadata - Stable ID: KFM-P18-INV-128 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 128 - Status label: PROPOSED - Carry-forward state: NEW - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should classify released maps by map type and purpose so thematic, topographic, cadastral, exploratory, and story-map outputs do not inherit the same authority posture.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should classify released maps by map type and purpose so thematic, topographic, cadastral, exploratory, and story-map outputs do not inherit the same authority posture. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps a thematic public overview from being mistaken for cadastral, regulatory, or boundary truth.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-POL-004.

Dependencies - PROPOSED: MapReleaseManifest - PROPOSED: LayerManifest - PROPOSED: Review profile registry

Tensions - PROPOSED: Mixed-purpose maps may need multiple authority labels.

Expansion directions - PROPOSED: Add map_purpose and map_type fields to map release metadata.

Open questions - NEEDS VERIFICATION: Which KFM outputs qualify as cadastral-like and therefore require stronger review?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 24, 25, 40; basis: thematic, topographic, cadastral, and mental-map discussion; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.10 KFM-P18-INV-130 — Spatial analysis outputs are question-scoped evidence

Idea card metadata - Stable ID: KFM-P18-INV-130 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 130 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Re-

lease, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM should label spatial analysis outputs by the question, assumptions, input layers, and model family they answer, not publish them as general-purpose truth.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should label spatial analysis outputs by the question, assumptions, input layers, and model family they answer, not publish them as general-purpose truth. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reduces misuse of buffers, overlays, interpolations, and geostatistical surfaces outside their intended analytical frame.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-006. - CONFIRMED baseline link: KFM-IDX-OBS-005.

Dependencies - PROPOSED: AnalysisManifest - PROPOSED: EvidenceBundle - PROPOSED: Model limitation labels

Tensions - PROPOSED: Users often want simple map answers even when the analysis is conditional.

Expansion directions - PROPOSED: Add question_scope and assumption_summary fields to derived analysis products.

Open questions - NEEDS VERIFICATION: Which analysis products require uncertainty surfaces?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 253, 271, 290; basis: GIS analysis and geostatistics/future GIS chapters in the GIS primer; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-006, KFM-IDX-OBS-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.11 KFM-P18-INV-152 — Public-payload scrub rule for API promotion

Idea card metadata - Stable ID: KFM-P18-INV-152 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 152 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should validate public and partner API payloads as if every returned field is exposed, regardless of whether the visible UI renders that field.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should validate public and partner API payloads as if every returned field is exposed, regardless of whether the visible UI renders that field. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents hidden response fields from bypassing sensitivity, rights, or exact-location policy through developer tools, third-party clients, or downstream integrations.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-002. - CONFIRMED baseline link: KFM-IDX-POL-004. - CONFIRMED baseline link: KFM-IDX-POL-007.

Dependencies - PROPOSED: Response schema registry - PROPOSED: Sensitivity policy - PROPOSED: Negative fixtures

Tensions - PROPOSED: Debug, reviewer, and steward payloads may need richer fields than public payloads.

Expansion directions - PROPOSED: Define a public-payload scrub validator and compare it against UI component rendering assumptions.

Open questions - NEEDS VERIFICATION: How are reviewer-only fields logged without becoming public API fields?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 24; basis: API data-leakage example where exact location remained in response payload despite not being surfaced by the client UI; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-002, KFM-IDX-POL-004, KFM-IDX-POL-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.12 KFM-P18-INV-161 — Model deployment as governed release boundary

Idea card metadata - Stable ID: KFM-P18-INV-161 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 161 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should treat model deployment as a governed release boundary with finite outcomes, evidence closure, rollback target, and post-release monitoring.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat model deployment as a governed release boundary with finite outcomes, evidence

closure, rollback target, and post-release monitoring. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents a trained model from entering a public map, API, or Focus Mode path merely because training completed.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-DOC-003. - CONFIRMED baseline link: KFM-IDX-OBS-003.

Dependencies - PROPOSED: ReleaseManifest - PROPOSED: Model card - PROPOSED: Monitoring plan

Tensions - PROPOSED: Some models may be used only internally for candidate generation and should not follow public-release rules.

Expansion directions - PROPOSED: Separate internal candidate-generation deployment from public claim-bearing deployment.

Open questions - NEEDS VERIFICATION: What model drift thresholds trigger ABSTAIN, DENY, or rollback?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 107; basis: machine-learning workflow section describing deployment after training, evaluation, and tuning; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-004, KFM-IDX-DOC-003, KFM-IDX-OBS-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.13 KFM-P18-INV-162 — Missing-data method provenance

Idea card metadata - Stable ID: KFM-P18-INV-162 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 162 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should record the missing-data handling method, missingness assumption, affected fields, and rationale as a transform on evidence-bearing analytical datasets.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record the missing-data handling method, missingness assumption, affected fields, and rationale as a transform on evidence-bearing analytical datasets. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It helps reviewers judge whether model outputs, indicators, and maps remain reliable after imputation, deletion, or advanced missing-data methods.

Related ideas - CONFIRMED baseline link: KFM-IDX-OBS-002. - CONFIRMED baseline link: KFM-IDX-POL-007.

Dependencies - PROPOSED: Transform receipt - PROPOSED: Data quality report - PROPOSED: Metric caveat field

Tensions - PROPOSED: Public readers may not understand missingness categories without concise explanations.

Expansion directions - PROPOSED: Build fixture comparisons for drop-row, mean/median imputation, KNN, and advanced methods.

Open questions - NEEDS VERIFICATION: Which missingness rates deny promotion versus require caveats?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 155; basis: data-analysis section concluding that missing-data technique depends on missingness, data type, and problem context; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-OBS-002, KFM-IDX-POL-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.14 KFM-P18-INV-164 — CNN feature-map provenance for image-derived evidence

Idea card metadata - Stable ID: KFM-P18-INV-164 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 164 - Status label: PROPOSED - Carry-forward state: NEW - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM should record model architecture, kernel/filter configuration class, input preprocessing, feature-map interpretation limits, and training evidence when CNN-derived image outputs support claims.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record model architecture, kernel/filter configuration class, input preprocessing, feature-map interpretation limits, and training evidence when CNN-derived image outputs support claims. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It is essential for remote-sensing, historic-image, or computer-vision layers where model output may be mistaken for direct observation.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-009. - CONFIRMED baseline link: KFM-IDX-AIG-007.

Dependencies - PROPOSED: Model card - PROPOSED: Remote-sensing source descriptor - PROPOSED: Image preprocessing receipt

Tensions - PROPOSED: Detailed CNN internals can be too complex for public users but still needed for reviewers.

Expansion directions - PROPOSED: Separate public explanation from technical model-provenance appendix.

Open questions - NEEDS VERIFICATION: When should CNN outputs be candidate evidence only rather than published assertions?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 200; basis: CNN section describing kernels/filters, sliding-window convolution, feature maps, and local spatial dependencies; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.15 KFM-P18-INV-176 — OCI digest-pinned geospatial artifact publication

Idea card metadata - Stable ID: KFM-P18-INV-176 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 176 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM map releases should support digest-pinned publication of PMTiles, GeoParquet, and COG artifacts through OCI-style artifact references when proof, catalog, and rollback metadata are present.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM map releases should support digest-pinned publication of PMTiles, GeoParquet, and COG artifacts through OCI-style artifact references when proof, catalog, and rollback metadata are present. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It gives map artifacts immutable addresses and rollback targets without treating storage tags as truth.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-004. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-SEC-008.

Dependencies - PROPOSED: TileArtifactManifest - PROPOSED: MapReleaseManifest - PROPOSED: Catalog record - PROPOSED: Signature verification

Tensions - PROPOSED: OCI tags are mutable unless the release pins digests.

Expansion directions - PROPOSED: Add OCI reference fields to tile/raster/vector artifact manifests only behind validation gates.

Open questions - NEEDS VERIFICATION: Which KFM release profiles are allowed to consume OCI-hosted geospatial artifacts?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 3, 10; basis: OCI/ORAS publication of PMTiles, GeoParquet, and COG artifacts with tags, digests, referrers, SBOM, and PROV; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-004, KFM-IDX-EVD-001, KFM-IDX-SEC-008 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.16 KFM-P18-INV-177 — OCI referrers as SBOM and PROV evidence links

Idea card metadata - Stable ID: KFM-P18-INV-177 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 177 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should treat OCI referrers for SBOM and PROV JSON-LD as evidence links that must be verified before public map artifacts are trusted.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat OCI referrers for SBOM and PROV JSON-LD as evidence links that must be verified before public map artifacts are trusted. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents PMTiles or COG files from being published as detached blobs without provenance and dependency context.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-003. - CONFIRMED baseline link: KFM-IDX-SEC-006. - CONFIRMED baseline link: KFM-IDX-OBS-005.

Dependencies - PROPOSED: Cosign or equivalent verification - PROPOSED: SBOM schema - PROPOSED: PROV schema - PROPOSED: Release gate

Tensions - PROPOSED: Referrer support varies by registry and requires operational verification.

Expansion directions - PROPOSED: Create fixture tests for missing SBOM, missing PROV, invalid attestation, and digest mismatch.

Open questions - NEEDS VERIFICATION: Which registry/referrer behavior is reliable enough for KFM release evidence?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 10, 13; basis: Cosign referrers linking SBOM and PROV JSON-LD to geospatial artifact releases; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-003, KFM-IDX-SEC-006, KFM-IDX-OBS-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.17 KFM-P18-INV-178 — MediaType and asset-role alignment across STAC, DCAT, and PROV

Idea card metadata - Stable ID: KFM-P18-INV-178 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 178 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM catalog closure should align mediaType, asset role, digest, and provenance fields across STAC, DCAT, and PROV for PMTiles, COG, and GeoParquet artifacts.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM catalog closure should align mediaType, asset role, digest, and provenance fields across STAC, DCAT, and PROV for PMTiles, COG, and GeoParquet artifacts. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It lets the UI and reviewers distinguish renderable tiles, raster assets, vector data, quicklooks, and metadata without guessing from filenames.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-001. - CONFIRMED baseline link: KFM-IDX-CAT-002. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: STAC profile - PROPOSED: DCAT distribution profile - PROPOSED: PROV activity - PROPOSED: Artifact manifest

Tensions - PROPOSED: Different catalog vocabularies can drift unless the mapping is tested as one closure unit.

Expansion directions - PROPOSED: Add a map-artifact asset-role matrix to catalog validators.

Open questions - NEEDS VERIFICATION: Should map artifacts have a dedicated KFM STAC extension or only profile-level required properties?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 10, 13; basis: STAC/DCAT/PROV mapping for GeoParquet, COG, and PMTiles with explicit media types and roles; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-001, KFM-IDX-CAT-002, KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.18 KFM-P18-INV-210 — 3D acquisition-method provenance for scan, photogrammetry, and CT

Idea card metadata - Stable ID: KFM-P18-INV-210 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 210 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM 3D assets should record acquisition method, processing workflow, resolution, scaling, georeferencing, and specialist attribution for CT, laser scan, photogrammetry, and image-based models.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM 3D assets should record acquisition method, processing workflow, resolution, scaling, georeferencing, and specialist attribution for CT, laser scan, photogrammetry, and image-based models. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes 3D evidence auditable and helps reviewers compare assets created by different acquisition paths.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-CAT-003.

Dependencies - PROPOSED: 3D acquisition receipt - PROPOSED: Processing provenance - PROPOSED: Georeference record - PROPOSED: Asset manifest

Tensions - PROPOSED: Different acquisition techniques carry different distortions and missing-data patterns.

Expansion directions - PROPOSED: Create 3DAssetAcquisitionReceipt with method, instrument/software, processing, scale, CRS, and uncertainty fields.

Open questions - NEEDS VERIFICATION: Which 3D metadata fields are mandatory for public release versus internal review?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 42, 43, 61; basis: CT scan, image-based 3D modeling, laser scanning, georeferencing, scaling, and full-resolution import into 3D GIS workflows; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-001, KFM-IDX-DAT-003, KFM-IDX-CAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.19 KFM-P18-INV-228 — Set-operator semantics for release deltas and intersections

Idea card metadata - Stable ID: KFM-P18-INV-228 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 228 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-003

Normalized statement - PROPOSED: KFM release-delta tooling should distinguish union, intersection, and difference semantics when comparing candidate, prior, and published record sets.

Detailed explanation - CONFIRMED: Source SRC-P18-003 (Advanced-SQL-Concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM release-delta tooling should distinguish union, intersection, and difference semantics when comparing candidate, prior, and published record sets. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It gives release reviewers an explicit language for additions, removals, retained records, and overlap checks.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-003. - CONFIRMED baseline link: KFM-IDX-CAT-006. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Prior release manifest - PROPOSED: Candidate manifest - PROPOSED: Stable record keys - PROPOSED: Delta report

Tensions - PROPOSED: Different SQL dialects expose different operator names and duplicate-handling behavior.

Expansion directions - PROPOSED: Add release_delta fixtures for added, removed, changed, retained, duplicate, and conflicted records.

Open questions - NEEDS VERIFICATION: Should duplicate handling be governed by release policy or query contract?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-003 — Advanced-SQL-Concepts.pdf; pages 21, 104; basis: UNION, UNION ALL, INTERSECT, EXCEPT/MINUS style operators combine, intersect, or difference datasets; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-003, KFM-IDX-CAT-006, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.20 KFM-P18-INV-233 — AI data ethics become release-gate evidence

Idea card metadata - Stable ID: KFM-P18-INV-233 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 233 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-004

Normalized statement - PROPOSED: KFM AI-related releases should include explicit privacy, bias, ownership, transparency, human-oversight, and misuse-review evidence before outputs become public or semi-public.

Detailed explanation - CONFIRMED: Source SRC-P18-004 (AI_Concepts_Using_Python.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM AI-related releases should include explicit privacy, bias, ownership, transparency, human-oversight, and misuse-review evidence before outputs become public or semi-public. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It converts generic responsible-AI language into a checkable release gate.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-005. - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: AI release review - PROPOSED: Source rights - PROPOSED: Bias audit - PROPOSED: Human oversight record

Tensions - PROPOSED: Ethics checklists can become performative if not tied to deny/abstain outcomes.

Expansion directions - PROPOSED: Add ai_ethics_review_receipt with reviewer, risk class, mitigations, release decision, and rollback trigger.

Open questions - NEEDS VERIFICATION: Which AI-assisted artifacts require ethics-board or steward review rather than maintainer-only approval?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-004 — AI_Concepts_Using_Python.pdf; pages 84, 357, 362, 366, 368; basis: AI data ethics include privacy, bias/fairness, ownership, transparency, ethics by design, human oversight, and documented review; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-005, KFM-IDX-POL-001, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.21 KFM-P18-INV-240 — Task-recorded geoprocessing workflows as replayable recipes

Idea card metadata - Stable ID: KFM-P18-INV-240 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 240 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED -

Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM GIS processing steps should be represented as replayable task recipes when a derived layer depends on multi-step geoprocessing.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM GIS processing steps should be represented as replayable task recipes when a derived layer depends on multi-step geoprocessing. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It turns manual GIS operations into auditable, reproducible workflow evidence.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: Workflow recipe - PROPOSED: Input layer manifest - PROPOSED: Tool parameter receipt - PROPOSED: Output digest

Tensions - PROPOSED: GUI-recorded tasks can still depend on unstated environment or version settings.

Expansion directions - PROPOSED: Add geoprocessing_task_receipt fields for tool names, parameters, input digests, software version, and output digest.

Open questions - NEEDS VERIFICATION: Which GIS workflows require task-recipe capture before derived release?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 75; basis: ArcGIS Task as a collection of steps or recipe for common workflow, using geoprocessing tools and summary statistics; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-004, KFM-IDX-EVD-001, KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.22 KFM-P18-INV-241 — Field Maps observations need attachment and GCP provenance

Idea card metadata - Stable ID: KFM-P18-INV-241 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 241 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM field observations should record mobile collection context, attachments, coordinate source, notes, and ground-control-point role where relevant.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM field observations should record mobile collection context, attachments, coordinate source, notes, and ground-control-point role where relevant. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes field-collected map evidence reviewable rather than treating mobile points as self-validating.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: FieldObservation schema - PROPOSED: Attachment manifest - PROPOSED: Coordinate method - PROPOSED: Consent/sensitivity policy

Tensions - PROPOSED: Photos and notes can expose sensitive locations, people, or infrastructure.

Expansion directions - PROPOSED: Add field_observation_receipt with device/app, coordinate_source, attachment_hashes, collector role, and sensitivity review.

Open questions - NEEDS VERIFICATION: What field-collection evidence is publishable versus steward-only?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 85, 86, 93, 96; basis: Field Maps feature layers, enabled attachments, RTK GPS coordinates, GCPs, notes, photos, and popup attributes; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-001, KFM-IDX-DAT-003, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.23 KFM-P18-INV-254 — Demographic enrichment in hazard workflows requires provenance and denominator receipts

Idea card metadata - Stable ID: KFM-P18-INV-254 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 254 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM hazard-exposure enrichment should record the demographic source, denominator, enrichment method, and missing-value filter before presenting who is affected.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea

should be treated as a governed design candidate: KFM hazard-exposure enrichment should record the demographic source, denominator, enrichment method, and missing-value filter before presenting who is affected. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents exposure maps from turning demographic enrichment into unsupported social claims.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Hazard layer - PROPOSED: demographic source descriptor - PROPOSED: aggregation denominator - PROPOSED: filter receipt

Tensions - PROPOSED: Commercial or hosted enrichment services may hide methodology, currency, or denominators.

Expansion directions - PROPOSED: Create `exposure_enrichment_receipt` with `source_version`, `geography_join`, `denominator`, `missing_count`, and `excluded_count`.

Open questions - NEEDS VERIFICATION: Should third-party demographic enrichment ever support authoritative public claims without independent source documentation?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 7, 8; basis: flood hazard areas, Enrich Layer analysis, summary statistics, socioeconomic characteristics, and missing-data filtering; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-POL-001, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.24 KFM-P18-INV-258 — Knowledge co-production and social learning as review outcomes

Idea card metadata - Stable ID: KFM-P18-INV-258 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 258 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM interactive GIS workshops should capture knowledge co-production and social-learning outcomes as review evidence, not merely as meeting summaries.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM interactive GIS workshops should capture knowledge co-production and social-learning outcomes as review evidence, not merely as meeting summaries. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release

state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes participatory learning traceable when it changes data interpretation, weights, or release decisions.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-MDP-003.

Dependencies - PROPOSED: Workshop record - PROPOSED: change decision - PROPOSED: participant role policy - PROPOSED: evidence update

Tensions - PROPOSED: Workshop evidence may be qualitative and privacy-sensitive.

Expansion directions - PROPOSED: Create workshop_review_receipt fields for issue, participants_by_role, artifacts_reviewed, changes_made, and sensitivity posture.

Open questions - NEEDS VERIFICATION: When should workshop-derived interpretations require steward approval before publication?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 83, 84, 85, 86, 87, 88; basis: interactive GIS-based approaches, knowledge co-production, social learning, shared understanding, and workshop discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-001, KFM-IDX-EVD-001, KFM-IDX-MDP-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.25 KFM-P18-INV-272 — 3D resolution dimensions become release and analysis constraints

Idea card metadata - Stable ID: KFM-P18-INV-272 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 272 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM 3D artifacts should record scalar, temporal, source, radiometric/spectral, and toolset resolution where those dimensions affect interpretation or release.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM 3D artifacts should record scalar, temporal, source, radiometric/spectral, and toolset resolution where those dimensions affect interpretation or release. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes 3D detail levels and acquisition capabilities explicit instead of treating all high-resolution models as equally reliable.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: 3D artifact manifest - PROPOSED: resolution profile - PROPOSED: tool capability record

Tensions - PROPOSED: High visual detail can exceed evidentiary certainty or public-release suitability.

Expansion directions - PROPOSED: Add `threed_resolution_profile` fields for `scale_range`, `temporal_events`, `source_formats`, `sensor_resolution`, `vertices`, `texture_size`, and `toolchain`.

Open questions - NEEDS VERIFICATION: Which resolution dimensions should trigger generalization or restricted access?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 8, 9, 42; basis: multi-resolution GIS, scalar resolution, temporal resolution, source resolution, toolset resolution, and high-resolution 3D model import; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-EVD-001, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.26 KFM-P18-INV-299 — Stakeholder MCA weights need decision provenance

Idea card metadata - Stable ID: KFM-P18-INV-299 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 299 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM multiple-criteria planning analyses should preserve stakeholder roles, criteria definitions, weights, constraints, and aggregation method as reviewable evidence.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM multiple-criteria planning analyses should preserve stakeholder roles, criteria definitions, weights, constraints, and aggregation method as reviewable evidence. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes planning recommendations auditable and exposes whose preferences shaped the output.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-004. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: criteria registry - PROPOSED: stakeholder role record - PROPOSED: MCA run receipt

Tensions - PROPOSED: A mathematically ranked output can obscure contested values and stakeholder power.

Expansion directions - PROPOSED: Add `criteria_weight_source` and `stakeholder_group` fields to MCA outputs.

Open questions - NEEDS VERIFICATION: How should KFM show disputed or minority stakeholder weights in public maps?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 238; basis: The sustainable urban GIS source presents stakeholder-based multiple-criteria analysis for designing cycle routes for different target populations.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-004, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.27 KFM-P18-INV-326 — WZDx GeoJSON as a governed transport-event source family

Idea card metadata - Stable ID: KFM-P18-INV-326 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 326 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should treat WZDx-style road-work and incident GeoJSON as a temporal transport-event source family that requires schema version, freshness, event namespace, catalog, and provenance checks before map preview.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat WZDx-style road-work and incident GeoJSON as a temporal transport-event source family that requires schema version, freshness, event namespace, catalog, and provenance checks before map preview. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents temporary road-event layers from bypassing transport evidence, time, and release controls.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-011. - CONFIRMED baseline link: KFM-IDX-PIP-004.

Dependencies - PROPOSED: WZDx schema/version - PROPOSED: event freshness policy - PROPOSED: STAC/PROV packaging - PROPOSED: PMTiles preview gate

Tensions - PROPOSED: Operational freshness can pressure publication speed, while KFM promotion requires reviewable evidence and rollback.

Expansion directions - PROPOSED: Add `transport_event_source_family` and `event_freshness` fields to layer manifests.

Open questions - NEEDS VERIFICATION: Which road-event feeds should be allowed as authoritative, contextual, or denied?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 88, 92, 235, 611, 621; basis: Master MapLibre v1.9 delta lists WZDx pipelines, event snapshots, PMTiles preview, STAC/PROV packaging, and schema/freshness validation requirements.; Phase 1 role: `candidate_synthesis_source_specialized_atlas`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-011, KFM-IDX-PIP-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.28 KFM-P18-INV-327 — Frictionless-style field validation before GeoJSON or TopoJSON tiling

Idea card metadata - Stable ID: KFM-P18-INV-327 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 327 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should validate GeoJSON and TopoJSON field types, geometry type, CRS, bbox, and manifest alignment before tile generation or catalog release.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should validate GeoJSON and TopoJSON field types, geometry type, CRS, bbox, and manifest alignment before tile generation or catalog release. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It catches broken or ambiguous map artifacts before they become public tiles or catalog records.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-007. - CONFIRMED baseline link: KFM-IDX-PIP-004.

Dependencies - PROPOSED: field schema - PROPOSED: geometry schema - PROPOSED: manifest entry - PROPOSED: catalog gate

Tensions - PROPOSED: Strict field validation can block exploratory layers unless candidate/nonpublication lanes are explicit.

Expansion directions - PROPOSED: Create a `pre_tile_geojson_topojson_validation` report for candidate map layers.

Open questions - NEEDS VERIFICATION: Should TopoJSON shared-arc checks be separate from generic geometry validation?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 78; basis: Master MapLibre v1.9 delta identifies Frictionless-style GeoJSON/TopoJSON gating as an early schema check before tile generation or catalog release.; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-007, KFM-IDX-PIP-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.29 KFM-P18-INV-328 — Energy and CO2 telemetry as a map-artifact promotion signal

Idea card metadata - Stable ID: KFM-P18-INV-328 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 328 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM map artifact promotion should allow sustainability telemetry such as energy_kwh and co2_kg to inform governance without turning carbon metrics into spatial truth claims.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM map artifact promotion should allow sustainability telemetry such as energy_kwh and co2_kg to inform governance without turning carbon metrics into spatial truth claims. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes expensive tile/raster builds auditable for sustainability and operational governance.

Related ideas - CONFIRMED baseline link: KFM-IDX-OBS-004. - CONFIRMED baseline link: KFM-IDX-EVD-006.

Dependencies - PROPOSED: RunReceipt - PROPOSED: energy telemetry - PROPOSED: CO2 estimate - PROPOSED: policy gate - PROPOSED: artifact provenance

Tensions - PROPOSED: Telemetry estimates may be approximate; treating them as exact measurements would overstate support.

Expansion directions - PROPOSED: Add sustainability_telemetry optional block to map build receipts and release summaries.

Open questions - NEEDS VERIFICATION: What telemetry precision is sufficient for governance but not misleading to public readers?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 5, 8; basis: Master MapLibre v1.9 delta describes promotion policies that may deny artifacts unless energy/CO2 telemetry and verified provenance are present.; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-OBS-004, KFM-IDX-EVD-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.30 KFM-P18-INV-330 — Sentinel-1 SAR layers need sovereignty, mask, and lineage controls

Idea card metadata - Stable ID: KFM-P18-INV-330 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 330 - Status label: PROPOSED - Carry-forward state: NEW - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM Sentinel-1 SAR-derived layers should carry sovereignty uncertainty, H3 sensitivity labels, mask_required flags, CARE/FAIR labels, and PROV lineage before public rendering.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM Sentinel-1 SAR-derived layers should carry sovereignty uncertainty, H3 sensitivity labels, mask_required flags, CARE/FAIR labels, and PROV lineage before public rendering. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It protects sensitive geography and steward obligations in derived remote-sensing products.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-004. - CONFIRMED baseline link: KFM-IDX-DAT-002.

Dependencies - PROPOSED: SAR source descriptor - PROPOSED: H3 sensitivity label - PROPOSED: mask receipt - PROPOSED: CARE/FAIR label - PROPOSED: PROV lineage

Tensions - PROPOSED: Coarse masks can reduce analytical utility; exact SAR-derived locations can expose sensitive contexts.

Expansion directions - PROPOSED: Add SAR_release_mask_profile and sovereignty_uncertainty fields to raster release manifests.

Open questions - NEEDS VERIFICATION: Which SAR use cases require DENY instead of generalized release?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 618, 621; basis: Master MapLibre v1.9 delta states Sentinel-1 workflows propagate h3_sensitive, mask_required, care_label, sovereignty_uncertainty, and PROV lineage.; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.31 KFM-P18-INV-346 — External ArcGIS web layer publication is separate from KFM promotion

Idea card metadata - Stable ID: KFM-P18-INV-346 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 346 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM should treat ArcGIS Online web-layer publication metadata and item names as external publishing context, not as proof that a layer is promoted, reviewed, or public-safe inside KFM.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat ArcGIS Online web-layer publication metadata and item names as external publishing context, not as proof that a layer is promoted, reviewed, or public-safe inside KFM. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents ordinary GIS sharing from being mistaken for governed publication.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-002. - CONFIRMED baseline link: KFM-IDX-CAT-001.

Dependencies - PROPOSED: external item id - PROPOSED: external metadata - PROPOSED: KFM release state - PROPOSED: PromotionDecision

Tensions - PROPOSED: Teams may see platform publication as sufficient because it makes a layer visible.

Expansion directions - PROPOSED: Add external_platform_publication_context to layer-source records.

Open questions - NEEDS VERIFICATION: Which external platform metadata fields should KFM import or ignore?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 8, 9; basis: Earth, Space, and Environmental Science Explorations with ArcGIS Pro shows publishing a web layer to ArcGIS Online with a unique name, summary, and tags.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-002, KFM-IDX-CAT-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.32 KFM-P18-INV-355 — Public-space social-interaction indicators need qualitative evidence links

Idea card metadata - Stable ID: KFM-P18-INV-355 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 355 - Status label: PROPOSED - Carry-forward state: NEW - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM urban social layers should link public-space social-interaction indicators to qualitative, observational, or participatory evidence rather than publishing an index as self-explanatory truth.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM urban social layers should link public-space social-interaction indicators to qualitative, observational, or participatory evidence rather than publishing an index as self-explanatory truth. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes social interaction claims inspectable and guards against treating a social index as a purely technical surface.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-POL-004.

Dependencies - PROPOSED: indicator definition - PROPOSED: field observation - PROPOSED: stakeholder review - PROPOSED: index method

Tensions - PROPOSED: Qualitative evidence may not fit simple numeric layer schemas.

Expansion directions - PROPOSED: Design mixed-method EvidenceBundle support for urban social indicators.

Open questions - NEEDS VERIFICATION: Which qualitative evidence forms should be admissible for public-space claims?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 17, 116; basis: GIS in Sustainable Urban Planning and Management includes a chapter on public spaces and social interaction in divided cities, using public spaces, social interaction, and index concepts.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.33 KFM-P18-INV-357 — Urban quality-of-life surfaces require indicator provenance

Idea card metadata - Stable ID: KFM-P18-INV-357 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 357 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from

CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM quality-of-life map layers should publish the indicator list, normalization method, spatial aggregation, and evidence sources that produced each composite surface.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM quality-of-life map layers should publish the indicator list, normalization method, spatial aggregation, and evidence sources that produced each composite surface. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Composite urban indicators are easy to consume but hard to audit unless their components and transformations are visible.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-004. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: indicator registry - PROPOSED: normalization method - PROPOSED: aggregation geography - PROPOSED: source descriptor

Tensions - PROPOSED: Composite scores can imply precision or fairness that the inputs do not support.

Expansion directions - PROPOSED: Add composite_indicator_receipt to planning-support layers.

Open questions - NEEDS VERIFICATION: How should indicator-weight uncertainty be shown in MapLibre layers?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 6, 14, 121; basis: GIS in Sustainable Urban Planning and Management describes urban quality-of-life analysis and GIS-based indicators in urban planning and policy contexts.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-004, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.34 KFM-P18-INV-364 — Trowel-edge 3D GIS as field-review support, not immediate publication

Idea card metadata - Stable ID: KFM-P18-INV-364 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 364 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-005

Normalized statement - PROPOSED: KFM should treat near-real-time 3D GIS field use as review support that can inform interpretation, while requiring later validation before public release.

Detailed explanation - CONFIRMED: Source SRC-P18-005 (Archaeological 3D GIS.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat near-real-time 3D GIS field use as review support that can inform interpretation, while requiring later validation before public release. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It preserves the usefulness of field interpretation without weakening promotion and review gates.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-DOC-002.

Dependencies - PROPOSED: field capture - PROPOSED: review note - PROPOSED: draft interpretation - PROPOSED: promotion gate

Tensions - PROPOSED: Fast field feedback can pressure teams to publish before validation.

Expansion directions - PROPOSED: Add draft_3d_review_state to field-derived 3D assets.

Open questions - NEEDS VERIFICATION: What validation threshold should move a field 3D model from WORK to PROCESSED?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-005 — Archaeological 3D GIS.pdf; pages 26, 62, 67; basis: Archaeological 3D GIS discusses deploying 3D GIS at the trowel's edge and using 3D models in support of real-time interpretation during field investigation.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-004, KFM-IDX-DOC-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.35 KFM-P18-INV-378 — Map metadata elements as minimum release disclosure

Idea card metadata - Stable ID: KFM-P18-INV-378 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 378 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM map-release records should require author or steward, publication date, purpose, projection, data sources, and scale/legend disclosure where relevant.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM map-release records should require author or steward, publication date, purpose, projection, data sources, and scale/legend disclosure where relevant. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture,

review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A map can look authoritative while hiding who made it, why, with which data, and at what representational scale.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-004. - CONFIRMED baseline link: KFM-IDX-DOC-007.

Dependencies - PROPOSED: MapReleaseManifest - PROPOSED: StyleManifest - PROPOSED: source ledger - PROPOSED: legend metadata

Tensions - PROPOSED: Public UI space is limited; some metadata may belong in an Evidence Drawer rather than on-map.

Expansion directions - PROPOSED: Create a minimum_public_map_metadata checklist.

Open questions - NEEDS VERIFICATION: Which metadata fields must be rendered visibly versus linked?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 59, 60; basis: A Primer of GIS lists essential and additional map elements, including legend, scale, title, author, date, purpose, projection, data sources, and gridlines.; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-004, KFM-IDX-DOC-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.36 KFM-P18-INV-391 — API lifecycle diagrams aligned to promotion states

Idea card metadata - Stable ID: KFM-P18-INV-391 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 391 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM APIs should document resource lifecycles in terms of governed states such as candidate, reviewed, released, corrected, withdrawn, or superseded.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM APIs should document resource lifecycles in terms of governed states such as candidate, reviewed, released, corrected, withdrawn, or superseded. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Evidence-facing APIs expose state transitions; documentation must not imply that POST/PUT/PATCH alone controls publication.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-004. - CONFIRMED baseline link: KFM-IDX-DOC-003.

Dependencies - PROPOSED: resource lifecycle - PROPOSED: promotion state - PROPOSED: HTTP verb - PROPOSED: DecisionEnvelope

Tensions - PROPOSED: HTTP method conventions can conflict with KFM promotion semantics unless documented.

Expansion directions - PROPOSED: Create lifecycle diagrams for ReleaseManifest, EvidenceBundle, and CorrectionNotice APIs.

Open questions - NEEDS VERIFICATION: How should KFM represent withdrawal and rollback over HTTP?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 35, 36, 38; basis: Designing Great Web APIs describes mapping resource lifecycles to HTTP verbs and mapping response codes during API design.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-004, KFM-IDX-DOC-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.37 KFM-P18-INV-413 — API versioning and contract-change notices as release duties

Idea card metadata - Stable ID: KFM-P18-INV-413 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 413 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM API contract changes should be versioned, documented, and tied to release/correction notices when they affect interpretation or client behavior.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM API contract changes should be versioned, documented, and tied to release/correction notices when they affect interpretation or client behavior. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: API contracts are part of the trust membrane; silent route or response changes can corrupt client-side evidence handling.

Related ideas - CONFIRMED baseline link: KFM-IDX-MDP-006. - CONFIRMED baseline link: KFM-IDX-PIP-009.

Dependencies - PROPOSED: API version registry - PROPOSED: change notice - PROPOSED: compatibility test - PROPOSED: client fixture

Tensions - PROPOSED: Too much version fragmentation can make API governance harder.

Expansion directions - PROPOSED: Link API contract changes to ReleaseManifest and rollback records.

Open questions - NEEDS VERIFICATION: Which response-field changes require a new major version?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 29, 35, 36; basis: Designing Great Web APIs covers resource ontology, URL relationships, HTTP verbs, response codes, and validating design through documentation.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-MDP-006, KFM-IDX-PIP-009 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.38 KFM-P18-INV-427 — Datum and coordinate-reference provenance for location claims

Idea card metadata - Stable ID: KFM-P18-INV-427 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 427 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM location-bearing claims should preserve coordinate reference system and datum provenance through intake, processing, tiling, and publication.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM location-bearing claims should preserve coordinate reference system and datum provenance through intake, processing, tiling, and publication. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A precise point can become misleading when its coordinate reference context is missing or silently transformed.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: SourceDescriptor - PROPOSED: geometry transform receipt - PROPOSED: CRS registry - PROPOSED: validator

Tensions - PROPOSED: Legacy sources may omit datum or use local conventions.

Expansion directions - PROPOSED: Create a CRS/datum provenance validator for geometry-bearing artifacts.

Open questions - NEEDS VERIFICATION: How should KFM handle sources whose datum is unknown but geometry is otherwise useful?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 53, 90, 94; basis: A Primer of GIS identifies coordinate sys-

tems and datums as basic geographic information concepts that influence how locations are represented and compared.; Phase 1 role: `candidate_synthesis_source`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-012, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.39 KFM-P18-INV-442 — Social vulnerability enrichment as a derived exposure layer

Idea card metadata - Stable ID: KFM-P18-INV-442 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 442 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-010

Normalized statement - PROPOSED: KFM hazard and planning maps should treat social vulnerability enrichment as a derived exposure layer with source, method, and equity caveats.

Detailed explanation - CONFIRMED: Source SRC-P18-010 (Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM hazard and planning maps should treat social vulnerability enrichment as a derived exposure layer with source, method, and equity caveats. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Exposure overlays can inform decisions but can also stigmatize communities if treated as raw truth.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-002. - CONFIRMED baseline link: KFM-IDX-DAT-010.

Dependencies - PROPOSED: hazard layer - PROPOSED: demographic source descriptor - PROPOSED: equity review - PROPOSED: EvidenceBundle

Tensions - PROPOSED: Demographic enrichment may use proprietary or generalized data.

Expansion directions - PROPOSED: Add `equity_caveat` and `enrichment_method` to exposure layer manifests.

Open questions - NEEDS VERIFICATION: Which social indicators are safe for public precision at neighborhood scale?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-010 — Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf; pages 3, 53, 69; basis: Earth, Space, and Environmental Science Explorations with ArcGIS Pro includes a chapter on assessing social vulnerability in areas impacted by Hurricane Florence.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-002, KFM-IDX-DAT-010 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.40 KFM-P18-INV-448 — OCI registry references for geospatial artifact provenance

Idea card metadata - Stable ID: KFM-P18-INV-448 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 448 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM map artifacts may reference OCI registry digests only when catalog, provenance, signature, and rollback metadata are present.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM map artifacts may reference OCI registry digests only when catalog, provenance, signature, and rollback metadata are present. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: OCI can strengthen reproducibility, but only if it remains downstream of KFM release governance.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-001. - CONFIRMED baseline link: KFM-IDX-SEC-006.

Dependencies - PROPOSED: TileArtifactManifest - PROPOSED: STAC/DCAT/PROV records - PROPOSED: Cosign attestation - PROPOSED: rollback manifest

Tensions - PROPOSED: Operational OCI support and referrers need future verification.

Expansion directions - PROPOSED: Add OCI artifact fields to the map artifact validation backlog.

Open questions - NEEDS VERIFICATION: Which artifact classes should be eligible for OCI-backed publication first?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 3, 4, 10, 11, 17; basis: Master MapLibre v1.9 records OCI/ORAS publication of PMTiles, GeoParquet, and COG artifacts with Cosign, STAC/DCAT/PROV mapping, digests, and referrers.; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-001, KFM-IDX-SEC-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.41 KFM-P18-INV-451 — Positional method provenance for location claims

Idea card metadata - Stable ID: KFM-P18-INV-451 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 451 - Status label: PROPOSED - Carry-forward state: NEW - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM location-bearing claims should record whether coordinates came from surveying, GPS/GNSS, digitization, or indirect measurement, together with the positional method and confidence relevant to the claim.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM location-bearing claims should record whether coordinates came from surveying, GPS/GNSS, digitization, or indirect measurement, together with the positional method and confidence relevant to the claim. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Coordinate values alone do not explain how location evidence was produced or how much confidence a reviewer should place in it.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-001. - CONFIRMED baseline link: KFM-IDX-DOC-008.

Dependencies - PROPOSED: source descriptor - PROPOSED: geometry record - PROPOSED: positional accuracy metadata - PROPOSED: EvidenceBundle

Tensions - PROPOSED: Detailed provenance can be costly for legacy sources with weak metadata.

Expansion directions - PROPOSED: Add a positional_method field to geometry/source fixtures before public spatial joins are treated as strong evidence.

Open questions - NEEDS VERIFICATION: Which legacy geographies can only support generalized method labels?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 150; basis: The GIS primer states that reliable geographic information depends on systematic location recording and identifies surveying, GPS, and digitization as generic ways of recording locations and characteristics.; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.42 KFM-P18-INV-474 — LayerCatalog search points to catalog records, not raw databases

Idea card metadata - Stable ID: KFM-P18-INV-474 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 474 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from

CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM layer-search UI should return cataloged layer records and released artifact references, not raw database handles or internal query access.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM layer-search UI should return cataloged layer records and released artifact references, not raw database handles or internal query access. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Search is a public trust surface; it must not become a backdoor into internal stores.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-CAT-001.

Dependencies - PROPOSED: LayerCatalog - PROPOSED: catalog record - PROPOSED: released layer - PROPOSED: EvidenceDrawerPayload

Tensions - PROPOSED: Search indexes can drift from release catalogs if not rebuilt deterministically.

Expansion directions - PROPOSED: Add no_raw_db_handle validation to layer-search fixtures.

Open questions - NEEDS VERIFICATION: How should stale search records be withdrawn after rollback?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 20, 24, 25; basis: The Master MapLibre atlas states that LayerCatalog search should point to catalog records, not raw DB, and reiterates no direct raw/canonical/internal store access for public UI.; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-001, KFM-IDX-CAT-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.10.43 KFM-P18-INV-494 — Published language for catalog and schema vocabulary

Idea card metadata - Stable ID: KFM-P18-INV-494 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 494 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: REL — Catalog Closure, Publication, Release, Rollback, and Recompile Discipline - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should maintain a published language for recurring catalog, schema, policy, evidence, release, and correction terms used across domain lanes.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded

this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should maintain a published language for recurring catalog, schema, policy, evidence, release, and correction terms used across domain lanes. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A published language reduces ambiguity when public APIs, docs, validators, and AI explanations use the same trust vocabulary.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-003. - CONFIRMED baseline link: KFM-IDX-MDP-003.

Dependencies - PROPOSED: published language - PROPOSED: schema vocabulary - PROPOSED: catalog profile - PROPOSED: documentation

Tensions - PROPOSED: Published language must evolve without freezing useful domain distinctions.

Expansion directions - PROPOSED: Add glossary-to-schema crosswalk fragments in Phase 7 front matter.

Open questions - NEEDS VERIFICATION: Which terms from Phase 3 need canonical glossary treatment first?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 43; basis: The DDD reference states that translations between bounded contexts require a common language when direct model translation is unsuitable.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-003, KFM-IDX-MDP-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11 API - Governed API, Resource Ontology, HTTP, and Service Contracts

Status: CONFIRMED source support for grouped Phase 3 ideas; PROPOSED category synthesis

Definition: Governed API design, resource ontology, HTTP semantics, endpoint exposure, documentation, and service contracts.

CONFIRMED: This chapter contains 47 Pass 18 idea cards assigned to API.

Idea ID	Title	Carry state	Status	Source IDs
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KFM-P18-INV-025 | APIs as explicit contracts at the trust membrane | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-040 | Design-first API process for governed interfaces | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-041 | HTTP lifecycle semantics for finite governed outcomes | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-042 | API security considered at design time | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-066 | Resource ontology before governed route implementation | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-095 | Published Language and open-host service for cross-lane integration | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-098 | HTTP response-code design should align with finite governed outcomes | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-099 | API documentation and prototypes are validation artifacts | NEW | PROPOSED | SRC-P18-006 |

KFM-P18-INV-100 | Scraping terms and robots constraints as acquisition gates | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-116 | Resource ontology for governed URL design | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-117 | API security and role constraints at design time | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-140 | Resource lifecycle to HTTP verb mapping as governed API evidence | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-141 | HTTP response-code map for finite trust outcomes | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-142 | API documentation and prototypes as validation surfaces | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-151 | Field-level authorization as API trust-membrane control | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-153 | API walkthrough and use-case validation as pre-implementation gates | NEW | PROPOSED | SRC-P18-006 |

KFM-P18-INV-154 | HTTP request and response fields as envelope metadata | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-199 | Third-party API provider stability as source-admission evidence | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-222 | Resource ontology for evidence API contract structure | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-223 | HTTP verb and response-code mapping to finite outcomes | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-224 | Wireframe-driven API validation for map and evidence workflows | NEW | PROPOSED | SRC-P18-006 |

KFM-P18-INV-236 | Published language and open-host services for governed APIs | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-247 | Stealth scraping and proxy evasion are denied by default | NEW | PROPOSED | SRC-P18-012 |

KFM-P18-INV-274 | API capability exposure requires business-purpose and trust-boundary review | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-276 | API documentation is the public contract surface | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-277 | Wireframe-driven API requirements expose map interaction gaps | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-278 | API model walkthroughs become validation evidence | NEW | PROPOSED | SRC-P18-006 |

KFM-P18-INV-279 | Resource ontology controls URL and relationship design | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-280 | HTTP response codes map finite trust outcomes | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-285 | Ajax and form scraping require request-state capture | NEW | PROPOSED | SRC-P18-012 |

KFM-P18-INV-301 | Self-contained API documentation as trust evidence for external consumers | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-302 | Wireframe-derived API call budgets for mobile map workflows | NEW | PROPOSED | SRC-P18-006 |

KFM-P18-INV-334 | API audience class as a contract and exposure field | NEW | PROPOSED | SRC-P18-006 |

KFM-P18-INV-336 | Scraping redirects and rate limits as source-adapter evidence | EXPANDED | PROPOSED | SRC-P18-012 |

KFM-P18-INV-350 | Published language as a public API and evidence vocabulary boundary | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-366 | Resource ontology before URL and lifecycle design | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-367 | HTTP response-code mapping should align with finite trust outcomes | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-390 | Design-first prototypes as API fail-fast gates | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-412 | API onboarding and developer-experience evidence | NEW | PROPOSED | SRC-P18-006 |

KFM-P18-INV-436 | API documentation prototypes as pre-implementation review gates | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-437 | HTTP verb constraints for governed resource mutation | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-441 | Open-host service boundary for governed public APIs | EXPANDED | PROPOSED | SRC-P18-009 |

KFM-P18-INV-460 | Resource ontology as governed API object map | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-461 | HTTP response codes mapped to finite KFM outcomes | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-486 | User-interface wireframes as API evidence tests | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-487 | Resource URL relationship topology for governed navigation | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-488 | HTTP verb constraints mapped to resource lifecycle | EXPANDED | PROPOSED | SRC-P18-006 |

8.11.1 KFM-P18-INV-025 — APIs as explicit contracts at the trust membrane

Idea card metadata - Stable ID: KFM-P18-INV-025 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 25 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should design governed APIs as contract surfaces before implementation, with authorization, resource lifecycle, and response semantics tied to evidence and policy.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should design governed APIs as contract surfaces before implementation, with authorization, resource lifecycle, and response semantics tied to evidence and policy. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A map-first UI can only remain trustworthy if API responses preserve finite outcomes, citations, policy decisions, and release state.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-001. - CONFIRMED baseline link: KFM-IDX-DOC-003. - CONFIRMED baseline link: KFM-IDX-AIG-004.

Dependencies - PROPOSED: OpenAPI/DTO contracts; RuntimeResponseEnvelope; policy pre/post checks

Tensions - PROPOSED: Generic API usability guidance must be constrained by KFM cite-or-abstain and fail-closed behavior.

Expansion directions - PROPOSED: Define governed API design checklist for evidence-bearing endpoints.

Open questions - NEEDS VERIFICATION: Which KFM API response codes map to DENY, ABSTAIN, ERROR, and ordinary HTTP statuses?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 7, 24, 29, 35, 37; basis: API as contract, design-first process, authorization, HTTP request/verb design; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-001, KFM-IDX-DOC-003, KFM-IDX-AIG-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.2 KFM-P18-INV-040 — Design-first API process for governed interfaces

Idea card metadata - Stable ID: KFM-P18-INV-040 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 40 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should design governed APIs from user activities, resource candidates, and documentation before implementation details harden.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should design governed APIs from user activities, resource candidates, and documentation before implementation details harden. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents public clients from depending on undocumented internal stores or accidental routes.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-006. - CONFIRMED baseline link: KFM-IDX-AIG-004.

Dependencies - PROPOSED: Activity model - PROPOSED: resource ontology - PROPOSED: OpenAPI/contract draft - PROPOSED: developer-facing documentation

Tensions - PROPOSED: Over-design can delay proof slices if not bounded to the smallest useful interface.

Expansion directions - PROPOSED: Create API activity maps for EvidenceBundle resolution and layer click workflows.

Open questions - NEEDS VERIFICATION: Which KFM API contracts need design-first review before the first proof slice?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 29, 33, 35; basis: design-first API process and resource-candidate discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-006, KFM-IDX-AIG-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.3 KFM-P18-INV-041 — HTTP lifecycle semantics for finite governed outcomes

Idea card metadata - Stable ID: KFM-P18-INV-041 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 41 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource

Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should map resource lifecycles, HTTP verbs, and response-code families to finite governed API outcomes without exposing raw policy internals.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should map resource lifecycles, HTTP verbs, and response-code families to finite governed API outcomes without exposing raw policy internals. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes API behavior predictable while preserving KFM policy and evidence distinctions.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-003. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: RuntimeResponseEnvelope - PROPOSED: HTTP status mapping - PROPOSED: error taxonomy - PROPOSED: policy decision separation

Tensions - PROPOSED: HTTP 4xx/5xx categories do not fully express KFM abstention or sensitivity denial.

Expansion directions - PROPOSED: Create a KFM API outcome-to-HTTP mapping table.

Open questions - NEEDS VERIFICATION: What exact HTTP status mapping should be used for ABSTAIN and DENY?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 37, 38; basis: HTTP verbs, requests, responses, and response-code families; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-003, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.4 KFM-P18-INV-042 — API security considered at design time

Idea card metadata - Stable ID: KFM-P18-INV-042 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 42 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should treat authentication, authorization, and data-leakage controls as API design requirements, not post-release patches.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat authentication, authorization, and data-leakage controls as API design requirements, not post-release patches. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It protects the trust membrane when internal interfaces become partner or public surfaces.

Related ideas - CONFIRMED baseline link: KFM-IDX-SEC-009. - CONFIRMED baseline link: KFM-IDX-POL-003.

Dependencies - PROPOSED: Authn/authz model - PROPOSED: data-leak review - PROPOSED: least-privilege endpoints - PROPOSED: public-surface policy

Tensions - PROPOSED: Security design must still be validated by implementation and runtime tests.

Expansion directions - PROPOSED: Add public_api_exposure_review to release gates.

Open questions - NEEDS VERIFICATION: Which internal KFM APIs are forbidden from becoming public without an ADR?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 23, 24; basis: API authentication, authorization, and data-leakage design discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-SEC-009, KFM-IDX-POL-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.5 KFM-P18-INV-066 — Resource ontology before governed route implementation

Idea card metadata - Stable ID: KFM-P18-INV-066 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 66 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should define API resource ontology, relationships, lifecycle verbs, response codes, documentation, and prototypes before hardening public governed API routes.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should define API resource ontology, relationships, lifecycle verbs, response codes, documentation, and prototypes before hardening public governed API routes. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and

rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, run-time routing, or release state.

Why it matters - PROPOSED: It keeps the API from exposing implementation stores or ad hoc policy state before resource meaning is settled.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-004. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: OpenAPI or equivalent contract - PROPOSED: finite outcome mapping

Tensions - PROPOSED: A developer-friendly API must still withhold raw or unpublished truth-path stores.

Expansion directions - PROPOSED: Create a governed_resource_ontology artifact before endpoint implementation.

Open questions - NEEDS VERIFICATION: How should ABSTAIN and DENY map to HTTP status codes without leaking sensitive reasons?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 1, 29, 33, 35, 36, 37; basis: API as contract, resource ontology, HTTP verbs, response codes, documentation and prototyping; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-004, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.6 KFM-P18-INV-095 — Published Language and open-host service for cross-lane integration

Idea card metadata - Stable ID: KFM-P18-INV-095 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 95 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should use published languages and open-host service patterns for shared cross-lane contracts instead of forcing every domain to translate every other domain directly.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should use published languages and open-host service patterns for shared cross-lane contracts instead of forcing every domain to translate every other domain directly. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It gives KFM a principled way to expose stable APIs and schemas across hydrology, ecology, settlement, hazards, and UI lanes.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-006. - CONFIRMED baseline link: KFM-IDX-CAT-001.

Dependencies - PROPOSED: Shared schema registry - PROPOSED: Governed API DTOs - PROPOSED: Published language docs

Tensions - PROPOSED: A shared language can freeze too much if it becomes a dumping ground.

Expansion directions - PROPOSED: Define which KFM contracts are published language versus lane-internal language.

Open questions - NEEDS VERIFICATION: Which object families need a published-language contract in Pass 18 categories?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 42, 43; basis: Open-host Service and Published Language definitions; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-006, KFM-IDX-CAT-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.7 KFM-P18-INV-098 — HTTP response-code design should align with finite governed outcomes

Idea card metadata - Stable ID: KFM-P18-INV-098 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 98 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should map API success and error response codes during design so they align with finite governed outcomes such as ANSWER, ABSTAIN, DENY, and ERROR.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should map API success and error response codes during design so they align with finite governed outcomes such as ANSWER, ABSTAIN, DENY, and ERROR. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents public clients from receiving ambiguous failures or treating policy denials as ordinary missing data.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-007. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: RuntimeResponseEnvelope - PROPOSED: OpenAPI/contract design - PROPOSED: Error taxonomy

Tensions - PROPOSED: HTTP status codes alone do not explain evidence or policy posture.

Expansion directions - PROPOSED: Create `response_code_to_kfm_outcome` mapping with citation and policy failure cases.

Open questions - NEEDS VERIFICATION: Which HTTP status should represent ABSTAIN without hiding that the system behaved correctly?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 9, 41, 42; basis: HTTP verbs, response code mapping, and design-phase error handling; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-007, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.8 KFM-P18-INV-099 — API documentation and prototypes are validation artifacts

Idea card metadata - Stable ID: KFM-P18-INV-099 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 99 - Status label: PROPOSED - Carry-forward state: NEW - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should treat API documentation and prototypes as validation artifacts for governed contract design, not as polish after implementation.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat API documentation and prototypes as validation artifacts for governed contract design, not as polish after implementation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes API contract errors, missing failure modes, and developer-experience gaps visible before implementation hardens.

Related ideas - CONFIRMED baseline link: KFM-IDX-MDP-001. - CONFIRMED baseline link: KFM-IDX-CAT-001.

Dependencies - PROPOSED: API contract docs - PROPOSED: Prototype fixtures - PROPOSED: Developer review checklist

Tensions - PROPOSED: Beautiful docs can mask incomplete policy and evidence behavior if examples are too happy-path.

Expansion directions - PROPOSED: Add `prototype_receipt` and `documented_failure_mode` coverage checks.

Open questions - NEEDS VERIFICATION: Should KFM require examples for ABSTAIN, DENY, and stale evidence in every public API doc?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 13, 16, 17, 18, 43; basis: API documentation, completeness, interactive docs, and prototyping validation; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.9 KFM-P18-INV-100 — Scraping terms and robots constraints as acquisition gates

Idea card metadata - Stable ID: KFM-P18-INV-100 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 100 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM should treat website terms, robots instructions, rate limits, and acquisition conduct as source-admission gates for any scraping fallback.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat website terms, robots instructions, rate limits, and acquisition conduct as source-admission gates for any scraping fallback. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps fallback acquisition from bypassing rights, operational burden, reproducibility, and public-source trust requirements.

Related ideas - CONFIRMED baseline link: KFM-IDX-SEC-006. - CONFIRMED baseline link: KFM-IDX-PIP-006.

Dependencies - PROPOSED: SourceDescriptor rights fields - PROPOSED: Acquisition policy - PROPOSED: Fetcher receipt

Tensions - PROPOSED: Scraping may be technically possible but legally, ethically, or operationally inappropriate.

Expansion directions - PROPOSED: Add robots_policy, terms_review, rate_limit_policy, user_agent, and scraping_allowed fields to source descriptors.

Open questions - NEEDS VERIFICATION: What evidence is enough to promote a scraped source out of quarantine?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 8, 10, 11, 17, 18; basis: scraping table of contents, robustness, persistence/tests, concurrency, rate limits, robots.txt, and terms of service; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-SEC-006, KFM-IDX-PIP-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.10 KFM-P18-INV-116 — Resource ontology for governed URL design

Idea card metadata - Stable ID: KFM-P18-INV-116 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 116 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM APIs should define resource ontology and relationships before URL design so routes reflect evidence, release, review, and correction objects.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM APIs should define resource ontology and relationships before URL design so routes reflect evidence, release, review, and correction objects. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It avoids route sprawl and keeps public interfaces aligned with KFM object families.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-001. - CONFIRMED baseline link: KFM-IDX-DOC-006.

Dependencies - PROPOSED: Resource ontology register - PROPOSED: OpenAPI contract

Tensions - PROPOSED: A route can look stable while encoding the wrong domain boundary.

Expansion directions - PROPOSED: Add resource_ontology.md and relation-to-route mappings.

Open questions - NEEDS VERIFICATION: Which KFM resources are top-level: claims, layers, bundles, releases, or domains?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 5, 20, 33, 35; basis: resource ontology, URL relationships, and resource lifecycle mapping discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-001, KFM-IDX-DOC-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.11 KFM-P18-INV-117 — API security and role constraints at design time

Idea card metadata - Stable ID: KFM-P18-INV-117 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 117 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from

CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM governed APIs should specify authentication, authorization, role scopes, and restricted-data behavior during API design, not after endpoints exist.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM governed APIs should specify authentication, authorization, role scopes, and restricted-data behavior during API design, not after endpoints exist. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps sensitive, unreleased, and policy-denied evidence from leaking through apparently ordinary endpoints.

Related ideas - CONFIRMED baseline link: KFM-IDX-SEC-001. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: Access-control policy - PROPOSED: API threat model - PROPOSED: Fixture tests

Tensions - PROPOSED: Late security design can leave public paths around raw or candidate data.

Expansion directions - PROPOSED: Add security-by-design checklist to API contract acceptance.

Open questions - NEEDS VERIFICATION: How should API docs expose DENY behavior without revealing restricted details?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 22, 23; basis: security at the start, authentication, role, and data restriction discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-SEC-001, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.12 KFM-P18-INV-140 — Resource lifecycle to HTTP verb mapping as governed API evidence

Idea card metadata - Stable ID: KFM-P18-INV-140 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 140 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should document how governed resource lifecycles map to HTTP verbs so read, review, correction, promotion, and withdrawal actions remain explicit.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category

before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should document how governed resource lifecycles map to HTTP verbs so read, review, correction, promotion, and withdrawal actions remain explicit. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents arbitrary endpoints from bypassing governed state transitions.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-002. - CONFIRMED baseline link: KFM-IDX-AIG-006.

Dependencies - PROPOSED: OpenAPI contract - PROPOSED: PromotionDecision - PROPOSED: CorrectionNotice

Tensions - PROPOSED: Some governed actions may require custom POST actions but still need policy checks.

Expansion directions - PROPOSED: Add lifecycle-to-route matrices for EvidenceBundle, ReleaseManifest, and CorrectionNotice APIs.

Open questions - NEEDS VERIFICATION: Which actions should be command resources rather than direct state mutation?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 35, 41; basis: mapping resource lifecycles to HTTP verbs in the web API design source; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-002, KFM-IDX-AIG-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.13 KFM-P18-INV-141 — HTTP response-code map for finite trust outcomes

Idea card metadata - Stable ID: KFM-P18-INV-141 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 141 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should bind HTTP status codes to finite trust outcomes so ANSWER, ABSTAIN, DENY, and ERROR are visible through API behavior as well as payload fields.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should bind HTTP status codes to finite trust outcomes so ANSWER, ABSTAIN, DENY, and ERROR are visible through API behavior as well as payload fields. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms

the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reduces ambiguity between missing evidence, forbidden access, failed validation, and system errors.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-003. - CONFIRMED baseline link: KFM-IDX-AIG-008.

Dependencies - PROPOSED: RuntimeResponseEnvelope - PROPOSED: DecisionEnvelope - PROPOSED: OpenAPI contract

Tensions - PROPOSED: HTTP status alone cannot carry all evidence and policy context.

Expansion directions - PROPOSED: Create an API outcome matrix for public, review, and admin surfaces.

Open questions - NEEDS VERIFICATION: Should ABSTAIN map to 200 with outcome payload or a distinct non-error response?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 36, 42; basis: mapping response codes in API endpoint design; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-003, KFM-IDX-AIG-008 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.14 KFM-P18-INV-142 — API documentation and prototypes as validation surfaces

Idea card metadata - Stable ID: KFM-P18-INV-142 - Pass: 18 - Phase 5 batch: 006 - Inventory sequence: 142 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 006. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should treat API documentation, mock payloads, and prototypes as validation surfaces that can reveal missing evidence, policy, and lifecycle fields before implementation.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat API documentation, mock payloads, and prototypes as validation surfaces that can reveal missing evidence, policy, and lifecycle fields before implementation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It lets reviewers catch trust-membrane gaps before route handlers or UI components harden them.

Related ideas - CONFIRMED baseline link: KFM-IDX-MDP-003. - CONFIRMED baseline link: KFM-IDX-AIG-006.

Dependencies - PROPOSED: OpenAPI examples - PROPOSED: Fixture registry - PROPOSED: Review console mocks

Tensions - PROPOSED: Prototype convenience can accidentally become production expectation.

Expansion directions - PROPOSED: Require prototype payloads to validate against the same schemas as production fixtures.

Open questions - NEEDS VERIFICATION: Which API prototypes must be included in release review?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 37, 43; basis: API design validation through documentation and prototyping; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-MDP-003, KFM-IDX-AIG-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.15 KFM-P18-INV-151 — Field-level authorization as API trust-membrane control

Idea card metadata - Stable ID: KFM-P18-INV-151 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 151 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should treat endpoint authorization and field-level data access as design-time contract requirements, not only runtime middleware settings.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat endpoint authorization and field-level data access as design-time contract requirements, not only runtime middleware settings. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It helps prevent a governed API from returning sensitive, unpublished, or role-inappropriate evidence fields even when a user is authenticated.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-POL-003. - CONFIRMED baseline link: KFM-IDX-SEC-009.

Dependencies - PROPOSED: API contract - PROPOSED: PolicyDecision - PROPOSED: Role and obligation model

Tensions - PROPOSED: Fine-grained field authorization can complicate cacheability and reusable DTOs.

Expansion directions - PROPOSED: Add fixture cases for role-specific hidden fields, revoked grants, and downstream Evidence Drawer payloads.

Open questions - NEEDS VERIFICATION: Which fields are universally public, role-scoped, embargoed, or never returned?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 23, 24; basis: API design security discussion of authentication, authorization, data leakage, and field/functional access rules; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-001, KFM-IDX-POL-003, KFM-IDX-SEC-009 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.16 KFM-P18-INV-153 — API walkthrough and use-case validation as pre-implementation gates

Idea card metadata - Stable ID: KFM-P18-INV-153 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 153 - Status label: PROPOSED - Carry-forward state: NEW - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should require API model walkthroughs against map screens, Evidence Drawer behavior, machine-to-machine use cases, and business-process diagrams before freezing governed API routes.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should require API model walkthroughs against map screens, Evidence Drawer behavior, machine-to-machine use cases, and business-process diagrams before freezing governed API routes. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes missing evidence resolution, authorization, workflow, and finite-outcome cases visible before implementation hardens an incomplete contract.

Related ideas - CONFIRMED baseline link: KFM-IDX-MDP-003. - CONFIRMED baseline link: KFM-IDX-DOC-003.

Dependencies - PROPOSED: API model - PROPOSED: UI wireframes - PROPOSED: Use-case fixtures

Tensions - PROPOSED: Design walkthroughs can become ceremony unless tied to executable fixtures.

Expansion directions - PROPOSED: Convert walkthrough findings into contract tests and route-level fixture matrices.

Open questions - NEEDS VERIFICATION: Which API design reviews must block implementation versus create backlog items?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 34; basis: design-first API process section naming API walkthroughs, use-case validation, and business-process diagrams; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.17 KFM-P18-INV-154 — HTTP request and response fields as envelope metadata

Idea card metadata - Stable ID: KFM-P18-INV-154 - Pass: 18 - Phase 5 batch: 007 - Inventory sequence: 154 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 007. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should record the HTTP method, route, request headers/body class, response code, response headers, and payload class in API receipts for trust-critical calls.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record the HTTP method, route, request headers/body class, response code, response headers, and payload class in API receipts for trust-critical calls. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It gives auditors enough transport-level context to distinguish retrieval, mutation, accepted processing, denial, abstention, and server failure cases.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-010. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-DOC-003.

Dependencies - PROPOSED: RuntimeResponseEnvelope - PROPOSED: RunReceipt - PROPOSED: API logging policy

Tensions - PROPOSED: Logging headers or bodies can expose secrets or sensitive data if not redacted.

Expansion directions - PROPOSED: Create a redacted API-call receipt profile for trust-membrane requests.

Open questions - NEEDS VERIFICATION: Which headers are safe to retain, hashed, redacted, or forbidden in receipts?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 37, 38; basis: HTTP primer listing request verb, URL, request header/body and response code/header/body; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-010, KFM-IDX-EVD-001, KFM-IDX-DOC-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.18 KFM-P18-INV-199 — Third-party API provider stability as source-admission evidence

Idea card metadata - Stable ID: KFM-P18-INV-199 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 199 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should treat third-party API longevity, versioning process, documentation quality, and provider dependency as SourceDescriptor review fields before relying on an external API.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat third-party API longevity, versioning process, documentation quality, and provider dependency as SourceDescriptor review fields before relying on an external API. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reduces brittle dependence on external services whose changes could break evidence pipelines or public clients.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-008. - CONFIRMED baseline link: KFM-IDX-SEC-011. - CONFIRMED baseline link: KFM-IDX-CAT-005.

Dependencies - PROPOSED: SourceDescriptor - PROPOSED: Version policy - PROPOSED: Fallback plan - PROPOSED: Source budget

Tensions - PROPOSED: A useful API can still be unsuitable as an authoritative or stable KFM source.

Expansion directions - PROPOSED: Add API-provider-risk fields to source admission and watcher design.

Open questions - NEEDS VERIFICATION: Which current KFM candidate sources rely on unstable third-party APIs rather than authoritative feeds?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 11, 43; basis: third-party API selection concerns including provider longevity, documented upgrade/versioning process, documentation, and design validation through documentation/prototyping; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-008, KFM-IDX-SEC-011, KFM-IDX-CAT-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.19 KFM-P18-INV-222 — Resource ontology for evidence API contract structure

Idea card metadata - Stable ID: KFM-P18-INV-222 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 222 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from

CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM governed APIs should use a resource ontology that names evidence, releases, layers, claims, reviews, and corrections consistently through URL/resource structure.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM governed APIs should use a resource ontology that names evidence, releases, layers, claims, reviews, and corrections consistently through URL/resource structure. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, run-time routing, or release state.

Why it matters - PROPOSED: It aligns API structure with KFM object families instead of exposing implementation tables or ad hoc routes.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-AIG-007. - CONFIRMED baseline link: KFM-IDX-CAT-005.

Dependencies - PROPOSED: API resource map - PROPOSED: Published language - PROPOSED: OpenAPI contract - PROPOSED: Governed API boundary

Tensions - PROPOSED: Poor resource naming can leak internal store concepts or flatten lifecycle distinctions.

Expansion directions - PROPOSED: Draft a governed API resource ontology for EvidenceBundle, ReleaseManifest, LayerManifest, ReviewRecord, and CorrectionNotice.

Open questions - NEEDS VERIFICATION: Which KFM object families deserve public resources versus internal-only routes?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 7, 20, 39; basis: API as contract; resource ontology for naming and typing resources and relationships through URL structure; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-001, KFM-IDX-AIG-007, KFM-IDX-CAT-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.20 KFM-P18-INV-223 — HTTP verb and response-code mapping to finite outcomes

Idea card metadata - Stable ID: KFM-P18-INV-223 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 223 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM API contracts should map resource lifecycle actions to HTTP verbs and response-code families while preserving finite KFM outcomes such as ANSWER, ABSTAIN, DENY, and ERROR.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM API contracts should map resource lifecycle actions to HTTP verbs and response-code families while preserving finite KFM outcomes such as ANSWER, ABSTAIN, DENY, and ERROR. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps protocol-level responses from obscuring policy, evidence, and validation decisions.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-007. - CONFIRMED baseline link: KFM-IDX-POL-005. - CONFIRMED baseline link: KFM-IDX-DOC-001.

Dependencies - PROPOSED: Runtime response envelope - PROPOSED: OpenAPI contract - PROPOSED: Policy decision - PROPOSED: Error taxonomy

Tensions - PROPOSED: HTTP 200 with an ABSTAIN body is different from HTTP 404 or HTTP 403; both layers need clarity.

Expansion directions - PROPOSED: Add API contract fixtures for ANSWER, ABSTAIN, DENY, ERROR, 2xx/4xx/5xx mapping, and problem-details fields.

Open questions - NEEDS VERIFICATION: How should KFM distinguish evidence absence, policy denial, invalid request, stale release, and server failure at API level?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 36, 38, 41; basis: HTTP verbs for retrieve/create/update/delete, response-code families, and resource lifecycle mapping to API actions; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-007, KFM-IDX-POL-005, KFM-IDX-DOC-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.21 KFM-P18-INV-224 — Wireframe-driven API validation for map and evidence workflows

Idea card metadata - Stable ID: KFM-P18-INV-224 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 224 - Status label: PROPOSED - Carry-forward state: NEW - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should validate governed API designs against map, Evidence Drawer, review-console, and Focus Mode wireframes before implementation hardens endpoints.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should validate governed API designs against map, Evidence Drawer, review-console, and Focus Mode wireframes before implementation hardens endpoints. - PROPOSED: Phase 3 scope note: Inventory-level

candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It ensures API contracts support real trust-visible workflows rather than only developer-oriented data access.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-AIG-007. - CONFIRMED baseline link: KFM-IDX-MDP-005.

Dependencies - PROPOSED: UI workflow model - PROPOSED: API prototype - PROPOSED: Documentation feedback - PROPOSED: Contract test

Tensions - PROPOSED: Wireframes can overfit to current UI ideas unless grounded in KFM object families and policy states.

Expansion directions - PROPOSED: Create API walkthrough fixtures for map click, evidence resolution, abstention, review, correction, and rollback workflows.

Open questions - NEEDS VERIFICATION: Which UI journeys are mandatory acceptance cases for governed API v1?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 30, 43; basis: UI wireframes drive API design, expose API gaps, and early documentation/prototyping validates API design through feedback; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.22 KFM-P18-INV-236 — Published language and open-host services for governed APIs

Idea card metadata - Stable ID: KFM-P18-INV-236 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 236 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM public and partner APIs should expose a documented published language for evidence, release, layer, review, and correction resources rather than leaking internal models.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM public and partner APIs should expose a documented published language for evidence, release, layer, review, and correction resources rather than leaking internal models. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It gives KFM stable integration contracts without freezing internal implementation or canonical stores as public APIs.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-AIG-007. - CONFIRMED baseline link: KFM-IDX-CAT-005.

Dependencies - PROPOSED: API resource ontology - PROPOSED: Published language spec - PROPOSED: OpenAPI contract - PROPOSED: Compatibility policy

Tensions - PROPOSED: A published language can accidentally become too broad and expose internal state.

Expansion directions - PROPOSED: Draft a KFM Published Language appendix for API objects, finite outcomes, citations, policy labels, and versioning.

Open questions - NEEDS VERIFICATION: Which internal object families should never be part of the public published language?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 42, 43; basis: open-host service defines a protocol for integration; published language provides a well-documented shared communication medium; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-001, KFM-IDX-AIG-007, KFM-IDX-CAT-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.23 KFM-P18-INV-247 — Stealth scraping and proxy evasion are denied by default

Idea card metadata - Stable ID: KFM-P18-INV-247 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 247 - Status label: PROPOSED - Carry-forward state: NEW - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM should deny stealth, proxy-evasion, user-agent disguise, or distributed scraping tactics unless a documented source agreement explicitly permits the behavior.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should deny stealth, proxy-evasion, user-agent disguise, or distributed scraping tactics unless a documented source agreement explicitly permits the behavior. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It aligns source acquisition with rights, source integrity, and fail-closed public-interest posture.

Related ideas - CONFIRMED baseline link: KFM-IDX-SEC-010. - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-PIP-008.

Dependencies - PROPOSED: Source terms review - PROPOSED: Connector policy - PROPOSED: Security review - PROPOSED: Deny log

Tensions - PROPOSED: Some technical scraping guides optimize for access rather than governance or rights.

Expansion directions - PROPOSED: Add connector_policy deny rules for user-agent spoofing, proxy rotation, stealth, and source-term violations.

Open questions - NEEDS VERIFICATION: Are there any KFM sources for which automated scraping is acceptable without an API and explicit terms review?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 49, 50, 64; basis: header disguise, proxy collection, and multi-server scraping are described as ways to avoid scraper detection; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.24 KFM-P18-INV-274 — API capability exposure requires business-purpose and trust-boundary review

Idea card metadata - Stable ID: KFM-P18-INV-274 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 274 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM API surfaces should expose capabilities only after their purpose, audience, contract, and trust-boundary consequence are documented.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM API surfaces should expose capabilities only after their purpose, audience, contract, and trust-boundary consequence are documented. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It aligns API design with KFM governance rather than treating routes as technical plumbing.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-006. - CONFIRMED baseline link: KFM-IDX-DOC-007. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: API design brief - PROPOSED: contract schema - PROPOSED: audience label - PROPOSED: trust-boundary review

Tensions - PROPOSED: A useful API can still bypass governance if it exposes raw or unreleased capability.

Expansion directions - PROPOSED: Add api_capability_review fields for audience, capability, allowed_data_states, prohibited_use, documentation_url, and finite_outcomes.

Open questions - NEEDS VERIFICATION: Which API capabilities should be internal-only even when technically safe?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 4, 5, 10; basis: APIs expose business capabilities, serve internal/external developers, act as contracts, and documentation shapes developer experience; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-006, KFM-IDX-DOC-007, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.25 KFM-P18-INV-276 — API documentation is the public contract surface

Idea card metadata - Stable ID: KFM-P18-INV-276 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 276 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM governed APIs should be designed as externally inspectable contracts because consumers do not have access to internal code, database diagrams, or private rationale.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM governed APIs should be designed as externally inspectable contracts because consumers do not have access to internal code, database diagrams, or private rationale. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes API clarity a governance and evidence-delivery obligation, not a developer-experience afterthought.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-004. - CONFIRMED baseline link: KFM-IDX-DOC-003. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: governed API contract - PROPOSED: developer-facing documentation - PROPOSED: EvidenceBundle response envelope

Tensions - PROPOSED: Internal implementation may be mature while the public contract is still ambiguous.

Expansion directions - PROPOSED: Add contract-readiness checks before exposing any new API surface.

Open questions - NEEDS VERIFICATION: Which API documents should be release-gating artifacts for KFM public clients?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 29; basis: The web API source explains that API consumers rely on design and documentation rather than source code or internal knowledge bases.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-004, KFM-IDX-DOC-003, KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.26 KFM-P18-INV-277 — Wireframe-driven API requirements expose map interaction gaps

Idea card metadata - Stable ID: KFM-P18-INV-277 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 277 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM map and review UI wireframes should be used to identify API call chains, missing operations, and mobile/network burdens before endpoint implementation.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM map and review UI wireframes should be used to identify API call chains, missing operations, and mobile/network burdens before endpoint implementation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It helps keep MapLibre, Evidence Drawer, and review-console APIs aligned with actual user tasks while avoiding ad hoc endpoints.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-AIG-005.

Dependencies - PROPOSED: UI task model - PROPOSED: API model - PROPOSED: mobile/offline budget

Tensions - PROPOSED: A wireframe can reveal needs but does not by itself prove source, evidence, or policy admissibility.

Expansion directions - PROPOSED: Create an API walkthrough artifact for each trust-visible UI flow.

Open questions - NEEDS VERIFICATION: How should KFM record wireframe-derived API requirements without making UI sketches canonical truth?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 30; basis: The API source states that web/mobile wireframes can drive API design by exposing interactions, gaps, and number of API calls, especially on unreliable mobile networks.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-AIG-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.27 KFM-P18-INV-278 — API model walkthroughs become validation evidence

Idea card metadata - Stable ID: KFM-P18-INV-278 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 278 - Status label: PROPOSED - Carry-forward state: NEW - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should treat API walkthroughs, use-case validation, and developer documentation review as evidence that a governed API model satisfies its intended users and tasks.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat API walkthroughs, use-case validation, and developer documentation review as evidence that a governed API model satisfies its intended users and tasks. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It creates a reviewable bridge between proposed endpoints and actual governed workflows.

Related ideas - CONFIRMED baseline link: KFM-IDX-MDP-002. - CONFIRMED baseline link: KFM-IDX-DOC-003.

Dependencies - PROPOSED: API model - PROPOSED: use-case inventory - PROPOSED: prototype feedback receipt

Tensions - PROPOSED: Walkthrough validation can confirm usability while implementation maturity remains UNKNOWN.

Expansion directions - PROPOSED: Add an APIModelValidationReport object for schema/API changes.

Open questions - NEEDS VERIFICATION: Should API walkthrough reports be required for internal-only governed API surfaces?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 34, 43; basis: The source recommends validating the API model through walkthroughs, use cases, documentation, prototypes, and early feedback.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.28 KFM-P18-INV-279 — Resource ontology controls URL and relationship design

Idea card metadata - Stable ID: KFM-P18-INV-279 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 279 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM API URL structures should be derived from a resource ontology that names resource families and relationships before path design.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM API URL structures should be derived from a resource ontology that names resource families and relationships before path design. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents endpoint sprawl and aligns API naming with KFM object-family boundaries.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-003. - CONFIRMED baseline link: KFM-IDX-CAT-001.

Dependencies - PROPOSED: resource ontology - PROPOSED: object-family register - PROPOSED: Directory Rules path review

Tensions - PROPOSED: Resource ontology may differ from repository directory placement; object meaning and file home are separate decisions.

Expansion directions - PROPOSED: Map API resources to KFM object families before endpoint naming.

Open questions - NEEDS VERIFICATION: Which object families deserve public API resources versus internal DTOs only?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 39; basis: The source defines an API ontology as the classification of resources and relationships, realized through URL structure.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-003, KFM-IDX-CAT-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.29 KFM-P18-INV-280 — HTTP response codes map finite trust outcomes

Idea card metadata - Stable ID: KFM-P18-INV-280 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 280 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM governed APIs should map success, abstention, denial, missing evidence, policy refusal, and system errors to documented response-code and envelope behavior.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate:

KFM governed APIs should map success, abstention, denial, missing evidence, policy refusal, and system errors to documented response-code and envelope behavior. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps public clients from interpreting every non-200 response as the same failure and preserves finite KFM outcomes.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-004. - CONFIRMED baseline link: KFM-IDX-GAT-005.

Dependencies - PROPOSED: DecisionEnvelope - PROPOSED: RuntimeResponseEnvelope - PROPOSED: OpenAPI contract

Tensions - PROPOSED: HTTP codes alone are insufficient unless the body envelope carries KFM truth/outcome labels.

Expansion directions - PROPOSED: Add endpoint matrices for ANSWER, ABSTAIN, DENY, ERROR, STALE, and NOT_FOUND cases.

Open questions - NEEDS VERIFICATION: Which KFM finite outcomes should share HTTP status codes but differ by envelope reason?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 41, 42; basis: The source explains mapping resource lifecycles to HTTP verbs and mapping success and error response codes during design.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-004, KFM-IDX-GAT-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.30 KFM-P18-INV-285 — Ajax and form scraping require request-state capture

Idea card metadata - Stable ID: KFM-P18-INV-285 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 285 - Status label: PROPOSED - Carry-forward state: NEW - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM connectors for Ajax or form-driven pages should capture form inputs, generated requests, JavaScript execution assumptions, and hidden data paths before admitting results.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM connectors for Ajax or form-driven pages should capture form inputs, generated requests, JavaScript execution assumptions, and hidden data paths before admitting results. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source

support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It preserves the actual acquisition path and avoids treating rendered browser content as if it were a simple static page fetch.

Related ideas - CONFIRMED baseline link: KFM-IDX-SRC-001. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: request-state receipt - PROPOSED: form parameter log - PROPOSED: JavaScript execution policy

Tensions - PROPOSED: Executing client-side code may change source access, terms, or security posture.

Expansion directions - PROPOSED: Add acquisition_path fields: static_html, form_submit, ajax_request, js_executed, api_endpoint_discovered.

Open questions - NEEDS VERIFICATION: When does scraping an Ajax backing endpoint become a separate API source requiring its own SourceDescriptor?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 55; basis: The scraping source explains that Ajax can issue additional server requests after page load and that client-side JavaScript can hide or encode data.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.31 KFM-P18-INV-301 — Self-contained API documentation as trust evidence for external consumers

Idea card metadata - Stable ID: KFM-P18-INV-301 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 301 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM governed APIs should be documented as if consumers cannot see source code, internal diagrams, or informal team knowledge.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM governed APIs should be documented as if consumers cannot see source code, internal diagrams, or informal team knowledge. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes API documentation a primary trust surface rather than an after-thought.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-006. - CONFIRMED baseline link: KFM-IDX-MDP-002.

Dependencies - PROPOSED: OpenAPI or equivalent contract - PROPOSED: public developer documentation - PROPOSED: citation/finite-outcome notes

Tensions - PROPOSED: Too much internal detail can leak implementation risk, while too little detail blocks audit and reuse.

Expansion directions - PROPOSED: Add documentation_completeness and consumer_visibility checks to governed API reviews.

Open questions - NEEDS VERIFICATION: Which API docs must be public, partner-only, or steward-only?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 29; basis: Web API source states API consumers often only have design and documentation, not source code or internal knowledge, and therefore API design must emphasize simplicity and clarity.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-006, KFM-IDX-MDP-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.32 KFM-P18-INV-302 — Wireframe-derived API call budgets for mobile map workflows

Idea card metadata - Stable ID: KFM-P18-INV-302 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 302 - Status label: PROPOSED - Carry-forward state: NEW - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM map and evidence workflows should use UI wireframes to estimate API calls, latency risk, and offline/degraded-state needs before route implementation.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM map and evidence workflows should use UI wireframes to estimate API calls, latency risk, and offline/degraded-state needs before route implementation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, run-time routing, or release state.

Why it matters - PROPOSED: It connects UX design to reliability, evidence access, and degraded-state planning.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-AIG-006.

Dependencies - PROPOSED: wireframe - PROPOSED: route sketch - PROPOSED: call count estimate - PROPOSED: degraded-state policy

Tensions - PROPOSED: Fine-grained APIs are easier to govern but may overload mobile workflows.

Expansion directions - PROPOSED: Add api_call_budget and degraded_mode fields to map workflow design notes.

Open questions - NEEDS VERIFICATION: What call-count threshold should trigger offline package or batch endpoint review?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 30; basis: Web API source says UI wireframes expose API gaps and the number of calls required, especially important for mobile applications where HTTP calls occur over unreliable networks.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.33 KFM-P18-INV-334 — API audience class as a contract and exposure field

Idea card metadata - Stable ID: KFM-P18-INV-334 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 334 - Status label: PROPOSED - Carry-forward state: NEW - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM APIs should classify each endpoint or resource as public, partner, steward, internal, or denied so API design aligns with exposure, documentation, and trust-membrane obligations.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM APIs should classify each endpoint or resource as public, partner, steward, internal, or denied so API design aligns with exposure, documentation, and trust-membrane obligations. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes API exposure a reviewable governance decision instead of an implicit route-level assumption.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-003. - CONFIRMED baseline link: KFM-IDX-MDP-002.

Dependencies - PROPOSED: API contract - PROPOSED: audience class - PROPOSED: documentation boundary - PROPOSED: access policy

Tensions - PROPOSED: Over-broad public classifications can leak data; over-restrictive classifications can block legitimate reuse.

Expansion directions - PROPOSED: Add api_audience_class to governed API route registry drafts.

Open questions - NEEDS VERIFICATION: How should endpoint audience class interact with release-state and EvidenceBundle sensitivity?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 1, 4; basis: Designing Great Web APIs frames APIs as contracts that share business capabilities and data across internal, partner, and public developer audiences.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.34 KFM-P18-INV-336 — Scraping redirects and rate limits as source-adaptor evidence

Idea card metadata - Stable ID: KFM-P18-INV-336 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 336 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-012

Normalized statement - PROPOSED: KFM scraping adapters should record redirects, rate limits, dynamic forms, and Ajax dependencies as source-adaptor risk evidence rather than treating scraped HTML as a clean source.

Detailed explanation - CONFIRMED: Source SRC-P18-012 (instant-web-scraping-with-java.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM scraping adapters should record redirects, rate limits, dynamic forms, and Ajax dependencies as source-adaptor risk evidence rather than treating scraped HTML as a clean source. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It preserves why a scraped source is brittle, high-risk, or unsuitable for promotion.

Related ideas - CONFIRMED baseline link: KFM-IDX-PIP-002. - CONFIRMED baseline link: KFM-IDX-SEC-009.

Dependencies - PROPOSED: source terms - PROPOSED: redirect log - PROPOSED: rate-limit log - PROPOSED: form/Ajax interaction receipt

Tensions - PROPOSED: Recording interactions can increase stored sensitive data if forms or session tokens are captured.

Expansion directions - PROPOSED: Add scraper_interaction_risk fields to source intake receipts.

Open questions - NEEDS VERIFICATION: Which scraping interactions should automatically quarantine the source?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-012 — instant-web-scraping-with-java.pdf; pages 1, 22, 46; basis: Instant Web Scraping with Java introduces redirects, IP rate limiters, forms, Ajax pages, and error handling as ordinary scraping challenges.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-PIP-002, KFM-IDX-SEC-009 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.35 KFM-P18-INV-350 — Published language as a public API and evidence vocabulary boundary

Idea card metadata - Stable ID: KFM-P18-INV-350 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 350 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM should use published language to stabilize public API terms for evidence, release, review, correction, and map-layer resources while insulating internal bounded-context vocabulary.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should use published language to stabilize public API terms for evidence, release, review, correction, and map-layer resources while insulating internal bounded-context vocabulary. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps public consumers from depending on unstable internal terms while preserving inspectable semantics.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-006. - CONFIRMED baseline link: KFM-IDX-MDP-002.

Dependencies - PROPOSED: published vocabulary - PROPOSED: internal model boundary - PROPOSED: API contract - PROPOSED: context map

Tensions - PROPOSED: A published language can ossify too early if adopted before domain review.

Expansion directions - PROPOSED: Add published_language_review to API contract chapters.

Open questions - NEEDS VERIFICATION: Which terms must remain public-stable across future passes?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 35, 36; basis: Domain-Driven Design Reference defines Published Language and Open-host Service as strategic context-mapping patterns.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-006, KFM-IDX-MDP-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.36 KFM-P18-INV-366 — Resource ontology before URL and lifecycle design

Idea card metadata - Stable ID: KFM-P18-INV-366 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 366 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource

Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should define resource ontology and lifecycle states before finalizing governed API URLs, verbs, response envelopes, and authorization checks.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should define resource ontology and lifecycle states before finalizing governed API URLs, verbs, response envelopes, and authorization checks. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents route naming from hardening before KFM knows the evidence, release, review, and correction resources being exposed.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-006. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: resource ontology - PROPOSED: lifecycle state - PROPOSED: HTTP verb mapping - PROPOSED: authorization policy

Tensions - PROPOSED: A too-rigid ontology can block domain growth.

Expansion directions - PROPOSED: Create an API resource ontology review step for evidence-facing APIs.

Open questions - NEEDS VERIFICATION: Which resource families should be stable public names in Pass 18?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 35, 36, 37; basis: Designing Great Web APIs frames API modeling-to-design transition around HTTP, URLs, resource ontology, lifecycles, verbs, and request structure.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-006, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.37 KFM-P18-INV-367 — HTTP response-code mapping should align with finite trust outcomes

Idea card metadata - Stable ID: KFM-P18-INV-367 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 367 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM governed APIs should map HTTP response codes to finite trust outcomes without collapsing DENY, ABSTAIN, ERROR, and ordinary malformed-request failures.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded

this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM governed APIs should map HTTP response codes to finite trust outcomes without collapsing DENY, ABSTAIN, ERROR, and ordinary malformed-request failures. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, run-time routing, or release state.

Why it matters - PROPOSED: It keeps developer-facing transport semantics aligned with KFM governance semantics.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-003. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: RuntimeResponseEnvelope - PROPOSED: PolicyDecision - PROPOSED: HTTP status map - PROPOSED: API documentation

Tensions - PROPOSED: HTTP status alone cannot carry the full reason, obligation, and citation state.

Expansion directions - PROPOSED: Add a response-code-to-outcome matrix to governed API docs.

Open questions - NEEDS VERIFICATION: Which errors are policy DENY versus technical ERROR?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 38, 39; basis: Designing Great Web APIs explains response codes including 204, 400, 401, 403, and 404 as part of API response design.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-003, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.38 KFM-P18-INV-390 — Design-first prototypes as API fail-fast gates

Idea card metadata - Stable ID: KFM-P18-INV-390 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 390 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM governed API changes should pass a design-first prototype or walkthrough gate before implementation hardens URLs, verbs, response shapes, and client expectations.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM governed API changes should pass a design-first prototype or walkthrough gate before implementation hardens URLs, verbs, response shapes, and client expectations. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms

the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, run-time routing, or release state.

Why it matters - PROPOSED: It catches mismatches between evidence workflow, UI needs, and public contract before routes become entrenched.

Related ideas - CONFIRMED baseline link: KFM-IDX-MDP-003. - CONFIRMED baseline link: KFM-IDX-DOC-006.

Dependencies - PROPOSED: API model - PROPOSED: prototype - PROPOSED: documentation draft - PROPOSED: review report

Tensions - PROPOSED: Prototype artifacts can be mistaken for implemented behavior.

Expansion directions - PROPOSED: Add `api_design_prototype_status` to API ADRs.

Open questions - NEEDS VERIFICATION: Which governed API changes require prototype review?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 24, 28, 37; basis: Designing Great Web APIs describes design-first process, API modeling, and validating design through documentation and prototyping.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-MDP-003, KFM-IDX-DOC-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.39 KFM-P18-INV-412 — API onboarding and developer-experience evidence

Idea card metadata - Stable ID: KFM-P18-INV-412 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 412 - Status label: PROPOSED - Carry-forward state: NEW - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM governed APIs should treat onboarding, documentation, examples, and prototype validation as evidence that external consumers can use the interface safely.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM governed APIs should treat onboarding, documentation, examples, and prototype validation as evidence that external consumers can use the interface safely. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A technically correct API can still cause misuse if consumers cannot understand resource meaning, status codes, citation duties, or policy outcomes.

Related ideas - CONFIRMED baseline link: KFM-IDX-MDP-003. - CONFIRMED baseline link: KFM-IDX-AIG-007.

Dependencies - PROPOSED: API contract - PROPOSED: documentation fixture - PROPOSED: example response - PROPOSED: consumer validation

Tensions - PROPOSED: Developer convenience must not bypass trust and policy boundaries.

Expansion directions - PROPOSED: Add `developer_experience_validation` to public API readiness checks.

Open questions - NEEDS VERIFICATION: What examples are safe to include without exposing sensitive or unreleased data?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 1, 7, 37; basis: Designing Great Web APIs frames APIs as contracts and emphasizes documentation, prototyping, and developer experience as part of API value.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.40 KFM-P18-INV-436 — API documentation prototypes as pre-implementation review gates

Idea card metadata - Stable ID: KFM-P18-INV-436 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 436 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM governed API changes should pass documentation/prototype review before implementation when they expose trust-bearing public behavior.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM governed API changes should pass documentation/prototype review before implementation when they expose trust-bearing public behavior. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Prototype review catches contract ambiguity before route code hardens into public authority.

Related ideas - CONFIRMED baseline link: KFM-IDX-MDP-002. - CONFIRMED baseline link: KFM-IDX-DOC-004.

Dependencies - PROPOSED: OpenAPI draft - PROPOSED: DTO schema - PROPOSED: review checklist - PROPOSED: consumer fixture

Tensions - PROPOSED: Prototypes can imply behavior that is not yet implemented.

Expansion directions - PROPOSED: Add a `docs_prototype_review` status to API change receipts.

Open questions - NEEDS VERIFICATION: Which API changes can skip prototype review as internal-only?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 5, 37, 43; basis: Designing Great Web APIs describes validating API design through documentation and prototyping.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-MDP-002, KFM-IDX-DOC-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.41 KFM-P18-INV-437 — HTTP verb constraints for governed resource mutation

Idea card metadata - Stable ID: KFM-P18-INV-437 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 437 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM API contracts should restrict mutating HTTP verbs to governed lifecycle actions and keep publication separate from ordinary create/update calls.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM API contracts should restrict mutating HTTP verbs to governed lifecycle actions and keep publication separate from ordinary create/update calls. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: HTTP method design can accidentally imply that publication is a simple write rather than a governed transition.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-002. - CONFIRMED baseline link: KFM-IDX-DOC-003.

Dependencies - PROPOSED: API contract - PROPOSED: PromotionDecision - PROPOSED: PolicyDecision - PROPOSED: route tests

Tensions - PROPOSED: HTTP conventions do not fully encode KFM trust outcomes.

Expansion directions - PROPOSED: Add verb-to-lifecycle assertions in API contract tests.

Open questions - NEEDS VERIFICATION: How should ABSTAIN be represented in HTTP without confusing it with server failure?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 33, 35, 36, 41; basis: Designing Great Web APIs maps resource lifecycles to HTTP verbs and response codes.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-002, KFM-IDX-DOC-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN:

Repository implementation status remains unverified.

8.11.42 KFM-P18-INV-441 — Open-host service boundary for governed public APIs

Idea card metadata - Stable ID: KFM-P18-INV-441 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 441 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-009

Normalized statement - PROPOSED: KFM public APIs should operate as open-host services exposing a published language while hiding internal canonical stores and lifecycle stages.

Detailed explanation - CONFIRMED: Source SRC-P18-009 (Domain-Driven Design Reference.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM public APIs should operate as open-host services exposing a published language while hiding internal canonical stores and lifecycle stages. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: The public API can be stable and useful without exposing RAW, WORK, QUARANTINE, or internal schema churn.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-004. - CONFIRMED baseline link: KFM-IDX-MDP-001.

Dependencies - PROPOSED: governed API - PROPOSED: PublishedLanguage - PROPOSED: trust membrane - PROPOSED: API docs

Tensions - PROPOSED: Published language can lag behind internal domain evolution.

Expansion directions - PROPOSED: Document open-host service rules in API governance fragments.

Open questions - NEEDS VERIFICATION: When should a breaking change create a new published language version?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-009 — Domain-Driven Design Reference.pdf; pages 3, 42, 43; basis: Domain-Driven Design Reference defines Open-host Service and Published Language as strategic design patterns.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-004, KFM-IDX-MDP-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.43 KFM-P18-INV-460 — Resource ontology as governed API object map

Idea card metadata - Stable ID: KFM-P18-INV-460 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 460 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED -

Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM API design should start from a resource ontology that names governed objects, their relationships, and their lifecycle semantics before URLs or handlers are finalized.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM API design should start from a resource ontology that names governed objects, their relationships, and their lifecycle semantics before URLs or handlers are finalized. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, run-time routing, or release state.

Why it matters - PROPOSED: API endpoints should expose KFM governed objects rather than accidental implementation tables.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-006. - CONFIRMED baseline link: KFM-IDX-DAT-001.

Dependencies - PROPOSED: object-family registry - PROPOSED: OpenAPI draft - PROPOSED: URL design - PROPOSED: domain language

Tensions - PROPOSED: A resource ontology can drift from domain language unless reviewed with stewards.

Expansion directions - PROPOSED: Build resource-ontology fragments before Phase 5 API-facing idea cards.

Open questions - NEEDS VERIFICATION: Which object families need public API resources versus internal-only records?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 33, 39; basis: Designing Great Web APIs includes building a resource ontology and defining URLs through relationships as part of detailed API design.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-006, KFM-IDX-DAT-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.44 KFM-P18-INV-461 — HTTP response codes mapped to finite KFM outcomes

Idea card metadata - Stable ID: KFM-P18-INV-461 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 461 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM governed APIs should map HTTP response codes to finite outcomes such as ANSWER, ABSTAIN, DENY, and ERROR without conflating transport failure with policy denial.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category

before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM governed APIs should map HTTP response codes to finite outcomes such as ANSWER, ABSTAIN, DENY, and ERROR without conflating transport failure with policy denial. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Users need to know whether a response failed technically, was denied by policy, or abstained for lack of evidence.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-003. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: governed API contract - PROPOSED: DecisionEnvelope - PROPOSED: error taxonomy - PROPOSED: client UI states

Tensions - PROPOSED: HTTP status alone cannot carry KFM evidence and policy semantics.

Expansion directions - PROPOSED: Add response-code mapping tests for DecisionEnvelope fixtures.

Open questions - NEEDS VERIFICATION: Which HTTP statuses should be reserved for policy DENY versus evidence ABSTAIN?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 35, 36, 38; basis: Designing Great Web APIs discusses mapping resource lifecycles to HTTP verbs, mapping response codes, and validating design through documentation and prototyping.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-003, KFM-IDX-POL-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.45 KFM-P18-INV-486 — User-interface wireframes as API evidence tests

Idea card metadata - Stable ID: KFM-P18-INV-486 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 486 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM governed API design should test Evidence Drawer, Focus Mode, review, and map-click workflows against wireframes before locking resource contracts.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM governed API design should test Evidence Drawer, Focus Mode, review, and map-click workflows against wireframes before locking resource contracts. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the

planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Trust-critical UI flows need APIs that expose evidence, policy, and release state, not just raw data access.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-004. - CONFIRMED baseline link: KFM-IDX-DAT-012.

Dependencies - PROPOSED: API design - PROPOSED: wireframe - PROPOSED: governed API - PROPOSED: Evidence Drawer

Tensions - PROPOSED: Wireframes can overfit to current UI concepts if not tied to durable contract tests.

Expansion directions - PROPOSED: Add wireframe_to_contract trace rows for key map and evidence work-flows.

Open questions - NEEDS VERIFICATION: Which public shell flows should become mandatory API acceptance tests?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 30; basis: The web API source states that interface wireframes can drive a top-down view of API usage and expose gaps in required API calls.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-004, KFM-IDX-DAT-012 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.46 KFM-P18-INV-487 — Resource URL relationship topology for governed navigation

Idea card metadata - Stable ID: KFM-P18-INV-487 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 487 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM API surfaces should expose relationships among claims, evidence bundles, catalog records, releases, and rollback targets through governed resource links.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM API surfaces should expose relationships among claims, evidence bundles, catalog records, releases, and rollback targets through governed resource links. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Navigable relationships make source support and correction lineage inspectable without giving clients direct access to internal stores.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-003. - CONFIRMED baseline link: KFM-IDX-AIG-004.

Dependencies - PROPOSED: resource ontology - PROPOSED: URL relationship - PROPOSED: OpenAPI - PROPOSED: catalog

Tensions - PROPOSED: Too many links can confuse clients and leak implementation structure.

Expansion directions - PROPOSED: Define link relation names for evidence, policy decision, release, correction, and rollback.

Open questions - NEEDS VERIFICATION: Which relations belong in public envelopes versus admin-only review payloads?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 33, 41; basis: The web API source describes resource ontology and defining URLs through relationships as API-design steps.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-CAT-003, KFM-IDX-AIG-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.11.47 KFM-P18-INV-488 — HTTP verb constraints mapped to resource lifecycle

Idea card metadata - Stable ID: KFM-P18-INV-488 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 488 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: API — Governed API, Resource Ontology, HTTP, and Service Contracts - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM public API contracts should mark which resources are read-only, review-only, or mutation-capable and map mutating actions to governed promotion or correction workflows.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM public API contracts should mark which resources are read-only, review-only, or mutation-capable and map mutating actions to governed promotion or correction workflows. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: HTTP verbs alone must not imply that a client can mutate canonical truth or bypass review.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-DOC-003.

Dependencies - PROPOSED: HTTP verb - PROPOSED: resource lifecycle - PROPOSED: read-only resource - PROPOSED: promotion gate

Tensions - PROPOSED: Administrative mutation endpoints may be necessary but must stay outside public paths.

Expansion directions - PROPOSED: Add lifecycle_capability metadata to API contract fragments.

Open questions - NEEDS VERIFICATION: How should ABSTAIN, DENY, and ERROR be represented across HTTP status and KFM outcome fields?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 41; basis: The web API source maps resource lifecycle requirements to HTTP verbs and notes that some resources are read-only or lack update/delete actions.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-001, KFM-IDX-DOC-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12 MAP - Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries

Status: CONFIRMED source support for grouped Phase 3 ideas; PROPOSED category synthesis

Definition: Map artifacts, tiles, rasters/vectors, MapLibre/renderer boundaries, style/layer registries, and map delivery controls.

CONFIRMED: This chapter contains 18 Pass 18 idea cards assigned to MAP.

Idea ID	Title	Carry state	Status	Source IDs
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KFM-P18-INV-045 | MapLibre renderer remains downstream of trust | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-050 | Deterministic tile and raster sidecar hashes | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-073 | WMS/WMTS/XYZ/PMTiles protocol registry controls external map services | NEW | PROPOSED | SRC-P18-039 |

KFM-P18-INV-074 | IIIF/Allmaps overlays require georeference and rights evidence | NEW | PROPOSED | SRC-P18-039 |

KFM-P18-INV-180 | Tile protocol enum for map-source governance | NEW | PROPOSED | SRC-P18-039 |

KFM-P18-INV-181 | Deterministic vector-tile builds from canonical GeoParquet | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-182 | PMTiles tilehash sidecars for offline verification | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-183 | COG internal-chunk verification manifests | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-184 | STAC-driven preview renderer as review evidence | NEW | PROPOSED | SRC-P18-039 |

KFM-P18-INV-275 | External map service protocols need attribution, terms, and protocol labels | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-331 | HydroGNSS products as uncertainty-aware public-safe raster context | NEW | PROPOSED | SRC-P18-039 |

KFM-P18-INV-351 | Map legends as evidence-disclosure surfaces | EXPANDED | PROPOSED | SRC-P18-002 |

KFM-P18-INV-400 | Vector-tile attribute whitelists as public-safety control | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-424 | Map artifact sustainability telemetry as non-truth operational evidence | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-425 | Allmaps and historic overlay georeference uncertainty | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-449 | IIIF and Allmaps historic overlays as rights-bound map context | NEW | PROPOSED | SRC-P18-039 |

KFM-P18-INV-450 | Preview-renderer screenshots as release-review evidence | NEW | PROPOSED | SRC-P18-039 |

KFM-P18-INV-475 | WMS and WMTS require external-source governance | EXPANDED | PROPOSED | SRC-P18-039 |

8.12.1 KFM-P18-INV-045 — MapLibre renderer remains downstream of trust

Idea card metadata - Stable ID: KFM-P18-INV-045 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 45 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MAP — Map Artifacts, Tiles,

Raster/Vector Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should preserve MapLibre as a downstream renderer and interaction runtime, never as the source of truth, policy, publication, citation, or AI authority.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should preserve MapLibre as a downstream renderer and interaction runtime, never as the source of truth, policy, publication, citation, or AI authority. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It reinforces the trust membrane for every map interaction and prevents visual layers from becoming canonical claims.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-DOC-007.

Dependencies - PROPOSED: LayerManifest - PROPOSED: StyleManifest - PROPOSED: EvidenceBundle resolution - PROPOSED: governed API click path

Tensions - PROPOSED: Users may still treat rendered pixels as truth if trust state is not visible.

Expansion directions - PROPOSED: Make `renderer_downstream=true` a required UI doctrine check.

Open questions - NEEDS VERIFICATION: Which visual states best signal released, stale, denied, and context-only layers?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 3, 30, 40; basis: executive determination and governance/trust membrane chapter; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-DOC-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12.2 KFM-P18-INV-050 — Deterministic tile and raster sidecar hashes

Idea card metadata - Stable ID: KFM-P18-INV-050 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 50 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MAP — Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should use deterministic PMTiles tilehash sidecars and COG internal tile/overview hashes for high-integrity map releases.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should use deterministic PMTiles tilehash sidecars and COG internal tile/overview hashes for high-integrity map releases. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It allows offline or client-side verification of map artifacts beyond a single file checksum.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-002. - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-003.

Dependencies - PROPOSED: PMTiles tilehash sidecar - PROPOSED: COG internal chunk manifest - PROPOSED: attestation predicate - PROPOSED: release manifest

Tensions - PROPOSED: High-integrity sidecars add build complexity and may be overkill for low-risk layers.

Expansion directions - PROPOSED: Classify artifact_integrity_level by public consequence and sensitivity.

Open questions - NEEDS VERIFICATION: Which releases require per-tile or per-chunk verification versus file-level checksum only?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 13, 14; basis: artifact canonicalization, PMTiles tilehash, COG sidecar, and attestation entries; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-002, KFM-IDX-DAT-003, KFM-IDX-EVD-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12.3 KFM-P18-INV-073 — WMS/WMTS/XYZ/PMTiles protocol registry controls external map services

Idea card metadata - Stable ID: KFM-P18-INV-073 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 73 - Status label: PROPOSED - Carry-forward state: NEW - Category: MAP — Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should register tile and map-service protocols with access, terms, attribution, freshness, and trust-state controls before they appear in MapLibre views.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should register tile and map-service protocols with access, terms, attribution, freshness, and trust-state controls before they appear in MapLibre views. - PROPOSED: Phase 3

scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: External map services can bypass provenance and rights review if embedded directly in styles or UI code.

Related ideas - CONFIRMED baseline link: KFM-IDX-CAT-007. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: ExternalServiceRegistry - PROPOSED: StyleManifest validator

Tensions - PROPOSED: External services are convenient but may be mutable, unavailable, or governed by incompatible terms.

Expansion directions - PROPOSED: Add protocol_type, attribution_required, cache_policy, freshness_policy, and deny_if_unverified fields.

Open questions - NEEDS VERIFICATION: Which external services are allowed only as contextual backdrops and never as evidence sources?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 11, 18; basis: tileProtocol enum for PMTiles/XYZ/WMTS/WMS and renderHints as governance-significant config; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12.4 KFM-P18-INV-074 — IIIF/Allmaps overlays require georeference and rights evidence

Idea card metadata - Stable ID: KFM-P18-INV-074 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 74 - Status label: PROPOSED - Carry-forward state: NEW - Category: MAP — Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should treat historic-map overlays from IIIF/Allmaps or similar systems as interpretive georeferenced artifacts requiring source rights, control points, and uncertainty metadata.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat historic-map overlays from IIIF/Allmaps or similar systems as interpretive georeferenced artifacts requiring source rights, control points, and uncertainty metadata. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Historic maps can be persuasive visual evidence, but their georeferencing, distortion, rights, and interpretive role need to be inspectable.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-006.

Dependencies - PROPOSED: HistoricMapOverlayManifest - PROPOSED: GCP provenance - PROPOSED: plugin allowlist

Tensions - PROPOSED: Overlay accuracy may vary by map, control points, and transformation method.

Expansion directions - PROPOSED: Add `georeference_method`, `control_point_count`, `residual_error`, `rights_status`, and `overlay_role` fields.

Open questions - NEEDS VERIFICATION: Should historic overlays default to contextual evidence rather than direct geometry evidence?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 13; basis: IIIF plus Allmaps historic map overlay and Story Node MapLibre panel entries; Phase 1 role: `candidate_synthesis_source_specialized_atlas`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12.5 KFM-P18-INV-180 — Tile protocol enum for map-source governance

Idea card metadata - Stable ID: KFM-P18-INV-180 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 180 - Status label: PROPOSED - Carry-forward state: NEW - Category: MAP — Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should require map source records to declare protocol class, such as PMTiles, XYZ, WMTS, or WMS, with access, attribution, and validation rules specific to that class.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should require map source records to declare protocol class, such as PMTiles, XYZ, WMTS, or WMS, with access, attribution, and validation rules specific to that class. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents external map services and static tile bundles from being handled as interchangeable URLs.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-007.

Dependencies - PROPOSED: LayerManifest - PROPOSED: SourceDescriptor - PROPOSED: Attribution review - PROPOSED: Access-term review

Tensions - PROPOSED: WMS/WMTS services can carry source terms and runtime availability risks that static PMTiles do not.

Expansion directions - PROPOSED: Define protocol-specific validation fixtures for PMTiles, XYZ, WMTS, and WMS.

Open questions - NEEDS VERIFICATION: Should unmanaged external WMS/WMTS layers ever be allowed on public KFM maps?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 11; basis: View specification render hints with tileProtocol values for pmtiles, xyz, wmts, and wms; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12.6 KFM-P18-INV-181 — Deterministic vector-tile builds from canonical GeoParquet

Idea card metadata - Stable ID: KFM-P18-INV-181 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 181 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MAP — Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM vector-tile builds should start from canonical processed inputs, pin tiler versions and flags, and record zoom, attribute, precision, and source-digest choices in release evidence.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM vector-tile builds should start from canonical processed inputs, pin tiler versions and flags, and record zoom, attribute, precision, and source-digest choices in release evidence. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes map tiles rebuildable and reviewable instead of one-off renderer artifacts.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-008. - CONFIRMED baseline link: KFM-IDX-PIP-009. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: Canonical GeoParquet - PROPOSED: Tiler configuration - PROPOSED: TileArtifactManifest - PROPOSED: Rebuild test

Tensions - PROPOSED: Tile simplification and attribute pruning can change meaning if not recorded.

Expansion directions - PROPOSED: Add tiler flags, zoom rules, precision, source digests, and attribute whitelist to tile build receipts.

Open questions - NEEDS VERIFICATION: Which tiler should be the default for each KFM artifact family?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 28; basis: GeoParquet to Tippecanoe/Planetiler to PMTiles pipeline with pinned tiler versions, flags, zoom strategy, and deterministic input requirements; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-008, KFM-IDX-PIP-009, KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12.7 KFM-P18-INV-182 — PMTiles tilehash sidecars for offline verification

Idea card metadata - Stable ID: KFM-P18-INV-182 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 182 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MAP — Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should allow PMTiles releases to carry sidecar tilehash manifests when offline or high-integrity verification requires more than a whole-file digest.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should allow PMTiles releases to carry sidecar tilehash manifests when offline or high-integrity verification requires more than a whole-file digest. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It supports verification of large static tile archives and helps diagnose partial corruption or cache faults.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-004. - CONFIRMED baseline link: KFM-IDX-SEC-008. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: PMTiles file - PROPOSED: Tilehash sidecar - PROPOSED: MapReleaseManifest - PROPOSED: Verification tool

Tensions - PROPOSED: Per-tile hashing adds complexity and may not be necessary for low-risk layers.

Expansion directions - PROPOSED: Define release classes where whole-file hash is sufficient versus tilehash required.

Open questions - NEEDS VERIFICATION: What tilehash granularity is practical for browser, CI, and offline package validation?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 13, 28; basis: PMTiles tilehash sidecars for byte ranges or tile indexes, with validation in artifact manifests; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-004, KFM-IDX-SEC-008, KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12.8 KFM-P18-INV-183 — COG internal-chunk verification manifests

Idea card metadata - Stable ID: KFM-P18-INV-183 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 183 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MAP — Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM raster releases should support COG internal tile and overview hash manifests when raster integrity, partial reads, or high-risk public use justify chunk-level verification.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM raster releases should support COG internal tile and overview hash manifests when raster integrity, partial reads, or high-risk public use justify chunk-level verification. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It gives raster and terrain-derived products an integrity model comparable to vector tile archives.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-004. - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-SEC-008.

Dependencies - PROPOSED: COG artifact - PROPOSED: COG chunk manifest - PROPOSED: Raster source descriptor - PROPOSED: Release validator

Tensions - PROPOSED: Chunk-level verification must not imply that the raster interpretation itself is authoritative.

Expansion directions - PROPOSED: Add COG chunk-hash fixtures for valid, missing, mismatched, and stale sidecars.

Open questions - NEEDS VERIFICATION: Which raster products need chunk-level verification rather than whole-file checksums?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 13, 28; basis: COG internal tile, overview, and byte-range sidecar hashes for raster release verification; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-004, KFM-IDX-DAT-003, KFM-IDX-SEC-008 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12.9 KFM-P18-INV-184 — STAC-driven preview renderer as review evidence

Idea card metadata - Stable ID: KFM-P18-INV-184 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 184 - Status label: PROPOSED - Carry-forward state: NEW - Category: MAP — Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should treat generated map previews as review evidence only when the renderer, style, camera, source artifact, and output digest are pinned.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat generated map previews as review evidence only when the renderer, style, camera, source artifact, and output digest are pinned. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It gives reviewers deterministic screenshots without turning screenshots into source truth.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-MDP-005.

Dependencies - PROPOSED: STAC item - PROPOSED: StyleManifest - PROPOSED: Renderer version - PROPOSED: Preview digest

Tensions - PROPOSED: A screenshot can be persuasive even when it is only a rendered derivative.

Expansion directions - PROPOSED: Record renderer inputs and link preview artifacts to Evidence Drawer and review decisions.

Open questions - NEEDS VERIFICATION: Should previews be stored in release proof packs or review-only artifacts?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 14, 19; basis: STAC-driven preview renderer using MapLibre/headless rendering with pinned style, source layers, camera, and output hash; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12.10 KFM-P18-INV-275 — External map service protocols need attribution, terms, and protocol labels

Idea card metadata - Stable ID: KFM-P18-INV-275 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 275 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MAP — Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM map-source registries should label WMS, WMTS, XYZ, PMTiles, and related protocols and attach attribution, terms, cache, and evidence limitations before rendering.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM map-source registries should label WMS, WMTS, XYZ, PMTiles, and related protocols and attach attribution, terms, cache, and evidence limitations before rendering. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents external map services from entering the UI as unlabeled or un-governed visual truth.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-CAT-001.

Dependencies - PROPOSED: LayerManifest - PROPOSED: source registry - PROPOSED: terms review - PROPOSED: protocol validator

Tensions - PROPOSED: External services may change content, terms, or availability without KFM control.

Expansion directions - PROPOSED: Add `external_map_service_profile` fields for `protocol`, `attribution`, `terms_url`, `cache_policy`, `freshness`, and `evidence_role`.

Open questions - NEEDS VERIFICATION: Should external WMS/WMTS layers be clickable EvidenceBundle sources or context-only overlays?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 102, 110; basis: `tileProtocol` enum includes `pmtiles`, `xyz`, `wmts`, and `wms`; external services need protocol and attribution/terms validation; Phase 1 role: `candidate_synthesis_source_specialized_atlas`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-POL-001, KFM-IDX-CAT-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12.11 KFM-P18-INV-331 — HydroGNSS products as uncertainty-aware public-safe raster context

Idea card metadata - Stable ID: KFM-P18-INV-331 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 331 - Status label: PROPOSED - Carry-forward state: NEW - Category: MAP — Map Artifacts, Tiles, Raster/Vector

Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM HydroGNSS-derived soil-moisture, inundation, freeze-thaw, and biomass-context layers should be time-aware, uncertainty-aware, QA-masked, and generalized before public use.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM HydroGNSS-derived soil-moisture, inundation, freeze-thaw, and biomass-context layers should be time-aware, uncertainty-aware, QA-masked, and generalized before public use. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps emerging environmental products useful while preventing model or sensor uncertainty from being hidden.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-001. - CONFIRMED baseline link: KFM-IDX-POL-002.

Dependencies - PROPOSED: QA mask - PROPOSED: time index - PROPOSED: uncertainty field - PROPOSED: dynamic H3 generalization - PROPOSED: release policy

Tensions - PROPOSED: Users may misread coarse or uncertain products as exact local observations.

Expansion directions - PROPOSED: Add HydroGNSS uncertainty and QA-mask fixtures to raster validators.

Open questions - NEEDS VERIFICATION: Which HydroGNSS variables are suitable for Kansas-first public layers?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 621, 627; basis: Master MapLibre v1.9 delta identifies HydroGNSS variables and says HydroGNSS governance uses dynamic H3 generalization and uncertainty masks.; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12.12 KFM-P18-INV-351 — Map legends as evidence-disclosure surfaces

Idea card metadata - Stable ID: KFM-P18-INV-351 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 351 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MAP — Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-002

Normalized statement - PROPOSED: KFM map legends should disclose not only symbols and colors, but also layer evidence class, release state, sensitivity transform, and uncertainty cues for each visible layer.

Detailed explanation - CONFIRMED: Source SRC-P18-002 (a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM map legends should disclose not only symbols and colors, but also layer evidence class, release state, sensitivity transform, and uncertainty cues for each visible layer. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A user cannot inspect a map claim when the legend hides whether a symbol represents observation, model, generalization, or reviewed release.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-POL-002.

Dependencies - PROPOSED: layer manifest - PROPOSED: style manifest - PROPOSED: legend schema - PROPOSED: EvidenceBundle class - PROPOSED: release state

Tensions - PROPOSED: A crowded legend can reduce readability if every governance cue is rendered visually at once.

Expansion directions - PROPOSED: Define a compact trust legend grammar for public KFM maps.

Open questions - NEEDS VERIFICATION: Which governance cues belong in a legend versus an Evidence Drawer?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-002 — a-primer-of-gis-fundamental-geographic-and-cartographic-concepts.pdf; pages 59, 200; basis: A Primer of GIS lists legend, scale, and orientation as essential map elements and ties symbol explanation to cartographic communication.; Phase 1 role: candidate_synthesis_source.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-012, KFM-IDX-POL-002 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12.13 KFM-P18-INV-400 — Vector-tile attribute whitelists as public-safety control

Idea card metadata - Stable ID: KFM-P18-INV-400 - Pass: 18 - Phase 5 batch: 016 - Inventory sequence: 400 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MAP — Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 016. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM vector-tile builds should whitelist tile attributes so public PMTiles/MVT/MLT outputs include only fields approved for release, size budgets, and evidence inspection.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM vector-tile builds should whitelist tile attributes so public PMTiles/MVT/MLT outputs include only fields approved for release, size budgets, and evidence inspection.

- PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5.
 - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Public tiles are easy to cache and inspect; unnecessary attributes can leak sensitive or unsupported data at every zoom.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-002. - CONFIRMED baseline link: KFM-IDX-SEC-009.

Dependencies - PROPOSED: TileArtifactManifest - PROPOSED: attribute allowlist - PROPOSED: sensitivity policy - PROPOSED: tile-size budget

Tensions - PROPOSED: Overly strict allowlists may remove fields needed for accessibility or Evidence Drawer linking.

Expansion directions - PROPOSED: Add tile_attribute_allowlist validation to PMTiles and MVT build gates.

Open questions - NEEDS VERIFICATION: Which fields are always required for click-to-evidence without leaking claim details?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 13, 34; basis: Master MapLibre Components-Functions-Features identifies attribute whitelisting as a way to keep vector tiles small and safer, and treats unneeded attributes as a leak and performance risk.; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-002, KFM-IDX-SEC-009 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12.14 KFM-P18-INV-424 — Map artifact sustainability telemetry as non-truth operational evidence

Idea card metadata - Stable ID: KFM-P18-INV-424 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 424 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MAP — Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM map artifact builds may record energy and CO2 telemetry as operational evidence without treating those telemetry values as claims about the mapped phenomenon.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM map artifact builds may record energy and CO2 telemetry as operational evidence without treating those telemetry values as claims about the mapped phenomenon. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Sustainability telemetry can guide build governance and release decisions, but it must not be confused with environmental truth contained in the layer itself.

Related ideas - CONFIRMED baseline link: KFM-IDX-OBS-005. - CONFIRMED baseline link: KFM-IDX-PIP-010.

Dependencies - PROPOSED: build telemetry - PROPOSED: ReleaseManifest - PROPOSED: policy gate - PROPOSED: observability dashboard

Tensions - PROPOSED: Telemetry tools and regional carbon estimates can be version-sensitive and uncertain.

Expansion directions - PROPOSED: Add sustainability_telemetry to map build receipts as optional operational metadata.

Open questions - NEEDS VERIFICATION: What thresholds should trigger review rather than automatic denial?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 3, 10, 25; basis: Master MapLibre Components-Functions-Features identifies energy and CO2 promotion policy and CI provenance telemetry for map artifact releases.; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-OBS-005, KFM-IDX-PIP-010 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12.15 KFM-P18-INV-425 — Allmaps and historic overlay georeference uncertainty

Idea card metadata - Stable ID: KFM-P18-INV-425 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 425 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MAP — Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM historic map overlays should expose georeference uncertainty, control-point provenance, and rights status before being used in Story Nodes or public maps.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM historic map overlays should expose georeference uncertainty, control-point provenance, and rights status before being used in Story Nodes or public maps. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Warped historic maps can be persuasive but may contain positional uncertainty and rights limits that differ from modern GIS layers.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-013. - CONFIRMED baseline link: KFM-IDX-POL-006.

Dependencies - PROPOSED: IIIF manifest - PROPOSED: GCP provenance - PROPOSED: overlay uncertainty field - PROPOSED: rights review

Tensions - PROPOSED: Historic overlays are useful precisely because they are approximate contextual evidence.

Expansion directions - PROPOSED: Add historic_overlay_uncertainty to StoryManifest and LayerManifest records.

Open questions - NEEDS VERIFICATION: What visual cue should communicate georeference uncertainty without hiding the overlay?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 13, 14, 16; basis: Master MapLibre Components-Functions-Features discusses IIIF/Allmaps overlays, georeference annotations, historic-map rights, and Story Node usage.; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-013, KFM-IDX-POL-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12.16 KFM-P18-INV-449 — IIIF and Allmaps historic overlays as rights-bound map context

Idea card metadata - Stable ID: KFM-P18-INV-449 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 449 - Status label: PROPOSED - Carry-forward state: NEW - Category: MAP — Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM historic-map overlays should preserve IIIF rights, georeferencing annotation provenance, and plugin governance before MapLibre display.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM historic-map overlays should preserve IIIF rights, georeferencing annotation provenance, and plugin governance before MapLibre display. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Historic overlays are powerful context but can carry rights, georeferencing, and interpretive uncertainty.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-009. - CONFIRMED baseline link: KFM-IDX-POL-003.

Dependencies - PROPOSED: IIIF manifest - PROPOSED: Allmaps annotation - PROPOSED: plugin allowlist - PROPOSED: Evidence Drawer

Tensions - PROPOSED: Overlay plugins may evolve independently of KFM validation.

Expansion directions - PROPOSED: Create a historic_overlay_manifest with rights and georeferencing hashes.

Open questions - NEEDS VERIFICATION: How should KFM disclose georeferencing error for warped historic maps?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 3, 4, 13, 36, 46; basis: Master MapLibre v1.9 records IIIF/Allmaps historic-map overlay workflows and emphasizes rights, georeference annotations, and Story Node use.; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12.17 KFM-P18-INV-450 — Preview-renderer screenshots as release-review evidence

Idea card metadata - Stable ID: KFM-P18-INV-450 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 450 - Status label: PROPOSED - Carry-forward state: NEW - Category: MAP — Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM release review should allow deterministic preview-renderer screenshots to serve as review evidence while preserving style, artifact, camera, and renderer digests.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM release review should allow deterministic preview-renderer screenshots to serve as review evidence while preserving style, artifact, camera, and renderer digests. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A screenshot is not truth, but it can document what reviewers saw at release time.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-006. - CONFIRMED baseline link: KFM-IDX-OBS-003.

Dependencies - PROPOSED: preview renderer - PROPOSED: style digest - PROPOSED: camera state - PROPOSED: review receipt

Tensions - PROPOSED: Visual output can differ by renderer version or device.

Expansion directions - PROPOSED: Add preview screenshot provenance to release review receipts.

Open questions - NEEDS VERIFICATION: Which pixel-diff thresholds are acceptable for public map review?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 14, 19, 20, 35, 36; basis: Master MapLibre v1.9 describes a STAC-driven preview renderer using MapLibre/headless rendering for review screenshots and Story/Focus evidence.; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.12.18 KFM-P18-INV-475 — WMS and WMTS require external-source governance

Idea card metadata - Stable ID: KFM-P18-INV-475 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 475 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: MAP — Map Artifacts, Tiles, Raster/Vector Delivery, and Renderer Boundaries - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM may use WMS or WMTS as map protocols only when the service has a SourceDescriptor, rights/attribution review, cache policy, and public-safety posture.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM may use WMS or WMTS as map protocols only when the service has a SourceDescriptor, rights/attribution review, cache policy, and public-safety posture. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: External map services can render quickly but can bypass KFM provenance if admitted only as URLs.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-003.

Dependencies - PROPOSED: external map service - PROPOSED: SourceDescriptor - PROPOSED: cache policy - PROPOSED: layer manifest

Tensions - PROPOSED: Service availability, terms, and attribution can change without KFM artifact rebuilds.

Expansion directions - PROPOSED: Create WMS/WMTS fixtures with rights, attribution, and cache policy gates.

Open questions - NEEDS VERIFICATION: Which external WMS/WMTS sources are safe for public use without mirrored artifacts?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 21, 46; basis: The Master MapLibre atlas records WMTS/WMS as allowed protocols while requiring SourceDescriptor, rights, attribution, cache policy, and source-governance validation.; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-POL-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.13 UIX - Trust-Visible UI, Evidence Drawer, Focus Mode, Story Nodes, and Review Surfaces

Status: CONFIRMED source support for grouped Phase 3 ideas; PROPOSED category synthesis

Definition: Trust-visible UI, Evidence Drawer, Focus Mode, Story Nodes, review surfaces, popups, and user-facing inspection flows.

CONFIRMED: This chapter contains 12 Pass 18 idea cards assigned to UIX.

Idea ID	Title	Carry state	Status	Source IDs
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KFM-P18-INV-048 | View registry blocks direct database handles in UI | NEW | PROPOSED | SRC-P18-039 |

KFM-P18-INV-049 | Preview renderer and Story Nodes as evidence-bearing outputs | NEW | PROPOSED | SRC-P18-039 |

KFM-P18-INV-063 | Bitemporal tables as correction-safe claim history | EXPANDED | PROPOSED | SRC-P18-007 |

KFM-P18-INV-115 | Wireframes as API contract discovery inputs | NEW | PROPOSED | SRC-P18-006 |

KFM-P18-INV-179 | View registry as a no-raw-database map boundary | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-185 | Style, camera, and renderer digests for visual regression | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-187 | Structured Focus Mode outputs for map context | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-250 | Story Node graph health and citation checks before release | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-329 | Route chunks and view registry hydration as governed UI evidence | NEW | PROPOSED | SRC-P18-039 |

KFM-P18-INV-333 | Map popups should launch evidence inspection rather than become evidence | EXPANDED | PROPOSED | SRC-P18-039 |

KFM-P18-INV-365 | UI wireframes as API requirement evidence | EXPANDED | PROPOSED | SRC-P18-006 |

KFM-P18-INV-499 | Style JSON version and digest for visual regression | EXPANDED | PROPOSED | SRC-P18-039 |

8.13.1 KFM-P18-INV-048 — View registry blocks direct database handles in UI

Idea card metadata - Stable ID: KFM-P18-INV-048 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 48 - Status label: PROPOSED - Carry-forward state: NEW - Category: UIX — Trust-Visible UI, Evidence Drawer, Focus Mode, Story Nodes, and Review Surfaces - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should use a view registry that points map views to governed catalog and artifact references rather than direct database handles or raw queries.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should use a view registry that points map views to governed catalog and artifact references rather than direct database handles or raw queries. - PROPOSED: Phase 3 scope note:

Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes the normal UI path consume governed released surfaces instead of internal stores.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-POL-003.

Dependencies - PROPOSED: ViewRegistry - PROPOSED: LayerCatalog - PROPOSED: governed API - PROPOSED: no-direct-DB policy

Tensions - PROPOSED: Developers may be tempted to add shortcuts for debugging or performance.

Expansion directions - PROPOSED: Add no_db_handle_in_public_view_registry policy tests.

Open questions - NEEDS VERIFICATION: What exception process, if any, allows internal review consoles to inspect non-public data?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 13, 20; basis: view registry and no-direct-DB / LayerCatalog notes; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.13.2 KFM-P18-INV-049 — Preview renderer and Story Nodes as evidence-bearing outputs

Idea card metadata - Stable ID: KFM-P18-INV-049 - Pass: 18 - Phase 5 batch: 002 - Inventory sequence: 49 - Status label: PROPOSED - Carry-forward state: NEW - Category: UIX — Trust-Visible UI, Evidence Drawer, Focus Mode, Story Nodes, and Review Surfaces - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 002. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM should treat preview screenshots and Story Nodes as evidence-bearing outputs tied to pinned styles, sources, cameras, renderer versions, and artifact digests.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat preview screenshots and Story Nodes as evidence-bearing outputs tied to pinned styles, sources, cameras, renderer versions, and artifact digests. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps screenshots and story panels from becoming detached illustrations with no proof trail.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-AIG-004.

Dependencies - PROPOSED: PreviewManifest - PROPOSED: style digest - PROPOSED: camera state - PROPOSED: renderer version - PROPOSED: Evidence Drawer link

Tensions - PROPOSED: Visual regressions may be hard to distinguish from legitimate style changes.

Expansion directions - PROPOSED: Add screenshot_digest and preview_source_manifest fields.

Open questions - NEEDS VERIFICATION: Should every public story map have a preview-render receipt?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 14; basis: preview renderer, Story Nodes, Focus Mode, style digest, and screenshot provenance entries; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.13.3 KFM-P18-INV-063 — Bitemporal tables as correction-safe claim history

Idea card metadata - Stable ID: KFM-P18-INV-063 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 63 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: UIX — Trust-Visible UI, Evidence Drawer, Focus Mode, Story Nodes, and Review Surfaces - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-007

Normalized statement - PROPOSED: KFM should use bitemporal or equivalent history models for claims whose real-world validity and recorded/release history can diverge.

Detailed explanation - CONFIRMED: Source SRC-P18-007 (developing-time-oriented-database-applications-in-sql.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should use bitemporal or equivalent history models for claims whose real-world validity and recorded/release history can diverge. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A claim can have one time when it was true in the world and another time when KFM learned, corrected, or released it.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-EVD-006.

Dependencies - PROPOSED: valid_time - PROPOSED: recorded_time/release_time - PROPOSED: Correction-Receipt

Tensions - PROPOSED: Bitemporal modeling increases complexity and must be applied where the evidence burden justifies it.

Expansion directions - PROPOSED: Define which KFM object families require bitemporal storage versus release snapshots only.

Open questions - NEEDS VERIFICATION: Should public APIs expose transaction/release history directly or through Evidence Drawer summaries?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-007 — developing-time-oriented-database-applications-in-sql.pdf; pages 11, 12, 277; basis: valid-time, transaction-time, bitemporal table, and temporal database design contents; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-001, KFM-IDX-EVD-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.13.4 KFM-P18-INV-115 — Wireframes as API contract discovery inputs

Idea card metadata - Stable ID: KFM-P18-INV-115 - Pass: 18 - Phase 5 batch: 005 - Inventory sequence: 115 - Status label: PROPOSED - Carry-forward state: NEW - Category: UIX — Trust-Visible UI, Evidence Drawer, Focus Mode, Story Nodes, and Review Surfaces - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 005. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM should use UI wireframes and map/evidence-drawer workflows to discover governed API contracts before implementation.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should use UI wireframes and map/evidence-drawer workflows to discover governed API contracts before implementation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It ensures APIs support evidence inspection, negative states, and public-user workflows rather than only database convenience.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-007. - CONFIRMED baseline link: KFM-IDX-MDP-002.

Dependencies - PROPOSED: OpenAPI draft - PROPOSED: UI flow registry - PROPOSED: Evidence Drawer contract

Tensions - PROPOSED: Wireframes can overfocus on happy paths unless DENY/ABSTAIN/ERROR flows are included.

Expansion directions - PROPOSED: Add wireframe_to_api_trace notes to contract reviews.

Open questions - NEEDS VERIFICATION: Should every API design review include at least one failure-state wireframe?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 5, 30; basis: user interface wireframes driving API design; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.13.5 KFM-P18-INV-179 — View registry as a no-raw-database map boundary

Idea card metadata - Stable ID: KFM-P18-INV-179 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 179 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: UIX — Trust-Visible UI, Evidence Drawer, Focus Mode, Story Nodes, and Review Surfaces - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM map view registries should expose cataloged datasource pointers and render hints, not raw database handles or ad hoc query instructions.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM map view registries should expose cataloged datasource pointers and render hints, not raw database handles or ad hoc query instructions. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps public map UI downstream of governed release artifacts rather than tied to canonical or internal stores.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-AIG-007. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: ViewRegistry schema - PROPOSED: LayerManifest - PROPOSED: Governed API - PROPOSED: Policy gate

Tensions - PROPOSED: Developers may prefer direct query handles for convenience; KFM needs deny-by-default validation.

Expansion directions - PROPOSED: Add no_db_handle and no_raw_query tests for every public view registry fixture.

Open questions - NEEDS VERIFICATION: Which existing UI registry surfaces, if any, already encode this boundary?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 11; basis: view registry design that forbids direct Neo4j/database handles and uses datasource pointers plus render hints; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-001, KFM-IDX-AIG-007, KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.13.6 KFM-P18-INV-185 — Style, camera, and renderer digests for visual regression

Idea card metadata - Stable ID: KFM-P18-INV-185 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 185 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: UIX — Trust-Visible UI, Evidence Drawer, Focus Mode, Story Nodes, and Review Surfaces - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM visual regression tests should pin style JSON digest, camera state, source layer names, renderer version, and artifact digests before comparing map outputs.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM visual regression tests should pin style JSON digest, camera state, source layer names, renderer version, and artifact digests before comparing map outputs. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes map appearance changes traceable instead of treating visual diffs as subjective UI polish.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-MDP-002. - CONFIRMED baseline link: KFM-IDX-EVD-004.

Dependencies - PROPOSED: StyleManifest - PROPOSED: Renderer pin - PROPOSED: Camera preset - PROPOSED: Visual-diff threshold

Tensions - PROPOSED: Renderer upgrades may change pixels without changing source truth.

Expansion directions - PROPOSED: Add visual-regression records to release receipts with renderer and style hashes.

Open questions - NEEDS VERIFICATION: Which visual diff thresholds preserve trust without blocking harmless antialiasing changes?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 19; basis: style JSON version, digest, source layers, camera, and renderer version used by preview renderer and visual regression checks; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-MDP-002, KFM-IDX-EVD-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.13.7 KFM-P18-INV-187 — Structured Focus Mode outputs for map context

Idea card metadata - Stable ID: KFM-P18-INV-187 - Pass: 18 - Phase 5 batch: 008 - Inventory sequence: 187 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: UIX — Trust-Visible UI, Evidence Drawer, Focus Mode, Story Nodes, and Review Surfaces - Category assignment status: PROPOSED assignment

derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 008. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM Focus Mode should emit structured map-context outputs that validate against a contract and fail closed to ABSTAIN, DENY, or ERROR when citations, policy, or schema checks fail.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM Focus Mode should emit structured map-context outputs that validate against a contract and fail closed to ABSTAIN, DENY, or ERROR when citations, policy, or schema checks fail. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents fluent map explanations from bypassing evidence, policy, or finite outcome rules.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-001. - CONFIRMED baseline link: KFM-IDX-AIG-002. - CONFIRMED baseline link: KFM-IDX-DOC-003.

Dependencies - PROPOSED: MapContextEnvelope - PROPOSED: CitationValidationReport - PROPOSED: PolicyDecision - PROPOSED: Focus Mode adapter

Tensions - PROPOSED: A valid schema does not guarantee a true claim; evidence closure still controls release.

Expansion directions - PROPOSED: Add schema-invalid, citation-missing, policy-denied, and stale-layer Focus Mode fixtures.

Open questions - NEEDS VERIFICATION: Which map context fields can Focus Mode see without exposing unreleased evidence?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 14; basis: LLM structured outputs parsed into Pydantic/JSON Schema with finite trust outcomes for map context; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-001, KFM-IDX-AIG-002, KFM-IDX-DOC-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.13.8 KFM-P18-INV-250 — Story Node graph health and citation checks before release

Idea card metadata - Stable ID: KFM-P18-INV-250 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 250 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: UIX — Trust-Visible UI, Evidence Drawer, Focus Mode, Story Nodes, and Review Surfaces - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM Story Nodes should validate graph references, catalog links, document IDs, citations, and Evidence Drawer payloads before narrative map panels are released.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM Story Nodes should validate graph references, catalog links, document IDs, citations, and Evidence Drawer payloads before narrative map panels are released. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps narrative map experiences from outrunning evidence and citation integrity.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-007. - CONFIRMED baseline link: KFM-IDX-CAT-005. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: StoryManifest - PROPOSED: Evidence Drawer payload - PROPOSED: Graph link validator - PROPOSED: Citation validator

Tensions - PROPOSED: Narrative coherence can make weak evidence feel stronger than it is.

Expansion directions - PROPOSED: Add `story_node_validation_report` for link resolution, citation coverage, graph-health status, policy state, and rollback target.

Open questions - NEEDS VERIFICATION: Which Story Node failures require DENY versus ABSTAIN or degraded display?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 31, 32; basis: Story Node and Evidence Drawer payloads link catalog, graph, documents, citations, and release/policy state with validation requirements; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-AIG-007, KFM-IDX-CAT-005, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.13.9 KFM-P18-INV-329 — Route chunks and view registry hydration as governed UI evidence

Idea card metadata - Stable ID: KFM-P18-INV-329 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 329 - Status label: PROPOSED - Carry-forward state: NEW - Category: UIX — Trust-Visible UI, Evidence Drawer, Focus Mode, Story Nodes, and Review Surfaces - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM map shells should hydrate route-specific UI chunks only after the view registry, render hints, evidence state, and access state are validated.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM map shells should hydrate route-specific UI chunks only after the

view registry, render hints, evidence state, and access state are validated. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps performance optimization subordinate to evidence and policy state.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-POL-003.

Dependencies - PROPOSED: view registry - PROPOSED: routeChunk - PROPOSED: lazyLoad - PROPOSED: render hints - PROPOSED: access policy

Tensions - PROPOSED: Lazy loading can hide missing evidence-state checks if route chunks are treated as independent UI code.

Expansion directions - PROPOSED: Add route_chunk_validation to UI release checks.

Open questions - NEEDS VERIFICATION: Which chunks are allowed to load before evidence drawer state is available?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 101, 114; basis: Master MapLibre v1.9 delta describes Vite routeChunk, lazyLoad, and view-registry validation for map routes.; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.13.10 KFM-P18-INV-333 — Map popups should launch evidence inspection rather than become evidence

Idea card metadata - Stable ID: KFM-P18-INV-333 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 333 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: UIX — Trust-Visible UI, Evidence Drawer, Focus Mode, Story Nodes, and Review Surfaces - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM map popups should be treated as navigation into STAC metadata, EvidenceBundle resolution, and Evidence Drawer review rather than as authoritative claim containers.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM map popups should be treated as navigation into STAC metadata, EvidenceBundle resolution, and Evidence Drawer review rather than as authoritative claim containers. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. -

NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents popup text from becoming an uncited public truth surface.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-007. - CONFIRMED baseline link: KFM-IDX-AIG-004.

Dependencies - PROPOSED: feature id - PROPOSED: STAC metadata - PROPOSED: EvidenceBundle resolver - PROPOSED: Evidence Drawer payload

Tensions - PROPOSED: Fast map interaction can tempt teams to summarize evidence in popups without full citations.

Expansion directions - PROPOSED: Add click_to_evidence_drawer tests for every public map layer.

Open questions - NEEDS VERIFICATION: What minimum fields can a popup show before evidence resolution completes?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 92; basis: Master MapLibre v1.9 delta says WZDx viewer clicks show STAC metadata and popup behavior should launch EvidenceBundle inspection.; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DOC-007, KFM-IDX-AIG-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.13.11 KFM-P18-INV-365 — UI wireframes as API requirement evidence

Idea card metadata - Stable ID: KFM-P18-INV-365 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 365 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: UIX — Trust-Visible UI, Evidence Drawer, Focus Mode, Story Nodes, and Review Surfaces - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-006

Normalized statement - PROPOSED: KFM API design should preserve UI wireframes, user activities, and participant roles as requirement evidence for governed API resources and routes.

Detailed explanation - CONFIRMED: Source SRC-P18-006 (Designing Great Web APIs.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM API design should preserve UI wireframes, user activities, and participant roles as requirement evidence for governed API resources and routes. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It connects public map and Evidence Drawer behavior to traceable API requirements instead of ad hoc route design.

Related ideas - CONFIRMED baseline link: KFM-IDX-MDP-003. - CONFIRMED baseline link: KFM-IDX-AIG-006.

Dependencies - PROPOSED: wireframe - PROPOSED: participant role - PROPOSED: activity model - PROPOSED: API model receipt

Tensions - PROPOSED: Wireframes can overfit current UI and miss non-UI clients.

Expansion directions - PROPOSED: Add requirements_artifact_ref to API contract records.

Open questions - NEEDS VERIFICATION: Which UI artifacts are stable enough to become API design evidence?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-006 — Designing Great Web APIs.pdf; pages 29, 30, 31, 34; basis: Designing Great Web APIs states that UI wireframes can drive a top-down view of how an API will be used and describes iterative API modeling and validation with requirements artifacts.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-MDP-003, KFM-IDX-AIG-006 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.13.12 KFM-P18-INV-499 — Style JSON version and digest for visual regression

Idea card metadata - Stable ID: KFM-P18-INV-499 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 499 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: UIX — Trust-Visible UI, Evidence Drawer, Focus Mode, Story Nodes, and Review Surfaces - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-039

Normalized statement - PROPOSED: KFM map-review screenshots should include style JSON version, style digest, camera state, and artifact digests so visual regression is evidence-bound.

Detailed explanation - CONFIRMED: Source SRC-P18-039 (Master MapLibre Components-Functions-Features.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM map-review screenshots should include style JSON version, style digest, camera state, and artifact digests so visual regression is evidence-bound. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A screenshot is not review evidence unless its render inputs can be reproduced.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: StyleManifest - PROPOSED: style digest - PROPOSED: preview renderer - PROPOSED: visual regression

Tensions - PROPOSED: Renderer nondeterminism can still produce small visual differences.

Expansion directions - PROPOSED: Add style_digest and camera_profile to preview-renderer receipts.

Open questions - NEEDS VERIFICATION: What pixel-diff threshold should trigger review rather than automatic failure?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-039 — Master MapLibre Components-Functions-Features.pdf; pages 19, 26; basis: The Master MapLibre atlas records a v1.9 idea that StyleManifest should store style JSON version, digest, and camera presets for review images.; Phase 1 role: candidate_synthesis_source_specialized_atlas.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14 PLN - Planning, Participation, Resilience, and Decision-Support Applications

Status: CONFIRMED source support for grouped Phase 3 ideas; PROPOSED category synthesis

Definition: Planning, participation, VGI, resilience, environmental decision support, urban indicators, equity, and stakeholder evidence.

CONFIRMED: This chapter contains 35 Pass 18 idea cards assigned to PLN.

Idea ID	Title	Carry state	Status	Source IDs
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KFM-P18-INV-010 | SDG-11-aligned settlement and infrastructure lens | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-011 | Planning-support implementation gap as KFM risk | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-012 | Interactive GIS as stakeholder dialogue surface | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-014 | Data-poor low-carbon transport scenarios | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-015 | Urban flood-risk and VGI shelter-analysis pattern | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-060 | Stakeholder preference weights as auditable planning evidence | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-062 | VGI evacuation-shelter preferences as reviewed community evidence | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-086 | Resettlement impacts as spatially situated temporal claims | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-201 | Planning-support implementation gap as adoption evidence | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-203 | Participatory site-suitability criteria and weight receipts | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-204 | Environmental quality indices need deprivation-context metadata | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-205 | Target-population multi-criteria route assessment | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-206 | VGI shelter preferences require authority separation | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-243 | TOD and accessibility indices need actual-versus-potential labels | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-244 | Urban water supply layers separate performance, preference, and policy | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-245 | Resettlement and displacement dynamics as temporal review evidence | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-257 | Collaborative planning weights and stakeholder roles need evidence records | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-259 | Street and public-space quality metrics split observation from perception | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-300 | VGI disaster-planning inputs require verification and freshness labels | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-320 | Data-poor planning support requires proxy and uncertainty disclosure | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-321 | Collaborative planning records stakeholder perspective and multidimensional criteria | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-322 | Stated choice experiments are preference evidence, not observed behavior | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-343 | SDG 11 city-concept labels as planning-policy lenses, not source truth | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-353 | Data-poor scenario analysis as bounded planning evidence | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-354 | Multi-criteria cycle-route assessment with target-population preferences | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-356 | Post-resettlement monitoring as a temporal equity claim family | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-420 | Indicator transferability caveat across global planning case studies | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-421 | Participatory planning evidence requires method and facilitation records | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-444 | Transit-oriented-development index as composite planning evidence | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-445 | VGI flood-shelter inputs as community-contributed evidence | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-468 | Data-poor scenario analysis carries explicit uncertainty | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-469 | Knowledge co-production method records for environmental health layers | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-470 | Stakeholder MCA weights as planning evidence | NEW | PROPOSED | SRC-P18-011 |

KFM-P18-INV-496 | Urban quality-of-life indicators as composite planning evidence | EXPANDED | PROPOSED | SRC-P18-011 |

KFM-P18-INV-497 | Stakeholder MCA weights as reviewable planning evidence | EXPANDED | PROPOSED | SRC-P18-011 |

8.14.1 KFM-P18-INV-010 — SDG-11-aligned settlement and infrastructure lens

Idea card metadata - Stable ID: KFM-P18-INV-010 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 10 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM can use sustainable, inclusive, compact/competitive, and resilient city concepts as analysis lenses for settlement and infrastructure domains.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM can use sustainable, inclusive, compact/competitive, and resilient city concepts as analysis lenses for settlement and infrastructure domains. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It offers a structured way to connect KFM settlement, infrastructure, transport, hazard, and equity layers.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-011. - CONFIRMED baseline link: KFM-IDX-DAT-012.

Dependencies - PROPOSED: Settlement/infrastructure domain lane; equity and hazard source descriptors

Tensions - PROPOSED: Global urban-planning cases need adaptation before Kansas-specific claims.

Expansion directions - PROPOSED: Draft an SDG-11-inspired but Kansas-specific settlement indicator matrix.

Open questions - NEEDS VERIFICATION: Which SDG-like indicators are appropriate for Kansas historical and contemporary contexts?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 14, 15, 16, 17; basis: urban planning, GIS, and SDG 11 framing; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.2 KFM-P18-INV-011 — Planning-support implementation gap as KFM risk

Idea card metadata - Stable ID: KFM-P18-INV-011 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 11 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM should track the gap between GIS planning-support potential and practical adoption as a product and governance risk.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should track the gap between GIS planning-support potential and practical adoption as a product and governance risk. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior; package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Adoption, usability, stewardship, and review workflows can fail even when data pipelines succeed.

Related ideas - CONFIRMED baseline link: KFM-IDX-OBS-001. - CONFIRMED baseline link: KFM-IDX-MDP-001.

Dependencies - PROPOSED: User research; review-console workflows; stakeholder role model

Tensions - PROPOSED: Buildability and institutional usability have different success metrics.

Expansion directions - PROPOSED: Add implementation_gap risk notes to UI and planning support chapters.

Open questions - NEEDS VERIFICATION: How will KFM measure whether a planning-support surface is adopted responsibly?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 16; basis: planning-support systems and implementation gap; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.3 KFM-P18-INV-012 — Interactive GIS as stakeholder dialogue surface

Idea card metadata - Stable ID: KFM-P18-INV-012 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 12 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM can support stakeholder dialogue through interactive GIS only when the dialogue surface preserves evidence, role, and review boundaries.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM can support stakeholder dialogue through interactive GIS only when the dialogue surface preserves evidence, role, and review boundaries. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Participation can improve interpretation, but ungoverned edits can weaken evidence and rights controls.

Related ideas - CONFIRMED baseline link: KFM-IDX-AIG-004. - CONFIRMED baseline link: KFM-IDX-MDP-004.

Dependencies - PROPOSED: Role-based review states; comment/correction receipts

Tensions - PROPOSED: Community input must not become unreviewed authoritative evidence.

Expansion directions - PROPOSED: Define a governed stakeholder annotation object.

Open questions - NEEDS VERIFICATION: Which stakeholder annotations are public, private, steward-only, or evidentiary?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 98; basis: interactive GIS and stakeholder dialogue; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.4 KFM-P18-INV-014 — Data-poor low-carbon transport scenarios

Idea card metadata - Stable ID: KFM-P18-INV-014 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 14 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM can support data-poor scenario analysis by making assumptions, uncertainty, and spatial/temporal scales first-class evidence fields.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM can support data-poor scenario analysis by making assumptions, uncertainty, and spatial/temporal scales first-class evidence fields. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: KFM will often face uneven historical or local data; transparent scenario assumptions are safer than silent gap filling.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-011. - CONFIRMED baseline link: KFM-IDX-PIP-001.

Dependencies - PROPOSED: ScenarioManifest; assumption ledger; model-output caveats

Tensions - PROPOSED: Data-poor does not mean evidence-free; assumptions need visible labels.

Expansion directions - PROPOSED: Add ScenarioManifest as a candidate schema family for planning analyses.

Open questions - NEEDS VERIFICATION: What minimum source support is needed before KFM can display scenario outputs?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 22, 23, 27, 29, 36, 39; basis: low-carbon transport scenario planning support; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN:

Repository implementation status remains unverified.

8.14.5 KFM-P18-INV-015 — Urban flood-risk and VGI shelter-analysis pattern

Idea card metadata - Stable ID: KFM-P18-INV-015 - Pass: 18 - Phase 5 batch: 001 - Inventory sequence: 15 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 001. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM can use volunteered geographic information and urban growth/flood-risk analysis only as reviewed, contextual, and uncertainty-labeled evidence.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM can use volunteered geographic information and urban growth/flood-risk analysis only as reviewed, contextual, and uncertainty-labeled evidence. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: VGI can expose lived reality, but it has sampling, privacy, and validation constraints.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-011. - CONFIRMED baseline link: KFM-IDX-POL-002. - CONFIRMED baseline link: KFM-IDX-PIP-004.

Dependencies - PROPOSED: VGI source-role registry; resilience analysis disclaimer; privacy filters

Tensions - PROPOSED: Disaster-related layers may be misused as emergency guidance.

Expansion directions - PROPOSED: Create VGI_contextual_only source role and shelter-analysis caveat template.

Open questions - NEEDS VERIFICATION: When should KFM deny VGI publication because of privacy or life-safety confusion?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 308, 312, 318, 321, 324, 326, 334; basis: urban flood-risk modeling and VGI evacuation-shelter analysis; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.6 KFM-P18-INV-060 — Stakeholder preference weights as auditable planning evidence

Idea card metadata - Stable ID: KFM-P18-INV-060 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 60 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience,

and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM should record stakeholder preference weights, target populations, and criteria when multiple-criteria planning analyses produce route or service recommendations.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should record stakeholder preference weights, target populations, and criteria when multiple-criteria planning analyses produce route or service recommendations. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Planning recommendations become more trustworthy when users can inspect whose preferences and criteria shaped the result.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: CriteriaWeightBundle - PROPOSED: StakeholderRole registry

Tensions - PROPOSED: Preference weights can encode policy choices; they should not masquerade as neutral GIS output.

Expansion directions - PROPOSED: Create a planning_criteria_receipt with weights, source, reviewer, and sensitivity fields.

Open questions - NEEDS VERIFICATION: How should conflicting stakeholder weights be represented without forcing a false consensus?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 7, 18; basis: multiple criteria analysis for cycle routes and target populations; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.7 KFM-P18-INV-062 — VGI evacuation-shelter preferences as reviewed community evidence

Idea card metadata - Stable ID: KFM-P18-INV-062 - Pass: 18 - Phase 5 batch: 003 - Inventory sequence: 62 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 003. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM should admit volunteered or social-media-derived planning evidence only with source-role labels, review, uncertainty, and official/non-official status separation.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should admit volunteered or social-media-derived planning evidence only with source-role labels, review, uncertainty, and official/non-official status separation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Community evidence can reveal useful local behavior, but it can be stale, biased, unverifiable, or unsafe if treated as official guidance.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-005. - CONFIRMED baseline link: KFM-IDX-DAT-013.

Dependencies - PROPOSED: VGIRewordRecord - PROPOSED: official_status field - PROPOSED: source freshness policy

Tensions - PROPOSED: VGI can be operationally valuable but must not become an emergency instruction surface.

Expansion directions - PROPOSED: Create a community_evidence_profile with role, confidence, freshness, and review state.

Open questions - NEEDS VERIFICATION: When should KFM show VGI as contextual evidence versus deny it from public maps?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 320, 321; basis: VGI for flood evacuation shelter planning and resident preferences; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-005, KFM-IDX-DAT-013 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.8 KFM-P18-INV-086 — Resettlement impacts as spatially situated temporal claims

Idea card metadata - Stable ID: KFM-P18-INV-086 - Pass: 18 - Phase 5 batch: 004 - Inventory sequence: 86 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 004. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM should model resettlement, displacement, and public-service access impacts as spatially situated temporal claims rather than one-time project labels.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should model resettlement, displacement, and public-service access impacts as spatially situated temporal claims rather than one-time project labels. - PROPOSED: Phase 3 scope note:

Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It supports correction-safe analysis of infrastructure projects whose benefits and harms vary by place, time, and household group.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-DAT-010.

Dependencies - PROPOSED: Temporal impact observation - PROPOSED: Community asset registry - PROPOSED: Policy sensitivity review

Tensions - PROPOSED: Infrastructure improvement narratives can hide localized harm.

Expansion directions - PROPOSED: Define displacement_event, resettlement_site, service_access_change, and livelihood_change claim roles.

Open questions - NEEDS VERIFICATION: How should KFM handle living-person resettlement data while preserving accountability?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 259, 275, 277; basis: development-induced displacement, social amenities, livelihood, community organization, and spatial impoverishment discussion; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-001, KFM-IDX-DAT-010 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.9 KFM-P18-INV-201 — Planning-support implementation gap as adoption evidence

Idea card metadata - Stable ID: KFM-P18-INV-201 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 201 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM should treat planning-support and GIS tooling adoption as an evidence and governance problem, not as automatic value created by adding map technology.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat planning-support and GIS tooling adoption as an evidence and governance problem, not as automatic value created by adding map technology. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps KFM from assuming that published layers or dashboards improve decisions without adoption, review, and usability evidence.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-001. - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-OBS-004.

Dependencies - PROPOSED: User research evidence - PROPOSED: Workflow adoption receipts - PROPOSED: Review-console feedback

Tensions - PROPOSED: A technically correct map can fail as a planning-support system if users cannot or do not use it.

Expansion directions - PROPOSED: Add adoption/readiness evidence fields to planning-support layer releases and review summaries.

Open questions - NEEDS VERIFICATION: What evidence should prove that a KFM planning tool is usable enough for public or steward workflows?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 16; basis: GIS planning support systems are promising but their promised utility still faces an implementation gap; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.10 KFM-P18-INV-203 — Participatory site-suitability criteria and weight receipts

Idea card metadata - Stable ID: KFM-P18-INV-203 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 203 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM suitability analyses should preserve criteria, rankings, stakeholder-group weights, and sensitivity-analysis settings as receipts before a site ranking supports a public claim.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM suitability analyses should preserve criteria, rankings, stakeholder-group weights, and sensitivity-analysis settings as receipts before a site ranking supports a public claim. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents a suitability map from hiding the value judgments and stakeholder assumptions behind it.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: Criteria registry - PROPOSED: Stakeholder weights - PROPOSED: Sensitivity analysis receipt - PROPOSED: Suitability layer manifest

Tensions - PROPOSED: Weights can encode policy choices that should not be mistaken for neutral measurement.

Expansion directions - PROPOSED: Add a suitability_receipt object for criteria, weights, alternatives, sensitivity runs, and reviewed purpose.

Open questions - NEEDS VERIFICATION: Which KFM domains may publish ranked suitability outputs, and which require steward-only access?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 42, 53; basis: collaborative planning framework with participatory site suitability, criteria, ranking, stakeholder weighting schemes, and sensitivity analysis; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-001, KFM-IDX-POL-001, KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.11 KFM-P18-INV-204 — Environmental quality indices need deprivation-context metadata

Idea card metadata - Stable ID: KFM-P18-INV-204 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 204 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM composite environmental quality or deprivation indices should disclose selected indicators, normalization, spatial units, weighting, and population context before they are used in planning narratives.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM composite environmental quality or deprivation indices should disclose selected indicators, normalization, spatial units, weighting, and population context before they are used in planning narratives. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes composite social-environmental claims inspectable and avoids reducing vulnerable communities to opaque scores.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Indicator registry - PROPOSED: Normalization method - PROPOSED: Spatial unit version - PROPOSED: Equity review

Tensions - PROPOSED: Composite indicators can hide unequal measurement quality across neighborhoods.

Expansion directions - PROPOSED: Add `index_component` and `equity_context` fields for composite spatial indicators.

Open questions - NEEDS VERIFICATION: Which indicator-weighting approaches are acceptable for KFM public-facing equity or deprivation layers?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 58, 64, 65; basis: environmental quality analysis in a multiple-deprivation context using environmental aspects, datasets, methodology, remote sensing, and spatial indicators; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-003, KFM-IDX-POL-001, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.12 KFM-P18-INV-205 — Target-population multi-criteria route assessment

Idea card metadata - Stable ID: KFM-P18-INV-205 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 205 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM route, access, and corridor analyses should preserve target-population preferences and multi-criteria assumptions when recommending or comparing route options.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM route, access, and corridor analyses should preserve target-population preferences and multi-criteria assumptions when recommending or comparing route options. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It avoids presenting route suitability as universal when it depends on population-specific needs and preferences.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Target-population profile - PROPOSED: MCA criteria - PROPOSED: Preference weights - PROPOSED: Route suitability layer

Tensions - PROPOSED: A route optimized for one population may disadvantage another.

Expansion directions - PROPOSED: Create `route_mca_receipt` fields for target population, criteria weights, constraints, and reviewed tradeoffs.

Open questions - NEEDS VERIFICATION: How should KFM handle conflicting route recommendations for different public groups?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 18, 238, 245, 246; basis: stakeholder-based multiple criteria analysis for designing cycle routes for different target populations, including preferences and criteria; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.13 KFM-P18-INV-206 — VGI shelter preferences require authority separation

Idea card metadata - Stable ID: KFM-P18-INV-206 - Pass: 18 - Phase 5 batch: 009 - Inventory sequence: 206 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 009. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM should treat volunteered geographic information about shelters, evacuation sites, or local preferences as community evidence that requires validation and cannot replace official emergency or planning authority.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat volunteered geographic information about shelters, evacuation sites, or local preferences as community evidence that requires validation and cannot replace official emergency or planning authority. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It lets KFM capture local knowledge without turning unverified crowd input into authoritative safety guidance.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-DAT-003.

Dependencies - PROPOSED: VGI source descriptor - PROPOSED: Official-source crosswalk - PROPOSED: Validation state - PROPOSED: No-life-safety disclaimer

Tensions - PROPOSED: VGI can be valuable for planning but unsafe if used as real-time emergency instruction.

Expansion directions - PROPOSED: Define VGIEvidenceRecord fields for contributor context, validation, official crosswalk, and release restrictions.

Open questions - NEEDS VERIFICATION: Which VGI classes can be public, and which must remain steward-review planning input?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 321, 326, 328; basis: VGI for residents' preferences around official and unofficial flood evacuation shelters and use in spatial planning; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-001, KFM-IDX-EVD-001, KFM-IDX-DAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.14 KFM-P18-INV-243 — TOD and accessibility indices need actual-versus-potential labels

Idea card metadata - Stable ID: KFM-P18-INV-243 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 243 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM transit, access, and development-potential indices should distinguish actual conditions from potential or scenario-modeled values.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM transit, access, and development-potential indices should distinguish actual conditions from potential or scenario-modeled values. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents scenario/planning potential from being read as observed present condition.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-001. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: Index schema - PROPOSED: Scenario label - PROPOSED: Method receipt - PROPOSED: Planning-use caveat

Tensions - PROPOSED: An index can guide planning while lacking direct evidentiary force as an observed fact.

Expansion directions - PROPOSED: Add index_kind enum values such as observed_actual, modeled_potential, scenario_projection, and policy_target.

Open questions - NEEDS VERIFICATION: Which public index labels are clear enough for non-expert users?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 291; basis: Actual and Potential TOD index maps and index values distinguish present and potential transit-oriented development conditions; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.15 KFM-P18-INV-244 — Urban water supply layers separate performance, preference, and policy

Idea card metadata - Stable ID: KFM-P18-INV-244 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 244 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM infrastructure and service layers should separate measured performance, household or stakeholder preference, and policy/administrative context.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM infrastructure and service layers should separate measured performance, household or stakeholder preference, and policy/administrative context. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents service maps from flattening lived experience, infrastructure reliability, and official planning posture into one number.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Service observation - PROPOSED: Preference survey evidence - PROPOSED: Policy source - PROPOSED: Infrastructure asset record

Tensions - PROPOSED: Preference evidence may be sensitive or unevenly sampled.

Expansion directions - PROPOSED: Create service_layer_evidence fields for performance_measure, preference_evidence, policy_context, and source_role.

Open questions - NEEDS VERIFICATION: How should KFM weight survey preference evidence against official infrastructure records?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 296; basis: urban water supply analysis separates performance, preferences, policies, coexisting infrastructure systems, and access/quality issues; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.16 KFM-P18-INV-245 — Resettlement and displacement dynamics as temporal review evidence

Idea card metadata - Stable ID: KFM-P18-INV-245 - Pass: 18 - Phase 5 batch: 010 - Inventory sequence: 245 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 010. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM settlement, infrastructure, and planning claims should record displacement, post-resettlement migration, social disarticulation, and changing household conditions when those dynamics affect interpretation.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM settlement, infrastructure, and planning claims should record displacement, post-resettlement migration, social disarticulation, and changing household conditions when those dynamics affect interpretation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It protects vulnerable-population claims from becoming static map labels divorced from temporal social effects.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Temporal household/context record - PROPOSED: Sensitivity review - PROPOSED: Planning evidence bundle

Tensions - PROPOSED: Precise household or community data can create privacy and stigma risk.

Expansion directions - PROPOSED: Add resettlement_context_receipt fields for time period, observed change, source role, privacy class, and release generalization.

Open questions - NEEDS VERIFICATION: When should displacement-related data be generalized, restricted, or denied?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 275, 277; basis: post-resettlement dynamics include household migration, social disarticulation, community organization, and access to economic opportunities; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-001, KFM-IDX-DAT-003, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.17 KFM-P18-INV-257 — Collaborative planning weights and stakeholder roles need evidence records

Idea card metadata - Stable ID: KFM-P18-INV-257 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 257 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: PLN — Planning, Participation, Re-

silience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM participatory planning outputs should record stakeholder roles, involvement level, criteria, weights, and consensus method when maps or indices reflect collaborative choices.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM participatory planning outputs should record stakeholder roles, involvement level, criteria, weights, and consensus method when maps or indices reflect collaborative choices. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It lets public users see when a planning layer reflects negotiated values rather than only measured conditions.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Stakeholder register - PROPOSED: criteria-weight receipt - PROPOSED: review record - PROPOSED: method note

Tensions - PROPOSED: Stakeholder weighting can be legitimate planning evidence but not neutral measurement.

Expansion directions - PROPOSED: Add `planning_weight_receipt` with `stakeholder_class`, `participation_level`, `criteria`, `weight`, `method`, and `dissent_notes`.

Open questions - NEEDS VERIFICATION: How should KFM expose dissent or minority viewpoints in collaborative planning outputs?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 29, 30, 31, 32; basis: collaborative planning, stakeholder roles, participation levels, intelligence/design/choice phases, and participatory site-suitability framework; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-001, KFM-IDX-DAT-003, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.18 KFM-P18-INV-259 — Street and public-space quality metrics split observation from perception

Idea card metadata - Stable ID: KFM-P18-INV-259 - Pass: 18 - Phase 5 batch: 011 - Inventory sequence: 259 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Imple-

mentation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 011. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM public-space and street-quality layers should distinguish measured built-environment attributes, expert audits, resident perceptions, and social-interaction interpretations.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM public-space and street-quality layers should distinguish measured built-environment attributes, expert audits, resident perceptions, and social-interaction interpretations. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It avoids presenting subjective or context-bound urban quality metrics as purely objective spatial facts.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-003. - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: Observation schema - PROPOSED: perception survey evidence - PROPOSED: audit method - PROPOSED: public caveat

Tensions - PROPOSED: Perception evidence can be affected by sampling bias and cultural context.

Expansion directions - PROPOSED: Add `public_space_metric` fields for `metric_kind`, `evidence_type`, `observer_group`, `survey_method`, and `confidence_label`.

Open questions - NEEDS VERIFICATION: How should KFM combine observed design features and resident perception without collapsing them?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 103, 183, 163; basis: public-space role, street quality as public space, divided-city context, and qualitative GIS/perception investigations; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.19 KFM-P18-INV-300 — VGI disaster-planning inputs require verification and freshness labels

Idea card metadata - Stable ID: KFM-P18-INV-300 - Pass: 18 - Phase 5 batch: 012 - Inventory sequence: 300 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 012. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM volunteered geographic information for disaster planning should carry contributor/source type, collection channel, verification status, freshness, and official-source comparison.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM volunteered geographic information for disaster planning should carry contributor/source type, collection channel, verification status, freshness, and official-source comparison. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It allows public participation without letting unverified or stale volunteered points become authoritative emergency guidance.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-004. - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: VGI source descriptor - PROPOSED: freshness policy - PROPOSED: official-source cross-check

Tensions - PROPOSED: Near-real-time volunteered data can be useful and also risky if treated as official or life-safety guidance.

Expansion directions - PROPOSED: Add VGI verification_status and not_emergency_alert_system fields to hazard/planning contexts.

Open questions - NEEDS VERIFICATION: Which VGI classes should be usable only for review and never public live mapping?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 321; basis: The sustainable urban GIS source discusses VGI from social media and mapping platforms for disaster response and flood-evacuation shelter planning.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-004, KFM-IDX-DAT-012, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.20 KFM-P18-INV-320 — Data-poor planning support requires proxy and uncertainty disclosure

Idea card metadata - Stable ID: KFM-P18-INV-320 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 320 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM planning-support tools built for data-poor contexts should record proxy data, assumptions, uncertainty, and limits of scenario analysis.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed

design candidate: KFM planning-support tools built for data-poor contexts should record proxy data, assumptions, uncertainty, and limits of scenario analysis. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It keeps scarce-data scenarios useful without overstating confidence.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: proxy data register - PROPOSED: scenario assumptions - PROPOSED: uncertainty note

Tensions - PROPOSED: Planning still needs decisions when evidence is incomplete.

Expansion directions - PROPOSED: Add data_poverty and proxy_source fields to planning-support receipts.

Open questions - NEEDS VERIFICATION: Which uncertainty labels are understandable to local planning audiences?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 17, 22; basis: Urban planning GIS source identifies a planning support tool for low-carbon transport scenario analysis in a data-poor context and frames it under sustainable transport planning.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-012, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.21 KFM-P18-INV-321 — Collaborative planning records stakeholder perspective and multidimensional criteria

Idea card metadata - Stable ID: KFM-P18-INV-321 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 321 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM collaborative planning layers should record stakeholder roles, criteria families, and how social, environmental, and economic dimensions were balanced.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM collaborative planning layers should record stakeholder roles, criteria families, and how social, environmental, and economic dimensions were balanced. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION:

Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It makes participation and weighting visible in planning-support outputs.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-DOC-007.

Dependencies - PROPOSED: stakeholder record - PROPOSED: criteria taxonomy - PROPOSED: weighting or deliberation receipt

Tensions - PROPOSED: Stakeholder inclusion can conflict with technical ranking simplicity.

Expansion directions - PROPOSED: Add participant_role and criteria_dimension fields to planning-support records.

Open questions - NEEDS VERIFICATION: How should KFM record disagreement among stakeholder groups?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 43, 44; basis: Urban planning GIS source describes collaborative planning for municipal housing, trans-disciplinary location evaluation, and a shift from planning for the public to planning with the public.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-001, KFM-IDX-DOC-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.22 KFM-P18-INV-322 — Stated choice experiments are preference evidence, not observed behavior

Idea card metadata - Stable ID: KFM-P18-INV-322 - Pass: 18 - Phase 5 batch: 013 - Inventory sequence: 322 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 013. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM preference-based planning analyses should label stated choice experiment outputs separately from observed behavior and policy decisions.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM preference-based planning analyses should label stated choice experiment outputs separately from observed behavior and policy decisions. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents modeled preferences from being mistaken for actual service use or official policy.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-012. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: preference method label - PROPOSED: attribute set - PROPOSED: sample and scenario description

Tensions - PROPOSED: Stated preferences can guide design but may diverge from real-world behavior.

Expansion directions - PROPOSED: Add evidence_kind = stated_preference | observed_behavior | policy_decision to planning records.

Open questions - NEEDS VERIFICATION: When can stated preference evidence support public prioritization maps?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 298; basis: Urban planning GIS source explains stated choice experiments as behavioral research using hypothetical attribute combinations, with advantages and realism limitations.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.23 KFM-P18-INV-343 — SDG 11 city-concept labels as planning-policy lenses, not source truth

Idea card metadata - Stable ID: KFM-P18-INV-343 - Pass: 18 - Phase 5 batch: 014 - Inventory sequence: 343 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 014. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM settlement and infrastructure analyses may use sustainable, inclusive, compact, competitive, and resilient city concepts as policy lenses, but should label them as interpretive frameworks rather than direct observations.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM settlement and infrastructure analyses may use sustainable, inclusive, compact, competitive, and resilient city concepts as policy lenses, but should label them as interpretive frameworks rather than direct observations. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It lets KFM use urban-planning concepts without overstating them as measured facts.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-013. - CONFIRMED baseline link: KFM-IDX-DOC-007.

Dependencies - PROPOSED: policy lens label - PROPOSED: indicator basis - PROPOSED: normative framework note - PROPOSED: review context

Tensions - PROPOSED: Normative labels can obscure whose goals and tradeoffs are represented.

Expansion directions - PROPOSED: Add `planning_policy_lens` to settlement/infrastructure evidence summaries.

Open questions - NEEDS VERIFICATION: Which KFM outputs should expose SDG-11 framing publicly?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 14, 16; basis: GIS in Sustainable Urban Planning and Management introduces sustainable, inclusive, compact, competitive, and resilient city concepts linked to SDG 11.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-013, KFM-IDX-DOC-007 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.24 KFM-P18-INV-353 — Data-poor scenario analysis as bounded planning evidence

Idea card metadata - Stable ID: KFM-P18-INV-353 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 353 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM should treat data-poor scenario planning outputs as bounded planning evidence with explicit assumptions, input scarcity labels, and non-authoritative model status.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should treat data-poor scenario planning outputs as bounded planning evidence with explicit assumptions, input scarcity labels, and non-authoritative model status. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It allows useful planning support without pretending sparse inputs can support the same claims as authoritative measurements.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-013. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: scenario definition - PROPOSED: assumption register - PROPOSED: input sufficiency label - PROPOSED: model output receipt

Tensions - PROPOSED: Scenario outputs may be politically influential even when evidentiary support is thin.

Expansion directions - PROPOSED: Define a `scenario_evidence_strength` field for planning layers.

Open questions - NEEDS VERIFICATION: How should KFM label planning scenarios that are useful but weakly measured?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 17, 22, 40, 57; basis: GIS in Sustainable Urban Planning and Management includes a planning support tool for low-carbon transport plans in a data-poor context and scenario-based low-carbon transport references.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.25 KFM-P18-INV-354 — Multi-criteria cycle-route assessment with target-population preferences

Idea card metadata - Stable ID: KFM-P18-INV-354 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 354 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM planning-support layers should preserve stakeholder or target-population preferences separately from computed route suitability scores in multi-criteria assessments.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM planning-support layers should preserve stakeholder or target-population preferences separately from computed route suitability scores in multi-criteria assessments. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents a computed suitability map from hiding whose preferences shaped the result.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-DAT-012.

Dependencies - PROPOSED: stakeholder group - PROPOSED: criteria weights - PROPOSED: route candidates - PROPOSED: decision receipt

Tensions - PROPOSED: Preferences can be sensitive or contested and must not be collapsed into a single neutral score.

Expansion directions - PROPOSED: Add criteria_weight_source and target_population labels to MCA outputs.

Open questions - NEEDS VERIFICATION: Can KFM publish preference-weighted route maps without steward review?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 18, 238; basis: GIS in Sustainable Urban Planning and Management identifies stakeholder-based assessment and multiple-criteria analysis for designing cycle routes for different target populations.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.26 KFM-P18-INV-356 — Post-resettlement monitoring as a temporal equity claim family

Idea card metadata - Stable ID: KFM-P18-INV-356 - Pass: 18 - Phase 5 batch: 015 - Inventory sequence: 356 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 015. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM should model resettlement and displacement-related urban planning evidence as temporal equity claims that track pre-event, relocation, and post-resettlement conditions separately.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should model resettlement and displacement-related urban planning evidence as temporal equity claims that track pre-event, relocation, and post-resettlement conditions separately. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: It prevents one-time map snapshots from erasing whether outcomes improved, worsened, or shifted over time.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-002. - CONFIRMED baseline link: KFM-IDX-DOC-005.

Dependencies - PROPOSED: valid time - PROPOSED: baseline condition - PROPOSED: post-event observation - PROPOSED: review state

Tensions - PROPOSED: Living-person and vulnerable-community data may require redaction, generalization, or denial.

Expansion directions - PROPOSED: Create a temporal equity indicator profile with sensitivity defaults.

Open questions - NEEDS VERIFICATION: What public-safe generalization is appropriate for displacement-related maps?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 18, 258; basis: GIS in Sustainable Urban Planning and Management includes post-resettlement socio-economic dynamics and identifies involuntary displacement, resettlement, and socio-economic dynamics.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.27 KFM-P18-INV-420 — Indicator transferability caveat across global planning case studies

Idea card metadata - Stable ID: KFM-P18-INV-420 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 420 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM planning indicators imported from global case studies should carry transferability caveats before being applied to Kansas contexts.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM planning indicators imported from global case studies should carry transferability caveats before being applied to Kansas contexts. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Indicators built for one city, data environment, or policy problem can mislead if transferred without context, scale, and population assumptions.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-013. - CONFIRMED baseline link: KFM-IDX-POL-004.

Dependencies - PROPOSED: indicator registry - PROPOSED: context field - PROPOSED: source case study - PROPOSED: Kansas adaptation review

Tensions - PROPOSED: Cross-context methods are useful precisely because they are adaptable.

Expansion directions - PROPOSED: Add transferability_notes to planning indicator definitions.

Open questions - NEEDS VERIFICATION: Which urban indicators are meaningful at Kansas county, municipality, tract, or neighborhood scales?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 14, 16, 17; basis: GIS in Sustainable Urban Planning and Management frames case studies across Global South and Global North contexts and links GIS to sustainable, inclusive, compact, competitive, and resilient city concepts.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-013, KFM-IDX-POL-004 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.28 KFM-P18-INV-421 — Participatory planning evidence requires method and facilitation records

Idea card metadata - Stable ID: KFM-P18-INV-421 - Pass: 18 - Phase 5 batch: 017 - Inventory sequence: 421 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived

from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 017. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM participatory or stakeholder-derived spatial evidence should record facilitation method, participant role, deliberation context, and dissent or uncertainty notes.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM participatory or stakeholder-derived spatial evidence should record facilitation method, participant role, deliberation context, and dissent or uncertainty notes. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Stakeholder evidence is valuable but not interchangeable with sensor or administrative data; method receipts preserve how the evidence was produced.

Related ideas - CONFIRMED baseline link: KFM-IDX-EVD-006. - CONFIRMED baseline link: KFM-IDX-POL-005.

Dependencies - PROPOSED: participant privacy policy - PROPOSED: method receipt - PROPOSED: review state - PROPOSED: consent/obligation field

Tensions - PROPOSED: Some participant details may need redaction while still preserving methodological transparency.

Expansion directions - PROPOSED: Create participatory_evidence_method fields for qualitative and collaborative GIS records.

Open questions - NEEDS VERIFICATION: How should KFM represent dissenting stakeholder positions without flattening them into one indicator?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 19, 83, 90; basis: GIS in Sustainable Urban Planning and Management discusses integrated and participatory planning, collaborative frameworks, stakeholder assessments, and social learning approaches.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-EVD-006, KFM-IDX-POL-005 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.29 KFM-P18-INV-444 — Transit-oriented-development index as composite planning evidence

Idea card metadata - Stable ID: KFM-P18-INV-444 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 444 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM planning-support layers that use composite indices should record component indicators, weights, and intended policy lens.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM planning-support layers that use composite indices should record component indicators, weights, and intended policy lens. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Composite indices can be useful but hide methodological choices unless components and weights are inspectable.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-010. - CONFIRMED baseline link: KFM-IDX-EVD-005.

Dependencies - PROPOSED: indicator registry - PROPOSED: weighting receipt - PROPOSED: planning layer manifest - PROPOSED: review note

Tensions - PROPOSED: Policy users may overread index scores as objective rankings.

Expansion directions - PROPOSED: Add `composite_index_receipt` to planning/equity layers.

Open questions - NEEDS VERIFICATION: Which KFM lanes should admit composite indices as public-facing evidence?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 7, 282, 284, 286, 288; basis: GIS in Sustainable Urban Planning and Management includes a chapter on planning for Transit Oriented Development using a TOD index.; Phase 1 role: `reference_source_candidate`.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.30 KFM-P18-INV-445 — VGI flood-shelter inputs as community-contributed evidence

Idea card metadata - Stable ID: KFM-P18-INV-445 - Pass: 18 - Phase 5 batch: 018 - Inventory sequence: 445 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 018. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM should label volunteered geographic information as community-contributed evidence with review, verification, and public-safety boundaries.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM should label volunteered geographic information as community-contributed evidence with review, verification, and public-safety boundaries. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should

remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: VGI can add local knowledge but must not be treated as authoritative emergency guidance without review.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-004. - CONFIRMED baseline link: KFM-IDX-DAT-010.

Dependencies - PROPOSED: source-role registry - PROPOSED: review workflow - PROPOSED: hazard policy - PROPOSED: EvidenceBundle

Tensions - PROPOSED: Community reports can be urgent yet unverified.

Expansion directions - PROPOSED: Define a VGI source_role with fail-closed public-safety rules.

Open questions - NEEDS VERIFICATION: Can KFM display unverified VGI as context without implying official status?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 7, 321, 322, 326; basis: GIS in Sustainable Urban Planning and Management includes a chapter on Volunteered Geographic Information for flood evacuation shelter planning in Jakarta.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-004, KFM-IDX-DAT-010 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.31 KFM-P18-INV-468 — Data-poor scenario analysis carries explicit uncertainty

Idea card metadata - Stable ID: KFM-P18-INV-468 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 468 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM planning-support outputs in data-poor contexts should be labeled as scenario analysis with assumptions, missing-data constraints, and decision-support limits.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM planning-support outputs in data-poor contexts should be labeled as scenario analysis with assumptions, missing-data constraints, and decision-support limits. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Scenario maps are useful, but they should not be published as observed facts where data is sparse.

Related ideas - CONFIRMED baseline link: KFM-IDX-DOC-007. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: scenario model - PROPOSED: assumption register - PROPOSED: planning evidence - PROPOSED: review note

Tensions - PROPOSED: Decision-makers may treat scenario outputs as predictions or authoritative measurements.

Expansion directions - PROPOSED: Add scenario_assumption and data_poverty fields to planning layers.

Open questions - NEEDS VERIFICATION: What threshold marks a layer as data-poor enough to require an ABSTAIN caveat?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 17, 22; basis: The urban planning GIS source includes a planning support tool for low-carbon transport plans in a data-poor context and frames scenario analysis as a planning method.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.32 KFM-P18-INV-469 — Knowledge co-production method records for environmental health layers

Idea card metadata - Stable ID: KFM-P18-INV-469 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 469 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM participatory environmental-health layers should record facilitation method, participant roles, issue framing, social-learning context, and how community inputs affected outputs.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM participatory environmental-health layers should record facilitation method, participant roles, issue framing, social-learning context, and how community inputs affected outputs. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Community-derived evidence needs method provenance so it is not flattened into anonymous map data.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-004. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: participatory session - PROPOSED: method note - PROPOSED: community input - PROPOSED: review state

Tensions - PROPOSED: Participation records can include privacy-sensitive or politically sensitive information.

Expansion directions - PROPOSED: Define public and steward-only fields for participatory-planning evidence.

Open questions - NEEDS VERIFICATION: How should KFM distinguish community preference from measured environmental burden?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 96; basis: The urban planning GIS source describes knowledge co-production and social learning on environmental health issues using interactive GIS-based approaches.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.33 KFM-P18-INV-470 — Stakeholder MCA weights as planning evidence

Idea card metadata - Stable ID: KFM-P18-INV-470 - Pass: 18 - Phase 5 batch: 019 - Inventory sequence: 470 - Status label: PROPOSED - Carry-forward state: NEW - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 019. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM multiple-criteria planning outputs should preserve stakeholder group definitions, criteria weights, scoring method, alternatives, and uncertainty in the resulting recommendation.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM multiple-criteria planning outputs should preserve stakeholder group definitions, criteria weights, scoring method, alternatives, and uncertainty in the resulting recommendation. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: A ranked planning option is only interpretable when the weights and stakeholders behind the ranking are visible.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-004. - CONFIRMED baseline link: KFM-IDX-POL-001.

Dependencies - PROPOSED: criteria table - PROPOSED: stakeholder group - PROPOSED: alternative route - PROPOSED: decision record

Tensions - PROPOSED: MCA results can appear objective while embedding contested priorities.

Expansion directions - PROPOSED: Add MCA criteria/weight fixtures to planning-support idea cards.

Open questions - NEEDS VERIFICATION: Who can approve stakeholder weights for semi-public release?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 238; basis: The urban planning GIS source presents stakeholder-based assessment and multiple criteria analysis for designing cycle routes for different target populations.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is NEW; no prior idea was treated as replaced or re-authored by this card.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.34 KFM-P18-INV-496 — Urban quality-of-life indicators as composite planning evidence

Idea card metadata - Stable ID: KFM-P18-INV-496 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 496 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM composite planning indicators should preserve component variables, weighting method, geography, time scope, and caveats rather than exposing only a final quality-of-life score.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM composite planning indicators should preserve component variables, weighting method, geography, time scope, and caveats rather than exposing only a final quality-of-life score. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Composite indicators are powerful but can obscure deprivation dimensions and methodological choices.

Related ideas - CONFIRMED baseline link: KFM-IDX-DAT-001. - CONFIRMED baseline link: KFM-IDX-CAT-003.

Dependencies - PROPOSED: spatial indicator - PROPOSED: quality of life - PROPOSED: composite metric - PROPOSED: planning layer

Tensions - PROPOSED: Public dashboards often demand simple scores even when evidence is multidimensional.

Expansion directions - PROPOSED: Define composite_indicator_receipt for planning-support layers.

Open questions - NEEDS VERIFICATION: Which KFM domain categories should be allowed to publish composite indices?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 6, 121; basis: The sustainable urban planning GIS source includes a chapter

on spatial variability of urban quality of life and repeatedly frames GIS as support for policy and planning indicators.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-DAT-001, KFM-IDX-CAT-003 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

8.14.35 KFM-P18-INV-497 — Stakeholder MCA weights as reviewable planning evidence

Idea card metadata - Stable ID: KFM-P18-INV-497 - Pass: 18 - Phase 5 batch: 020 - Inventory sequence: 497 - Status label: PROPOSED - Carry-forward state: EXPANDED - Category: PLN — Planning, Participation, Resilience, and Decision-Support Applications - Category assignment status: PROPOSED assignment derived from CONFIRMED Phase 3 inventory; subject to Phase 5 card-level review - Source support label: CONFIRMED - Implementation maturity: UNKNOWN — no repo/runtime proof was verified in Phase 5 Batch 020. - Source IDs: SRC-P18-011

Normalized statement - PROPOSED: KFM multiple-criteria planning outputs should keep stakeholder weights, criteria definitions, and scenario assumptions reviewable as evidence objects.

Detailed explanation - CONFIRMED: Source SRC-P18-011 (GIS in Sustainable Urban Planning and Management.pdf) supports the source concept recorded for this idea in the Phase 3 inventory. - CONFIRMED: Phase 3 inventory recorded this idea with source support label CONFIRMED and Phase 4 assigned it to a dependency-ordered category before Phase 5 card writing. - PROPOSED: In KFM, this idea should be treated as a governed design candidate: KFM multiple-criteria planning outputs should keep stakeholder weights, criteria definitions, and scenario assumptions reviewable as evidence objects. - PROPOSED: Phase 3 scope note: Inventory-level candidate record; full idea-card prose deferred to Phase 5. - PROPOSED: Public or semi-public use should remain downstream of evidence, source role, policy posture, review state, release state, correction path, and rollback target. - UNKNOWN: No current-session repository evidence proves that this idea is already implemented, tested, released, or wired to a public KFM surface. - NEEDS VERIFICATION: Source support confirms the planning concept; it does not prove mounted KFM repo behavior, package versions, source rights, runtime routing, or release state.

Why it matters - PROPOSED: Planning recommendations depend on who weighted the criteria and how; those choices must be visible for governance.

Related ideas - CONFIRMED baseline link: KFM-IDX-POL-001. - CONFIRMED baseline link: KFM-IDX-EVD-001.

Dependencies - PROPOSED: multiple criteria analysis - PROPOSED: stakeholder weights - PROPOSED: review packet - PROPOSED: planning decision

Tensions - PROPOSED: Stakeholder weights may be sensitive or politically contested.

Expansion directions - PROPOSED: Add MCA_weight_receipt to planning and infrastructure expansion agenda.

Open questions - NEEDS VERIFICATION: When should stakeholder inputs be anonymized, attributed, or denied for public release?

Future work - PROPOSED: Draft or update the relevant fragment, fixture, validator, or manifest in a later phase without bypassing lifecycle, policy, evidence, release, correction, or rollback controls.

Source attribution - CONFIRMED source attribution: SRC-P18-011 — GIS in Sustainable Urban Planning and Management.pdf; pages 7, 18; basis: The sustainable urban planning GIS source lists stakeholder-based assessment and multiple criteria analysis for designing cycle routes as planning-support methods.; Phase 1 role: reference_source_candidate.

Carry-forward note - CONFIRMED: Carry-forward state is EXPANDED; this card extends KFM-IDX-POL-001, KFM-IDX-EVD-001 with Pass 18 source structure and does not replace or inline prior-pass prose.

Phase 5 self-check - CONFIRMED: Required schema fields are present in this Markdown card. - CONFIRMED: No prior-pass body text is inlined. - CONFIRMED: Narrative claims use explicit truth labels. - UNKNOWN: Repository implementation status remains unverified.

9 Cross-Cutting Themes

- **CONFIRMED:** This fragment is derived from persisted Phase 3, Phase 4, and Phase 5 artifacts on disk.
- **PROPOSED:** The themes below are synthesis structure for the final dossier, not implementation claims.

9.1 P18-XCT-001 — Representation is a governed claim surface, not neutral display

- **PROPOSED** synthesis from **CONFIRMED** Phase 5 cards: KFM should treat map layers, styles, tiles, and UI states as choice-bearing representations that must expose scale, intended use, source support, policy posture, and correction state.
- **PROPOSED:** Why it matters: Public trust can be weakened when a map looks authoritative while hiding omissions, generalization, stale state, or renderer limitations.
- **PROPOSED:** Expansion direction: Add `representation_intent`, `intended_use`, `prohibited_use`, `scale_limit`, and `stale_state` checks to layer/release review fragments.
- **CONFIRMED:** Category support: REP, MAP, UIX.
- **CONFIRMED:** Source support IDs in card index: SRC-P18-002, SRC-P18-039.
- **CONFIRMED:** Representative idea IDs: KFM-P18-INV-001, KFM-P18-INV-002, KFM-P18-INV-004, KFM-P18-INV-045, KFM-P18-INV-048, KFM-P18-INV-049, KFM-P18-INV-050, KFM-P18-INV-051, KFM-P18-INV-052, KFM-P18-INV-055.

9.2 P18-XCT-002 — Temporal structure is a first-class evidence dimension

- **PROPOSED** synthesis from **CONFIRMED** Phase 5 cards: KFM should keep valid time, observation time, transaction or retrieval time, release time, and correction time distinct whenever the distinction changes the claim.
- **PROPOSED:** Why it matters: Historical, environmental, planning, and ownership claims can be wrong if temporal validity is collapsed into a single date field.
- **PROPOSED:** Expansion direction: Create a temporal-support checklist for source descriptors, canonical records, release manifests, and public query responses.
- **CONFIRMED:** Category support: MOD, REL, ANA.
- **CONFIRMED:** Source support IDs in card index: SRC-P18-007, SRC-P18-003.
- **CONFIRMED:** Representative idea IDs: KFM-P18-INV-021, KFM-P18-INV-022, KFM-P18-INV-023, KFM-P18-INV-026, KFM-P18-INV-027, KFM-P18-INV-028, KFM-P18-INV-029, KFM-P18-INV-030, KFM-P18-INV-063, KFM-P18-INV-064.

9.3 P18-XCT-003 — Source admission and connector behavior must be policy-aware

- **PROPOSED** synthesis from **CONFIRMED** Phase 5 cards: KFM should admit external data through source descriptors, error handling, rate and rights controls, extraction receipts, and fail-closed validators before any artifact can inform publication.
- **PROPOSED:** Why it matters: Connectors and scrapers can silently change, misread pages, violate terms, or lose provenance unless they are treated as governed intake systems.
- **PROPOSED:** Expansion direction: Write source-admission gates that require `source_role`, `rights_spdx` or `NOASSERTION`, `extraction_method`, `retry/error` policy, and quarantine outcome.
- **CONFIRMED:** Category support: SRC, EVD, POL, VAL.
- **CONFIRMED:** Source support IDs in card index: SRC-P18-012, SRC-P18-006, SRC-P18-039.
- **CONFIRMED:** Representative idea IDs: KFM-P18-INV-005, KFM-P18-INV-008, KFM-P18-INV-036, KFM-P18-INV-037, KFM-P18-INV-038, KFM-P18-INV-039, KFM-P18-INV-044, KFM-P18-INV-046, KFM-P18-INV-056, KFM-P18-INV-067.

9.4 P18-XCT-004 — Analysis and ML outputs remain interpretive derivatives

- **PROPOSED** synthesis from **CONFIRMED** Phase 5 cards: KFM should use statistics, indicators, clustering, regression, model evaluation, and AI-assisted synthesis as bounded interpretation over evidence, not as replacement evidence.

- PROPOSED: Why it matters: Analytical confidence, training/test choices, missing-data treatment, and indicator construction can change conclusions without changing the underlying evidence.
- PROPOSED: Expansion direction: Add `model_or_indicator_card` records with inputs, assumptions, validation metrics, intended use, prohibited use, and `evidence_refs`.
- CONFIRMED: Category support: ANA, VAL, EVD.
- CONFIRMED: Source support IDs in card index: SRC-P18-004, SRC-P18-003, SRC-P18-011.
- CONFIRMED: Representative idea IDs: KFM-P18-INV-003, KFM-P18-INV-013, KFM-P18-INV-018, KFM-P18-INV-021, KFM-P18-INV-022, KFM-P18-INV-024, KFM-P18-INV-026, KFM-P18-INV-027, KFM-P18-INV-028, KFM-P18-INV-029.

9.5 P18-XCT-005 — Field, remote-sensing, and 3D capture need evidentiary humility

- PROPOSED synthesis from CONFIRMED Phase 5 cards: KFM should distinguish captured geometry, interpreted features, archaeological or environmental hypotheses, and public-safe derived products.
- PROPOSED: Why it matters: 3D models, sensor scenes, drone/field captures, and remote-sensing layers can be compelling but still require review, sensitivity controls, uncertainty notes, and rights checks.
- PROPOSED: Expansion direction: Create `capture_record`, `interpretation_record`, and `public_derivative` links in the evidence model, with generalized geometry as a separate release artifact.
- CONFIRMED: Category support: FIE, REP, POL.
- CONFIRMED: Source support IDs in card index: SRC-P18-005, SRC-P18-010.
- CONFIRMED: Representative idea IDs: KFM-P18-INV-006, KFM-P18-INV-007, KFM-P18-INV-016, KFM-P18-INV-017, KFM-P18-INV-019, KFM-P18-INV-020, KFM-P18-INV-053, KFM-P18-INV-057, KFM-P18-INV-059, KFM-P18-INV-079.

9.6 P18-XCT-006 — APIs and renderers are trust membranes, not truth stores

- PROPOSED synthesis from CONFIRMED Phase 5 cards: KFM should expose public value through governed resources, finite outcomes, released artifacts, and evidence-backed payloads rather than raw stores, direct model output, or renderer state.
- PROPOSED: Why it matters: A browser map, API endpoint, popup, or Focus Mode answer can mislead if it bypasses `EvidenceBundle` resolution, policy checks, release state, or rollback targets.
- PROPOSED: Expansion direction: Align API resource ontology, `LayerManifest`, `Evidence Drawer` payload, and `MapReleaseManifest` around the same evidence and policy envelope.
- CONFIRMED: Category support: API, MAP, UIX, REL.
- CONFIRMED: Source support IDs in card index: SRC-P18-006, SRC-P18-039.
- CONFIRMED: Representative idea IDs: KFM-P18-INV-009, KFM-P18-INV-023, KFM-P18-INV-025, KFM-P18-INV-040, KFM-P18-INV-041, KFM-P18-INV-042, KFM-P18-INV-045, KFM-P18-INV-047, KFM-P18-INV-048, KFM-P18-INV-049.

9.7 P18-XCT-007 — Planning support requires participatory, equity, and scenario context

- PROPOSED synthesis from CONFIRMED Phase 5 cards: KFM planning outputs should state who the decision support is for, which communities and constraints are represented, and where indicators or scenarios are incomplete.
- PROPOSED: Why it matters: Sustainable planning, social vulnerability, accessibility, hazard exposure, and urban-service indicators can become policy-significant when exposed as maps.
- PROPOSED: Expansion direction: Add `planning_decision_context` fragments that capture stakeholder scope, equity lens, scenario assumptions, and review state.
- CONFIRMED: Category support: PLN, ANA, POL.
- CONFIRMED: Source support IDs in card index: SRC-P18-011, SRC-P18-010.
- CONFIRMED: Representative idea IDs: KFM-P18-INV-010, KFM-P18-INV-011, KFM-P18-INV-012, KFM-P18-INV-014, KFM-P18-INV-015, KFM-P18-INV-060, KFM-P18-INV-062, KFM-P18-INV-086, KFM-P18-INV-201, KFM-P18-INV-203.

9.8 P18-XCT-008 — Validation, observability, and rollback are part of publication truth

- PROPOSED synthesis from CONFIRMED Phase 5 cards: KFM should treat validation results, service SLOs, visual regression, run receipts, catalog closure, and rollback references as publication-supporting objects.
- PROPOSED: Why it matters: A claim can be evidence-supported but not safely published if the artifact cannot be verified, rendered, corrected, or rolled back.
- PROPOSED: Expansion direction: Add release validation fragments that tie tests, telemetry, checksums, catalog closure, and rollback cards to the final promotion decision.
- CONFIRMED: Category support: VAL, REL, EVD.
- CONFIRMED: Source support IDs in card index: SRC-P18-039, SRC-P18-003, SRC-P18-006.
- CONFIRMED: Representative idea IDs: KFM-P18-INV-009, KFM-P18-INV-023, KFM-P18-INV-037, KFM-P18-INV-038, KFM-P18-INV-046, KFM-P18-INV-047, KFM-P18-INV-058, KFM-P18-INV-067, KFM-P18-INV-075, KFM-P18-INV-078.

10 Overlaps and Gaps

- **CONFIRMED:** This fragment is based on the 500-card Phase 5 index and Phase 4 dependency categories.
- **PROPOSED:** Overlaps are synthesis groupings that identify where future implementation or final prose should avoid duplicate authority.

10.1 P18-OVL-001 — Source admission overlaps with evidence receipts

- **PROPOSED synthesis:** Source descriptors, connector behavior, extraction logs, and run receipts need one joined review surface so intake evidence does not split from validation evidence.
- **PROPOSED: Risk:** Parallel source and receipt homes would weaken auditability.
- **CONFIRMED:** Category support: SRC, EVD, VAL.
- **CONFIRMED:** Representative idea IDs: KFM-P18-INV-008, KFM-P18-INV-036, KFM-P18-INV-037, KFM-P18-INV-038, KFM-P18-INV-039, KFM-P18-INV-044, KFM-P18-INV-046, KFM-P18-INV-067.

10.2 P18-OVL-002 — Temporal modeling overlaps with release and correction

- **PROPOSED synthesis:** Temporal database design, release versioning, rollback, and correction notices all need shared language for what changed, when it was true, when it was observed, and when KFM published it.
- **PROPOSED: Risk:** Single-date release notes can erase valid-time and transaction-time distinctions.
- **CONFIRMED:** Category support: MOD, REL.
- **CONFIRMED:** Representative idea IDs: KFM-P18-INV-009, KFM-P18-INV-023, KFM-P18-INV-043, KFM-P18-INV-047, KFM-P18-INV-058, KFM-P18-INV-075, KFM-P18-INV-092, KFM-P18-INV-093.

10.3 P18-OVL-003 — Renderer governance overlaps with API contract design

- **PROPOSED synthesis:** MapLibre, PMTiles/COG delivery, Evidence Drawer payloads, and API resource ontology are separable systems but need matching identifiers, finite outcomes, and evidence references.
- **PROPOSED: Risk:** A visually correct layer can still be governance-invalid if the API contract cannot resolve evidence and policy state.
- **CONFIRMED:** Category support: API, MAP, UIX.
- **CONFIRMED:** Representative idea IDs: KFM-P18-INV-025, KFM-P18-INV-040, KFM-P18-INV-041, KFM-P18-INV-042, KFM-P18-INV-045, KFM-P18-INV-048, KFM-P18-INV-049, KFM-P18-INV-050.

10.4 P18-OVL-004 — Analytics overlaps with planning and policy exposure

- **PROPOSED synthesis:** Indicators, machine-learning outputs, scenario scores, and planning support become policy-significant when they appear in public maps or decision-support reports.
- **PROPOSED: Risk:** Model outputs can be misread as authoritative determinations rather than bounded analyses.
- **CONFIRMED:** Category support: ANA, PLN, POL.
- **CONFIRMED:** Representative idea IDs: KFM-P18-INV-003, KFM-P18-INV-005, KFM-P18-INV-010, KFM-P18-INV-011, KFM-P18-INV-012, KFM-P18-INV-013, KFM-P18-INV-014, KFM-P18-INV-015.

10.5 P18-OVL-005 — 3D/field capture overlaps with archaeology, hazards, and sensitivity

- **PROPOSED synthesis:** 3D GIS, field maps, drone-like capture, remote sensing, and archaeological interpretation all require public-safe geometry, source role, and review-state separation.
- **PROPOSED: Risk:** Precise or compelling geometry can expose sensitive places or overstate interpretation.
- **CONFIRMED:** Category support: FIE, POL, REP.
- **CONFIRMED:** Representative idea IDs: KFM-P18-INV-001, KFM-P18-INV-002, KFM-P18-INV-004, KFM-P18-INV-005, KFM-P18-INV-006, KFM-P18-INV-007, KFM-P18-INV-016, KFM-P18-INV-017.

11 Gaps

11.1 P18-GAP-001 — Mounted repo behavior remains unverified

- UNKNOWN implementation depth: No Phase 6 artifact can confirm actual KFM route names, CI behavior, source registry files, runtime logs, dashboards, deployed services, or release state.
- PROPOSED: Needed action: Verify against a mounted repo, tests, workflows, emitted receipts, and release artifacts before implementation claims are upgraded.

11.2 P18-GAP-002 — Current source rights and version-sensitive facts were not rechecked in Phase 6

- NEEDS VERIFICATION: Source PDFs and Phase 1 extraction support idea synthesis, but operational endpoints, package versions, source terms, and current API behavior require fresh official-source verification before use.
- PROPOSED: Needed action: Create a source-rights and source-version verification backlog before publication or connector activation.

11.3 P18-GAP-003 — Five hundred cards need later prioritization into buildable epics

- PROPOSED gap from synthesis: The card inventory is intentionally broad and publication-grade; implementation would need grouping into reversible PRs or design epics rather than attempting all ideas at once.
- PROPOSED: Needed action: Derive a small number of phaseable epics in Phase 7/8 and preserve the full inventory as `baseline_for_next_pass`.

11.4 P18-GAP-004 — Schema-home and path placement must remain doctrine-governed

- NEEDS VERIFICATION: Phase 6 fragments do not create repo paths; future file-home choices must consult Directory Rules, current repo evidence, and ADRs.
- PROPOSED: Needed action: Treat any future path as PROPOSED until Directory Rules and mounted repo evidence agree.

11.5 P18-GAP-005 — Source adaptations from textbooks need KFM-specific acceptance criteria

- PROPOSED gap from synthesis: Several sources are conceptual or instructional references; their transfer into KFM requires explicit acceptance criteria, not direct adoption as doctrine.
- PROPOSED: Needed action: For each reference-derived idea, write a KFM acceptance note that states the source support, adaptation, and limits.

12 Weakly Supported Material

- **CONFIRMED:** This fragment identifies material that must remain bounded in the final dossier.
- **PROPOSED:** The purpose is to prevent fluent synthesis from upgrading weak support into publication claims.

12.1 P18-WEAK-001

- **UNKNOWN: Material:** Actual repository implementation, routes, packages, tests, workflows, dashboards, logs, deployment state, branch protections, and release artifacts.
- **UNKNOWN: Reason:** No mounted repo/runtime evidence was available in the generated artifacts.
- **PROPOSED: Handling:** Keep as UNKNOWN in final dossier; do not promote to CONFIRMED.

12.2 P18-WEAK-002

- **NEEDS VERIFICATION: Material:** Operational source facts, API endpoints, source terms, licenses, quotas, package versions, and tool behavior.
- **NEEDS VERIFICATION: Reason:** Phase 1 extracted PDFs; Phase 6 did not perform current official-source verification.
- **PROPOSED: Handling:** Route to source verification backlog before source activation or public release.

12.3 P18-WEAK-003

- **PROPOSED: Material:** KFM-specific schema fields, validator names, source descriptor fields, release gates, and API/UI contracts inferred from source concepts.
- **PROPOSED: Reason:** These are synthesis and design candidates, not implementation proof.
- **PROPOSED: Handling:** Keep as PROPOSED until schemas, validators, fixtures, and tests are created and inspected.

12.4 P18-WEAK-004

- **DENY / NEEDS VERIFICATION: Material:** Exact archaeological locations, rare species locations, living-person data, DNA/genomic information, and critical infrastructure exposure.
- **DENY / NEEDS VERIFICATION: Reason:** KFM publication doctrine requires fail-closed treatment when sensitivity, rights, or public-safety posture is unclear.
- **PROPOSED: Handling:** Use quarantine, redaction, generalization, staged access, or denial until policy and review support release.

12.5 P18-WEAK-005

- **PROPOSED: Material:** Model-derived classifications, suitability scores, scenario rankings, and AI-generated explanations.
- **PROPOSED: Reason:** Analysis and AI sources support workflows and methods but do not turn outputs into root truth.
- **PROPOSED: Handling:** Require evidence_refs, model cards, validation metrics, intended-use labels, and citation validation.

12.6 P18-WEAK-006

- **NEEDS VERIFICATION: Material:** Machine extraction quality for long PDFs and page/slice indexes.
- **NEEDS VERIFICATION: Reason:** Phase 1 reported extraction succeeded, but extraction quality remains machine-text quality and may not capture all visual/table nuances.
- **PROPOSED: Handling:** For high-stakes claims, re-check relevant source pages, tables, and rendered images before final publication.

13 Expansion Agenda

- **CONFIRMED:** This agenda is derived from Phase 3 inventory, Phase 4 categories, Phase 5 cards, and Phase 6 synthesis.
- **PROPOSED:** The agenda is a prioritization aid for final prose and future passes, not a claim that implementation work has been done.

13.1 P18-AG-001 — Finalize source-ledger and rights-verification backlog

- **PROPOSED:** Priority: P0.
- **PROPOSED:** Rationale: The final dossier needs cite-or-abstain posture and cannot elevate operational source facts without rights and version checks.
- **CONFIRMED:** Inputs: phase_1_source_intake.jsonl, ideas_inventory.jsonl.
- **PROPOSED:** Outputs: source_ledger_fragment, verification_backlog.

13.2 P18-AG-002 — Package 500 cards into category chapters without re-rendering the PDF yet

- **PROPOSED:** Priority: P0.
- **PROPOSED:** Rationale: Phase 5 generated card files; Phase 8 must concatenate from disk rather than inline or one-shot render.
- **CONFIRMED:** Inputs: phase5_card_index.jsonl, phase4_category_assignments.jsonl.
- **PROPOSED:** Outputs: detailed_idea_chapter_fragments.

13.3 P18-AG-003 — Derive buildable epics from the 14-category map

- **PROPOSED:** Priority: P1.
- **PROPOSED:** Rationale: Implementation planning needs reversible increments rather than a 500-item undifferentiated backlog.
- **CONFIRMED:** Inputs: phase4_category_map.json, phase6_cross_cutting_themes.json.
- **PROPOSED:** Outputs: epic_backlog_fragment.

13.4 P18-AG-004 — Create representation and fitness-for-use acceptance criteria

- **PROPOSED:** Priority: P1.
- **PROPOSED:** Rationale: GIS/cartographic sources heavily support representation choice, purpose, scale, and audience as governance concerns.
- **CONFIRMED:** Inputs: REP cards, MAP cards, UIX cards.
- **PROPOSED:** Outputs: representation_governance_checklist.

13.5 P18-AG-005 — Create temporal-support and correction-support criteria

- **PROPOSED:** Priority: P1.
- **PROPOSED:** Rationale: Temporal database and release/correction ideas need shared acceptance criteria before publication claims are made.
- **CONFIRMED:** Inputs: MOD cards, REL cards.
- **PROPOSED:** Outputs: temporal_release_checklist.

13.6 P18-AG-006 — Define public-safe policy profiles for sensitive domains

- **PROPOSED:** Priority: P2.
- **PROPOSED:** Rationale: Archaeology, biodiversity, living-person, DNA/genomic, and infrastructure-adjacent material must fail closed when sensitivity is unclear.
- **CONFIRMED:** Inputs: POL cards, FIE cards, domain doctrine.
- **PROPOSED:** Outputs: sensitive_material_register_fragment.

13.7 P18-AG-007 — Prepare final PDF QA and self-check plan

- PROPOSED: Priority: P2.
- PROPOSED: Rationale: Phase 8 requires no cut-off text, no placeholders, no broken cross-refs, embedded fonts, TOC, page numbers, and exactly one PDF in out/.
- CONFIRMED: Inputs: all fragments, phase receipts.
- PROPOSED: Outputs: phase8_self_check_plan.

14 Open Questions Seed Fragment

- CONFIRMED: The questions below are sampled from persisted Phase 3 inventory records and retain NEEDS VERIFICATION posture.

Category	Idea ID	Question
REP	KFM-P18-INV-001	NEEDS VERIFICATION: Where should representation-purpose metadata live relative to catalog and layer manifests?
REP	KFM-P18-INV-002	NEEDS VERIFICATION: Should fitness-for-use be validated at layer level, release level, or both?
ANA	KFM-P18-INV-003	NEEDS VERIFICATION: How should KFM name records that are both feature state and event consequence?
REP	KFM-P18-INV-004	NEEDS VERIFICATION: Should transform receipts be required for all cross-source overlays?
POL	KFM-P18-INV-005	NEEDS VERIFICATION: Which releases require cultural/steward review beyond normal policy gates?
FIE	KFM-P18-INV-006	NEEDS VERIFICATION: What aggregation threshold is sufficient for public demographic overlays?
FIE	KFM-P18-INV-007	NEEDS VERIFICATION: How should KFM verify collector identity and device time for field observations?
SRC	KFM-P18-INV-008	NEEDS VERIFICATION: Which scenario metadata is mandatory before a map layer can be public?
REL	KFM-P18-INV-009	NEEDS VERIFICATION: How should KFM catalog externally hosted layers that are not KFM-published?
PLN	KFM-P18-INV-010	NEEDS VERIFICATION: Which SDG-like indicators are appropriate for Kansas historical and contemporary contexts?
PLN	KFM-P18-INV-011	NEEDS VERIFICATION: How will KFM measure whether a planning-support surface is adopted responsibly?
PLN	KFM-P18-INV-012	NEEDS VERIFICATION: Which stakeholder annotations are public, private, steward-only, or evidentiary?
ANA	KFM-P18-INV-013	NEEDS VERIFICATION: Should indicator weights require review separate from data-source review?
PLN	KFM-P18-INV-014	NEEDS VERIFICATION: What minimum source support is needed before KFM can display scenario outputs?
PLN	KFM-P18-INV-015	NEEDS VERIFICATION: When should KFM deny VGI publication because of privacy or life-safety confusion?
FIE	KFM-P18-INV-016	NEEDS VERIFICATION: Which 3D archaeology outputs are public, generalized, steward-only, or denied?
FIE	KFM-P18-INV-017	NEEDS VERIFICATION: Should 2.5D scenes be prohibited from supporting subsurface or vertical claims?
ANA	KFM-P18-INV-018	NEEDS VERIFICATION: What citation burden should an interpretive 3D reconstruction carry?
FIE	KFM-P18-INV-019	NEEDS VERIFICATION: Which field analyses require steward review before leaving WORK/QUARANTINE?
FIE	KFM-P18-INV-020	NEEDS VERIFICATION: What method metadata is enough to make a derived spatial analysis reproducible?
ANA	KFM-P18-INV-021	NEEDS VERIFICATION: Which KFM object families require full bitemporal treatment versus simpler time stamps?
ANA	KFM-P18-INV-022	NEEDS VERIFICATION: How should KFM represent fuzzy, approximate, or contested valid-time intervals?
REL	KFM-P18-INV-023	NEEDS VERIFICATION: Should every public report cite a release manifest and query hash?
ANA	KFM-P18-INV-024	NEEDS VERIFICATION: Which terms require global KFM definitions versus lane-specific meanings?
API	KFM-P18-INV-025	NEEDS VERIFICATION: Which KFM API response codes map to DENY, ABSTAIN, ERROR, and ordinary HTTP statuses?
ANA	KFM-P18-INV-026	NEEDS VERIFICATION: Which KFM panels require window-query receipts or stored analytical view definitions?
ANA	KFM-P18-INV-027	NEEDS VERIFICATION: Should network traversal live in SQL views, graph projections, or both?

Category	Idea ID	Question
ANA	KFM-P18-INV-028	NEEDS VERIFICATION: When does a query-stage become a pipeline transform that needs its own RunReceipt?
ANA	KFM-P18-INV-029	NEEDS VERIFICATION: Which join types require source-role approval before promotion?
ANA	KFM-P18-INV-030	NEEDS VERIFICATION: Should derived indicators expose NULL counts and denominator definitions by default?
ANA	KFM-P18-INV-031	NEEDS VERIFICATION: What AI input manifests are required before any Focus Mode or model-assisted workflow is allowed?
ANA	KFM-P18-INV-032	NEEDS VERIFICATION: Which model families are allowed on public KFM surfaces, and which must remain internal review aids?
ANA	KFM-P18-INV-033	NEEDS VERIFICATION: Which evaluation thresholds trigger ABSTAIN rather than ANSWER or publish?
ANA	KFM-P18-INV-034	NEEDS VERIFICATION: What bias-review evidence is sufficient for each KFM domain lane?
ANA	KFM-P18-INV-035	NEEDS VERIFICATION: Should AI feature sources use the same SourceDescriptor schema as non-AI sources or an extension?
SRC	KFM-P18-INV-036	NEEDS VERIFICATION: Which source classes may be scraped at all, and which require manual steward approval?
EVD	KFM-P18-INV-037	NEEDS VERIFICATION: Should failed scrape attempts be cataloged as retrieval episodes or only retained as run receipts?
VAL	KFM-P18-INV-038	NEEDS VERIFICATION: What is the minimum no-network fixture set for scrape-based connectors?
SRC	KFM-P18-INV-039	NEEDS VERIFICATION: Which interactive acquisition methods are acceptable when the source is public but lacks an API?
API	KFM-P18-INV-040	NEEDS VERIFICATION: Which KFM API contracts need design-first review before the first proof slice?
API	KFM-P18-INV-041	NEEDS VERIFICATION: What exact HTTP status mapping should be used for ABSTAIN and DENY?
API	KFM-P18-INV-042	NEEDS VERIFICATION: Which internal KFM APIs are forbidden from becoming public without an ADR?
MOD	KFM-P18-INV-043	NEEDS VERIFICATION: Which KFM object families are shared kernel versus published language?
SRC	KFM-P18-INV-044	NEEDS VERIFICATION: When should KFM conform to an upstream standard versus wrap it with an anticorruption adapter?
MAP	KFM-P18-INV-045	NEEDS VERIFICATION: Which visual states best signal released, stale, denied, and context-only layers?
VAL	KFM-P18-INV-046	NEEDS VERIFICATION: Should all public map artifacts require OCI references or only high-significance releases?
REL	KFM-P18-INV-047	NEEDS VERIFICATION: Which asset roles are required for each KFM map layer type?
UIX	KFM-P18-INV-048	NEEDS VERIFICATION: What exception process, if any, allows internal review consoles to inspect non-public data?
UIX	KFM-P18-INV-049	NEEDS VERIFICATION: Should every public story map have a preview-render receipt?
MAP	KFM-P18-INV-050	NEEDS VERIFICATION: Which releases require per-tile or per-chunk verification versus file-level checksum only?
REP	KFM-P18-INV-051	NEEDS VERIFICATION: Which layer types require a full transform chain versus a simpler CRS declaration?
REP	KFM-P18-INV-052	NEEDS VERIFICATION: Should geometry quality be attached to every feature or inherited from dataset-level metadata?
FIE	KFM-P18-INV-053	NEEDS VERIFICATION: Which remote-sensing outputs require human review before public release?
ANA	KFM-P18-INV-054	NEEDS VERIFICATION: Can the same EvidenceBundle safely cover multiple spatial model families?
REP	KFM-P18-INV-055	NEEDS VERIFICATION: Which administrative boundaries are stable enough for basemap use and which require full assertion handling?
POL	KFM-P18-INV-056	NEEDS VERIFICATION: Should KFM deny or abstain from live operational aviation guidance?
FIE	KFM-P18-INV-057	NEEDS VERIFICATION: Which hazard contexts are safe for public use without life-safety implications?

Category	Idea ID	Question
REL	KFM-P18-INV-058	NEEDS VERIFICATION: How should KFM label vulnerability maps that rely on outdated Census geography or attributes?
FIE	KFM-P18-INV-059	NEEDS VERIFICATION: Should climate scenario layers be excluded from ordinary factual search results unless labeled as scenario?
PLN	KFM-P18-INV-060	NEEDS VERIFICATION: How should conflicting stakeholder weights be represented without forcing a false consensus?
ANA	KFM-P18-INV-061	NEEDS VERIFICATION: Can TOD-style indicators be reused for Kansas frontier settlement without importing urban-only assumptions?
PLN	KFM-P18-INV-062	NEEDS VERIFICATION: When should KFM show VGI as contextual evidence versus deny it from public maps?
UIX	KFM-P18-INV-063	NEEDS VERIFICATION: Should public APIs expose transaction/release history directly or through Evidence Drawer summaries?
ANA	KFM-P18-INV-064	NEEDS VERIFICATION: Which map popups should expose temporal query class versus only a simplified date label?
ANA	KFM-P18-INV-065	NEEDS VERIFICATION: Which KFM object families are true aggregates versus cross-domain projections?
API	KFM-P18-INV-066	NEEDS VERIFICATION: How should ABSTAIN and DENY map to HTTP status codes without leaking sensitive reasons?
VAL	KFM-P18-INV-067	NEEDS VERIFICATION: Should KFM globally deny proxy rotation and undercover scraping unless a steward exception exists?
SRC	KFM-P18-INV-068	NEEDS VERIFICATION: Should INTERSECT/EXCEPT-derived claims receive a different confidence label than direct-source claims?
ANA	KFM-P18-INV-069	NEEDS VERIFICATION: Which classifications need steward approval before public release?
ANA	KFM-P18-INV-070	NEEDS VERIFICATION: Should imputed features be eligible for consequential public claims without separate caveats?
ANA	KFM-P18-INV-071	NEEDS VERIFICATION: What model classes should default to ABSTAIN for insufficient evaluation?
SRC	KFM-P18-INV-072	NEEDS VERIFICATION: What evidence is sufficient to promote MLT from pilot to default for any layer family?
MAP	KFM-P18-INV-073	NEEDS VERIFICATION: Which external services are allowed only as contextual backdrops and never as evidence sources?
MAP	KFM-P18-INV-074	NEEDS VERIFICATION: Should historic overlays default to contextual evidence rather than direct geometry evidence?
REL	KFM-P18-INV-075	NEEDS VERIFICATION: What field-captured materials require quarantine even after redaction?
REP	KFM-P18-INV-076	NEEDS VERIFICATION: Should scale limits be enforced at UI zoom levels, evidence statements, or both?
REP	KFM-P18-INV-077	NEEDS VERIFICATION: Which symbol changes require governance review versus normal design review?
VAL	KFM-P18-INV-078	NEEDS VERIFICATION: How should KFM distinguish observed process, modeled process, and illustrative animation?
FIE	KFM-P18-INV-079	NEEDS VERIFICATION: Which damage estimates are admissible as evidence versus contextual summaries?
FIE	KFM-P18-INV-080	NEEDS VERIFICATION: How should KFM label modeled future fire context versus observed fire history?

15 Final Self-Check

CONFIRMED: Phase 8 assembled this dossier from a single concatenated Markdown file on disk: `/mnt/data/kfm_pass_18/build/KFM_Pass_18_concatenated.md`.

CONFIRMED: The prior Pass 17 body is not inlined. The baseline is referenced through compact sidecars and Appendix A.

CONFIRMED: Phase 8 pre-render validation found 500 Phase 3 inventory records, 500 Phase 5 card files, 14 Phase 4 categories, and 83 prior baseline ideas.

CONFIRMED: Every prior baseline idea has a carry state in `prior_index.json`; missing carry-state count is 0.

CONFIRMED: Truth-label tokens are present in all 500 Phase 5 card files; card files missing truth-label tokens: 0.

CONFIRMED: Placeholder-token scan returned no matches in the Phase 6/7 fragments or Phase 5 cards.

CONFIRMED: Phase 8 wrote only final-phase publication artifacts to `out/`: exactly one PDF is required there, with JSON receipt and baseline sidecar allowed as plumbing.

CONFIRMED: Final PDF rendering, font embedding, page count, text extraction, and sample render inspection are recorded in `out/receipt.json` after the render.

UNKNOWN: Actual KFM repository implementation maturity, route names, package manager, CI behavior, runtime behavior, branch state, deployment state, dashboards, and public release state remain unverified.

NEEDS VERIFICATION: High-stakes source facts, rights, source endpoint behavior, package versions, exact archaeological locations, rare-species locations, living-person data, DNA/genomic data, and infrastructure exposure require separate governed verification before public or semi-public release.

16 Appendix A - Prior Baseline Reference

CONFIRMED: Prior baseline: Pass 17, represented by `prior_pass_ref.json` and `prior_index.json`.

CONFIRMED: Prior baseline source ID: SRC-P18-033.

CONFIRMED: Phase 2 parsed 83 prior ideas and 10 prior categories from the baseline.

CONFIRMED: Phase 2 reported no duplicate prior idea IDs and no missing carry-state values.

CONFIRMED: The prior-pass body is not copied into Pass 18; it is represented by compact sidecars and validation receipts.

UNKNOWN: Any implementation status implied by Pass 17 remains unverified unless separately checked against current repository evidence.

16.1 Prior baseline category count reference

- **CONFIRMED:** DOC - Doctrinal Foundations and Invariants: 8 prior ideas; carry-forward state: RETAINED sidecar baseline.
- **CONFIRMED:** PIP - Pipelines, Promotion and Replay Discipline: 10 prior ideas; carry-forward state: RETAINED sidecar baseline.
- **CONFIRMED:** CAT - Cataloging, STAC / DCAT / PROV-O and Identifier Determinism: 7 prior ideas; carry-forward state: RETAINED sidecar baseline.
- **CONFIRMED:** EVD - Evidence, Receipts and Attestation: 7 prior ideas; carry-forward state: RETAINED sidecar baseline.
- **CONFIRMED:** POL - Policy-as-Code, Governance Gates and Sensitivity: 7 prior ideas; carry-forward state: RETAINED sidecar baseline.
- **CONFIRMED:** SEC - Supply-Chain, Dependency-Confusion and Security Hardening: 12 prior ideas; carry-forward state: RETAINED sidecar baseline.
- **CONFIRMED:** AIG - AI / LLM Governance and Retrieval: 8 prior ideas; carry-forward state: RETAINED sidecar baseline.
- **CONFIRMED:** DAT - Domains, Data Formats and Visualization: 13 prior ideas; carry-forward state: RETAINED sidecar baseline.
- **CONFIRMED:** OBS - Observability, Telemetry and Sustainability: 5 prior ideas; carry-forward state: RETAINED sidecar baseline.
- **CONFIRMED:** MDP - Documentation Discipline and Markdown Determinism: 6 prior ideas; carry-forward state: RETAINED sidecar baseline.